

The Riemann–Siegel Remainder as a Fractional Summand

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Abstract

Starting from the Riemann–Siegel decomposition of $\zeta(s)$ given in Siegel’s 1932 paper, we introduce a change of variable, $t = I(T)$, that replaces Siegel’s coupled pair “imaginary part t / summation index m ” with one real index T . We then split Siegel’s single remainder integral R into two exact pieces, R_{1ps} and R_{2ps} , and prove that $R = R_{1ps} + R_{2ps}$, so that

$$\zeta = \Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}.$$

Our central observation is that each of these remainders is nothing more than *one additional, fractional partial summand* appended to its Dirichlet sum:

$$\zeta(s) = \sum_{n=1}^m n^{-s} + \hat{d}_1 (m+1)^{-s} + \chi(s) \sum_{n=1}^m n^{s-1} + \hat{d}_2 \chi(s) (m+1)^{s-1},$$

with $\hat{d}_1, \hat{d}_2$ real numbers (always positive on the critical line), the fractions of those two summands that are used. As a corollary, when $\sigma = \frac{1}{2}$ one has $d_1 = d_2$ (equivalently $\hat{d}_1 = \hat{d}_2$); this fact is formally verified in Lean. Also, with this rescaling the remainder terms are nearly periodic in T with period one, converging to a fixed waveform in the fractional part of T . This decomposition of Siegel’s R was discovered through experimental mathematics using a spiral visualization of the partial sums, described later in the paper. We also discuss a number of other observations, including what we call the yin yang curves, the zero counting function, and ovals of equal length leg loci.

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1 Introduction

Siegel published his study of Riemann’s unpublished papers in 1932 [20]. Working through the Nachlaß at The Institute for Advanced Study he recovered, from calculations Riemann never published, an asymptotic evaluation of ζ by the saddle-point method. Thus he found the famous Riemann–Siegel formula. That formula is the starting point of this paper.

This paper makes three claims, in increasing order of importance.

1. **A reparameterization.** We introduce a change of variable $t = I(T)$ that collapses Siegel’s two roles for t (the imaginary part of the input and the index of summation) into a single real “index” T . With this substitution the Riemann–Siegel formula is unchanged in value; it is the *same* formula, viewed through a different variable, T , instead of t .
2. **A decomposition of the remainder.** The Riemann–Siegel formula states that $\zeta = \Sigma_1 + \Sigma_2 + R$, where Σ_1, Σ_2 are finite Dirichlet sums and R is a remainder integral. In this paper we define two exact remainder terms, R_{1ps} and R_{2ps} , and prove

$$R = R_{1ps} + R_{2ps}, \quad \text{hence} \quad \zeta = \Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}. \quad (1)$$

Equation (1) is our main result.

3. **An interpretation.** The two new remainders are not opaque integrals: each is exactly *one more summand* of its Dirichlet sum, namely the $(m+1)$ st term, scaled by a real “fractional” weight $\hat{d}_1$ or $\hat{d}_2$, always positive on the critical line and positive elsewhere outside narrow windows around the poles of §8.4, as proved in §8.5. In this sense the remainder is simply the partial sum carried one fractional step further.

We show that on the critical line $\sigma = \frac{1}{2}$ the two weights coincide, $d_1 = d_2$; we give a proof in §8 and we provide the proof formally verified in Lean at the end of the paper.

Origin of the result. The decomposition (1) was not found analytically. It emerged from experimental mathematics: plotting the partial sums of ζ as a sequence of Euler-like spirals in the complex plane and studying their geometry with a visualization tool we have developed (and published) over several years. These spirals have precedent. Erickson [8] plots the partial sums, identifies the shapes as Cornu spirals, labels their centers C_n , finds $C_0 = \zeta(s)$, and traces the symmetry he sees among the C_n back to the approximate functional equation. Nickel [16, 17] studies the same Argand diagram of the steps n^{-s} geometrically, pairing each early step with a conjugate region of linked Euler (Cornu) spirals, recovering the functional equation from that symmetry and locating zeros where two vectors point in opposite directions with equal magnitudes, which happens exactly when $\sigma = \frac{1}{2}$. Kapitonen [11] names the polyline of steps the Riemann spiral and works with its signed radius of curvature, its reverse points and its inflection points. That geometric story (the spirals, a local coordinate frame, and what we call the “yin” and “yang” curves that actually produce R_{1ps} and R_{2ps}) is deferred to §9 and §11, so that the reader sees the clean algebraic result first and its experimental origin second.

Roadmap. §2 recalls Siegel’s 1932 decomposition. §3 introduces the mapping $I(T)$ and shows the reparameterized formula is identical to the Riemann–Siegel formula. §4 defines R_{1ps}, R_{2ps} and proves (1). §5 establishes the “one more partial summand” interpretation, and §6 recasts each partial sum together with its remainder as a product of homogeneous transformation matrices,

one per link, the remainder entering as one extra fractional link. §7 discusses other remainders. §8 studies the weight d_1 as a function of the index: its handoff jumps, two exactly continuous coordinates built from it, and zeta-free closed-form approximations on and off the critical line. §9 tells the experimental/geometric story behind the result. §10 collects the $I(T)$ functions, including the bisector-frame origin of $I(T)$ and variants for L -functions. §11 develops the yin and yang curves from which the weights d_1 and d_2 were derived, and proves their common limit curve as $T \rightarrow \infty$. §12 extends those curves and the weights d_1, d_2 from the bisector pair to every link, and writes the crossing walks Σ_{1x} and Σ_{2x} . §13 turns the leg angles ϑ_1 and ϑ_2 into a zero-counting function on the critical line. §14 collects further observations. §15 reviews the connection of this paper to other literature. §16 lists statements we would like to prove but have not. A glossary of the geometric vocabulary, with a short dictionary to names used by other authors, follows as §17.

2 Siegel's 1932 decomposition

We begin with the Riemann–Siegel decomposition of $\zeta(s)$ into two finite Dirichlet sums plus a remainder integral, as it appears in Siegel's 1932 paper [20] (his equation 13):

$$\zeta(s) = \sum_{n=1}^m n^{-s} + \frac{(2\pi)^s}{2\Gamma(s) \cos(\frac{\pi s}{2})} \sum_{n=1}^m n^{s-1} + \frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s) (e^{2\pi i s} - 1)} \int_{C_2} g(x) dx, \quad (2)$$

where the path C_2 consists of two half-lines meeting at the point $\eta(1 - \frac{\epsilon}{2})$ for a small fixed $\epsilon > 0$, one half-line passing through the saddle point $\eta = +\sqrt{t/2\pi}$ and the other through $-(m + \frac{1}{2})$, and g is defined below.

It is convenient to name the three pieces. Recall the completion factor

$$\chi(s) := 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s) = \frac{(2\pi)^s}{2\Gamma(s) \cos(\frac{\pi s}{2})}, \quad (3)$$

so that, writing $s = \sigma + it$ and taking the first m terms,

$$\Sigma_1(s) := \sum_{n=1}^m n^{-s}, \quad (4)$$

$$\Sigma_2(s) := \frac{(2\pi)^s}{2\Gamma(s) \cos(\frac{\pi s}{2})} \sum_{n=1}^m n^{s-1} = \chi(s) \sum_{n=1}^m n^{s-1}, \quad (5)$$

$$R(s) := \frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s) (e^{2\pi i s} - 1)} \int_{C_2} g(x) dx, \quad (6)$$

where

$$g(x) := x^{s-1} \frac{e^{-2\pi i m x}}{e^{2\pi i x} - 1}. \quad (7)$$

With these names, (2) reads

$$\zeta(s) = \Sigma_1 + \Sigma_2 + R. \quad (8)$$

In Siegel's formulation m is the index of summation, and to minimize the remainder he takes

$$m = \left\lfloor \sqrt{\frac{t}{2\pi}} \right\rfloor. \quad (9)$$

Thus t plays two roles at once: it is the imaginary part of the input s , and, through (9), it also fixes the number of terms m . In this way m depends on t . The next section reverses the dependency.

3 The same formula, reparameterized: The $I(T)$ mapping

We introduce a change-in-variable mapping for t :

$$t = \frac{\pi(2T+1)}{\log(\frac{1}{T}+1)} = I(T), \quad m = \lfloor T \rfloor. \quad (10)$$

Here T is a real number, while $m = \lfloor T \rfloor$ is an integer. So now m and t depend on T , the reverse of Siegel's (9). The single function I therefore determines both the imaginary part of the input and the integer index of summation; for this reason we call T “the index” even though it is real.

So now $s = \sigma + it$ is the same as

$$s = \sigma + iI(T), \quad (11)$$

and we carry σ and T as two separate arguments of ζ . Nothing about the *value* of ζ changes; we have only renamed the variable that generates t and m . In this notation Siegel's decomposition (8) becomes

$$\zeta(\sigma, T) = \Sigma_1 + \Sigma_2 + R, \quad (12)$$

where, with $m = \lfloor T \rfloor$,

$$\Sigma_1(\sigma, T) = \sum_{n=1}^m \frac{1}{n^{\sigma+iI(T)}}, \quad (13)$$

$$\Sigma_2(\sigma, T) = \chi(\sigma + iI(T)) \sum_{n=1}^m \frac{1}{n^{1-\sigma-iI(T)}}. \quad (14)$$

Using the identity

$$\frac{(2\pi)^s e^{\frac{\pi i s}{2}}}{\Gamma(s)(e^{2\pi i s} - 1)} = \frac{\chi(s)}{e^{i\pi s} - 1}, \quad (15)$$

the remainder becomes

$$R(\sigma, T) = \frac{\chi(\sigma + iI(T))}{e^{i\pi(\sigma+iI(T))} - 1} \int_{C_2} g(x) dx. \quad (16)$$

The point of this section is only that (12) is (8): the reparameterization is an exact identity, not an approximation. Everything that follows takes place inside this reindexed but otherwise unchanged Riemann–Siegel formula.

Remark 3.1. The derivation of $I(T)$ and its approximations and inverse are developed in §10.1. The context *behind* the mapping is not needed to state or use (12).

4 Decomposing the remainder: $R = R_{1ps} + R_{2ps}$

We now split Siegel's single remainder R into two exact pieces, named R_{1ps} and R_{2ps} (the subscript “ ps ” distinguishes them from the approximate remainders of other authors). Define

$$R_{1ps}(\sigma, T) := d_1(\sigma, T) \cdot \lfloor T \rfloor^{-iI(T)}, \quad (17)$$

$$R_{2ps}(\sigma, T) := d_2(\sigma, T) \cdot \lfloor T \rfloor^{iI(T)} \cdot \frac{\chi(\sigma + iI(T))}{|\chi(\sigma + iI(T))|}, \quad (18)$$

with real weights

$$d_1(\sigma, T) := |R| \frac{\sin(\omega - \arg R + \arg \chi)}{\sin(2\omega + \arg \chi)}, \quad (19)$$

$$d_2(\sigma, T) := |R| \frac{\sin(\omega + \arg R)}{\sin(2\omega + \arg \chi)}, \quad (20)$$

where $\arg R = \arg R(\sigma, T)$, $\arg \chi = \arg \chi(\sigma + iI(T))$, and

$$\omega(T) := I(T) \log(\lfloor T + 1 \rfloor). \quad (21)$$

Geometrically, ω is the *absolute* direction angle, measured from the positive real axis, of the next summand of the first Dirichlet sum: with $t = I(T)$ and $m = \lfloor T \rfloor$, the summand $n^{-s} = n^{-\sigma} e^{-it \ln n}$ rotates by $\theta_n = -t \ln \frac{n+1}{n}$ relative to its predecessor, and these relative rotations telescope,

$$\theta_1 + \theta_2 + \cdots + \theta_m = -t \ln \frac{2}{1} - t \ln \frac{3}{2} - \cdots - t \ln \frac{m+1}{m} = -t \ln(m+1) = -\omega, \quad (22)$$

so the $(m+1)$ st summand points along $e^{-i\omega}$.

Note: both d_1 and d_2 are real, for the reason given just below, and on the critical line they are always positive, which is Theorem 8.3 of §8.5. Writing $\psi = \arg \chi(\sigma + iI(T))$, the definitions (17)–(18) express R as a combination of two fixed *unit* directions,

$$R = d_1 e^{-i\omega} + d_2 e^{i(\omega+\psi)}, \quad (23)$$

namely the phases of the next summand of each Dirichlet sum. An alternative route to d_1 and d_2 would be to start from (23) and require $d_1, d_2 \in \mathbb{R}$: that makes it a 2×2 real linear system, its real and imaginary parts, whose solution by Cramer's rule is exactly the pair of sine expressions (19)–(20), real by inspection. They are positive exactly when R lies in the cone spanned by the two directions $e^{-i\omega}$ and $e^{i(\omega+\psi)}$, so that both numerators carry the same sign as the common denominator $\sin(2\omega + \psi)$. On the critical line R never leaves the cone: there $d_1 = d_2$ (Corollary 8.1) and the common value stays strictly inside the link, the normalized fraction $\lceil T \rceil^\sigma d_1$ sweeping $\approx [0.23, 0.78]$ in every unit interval (§8); we verify $d_1 = d_2 > 0$ at every one of 30,562 samples of a dense grid over $1 \leq T \leq 30$, including magnified windows around the near-parallel instants at fractional parts $\approx \frac{1}{4}, \frac{3}{4}$, and in unit windows at $T = 100, 300$, and 1000 (`check_d1_positive_critical.py`).¹ Off the critical line the weights are positive outside a narrow window around each pole of §8.4; inside such a window R exits the cone and one of the two weights turns negative (Remark 14.3, where $d_1 = -d_2$ exactly at a pole). That those windows have width $O(1/T)$ is Theorem 8.5, in the same subsection. Geometrically, positivity is the statement that each fractional summand points *forward* along its partial sum, a positive fraction of the next term.

Theorem 4.1. *For all σ and T ,*

$$R_{1ps} + R_{2ps} = R.$$

Consequently,

$$\zeta = \Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}.$$

¹Every script named in this paper, both the numerical checks and the ones that generate the figures, is in the directory `papers/my main paper/rewrite_v7/` of the Zest repository, <https://github.com/paul-stahura/zest>.

Proof. Write $\phi = \arg R$, $\psi = \arg \chi$, and $\omega = \omega(T)$, so that $R = |R|e^{i\phi}$ and, for non-integer T , $\lceil T \rceil^{iI(T)} = e^{i\omega}$ and $\chi/|\chi| = e^{i\psi}$.

Step 1 (substitute). From (17)–(20),

$$R_{1ps} = |R| \frac{\sin(\omega - \phi + \psi)}{\sin(2\omega + \psi)} e^{-i\omega}, \quad R_{2ps} = |R| \frac{\sin(\omega + \phi)}{\sin(2\omega + \psi)} e^{i\omega} e^{i\psi}.$$

Step 2 (add).

$$R_{1ps} + R_{2ps} = \frac{|R|}{\sin(2\omega + \psi)} \left[e^{-i\omega} \sin(\omega - \phi + \psi) + e^{i(\omega + \psi)} \sin(\omega + \phi) \right].$$

Step 3 (expand via $\sin \theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$).

$$\begin{aligned} e^{-i\omega} \sin(\omega - \phi + \psi) &= \frac{e^{i(\psi - \phi)} - e^{i(-2\omega + \phi - \psi)}}{2i}, \\ e^{i(\omega + \psi)} \sin(\omega + \phi) &= \frac{e^{i(2\omega + \psi + \phi)} - e^{i(\psi - \phi)}}{2i}. \end{aligned}$$

Step 4 (add the expansions). The $e^{i(\psi - \phi)}$ terms cancel, leaving

$$\frac{1}{2i} \left[e^{i(2\omega + \psi + \phi)} - e^{i(\phi - \psi - 2\omega)} \right].$$

Step 5 (factor $e^{i\phi}$).

$$= \frac{e^{i\phi}}{2i} \left[e^{i(2\omega + \psi)} - e^{-i(2\omega + \psi)} \right] = e^{i\phi} \sin(2\omega + \psi).$$

Step 6 (substitute back).

$$R_{1ps} + R_{2ps} = \frac{|R|}{\sin(2\omega + \psi)} \cdot e^{i\phi} \sin(2\omega + \psi) = |R|e^{i\phi} = R.$$

Integer T . At integer T one has $\lceil T \rceil^{iI(T)} = -e^{i\omega}$, and the factor $(-1)^{\lceil T \rceil}$ in R shifts $\phi = \arg R$ by π . That π shift flips the signs of both sine terms in d_1, d_2 , while the -1 contributes a further sign; the two sign flips cancel, so $R = R_{1ps} + R_{2ps}$ holds for all T . $\square$

A formalization of the algebra of Theorem 4.1 in Lean is given in Appendix A. It assumes no axioms of its own; the appendix records what its three hypotheses are and what is left outside its scope. Since $R = R_{1ps} + R_{2ps}$, we have in particular proved the exact formula

$$\zeta(\sigma, T) = \Sigma_1 + R_{1ps} + R_{2ps} + \Sigma_2. \quad (24)$$

Figure 1 is the theorem drawn: as T crosses a unit interval the two remainders trace their own paths in the plane, and the path of $\frac{1}{2}R$ is the average of them, point by point, which is $R_{1ps} + R_{2ps} = R$ read as a midpoint.

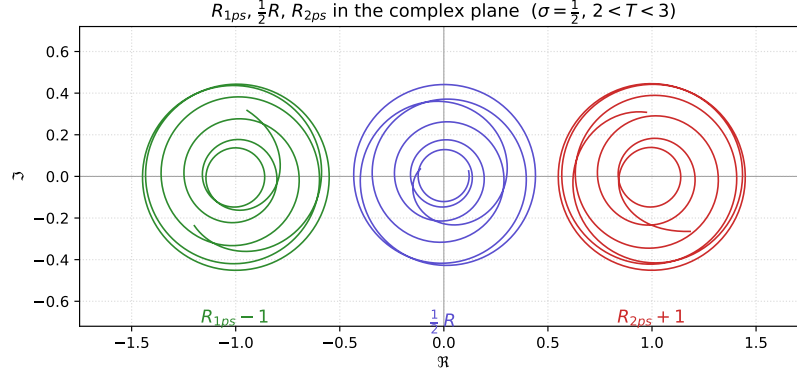


Figure 1: $R_{1ps}(\sigma, T) - 1$ (green, left), $\frac{1}{2}R(\sigma, T)$ (violet, middle), $R_{2ps}(\sigma, T) + 1$ (red, right), for $\sigma = \frac{1}{2}$ and $2 < T < 3$. The middle trajectory is the average of the left and right trajectories (the unit shifts cancel), since $R_{1ps} + R_{2ps} = R$. Generated by `fig_remainder_average.py`.

	Last summand	Next summand	Fractional amount	Partial summand
R_{1ps}	$\frac{1}{[T]^{\sigma+iI(T)}}$	$\frac{1}{[T]^{\sigma+iI(T)}}$	$[T]^\sigma d_1$	$d_1 [T]^{-iI(T)}$
R_{2ps}	$\frac{\chi(\sigma + iI(T))}{[T]^{1-\sigma-iI(T)}}$	$\frac{\chi(\sigma + iI(T))}{[T]^{1-\sigma-iI(T)}}$	$\frac{[T]^{1-\sigma}}{ \chi(\sigma + iI(T)) } d_2$	$d_2 [T]^{iI(T)} \frac{\chi(\sigma + iI(T))}{ \chi(\sigma + iI(T)) }$

Table 1: Each remainder is the $(m + 1)$ st (“next”) summand of its Dirichlet sum, scaled by a real fractional amount. The final column is the resulting *partial summand* = (fractional amount) $\times$ (next summand), which simplifies to R_{1ps} and R_{2ps} respectively.

5 The remainders are “one more partial summand”

Theorem 4.1 is more than an algebraic identity. Unlike Siegel’s integral R (or the approximate remainders R_{ak} of Kuznetsov), which bear no resemblance to the terms of the Dirichlet sums, each of R_{1ps} and R_{2ps} is *exactly the next term* of its partial sum, scaled by a real number (positive in the sense of the note after (23)). We call such a term a *fractional summand*.

Concretely, the last term actually included in Σ_1 is the $\lceil T \rceil$ -th term, and the very next term would be the $\lceil T \rceil$ -th. The remainder R_{1ps} lies along that next term, shortened by a real factor; and similarly for R_{2ps} along the next term of Σ_2 . Table 1 makes this precise.

Reading (24) in this light, and writing $s = \sigma + iI(T)$ and $m = \lceil T \rceil$, the zeta function takes the clean “one more summand” form

$$\zeta(s) = \sum_{n=1}^m n^{-s} + \hat{d}_1 (m+1)^{-s} + \chi(s) \sum_{n=1}^m n^{s-1} + \hat{d}_2 \chi(s) (m+1)^{s-1} \quad (25)$$

with $\hat{d}_1, \hat{d}_2$ reals, positive on the critical line (and off it outside the pole windows of §8.4). In words: each finite Dirichlet sum, carried exactly one fractional term past its cutoff, reproduces ζ with no remainder terms.

The coefficients here are the *fractional amounts* of Table 1, not the weights d_1, d_2 of (17)–(18). The hat is what distinguishes them, and the two differ by the length of the link they sit on:

$$\hat{d}_1 = \lceil T \rceil^\sigma d_1, \quad \hat{d}_2 = \frac{\lceil T \rceil^{1-\sigma}}{|\chi|} d_2. \quad (26)$$

The unhatted weight multiplies a *unit* vector, so $|R_{1ps}| = d_1$ is an absolute length, whereas $\hat{d}_1$ multiplies the full summand $(m+1)^{-s}$ and is therefore the dimensionless fraction of that summand which is used. At $\sigma = \frac{1}{2}$, $T = 6.18$ for instance, $d_1 = 0.0875$ while $\hat{d}_1 = \sqrt{7} d_1 = 0.2315$; see Figure 2.

6 Summands as links and joints

Equation (24) presents ζ as the sum of the four terms $\Sigma_1 + R_{1ps} + \Sigma_2 + R_{2ps}$, two of which are partial sums. Read geometrically, each summand of a partial sum is one *link* of a planar chain, and each remainder R_{1ps}, R_{2ps} is one additional *fractional* link. One can think of $\hat{d}_1$ (and likewise $\hat{d}_2$) as the percentage of the way along that additional link at which the chain stops: the link taken at its full length would be the whole $(m+1)$ st summand, and R_{1ps} is that fraction $\hat{d}_1$ of it. Write $s = \sigma + iI(T)$, $t = I(T)$, $m = \lceil T \rceil$, and $\omega = t \ln(m+1)$, and denote two complex numbers by

$$B_1 = \Sigma_1 + R_{1ps}, \quad B_2 = \Sigma_2 + R_{2ps}, \quad \zeta = B_1 + B_2. \quad (27)$$

Each remainder is a single extra link appended to its partial sum, pointing in the direction of the $(m+1)$ -th summand with a real length d_1 or d_2 (the fractional summands of §5):

$$R_{1ps} = d_1 \lceil T \rceil^{-iI(T)} = d_1 e^{-i\omega}, \quad R_{2ps} = d_2 \lceil T \rceil^{iI(T)} \frac{\chi}{|\chi|} = d_2 e^{i(\omega + \arg \chi)}, \quad (28)$$

with d_1, d_2 as in (19)–(20). The first equality in each is the definition (17)–(18); the second uses that, for non-integer T , $\lceil T \rceil = \lfloor T + 1 \rfloor$, so $\lceil T \rceil^{\pm iI(T)} = e^{\pm iI(T) \log \lceil T \rceil} = e^{\pm i\omega}$ by the definition $\omega = I(T) \log \lceil T + 1 \rceil$, together with $\chi/|\chi| = e^{i \arg \chi}$.

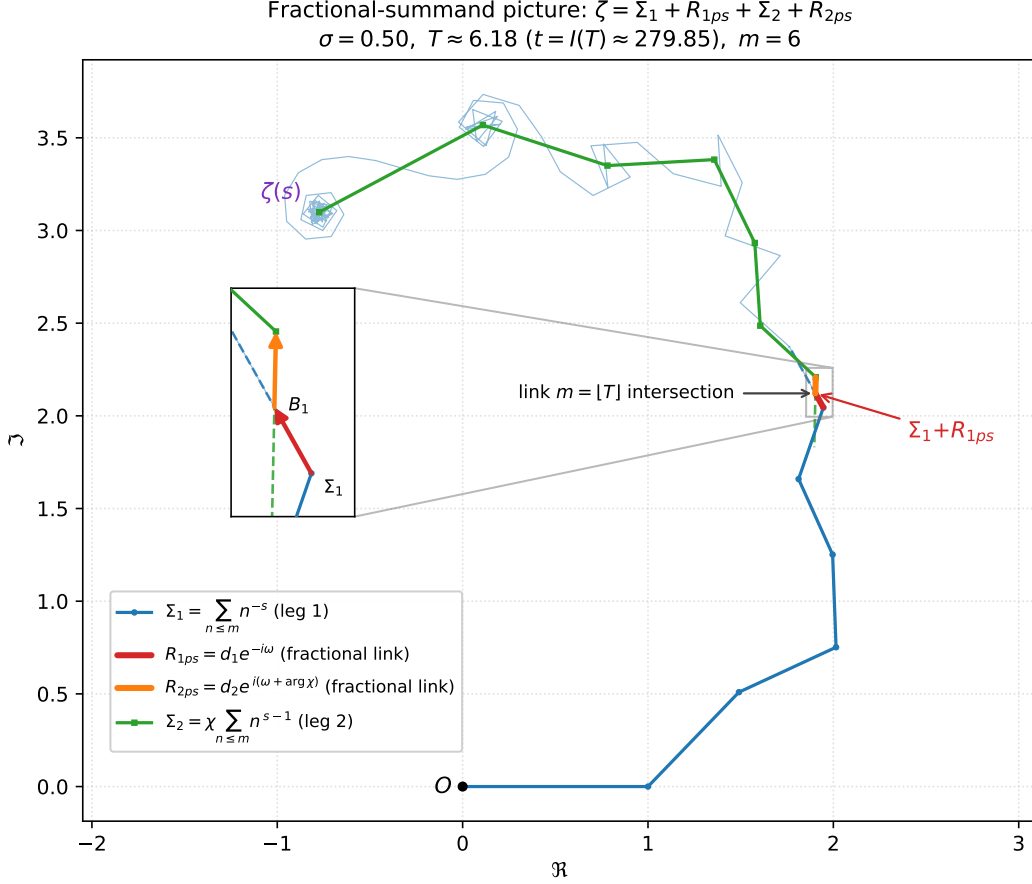


Figure 2: The fractional-summand picture: pictorial (spiral) representation of the four pieces of (25) in the complex plane for $\sigma = 0.5$ and $T \approx 6.18$ (equivalently $t = I(T) \approx 279.85$, $m = 6$). The blue path is the partial-sum spiral $\Sigma_1 = \sum_{n \leq m} n^{-s}$; the green path, laid tip-to-tail from the point $\Sigma_1 + R_{1ps}$, is $\Sigma_2 = \chi \sum_{n \leq m} n^{s-1}$. The short red and orange segments $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg \chi)}$ lie along link m of each spiral (the faint dashed vectors show the full next summand, of which each remainder is a “fractional” part), so that the four pieces add up to the point $\zeta(s)$. Underneath, the thin faint blue path continues the Σ_1 chain past m out to link $\lfloor I(T)/\pi \rfloor = 89$, winding into its spiral near ζ . On the critical line $d_1 = d_2$, so the two remainders have equal length. The fractional links are short here ($d_1 = d_2 \approx 0.087$), so the inset zooms in on them. Generated by `fig1_spiral_summands.py`.

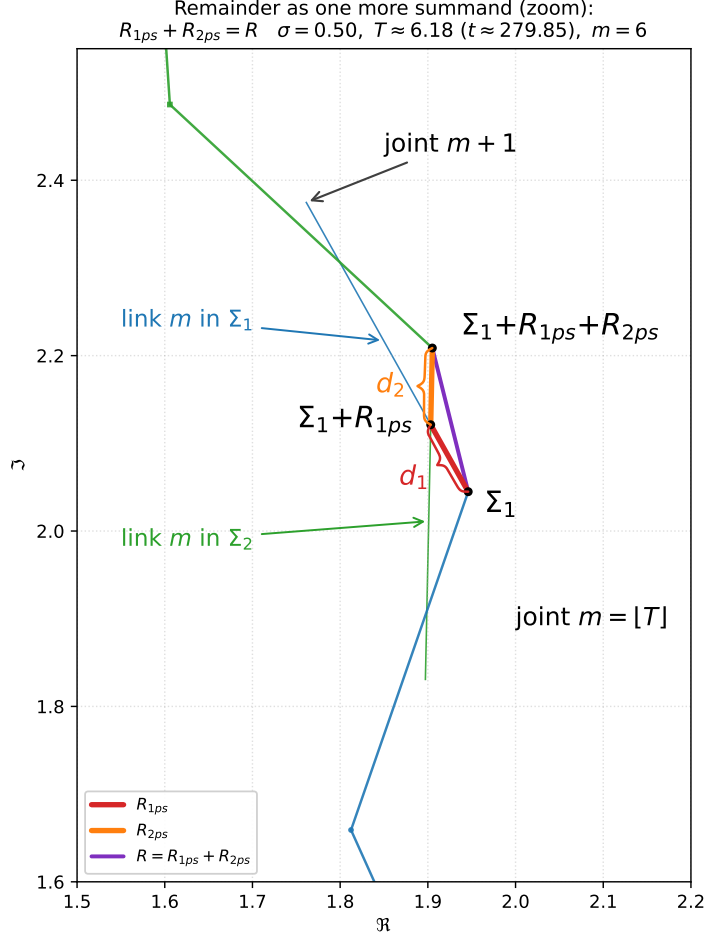


Figure 3: Zoom into the joint region of Figure 2 (imaginary part ≈ 1.5 – 2.6), showing the theorem $R_{1ps} + R_{2ps} = R$ as a small triangle. Starting from the joint Σ_1 (end of the sum-1 partial sum), the red link R_{1ps} reaches $\Sigma_1 + R_{1ps}$ and the orange link R_{2ps} reaches $\Sigma_1 + R_{1ps} + R_{2ps}$; the straight purple vector from Σ_1 to that last joint is Siegel’s remainder $R = R_{1ps} + R_{2ps}$ (see the legend). The curly braces labeled d_1 and d_2 span the two fractional links, marking their lengths, i.e. how far each reaches from its own joint. The thin blue and green lines show the entire extent each fractional summand would have if it were not shortened to the lengths d_1 and d_2 : each extends its fractional link to the full $(m + 1)$ st summand. The faint blue and green chains are the neighboring links of the two partial sums, arriving at and departing from the joint region. Generated by `fig2_remainder_zoom.py`.

6.1 Sum form

Splitting ζ into its partial sum plus this last fractional link,

$$B_1 = \sum_{n=1}^m n^{-s} + d_1 e^{-i\omega}, \quad (29)$$

$$B_2 = \chi(s) \sum_{n=1}^m n^{s-1} + d_2 e^{i(\omega + \arg \chi)}. \quad (30)$$

Componentwise. Writing out real and imaginary parts,

$$\Re B_1 = \sum_{n=1}^m \frac{\cos(t \ln n)}{n^\sigma} + d_1 \cos(t \ln(m+1)), \quad (31)$$

$$\Im B_1 = - \sum_{n=1}^m \frac{\sin(t \ln n)}{n^\sigma} - d_1 \sin(t \ln(m+1)), \quad (32)$$

$$\Re B_2 = \sum_{n=1}^m \frac{|\chi| \cos(\arg \chi + t \ln n)}{n^{1-\sigma}} + d_2 \cos(\arg \chi + t \ln(m+1)), \quad (33)$$

$$\Im B_2 = \sum_{n=1}^m \frac{|\chi| \sin(\arg \chi + t \ln n)}{n^{1-\sigma}} + d_2 \sin(\arg \chi + t \ln(m+1)). \quad (34)$$

6.2 Product form

This subsection is a tangent, and a reader in a hurry can skip to §7: nothing later rests on the matrix form, which is motivated by an analogy with multi-jointed planar robot manipulators. The one thing carried forward is the numbering of links and joints fixed in the next paragraph, which the rest of the paper uses throughout.

The goal of this subsection is to write each link as a homogeneous coordinate transformation matrix, and then to group the links into a single transformation matrix, so that ζ takes the form of a matrix product with four components: the two sums Σ_1, Σ_2 and the two partial-summand remainders R_{1ps}, R_{2ps} .

Each summand is a link in the plane. We number the links of each chain from zero: link n runs from joint n to joint $n+1$ (joint n being the value of the partial sum after n summands), so link n represents the $(n+1)$ st summand. The m summands of a partial sum are thus links $0, \dots, m-1$, and each remainder is the *fractional link* m . We encode one link by the 3×3 matrix with link length l and rotation θ from the previous link,²

$$M(\theta, l) = \begin{pmatrix} \cos \theta & -\sin \theta & l \cos \theta \\ \sin \theta & \cos \theta & l \sin \theta \\ 0 & 0 & 1 \end{pmatrix}, \quad (35)$$

²This is the planar case of the Denavit–Hartenberg convention of robot kinematics [7], in which each link of a serial chain is encoded by the 4×4 homogeneous transformation $\text{Rot}_z(\theta_i) \text{Trans}_z(d_i) \text{Trans}_x(a_i) \text{Rot}_x(\alpha_i)$ built from four parameters: the joint angle θ_i , the joint offset d_i , the link length a_i and the link twist α_i . A planar chain has $d_i = \alpha_i = 0$, which leaves the two parameters (θ_i, a_i) and the 3×3 matrix (35); (42) is then the forward-kinematic product of the chain. One collision of names to watch: the Denavit–Hartenberg offset d_i has nothing to do with the weights d_1, d_2 of (19)–(20).

where the accumulated position is the top two entries of the last column. The link matrix rotates by θ and then translates forward l along the new heading. Its inverse is again a link-shaped matrix, of a second type that translates *back* l along the current heading and then undoes the rotation:

$$M(\theta, l)^{-1} = \begin{pmatrix} \cos \theta & \sin \theta & -l \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (36)$$

Forward links build chain 1; inverses are what let chain 2 be walked backwards.

Chain 1. Collect the m summand links of Σ_1 into a single product and give the fractional link its own name:

$$P_{\Sigma_1} = \prod_{n=0}^{m-1} M(\theta_n, (n+1)^{-\sigma}), \quad M_{R_{1ps}} = M(-t \ln \frac{m+1}{m}, d_1), \quad (37)$$

where link n carries the two parameters θ_n , its rotation from the previous link, and $l_n = (n+1)^{-\sigma}$, its length: as in (22), $\theta_0 = 0$ and $\theta_n = -t \ln \frac{n+1}{n}$ for $n \geq 1$. Written out, with the first two factors and the last two shown, the chain that reaches B_1 is

$$P_{\Sigma_1} M_{R_{1ps}} = \underbrace{\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link 0}} \underbrace{\begin{pmatrix} \cos \theta_1 & -\sin \theta_1 & \frac{\cos \theta_1}{2^\sigma} \\ \sin \theta_1 & \cos \theta_1 & \frac{\sin \theta_1}{2^\sigma} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link 1}} \cdots \underbrace{\begin{pmatrix} \cos \theta_{m-1} & -\sin \theta_{m-1} & \frac{\cos \theta_{m-1}}{m^\sigma} \\ \sin \theta_{m-1} & \cos \theta_{m-1} & \frac{\sin \theta_{m-1}}{m^\sigma} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } m-1} \underbrace{\begin{pmatrix} \cos \theta_m & -\sin \theta_m & d_1 \cos \theta_m \\ \sin \theta_m & \cos \theta_m & d_1 \sin \theta_m \\ 0 & 0 & 1 \end{pmatrix}}_{\text{fractional link } m}, \quad (38)$$

the leading factor being the unit step $1^{-s} = 1$ along the real axis, where $\theta_0 = 0$ and $l_0 = 1$ collapse (35) to a pure translation, and the trailing factor being $M_{R_{1ps}}$, whose rotation $\theta_m = -t \ln \frac{m+1}{m}$ continues the same pattern while its length is d_1 rather than a power of $m+1$. The product carries the frame from the origin to the bisector point $B_1 = \Sigma_1 + R_{1ps}$, whose coordinates are the top two entries of the last column, and it arrives with accumulated rotation $-t \ln(m+1) = -\omega$ by the telescoping (22).

Chain 2. The second chain is traversed tip-first, exactly as Figure 2 draws leg 2: the fractional link R_{2ps} leaves B_1 , and the summand links follow in descending order. Walking a link backwards is exactly its inverse (36); regrouping each rotation with the backward translation that follows it returns everything to the forward type (35), with negated parameters and each rotation paired with the preceding link's length. Two bookkeeping factors appear at the ends of the backward walk: a leading pure translation $M(0, -d_2)$, and a trailing pure rotation $M(-\arg \chi, 0)$, the functional-equation factor χ peeled off on its own. The leading translation absorbs the zero-length joint $M(2\omega + \arg \chi + \pi, 0)$ that must sit between the two chains (a rotation followed by a translation is a single link), defining the backward fractional link

$$M_{R_{2ps}} := M(2\omega + \arg \chi + \pi, 0) M(0, -d_2) = M(2\omega + \arg \chi + \pi, -d_2); \quad (39)$$

the backward summand links collect into

$$P_{\Sigma_2} := \prod_{n=m}^1 M(\theta_n, -|\chi| n^{\sigma-1}) \quad (n \text{ descending}, \theta_n = -t \ln \frac{n+1}{n}); \quad (40)$$

and the trailing rotation is simply dropped, since a post-multiplied zero-length joint never moves the position column, only the final orientation.

Written out as (38) was, with the first two factors and the last two shown and with $\psi = \arg \chi$ as before, the second chain is

$$\begin{aligned}
M_{R_{2ps}} P_{\Sigma_2} = & \underbrace{\begin{pmatrix} \cos(2\omega + \psi + \pi) & -\sin(2\omega + \psi + \pi) & -d_2 \cos(2\omega + \psi + \pi) \\ \sin(2\omega + \psi + \pi) & \cos(2\omega + \psi + \pi) & -d_2 \sin(2\omega + \psi + \pi) \\ 0 & 0 & 1 \end{pmatrix}}_{\text{fractional link } m} \underbrace{\begin{pmatrix} \cos \theta_m - \sin \theta_m - \frac{|\chi| \cos \theta_m}{m^{1-\sigma}} \\ \sin \theta_m & \cos \theta_m - \frac{|\chi| \sin \theta_m}{m^{1-\sigma}} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link } m-1} \cdots \\
& \cdots \underbrace{\begin{pmatrix} \cos \theta_2 - \sin \theta_2 - \frac{|\chi| \cos \theta_2}{2^{1-\sigma}} \\ \sin \theta_2 & \cos \theta_2 - \frac{|\chi| \sin \theta_2}{2^{1-\sigma}} \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link 1}} \underbrace{\begin{pmatrix} \cos \theta_1 - \sin \theta_1 - |\chi| \cos \theta_1 \\ \sin \theta_1 & \cos \theta_1 - |\chi| \sin \theta_1 \\ 0 & 0 & 1 \end{pmatrix}}_{\text{link 0}},
\end{aligned} \tag{41}$$

the leading rotation being the merged joint of (39). The links run backwards, from the fractional link m down to link 0, each brace naming the link whose length that factor carries; the rotation it carries is the θ_n of the link after it, which is the regrouping. Its last column is not B_2 but $e^{i\omega} B_2$, leg 2 read in the frame chain 1 leaves behind, whose orientation is $-\omega$; multiplying on the left by $P_{\Sigma_1} M_{R_{1ps}}$ turns that back onto world axes and adds B_1 , which is (42). All three columns agree to 28 digits at $\sigma = \frac{1}{2}$, $T = 6.18$ (`check_chain2_matrix.py`).

The full 3×3 equation for ζ . With these four names the two chains combine into one serial product, in a single link type:

$$Z = P_{\Sigma_1} M_{R_{1ps}} M_{R_{2ps}} P_{\Sigma_2} = \begin{pmatrix} \cos \alpha_\zeta & -\sin \alpha_\zeta & \Re \zeta \\ \sin \alpha_\zeta & \cos \alpha_\zeta & \Im \zeta \\ 0 & 0 & 1 \end{pmatrix}, \quad \alpha_\zeta = \pi + \arg \chi. \tag{42}$$

The last column is ζ and the rotation block records α_ζ , discussed below. The factors line up one to one with the terms of (24), repeated here for comparison:

$$\zeta(\sigma, T) = \Sigma_1 + R_{1ps} + R_{2ps} + \Sigma_2. \tag{24}$$

Figure 4 draws (42) with one arrow per factor.

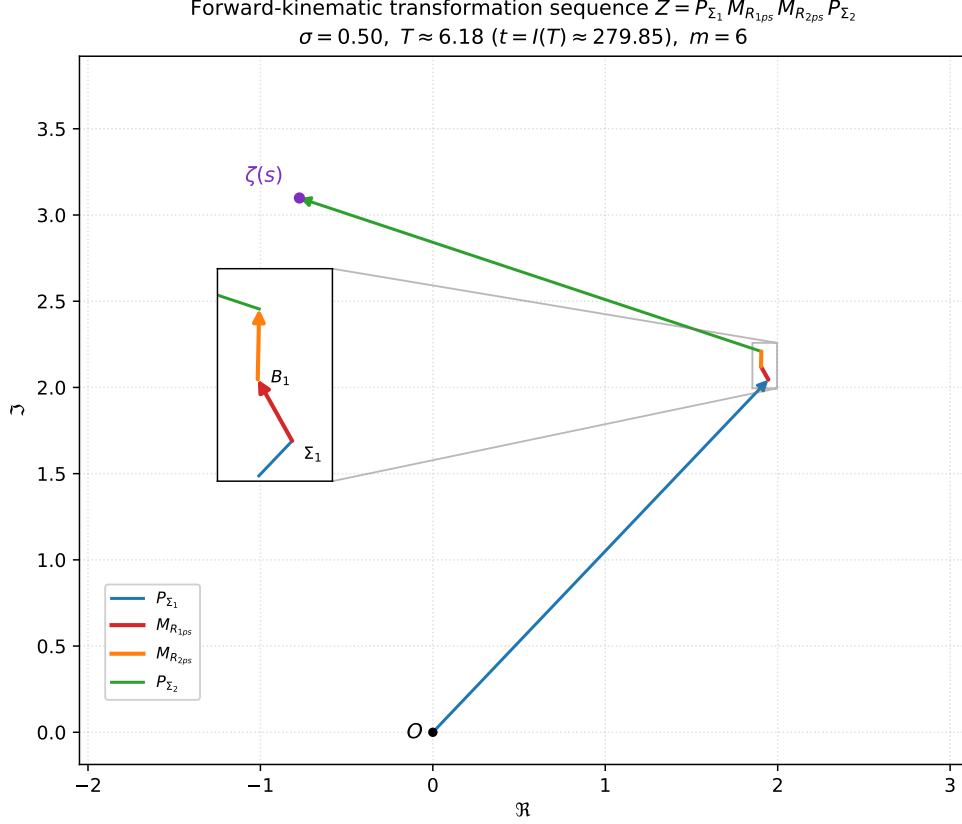


Figure 4: The serial chain of (42) at $\sigma = \frac{1}{2}$, $T = 6.18$ ($t = I(T) \approx 279.85$, $m = 6$), one arrow for the net displacement of each matrix factor: P_{Σ_1} (blue, O to Σ_1), $M_{R_{1ps}}$ (red, to B_1), $M_{R_{2ps}}$ (orange, to $B_1 + R_{2ps}$; its rotation parameter carries the merged joint $2\omega + \arg \chi + \pi$), and P_{Σ_2} (green, to ζ). The four arrows are the four terms of $\zeta = \Sigma_1 + R_{1ps} + R_{2ps} + \Sigma_2$; the two fractional links are short here ($d_1 = d_2 \approx 0.087$), so the inset zooms in on them. Colors match Figure 2. Generated by `fig.zeta_chain.py`.

Three features of (42) deserve comment. First, the rotation $2\omega + \arg \chi + \pi$ carried by $M_{R_{2ps}}$ has three jobs at once: $+\omega$ cancels chain 1's accumulated rotation $-t \ln(m+1) = -\omega$, a further $+(\omega + \arg \chi)$ pre-rotates against the counter-rotation of the backward-walked chain 2, and the final $+\pi$ rotates the backward translations to point forward again; together $\omega + (\omega + \arg \chi) + \pi = 2\omega + \arg \chi + \pi$. Second, the links of P_{Σ_2} carry chain 1's very rotation parameters θ_n : in this form the two chains use identical rotations and differ only in their lengths, $(n+1)^{-\sigma}$ forward versus $-|\chi|n^{\sigma-1}$ backward. Third, the net displacements of the four factors P_{Σ_1} , $M_{R_{1ps}}$, $M_{R_{2ps}}$, P_{Σ_2} are exactly the four terms Σ_1 , $R_{1ps} = d_1 e^{-i\omega}$, $R_{2ps} = d_2 e^{i(\omega + \arg \chi)}$, Σ_2 of (24), and the final orientation $\alpha_\zeta = \pi + \arg \chi$ is the direction of $-\chi/|\chi|$: the trailing rotation $M(-\arg \chi, 0)$ (the functional-equation factor χ , peeled off automatically when the base link of chain 2 is regrouped) was discarded, so χ 's direction is left showing in the orientation block, and Z packages the pair $(\zeta, \arg \chi)$ in one matrix. (Appending the discarded $M(-\arg \chi, 0)$ would leave the position column untouched and make the orientation the constant π .)

7 Other remainders

This section is about three different remainders that are all equal to each other, exactly in two of the three cases and to high accuracy in the third.

7.1 Kuznetsov's remainders

Our remainders R_{1ps}, R_{2ps} are not the only way to split Siegel's R into components. It is worth contrasting them with the remainder of Alexey Kuznetsov [12], whose paper appeared on the arXiv in March 2025 and in print the year after. It gives a simple and accurate algorithm for computing ζ , which we have found useful in our own numerical work. There are two differences. The first is one of *kind*: Kuznetsov's remainder is an *approximation* to R , whereas $R_{1ps} + R_{2ps} = R$ is an *exact* identity (Theorem 4.1). The second is one of *direction*: neither of his two halves lies along the next summand of its Dirichlet sum, while each of ours is exactly that summand, shortened.

Where his pair comes from is worth a word, since Siegel has one remainder and Kuznetsov has two. Siegel's §2 opens at his equation 8, one Dirichlet sum plus one contour integral, and the residue theorem carries that into his equation 13, our (2): still one integral, now standing beside both sums. One integral, one remainder, and that single R is what we cut in two. Kuznetsov opens instead at the symmetric formula of Siegel's §3, his equation 56, the one Siegel says makes the functional equation evident. It carries two integrals, one in s and its reflection in $1 - s$ weighted by χ , so his two halves arrive with the starting point rather than from a later choice. Gabcke's rederivation [9] begins in the same place, and Kuznetsov reaches it through the quadrature forms of Galway [10] and Arias de Reyna [1] rather than from Siegel directly.

Writing Kuznetsov's approximate remainder as R_{ak} (the subscript “*ak*” for Alexey Kuznetsov, and to distinguish it from Siegel's exact, unsubscripted R), his method gives

$$R \sim R_{ak} = -\frac{1}{2}(-1)^{\lfloor T \rfloor} \left(I_1(\sigma, T) + \chi(\sigma + iI(T)) I_2(\sigma, T) \right), \quad (43)$$

where I_1, I_2 are the two integrals of Kuznetsov's construction, each evaluated as a short finite series in precomputed coefficients $\omega_0, \omega_1[n], \lambda[n]$ (his notation; this ω_0 is unrelated to the direction angle $\omega(T)$ of §4); in some of our calculations we use 8 such coefficients, which already gives several digits of accuracy across the range of interest. Just as we split R into two pieces, the two terms of (43) define Kuznetsov's two half-remainders

$$R_{ak} = R_{1ak} + R_{2ak}, \quad R_{1ak} = -\frac{1}{2}(-1)^{\lfloor T \rfloor} I_1, \quad R_{2ak} = -\frac{1}{2}(-1)^{\lfloor T \rfloor} \chi I_2, \quad (44)$$

so that, in parallel with (24),

$$\zeta \sim \Sigma_1 + R_{1ak} + \Sigma_2 + R_{2ak}.$$

The split (44) mirrors the functional equation: R_{1ak} pairs with the first Dirichlet sum Σ_1 and R_{2ak} carries the functional-equation factor χ alongside Σ_2 . Figure 5 compares the two routes from Σ_1 to $\Sigma_1 + R$: the exact fractional links on one side and Kuznetsov's dashed pair on the other.

This pairing is exactly Siegel's: in his 1932 paper [20] Siegel writes ζ as $f(s)$ plus its reflected counterpart, and the first side is

$$f(s) \sim \Sigma_1 + R_{1ak}, \quad (45)$$

i.e. the first partial sum completed by an approximation to the first half of the remainder. The sign is $\sim$ rather than $=$ because R_{1ak} is the quantity $-\frac{1}{2}(-1)^{\lfloor T \rfloor} I_1$ of (44), with I_1 evaluated as a truncated series; read the other way round, as the *definition* $R_{1ak} := f(s) - \Sigma_1$, the relation is an identity and the approximation moves into the numerical evaluation of R_{1ak} .

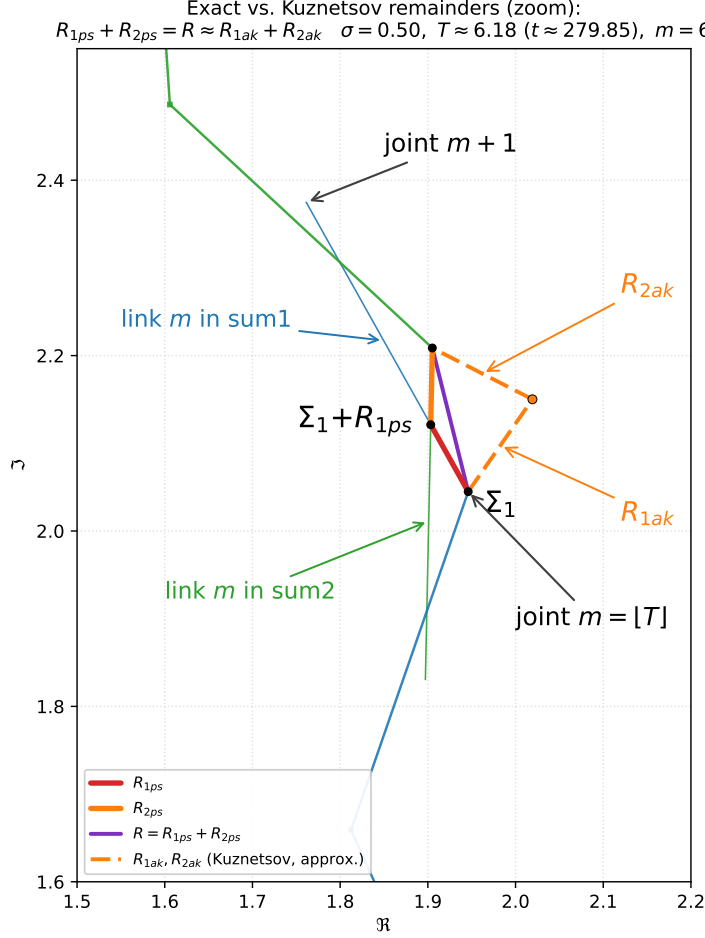


Figure 5: Exact versus Kuznetsov remainders, in the same zoom region as Figure 3. The exact fractional summands R_{1ps} (red) and R_{2ps} (orange) run from Σ_1 to $\Sigma_1 + R$ on one side of the resultant R (purple); Kuznetsov’s approximate half-remainders R_{1ak}, R_{2ak} are drawn as the orange dashed pair. Since $R_{1ak} + R_{2ak} \sim R$ (here to within 10^{-14}), the dashed path lands on $\Sigma_1 + R$; unlike the exact links, the ak links do not lie along a summand direction. Computed with 8 Kuznetsov coefficients and generated by `fig4_kuznetsov_zoom.py`.

The form of (45) is Siegel’s own, and it is worth recording where in his paper to look for it. He defines f in his §3, equation 58, as the contour integral

$$f(s) = \int_{0 \swarrow 1} \frac{x^{-s} e^{\pi i x^2}}{e^{\pi i x} - e^{-\pi i x}} dx, \quad (46)$$

where his symbol $0 \swarrow 1$ marks a straight path crossing the real axis between 0 and 1, running from upper right to lower left parallel to the bisector of the first and third quadrants; it is the reflection in the real axis of the path $0 \searrow 1$ carrying the integral Φ of his §1. The Gaussian factor $e^{\pi i x^2}$ decays in both directions along that line, so (46) converges. He then develops f asymptotically in his §4, where his equation 84 reads

$$f(s) = \sum_{n=1}^{m_1} n^{-s} + O(|s|/2\pi e)^{-\sigma/2} \quad (\sigma \geq 0, t > 0), \quad (47)$$

with $m_1 = [\Re\eta_1 - \Im\eta_1]$ and $\eta_1 = \sqrt{s/2\pi i}$, which to leading order is Siegel's cutoff $\lfloor \sqrt{t/2\pi} \rfloor$ of (9). Earlier in the same section he states it in words as well: developing f asymptotically yields “as a principal term a sum of $\lfloor \sqrt{t/2\pi} \rfloor$ summands, that is $\sum_{n=1}^m n^{-s}$ ”. So “first partial sum plus a remainder” is Siegel's own description of f , and it confirms that $\sim$ is the right sign in (45): at $\sigma = \frac{1}{2}$ the error term of (47) is only $O(t^{-1/4})$. One caution on the cutoff: because $\sqrt{t/2\pi} = T + \frac{1}{2} - \frac{1}{24T} + O(T^{-2})$ by (100), Siegel's m_1 is one more than our $m = \lfloor T \rfloor$ whenever $\{T\} > \frac{1}{2} + \frac{1}{24T}$. This does not disturb (45), since R_{1ak} carries the cutoff as a parameter and re-centers with it; it is Siegel's choice, not ours, that makes the remainder smallest. Equation numbers cited here are those of Siegel's paper as reprinted in his collected works, the numbering carried over by the Barkan–Sklar translation [21].³

Neither Siegel nor Kuznetsov names the other side, but it is $\Sigma_2 + R_{2ak}$, and adding the two recovers ζ to the same accuracy. The contrast with our result is that $\Sigma_1 + R_{1ps}$ and $\Sigma_2 + R_{2ps}$ are exact however they are read, and, moreover, each remainder there is a single fractional summand (§5).

7.2 The three remainders naming convention summary

There are a number of remainders in play, so to keep them straight we have named them with subscripts. Siegel's remainder integral is written R when unsplit; each named split below recovers that same R (exactly, or approximately for the Kuznetsov form):

1. Riemann–Siegel half-split.

$$R = R_{rs} = R_{1rs} + R_{2rs}, \quad R_{1rs} = R_{2rs} = \frac{1}{2}R.$$

Here R_{1rs} is simply half the remainder integral.

2. Partial-summand split (§4–§5).

$$R = R_{ps} = R_{1ps} + R_{2ps}.$$

Here R_{1ps} is a partial next summand: the $(\lceil T \rceil)$ -th Dirichlet term of Σ_1 , shortened by the real weight d_1 .

3. Kuznetsov / Siegel f split (§7.1).

$$R \sim R_{ak} = R_{1ak} + R_{2ak}.$$

Here R_{1ak} is Siegel's $f(s)$ minus the first partial sum: $R_{1ak} = f(s) - \Sigma_1$ (approximately, once R_{1ak} is evaluated numerically).

All three are measured from the same pair of joints, both at index $m = \lfloor T \rfloor$: each $R_{1\bullet}$ leaves joint m of the Σ_1 chain, which is the partial sum Σ_1 itself, and each $R_{2\bullet}$ arrives at joint m of the Σ_2 chain, which is the point $\Sigma_1 + R$. A split therefore chooses nothing but the point $\Sigma_1 + R_{1\bullet}$ at which the two halves meet, its own bisector point in the sense of §9.1; the anchoring joints are common to all three.

³One incidental finding while checking (45) against the source: the translation's equation 83 prints the sum as $\sum n^{s-1}$, while equation 84, derived from it, has $\sum n^{-s}$. The numerics say n^{-s} is correct, so equation 83 carries a typo in the Barkan–Sklar translation.

7.3 Scaling the remainders with the fractional part of T unchanged

Write $T = \lfloor T \rfloor + \{T\}$ for the integer and fractional parts of the index. Empirically, the *shape* of the remainder figure depends almost entirely on the fractional part $\{T\}$. By shape we mean the relative angles among R , R_{1ps} , R_{2ps} , R_{1ak} , and R_{2ak} once the configuration is rotated into the frame in which R lies along the positive real axis. The integer part $\lfloor T \rfloor$ mainly retunes the overall size: with $n = \lfloor T \rfloor$ the lengths scale like $n^{-\sigma}$ times a slowly varying amplitude. Figure 6 makes this visible. Each panel shows only the remainders, drawn in the R -frame (rotated so R lies on the real axis and translated so the midpoint of R sits at the origin), with a common axis scale. The top row holds $\{T\} = 0.18$ at $T = 6.18, 50.18$, and 100.18 ; the bottom row holds $\{T\} = 0.72$ at $T = 6.72, 50.72$, and 100.72 . Across a row the figure shrinks but keeps its shape; across a column (same $\lfloor T \rfloor$, different $\{T\}$) the shape changes.

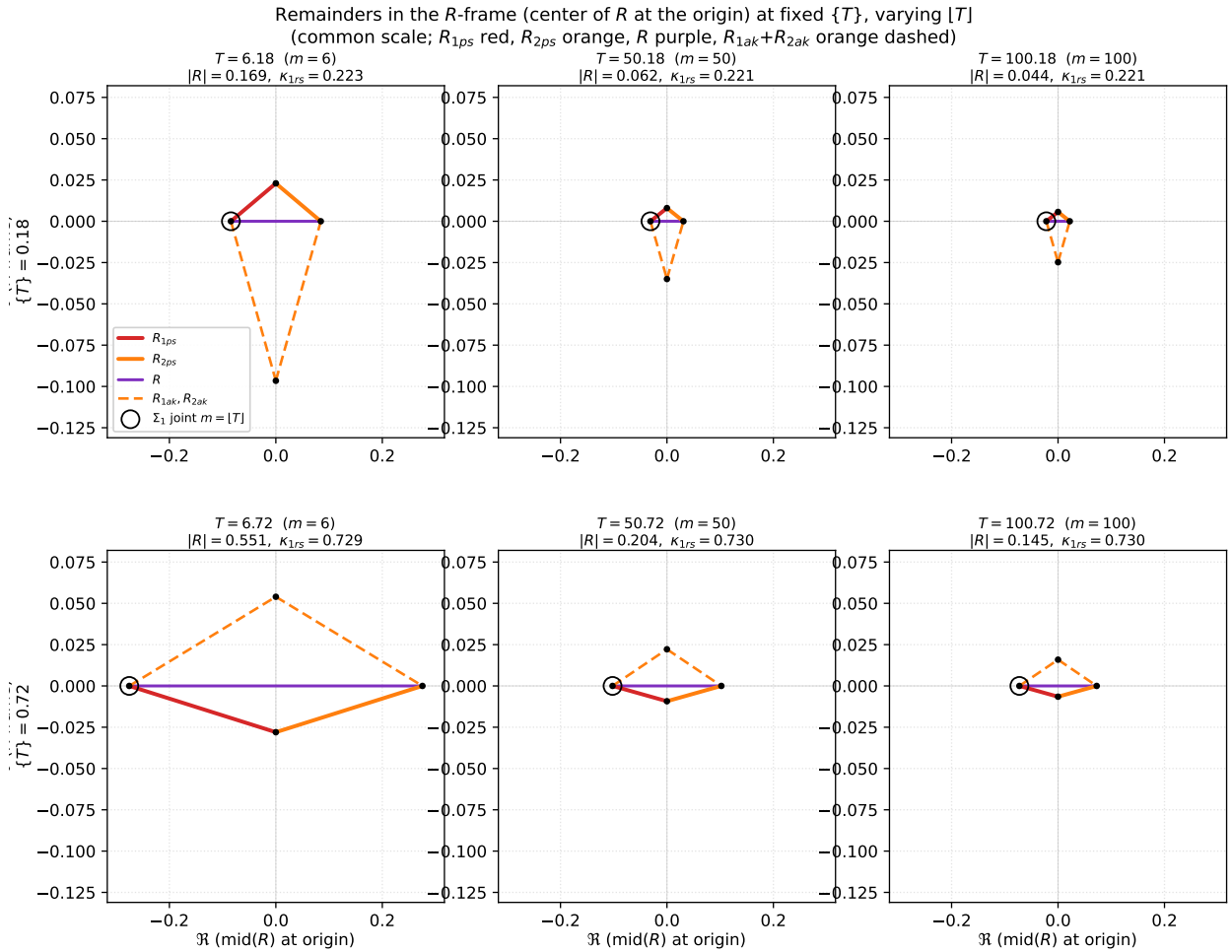


Figure 6: Remainders in the R -frame at fixed fractional part $\{T\}$, varying $\lfloor T \rfloor$. The frame is centered on the midpoint of R (so R runs from $-|R|/2$ to $+|R|/2$ on the real axis). Top row: $\{T\} = 0.18$ at $T = 6.18, 50.18, 100.18$. Bottom row: $\{T\} = 0.72$ at $T = 6.72, 50.72, 100.72$. Colors match Figure 5: R_{1ps} red, R_{2ps} orange, R purple, R_{1ak} and R_{2ak} orange dashed. The large open circle marks the left endpoint of R , which is the anchoring joint Σ_1 at $m = \lfloor T \rfloor$ that all three splits leave from. All six panels share the same axis scale. Generated by `fig_remainder_scale_panels.py`.

That figure holds $\{T\}$ fixed and climbs the strip. The complementary view holds the chain fixed and lets T sweep a whole unit, which is Figure 7: the same R -frame in sixteen snapshots from $T = 4$ to $T = 5$ at $m = 4$, the sweep that Figure 34 will draw for the neighboring links. Two things show up there that a fixed $\{T\}$ cannot show. First, on the critical line both splits are symmetric, $|R_{1ps}| = |R_{2ps}|$ exactly and likewise for the ak pair, so both apexes ride the perpendicular bisector of R and the state of a split is the single number giving the height of its apex above the chord. Second, the two splits move quite differently: the ak apex marches steadily from below R to well above it, while the ps apex swings down onto the chord, crosses to the far side, and comes back, touching the chord exactly at the two heights where $\sin(2\omega + \arg \chi)$ vanishes (§8.4). There the cone of (23) has collapsed to a line, and on the critical line the split collapses with it into the plain halving $R_{1ps} = R_{2ps} = \frac{1}{2}R$: the d_1 pole of (19) is removable at $\sigma = \frac{1}{2}$, its numerator vanishing with its denominator, which is why the sweep passes through those heights smoothly. Off the line it is a genuine pole: at $\sigma = 0.6$ and $\sigma = 0.75$, d_1 runs through infinity and changes sign there, and that is what the bands of §8.4 are.

The remainder splits in the R -frame, $T = 4 \rightarrow 5$ at $\sigma = 1/2$ (chain length $m = 4$, common scale)

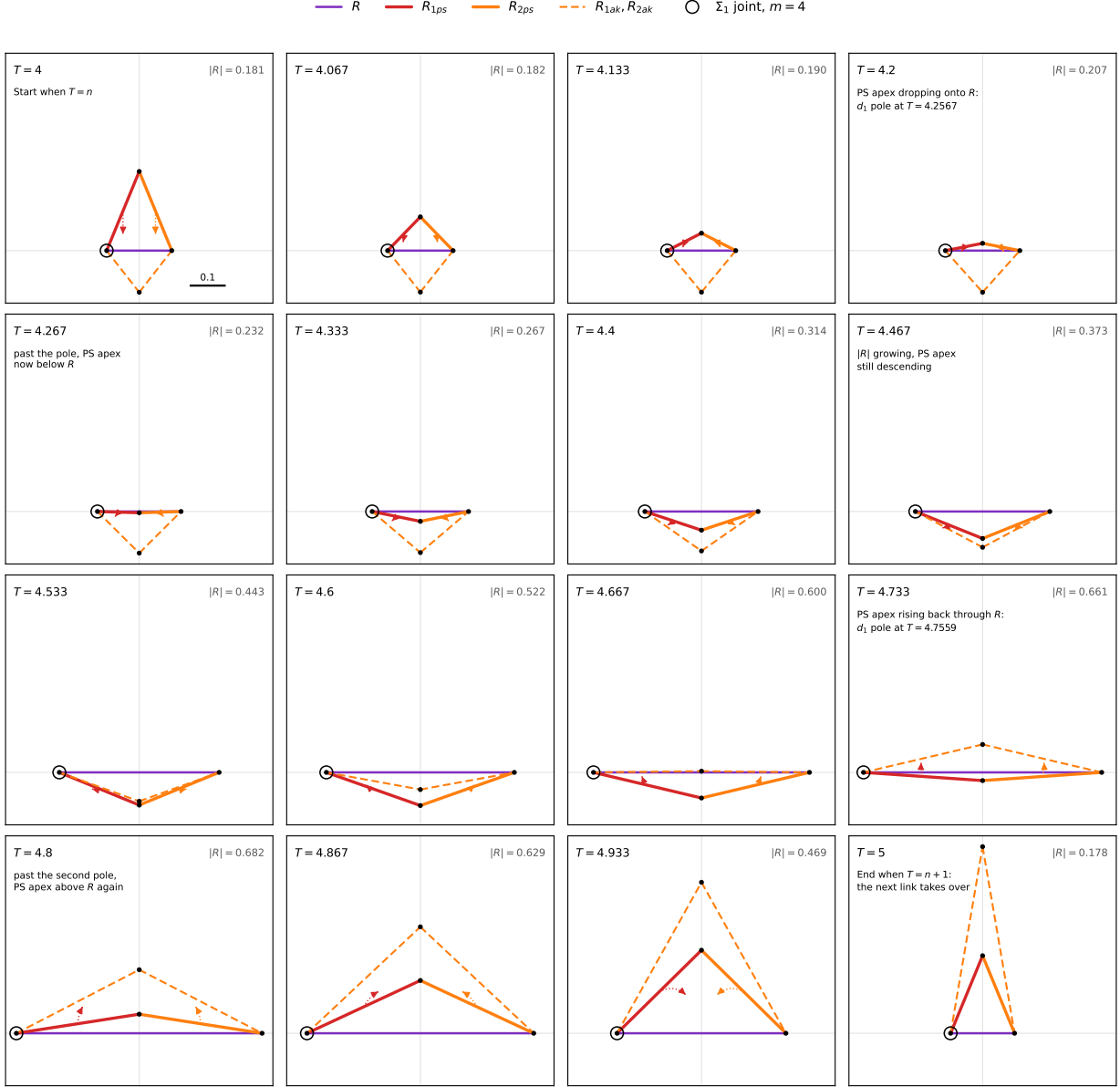


Figure 7: The remainder splits in the R -frame as T sweeps one unit, $T = 4 \rightarrow 5$ at $\sigma = \frac{1}{2}$, sixteen snapshots read left to right from the top left, the companion of Figure 34 for the remainders. The frame and the colors are those of Figure 6: R purple along the real axis with its midpoint at the origin, R_{1ps} red and R_{2ps} orange from the Σ_1 joint (open circle) to the far end of R , R_{1ak} and R_{2ak} orange dashed, all sixteen panels to one scale, with $|R|$ given in each. The chain length is held at $m = 4$ throughout, as Figure 34 holds the bisector link constant, so $T = 5$ is the instant the handoff happens. The dotted arrow on R_{1ps} and on R_{2ps} runs from the middle of that segment to the middle of it as the next panel depicts it, tracing the motion between panels. Both apexes stay on the perpendicular bisector of R , the two halves of each split being equal in length on the critical line. The ps apex crosses the chord between the fourth and fifth panels and again between the twelfth and thirteenth, at the two heights of (67) where the split is the plain halving; the ak apex crosses once, between the tenth and eleventh. Generated by `fig_remainder_frame_sweep.py`.

To make the scaling precise, factor out the leading $n^{-\sigma}$ from each half-remainder length by writing

$$|R_{1\bullet}(\sigma, T)| = \kappa_{1\bullet}(\sigma, T) n^{-\sigma}, \quad n = \lceil T \rceil,$$

for the three first halves $R_{1\bullet} \in \{R_{1rs}, R_{1ps}, R_{1ak}\}$ of §7.2. Each split is then on the same footing: $R = R_{1rs} + R_{2rs} = R_{1ps} + R_{2ps} \sim R_{1ak} + R_{2ak}$, and for the rs half-split one has $R_{1rs} = R_{2rs} = \frac{1}{2}R$ so $\kappa_{2rs} = \kappa_{1rs}$. The three amplitudes admit the closed forms

$$\begin{aligned} \kappa_{1rs} &= \frac{1}{2} n^\sigma |R| \approx \frac{1}{4} n^\sigma |I_1(s) + \chi(s) I_2(s)|, \\ \kappa_{1ps} &= d_1 n^\sigma = 2\kappa_{1rs} \frac{\sin(\omega - \arg R + \psi)}{\sin(2\omega + \psi)}, \\ \kappa_{1ak} &= \frac{1}{2} n^\sigma |I_1(s)|, \end{aligned} \tag{48}$$

with $s = \sigma + iI(T)$, $\omega = I(T) \log n$, $\psi = \arg \chi(s)$, and I_1, I_2 the Kuznetsov integrals of §7.1; the “ $\approx$ ” in the first line holds when R is evaluated by Kuznetsov’s method (43), and the second equality in the κ_{1ps} line is exact with κ_{1rs} read as $\frac{1}{2}n^\sigma |R|$. Consequently, at fixed σ and fixed $\{T\} = f$, the passage from $T_1 = N_1 + f$ to $T_2 = N_2 + f$ multiplies every half-remainder length by

$$\frac{|R_{1\bullet}(\sigma, T_2)|}{|R_{1\bullet}(\sigma, T_1)|} = \left(\frac{N_1 + 1}{N_2 + 1} \right)^\sigma \frac{\kappa_{1\bullet}(\sigma, T_2)}{\kappa_{1\bullet}(\sigma, T_1)}. \tag{49}$$

When the fractional part is held fixed the ratio of κ ’s is nearly 1, so the leading factor $((N_1 + 1)/(N_2 + 1))^\sigma$ already accounts for almost all of the size change. Figure 8 checks this in the R -frame: the left panel is $T = 6.18$; the right panel is $T = 50.18$ with all remainder vectors scaled by the exact factor $\lambda = |R|_{6.18}/|R|_{50.18}$ from (49) (equivalently $\kappa_{1rs,1}/\kappa_{1rs,2}$ times the $n^{-\sigma}$ ratio). The two panels nearly coincide.

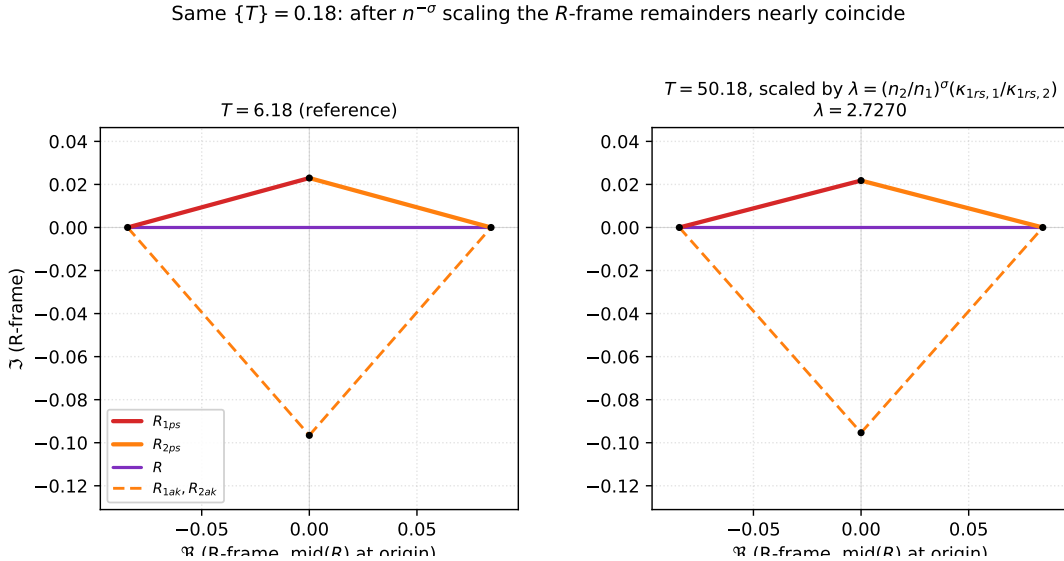


Figure 8: Same fractional part $\{T\} = 0.18$: R -frame remainders at $T = 6.18$ (left) and at $T = 50.18$ after scaling by $\lambda = (n_2/n_1)^\sigma (\kappa_{1rs,1}/\kappa_{1rs,2})$ (right). The scaled figure matches the reference to high accuracy ($|R_{1ps}|$ agrees to about 0.3%). Generated by `fig_remainder_scale_panels.py`.

If one wants exactness rather than a close empirical match, divide by R itself. By (19) the modulus of R cancels identically, leaving

$$\frac{R_{1ps}}{R} = \frac{\sin(\omega - \arg R + \arg \chi)}{\sin(2\omega + \arg \chi)} e^{-i(\omega + \arg R)}, \quad (50)$$

a function of the angles alone, exact at every T and every σ : the size is then gone exactly, not almost. On the critical line the equal legs of Corollary 9.1 force $\text{Re}(R_{1ps}/R) = \frac{1}{2}$, so the shape is carried by the single number $\text{Im}(R_{1ps}/R)$, and that number is a function of $\{T\}$ up to $O(1/T)$, which is the content of Theorem 8.8.

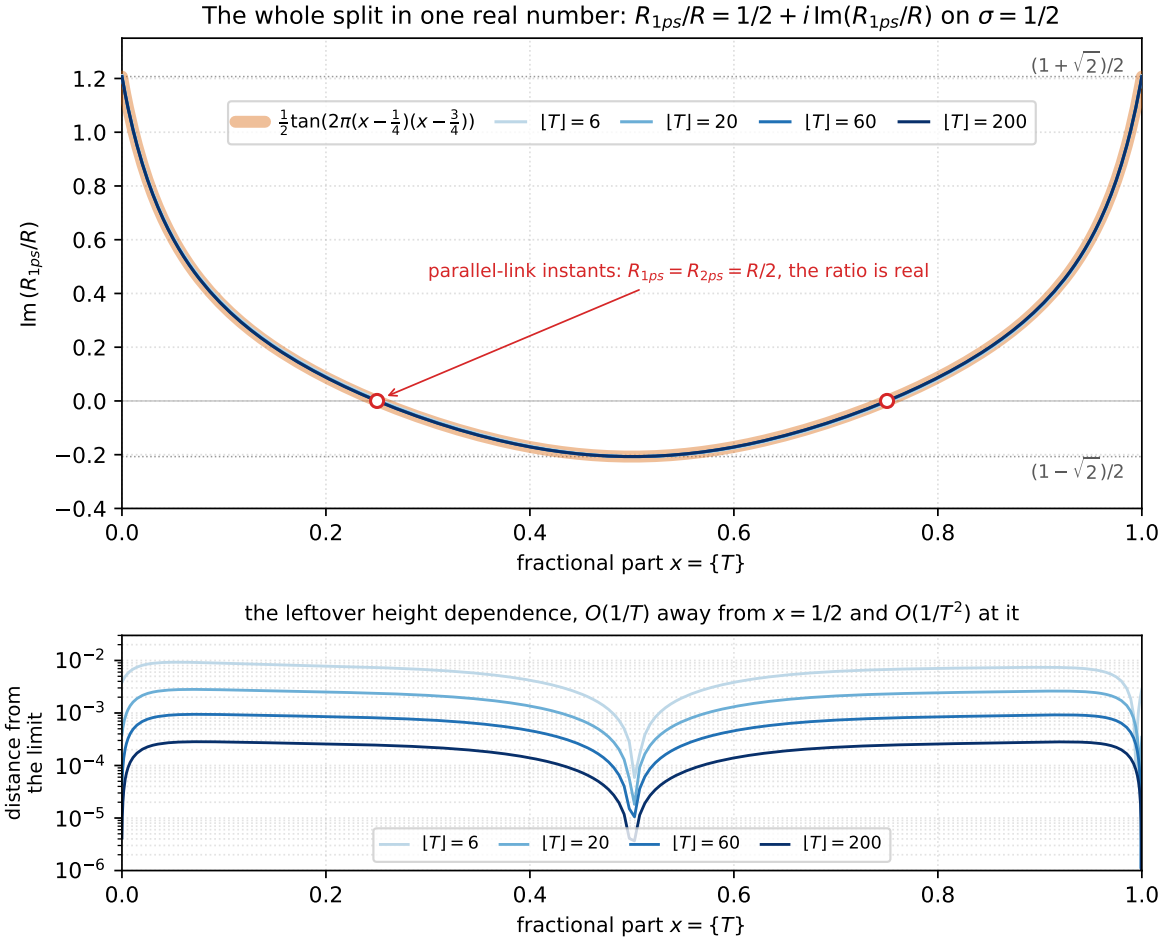


Figure 9: The split after division by R , on the critical line. Top: $\text{Im}(R_{1ps}/R)$ against $x = \{T\}$ at $[T] = 6, 20, 60, 200$, each computed through ζ and the Cramer solution (19), laid over the limit $\frac{1}{2} \tan(2\pi(x - \frac{1}{4})(x - \frac{3}{4}))$ (thick orange). The limit falls from $(1 + \sqrt{2})/2$ at the integers to $(1 - \sqrt{2})/2$ at $x = \frac{1}{2}$, a swing of exactly $\sqrt{2}$, and vanishes at the two parallel-link instants (circles; $x = 0.250156$ and 0.750155 at $[T] = 200$), where $R_{1ps} = R_{2ps} = \frac{1}{2}R$. Bottom: distance from the limit, $O(1/T)$ in general and $O(1/T^2)$ at $x = \frac{1}{2}$. Generated by `fig_ratio_shape.py`.

Figure 9 dissects that single number. On the critical line $\arg R = \frac{1}{2} \arg \chi$ and $\chi = e^{-2i\vartheta}$, so $\omega - \arg R + \arg \chi = \omega - \vartheta$ and $2\omega + \arg \chi = 2(\omega - \vartheta)$, the π ambiguity in $\arg R$ flipping the sine

and the exponential of (50) together. That equation therefore collapses to

$$\frac{R_{1ps}}{R} = \frac{e^{-i(\omega-\vartheta)}}{2 \cos(\omega-\vartheta)} = \frac{1}{2} - \frac{i}{2} \tan(\omega-\vartheta), \quad (51)$$

which is the exact form $d_1 = e^{i\vartheta} R/2 \cos(\omega-\vartheta)$ of §8.2 divided by R . Three things follow, and the figure shows all three. First, the real part is $\frac{1}{2}$ identically, which is Corollary 9.1 once more; the plotted values were computed the long way, through ζ and (19), and at all 964 samples their real parts agree with $\frac{1}{2}$ to thirty digits. Second, everything that is left is the one angle $\omega-\vartheta$ modulo π , and that angle tends to $-\beta$, where $\beta = 2\pi(x - \frac{1}{4})(x - \frac{3}{4})$ and $x = \{T\}$; this is the same reduction that turns $2 \cos(\omega-\vartheta)$ into $2 \cos \beta$ in the proof of Theorem 8.8. So $\text{Im}(R_{1ps}/R) \rightarrow \frac{1}{2} \tan \beta$, which is the second of the two tangents in the limit waveform of §8.1: that waveform is $\tan(2\pi x)$ times this very quantity. The four heights lie on that limit to within 9.3×10^{-3} , 2.8×10^{-3} , 9.4×10^{-4} and 2.8×10^{-4} at $\lfloor T \rfloor = 6, 20, 60, 200$, shrinking in proportion to the index, which is the $O(1/T)$ rate of the theorem seen directly. Third, the range is fixed by β , which runs from $\frac{3\pi}{8}$ at the integers to $-\frac{\pi}{8}$ at $x = \frac{1}{2}$: the curve therefore lies between $(1 + \sqrt{2})/2$ and $(1 - \sqrt{2})/2$, swinging by exactly $\sqrt{2}$, and the tangent stays a full eighth-turn clear of its pole at $\frac{\pi}{2}$, which is the boundedness of d_1 on the line read as an angle. It crosses zero where β does, at the parallel-link instants: there $\sin(2\omega + \arg \chi)$ vanishes while $\cos(\omega-\vartheta)$ does not, so the pole of the Cramer solution (19) is removable on the critical line and the two halves are each exactly $\frac{1}{2}R$.

8 The positive real function d_1 and its periodicity

In this section we show the periodicity and positivity of d_1 and d_2 . The proofs are long and technical; the reader may want to skip this section on a first reading.

The weights d_1 and d_2 of §4 are the quantitative heart of the two remainders: $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg \chi)}$ are fixed in direction, so everything the remainders do is carried by the two real functions $d_1(\sigma, T)$ and $d_2(\sigma, T)$. This section studies d_1 as a function of the index T , and its first half is one long act of normalization, in which d_1 is periodic at every step and the steps remove everything else. As it stands, d_1 repeats an arch on every unit of T , but the arches decay with the shrinking links. Reading the fraction of the link instead of the distance takes out the decay, yet jumps remain at the integer handoffs, because there the fraction is re-measured along the next link, which is shorter than the last and oriented the other way. The fold relation $d_1^- + d_1^+ = n^{-\sigma}$ is exact, and that exactness absorbs the jumps: measured in one fixed unit, the coordinate h of (54) has no jumps at all, at the price of flattening again. Re-zooming smoothly gives p of (55), continuous with steady amplitude, whose arches alternate, each nearly the vertical flip of its neighbor. Flipping alternate arches and pinning each to begin and end at zero yields the pinned waveform (60), one continuous periodic curve. With the normalization in hand we approximate: d_1 on the critical line (§8.2), then off it (§8.3), where a new feature appears, the poles of d_1 and d_2 , located in §8.4. The numerics then become theorems: d_1 and d_2 are positive, on the critical line without exception and off it outside the narrow pole windows of $\{T\}$ (§8.5); the pinned waveform has an exact closed form in the limit (§8.6); and uniform two-sided bounds hold at every height (§8.7).

8.1 The exact formulas

Corollary 8.1. *When $\sigma = \frac{1}{2}$ the two fractional weights coincide, $d_1 = d_2$, and hence $|R_{1ps}| = |R_{2ps}|$: the two fractional summands are always equal in length on the critical line.*

This follows from the definitions (19)–(20) together with the functional symmetry of χ on the critical line. Explicitly, when $\sigma = \frac{1}{2}$ the functional equation forces $|\chi| = 1$ and $\arg R = \frac{1}{2} \arg \chi$ modulo π (proved in Appendix B). Writing $\psi = \arg \chi$ and substituting $\arg R = \frac{\psi}{2}$ into (19)–(20), the two numerators collapse to the *same* value:

$$d_1 = |R| \frac{\sin(\omega - \frac{\psi}{2} + \psi)}{\sin(2\omega + \psi)} = |R| \frac{\sin(\omega + \frac{\psi}{2})}{\sin(2\omega + \psi)}, \quad d_2 = |R| \frac{\sin(\omega + \frac{\psi}{2})}{\sin(2\omega + \psi)}, \quad (52)$$

so $d_1 = d_2$. Equivalently, setting $d_1 = d_2$ in (23) gives $R = d_1 (e^{-i\omega} + e^{i(\omega+\psi)}) = 2d_1 \cos(\omega + \frac{\psi}{2}) e^{i\psi/2}$, which indeed has argument $\frac{\psi}{2}$ (or $\frac{\psi}{2} + \pi$, where the cosine is negative), consistent with $\arg R = \frac{1}{2} \arg \chi$ modulo π . We have verified this in Lean; the proof is recorded in Appendix B. We stress that this critical-line equality is a pleasant consequence of the decomposition, not its motivation.

Since $d_1 = d_2$ on the critical line, a single curve there tells the whole story of both remainders, and it is worth drawing. Figure 10 plots $d_1(\frac{1}{2}, T)$ for $1 \leq T \leq 7$, and the graph is unexpectedly tame: as a function of the height t the zeta landscape is famously erratic, but viewed through the index T the weight d_1 oscillates once per unit of T , every unit interval the same shape, with only two blemishes: the amplitude decays slowly, and the curve jumps at each integer T , where $m = \lfloor T \rfloor$ increments and the fractional summands hand off to the next link. The bottom panel shows that a single normalization (by the length of the link that carries d_1) removes the decay and most of the jumps at once, leaving a curve that looks nearly *periodic* in T . That is a strong hint that d_1 is, at heart, a fixed waveform in the fractional part of T dressed up by the link geometry. The rest of this section chases that hint down, first by removing the jumps exactly, then by identifying the waveform in closed form.

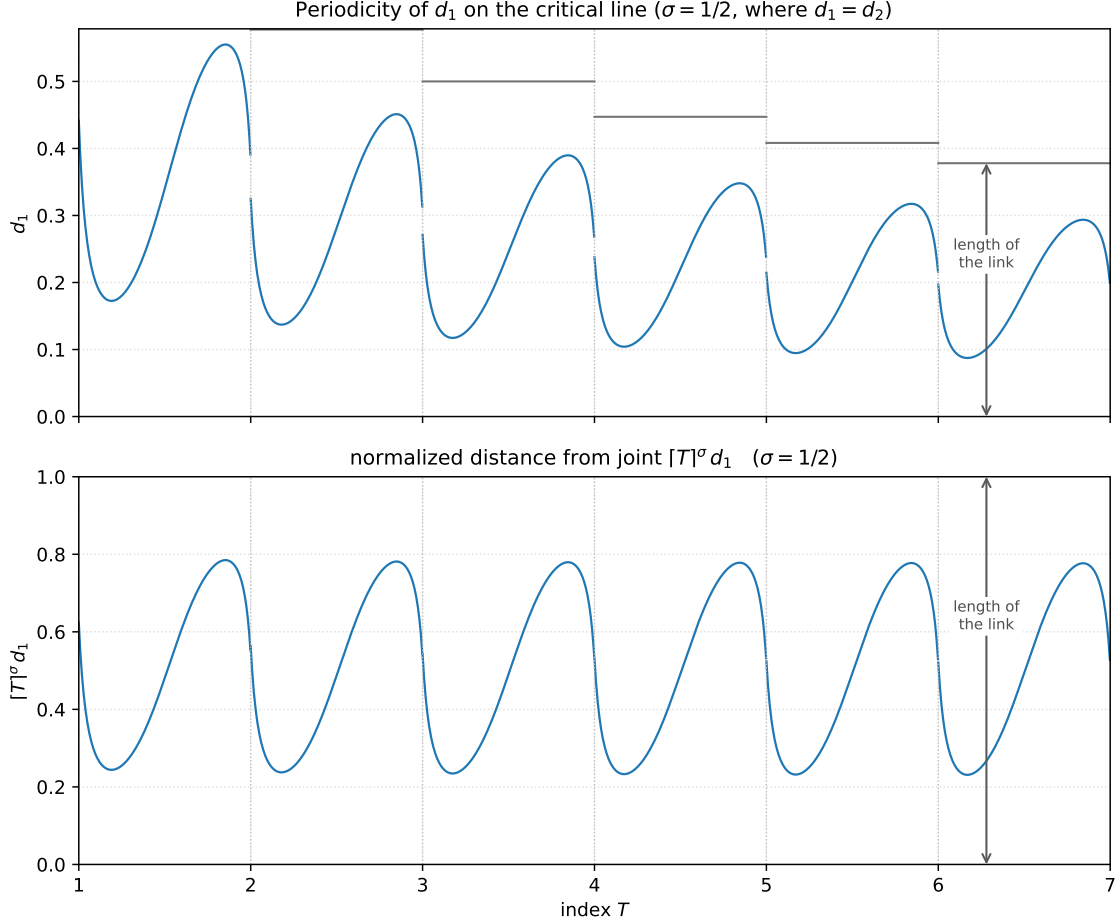


Figure 10: Top: the common fractional weight $d_1 = d_2$ on the critical line, plotted against the index T for $1 \leq T \leq 7$. The curve oscillates once per unit of T and jumps at each integer T , where $m = \lfloor T \rfloor$ increments and the fractional summands hand off to the next link; the amplitude decays slowly as T grows. The gray step is the length $[T]^{-\sigma}$ of the link that carries the fractional summand, which d_1 never exceeds; the panel keeps its original top, so the steps for $1 \leq T < 3$ lie at or above the frame. The double-headed arrow on $6 < T < 7$ spans that link length. Bottom: the same formula for d_1 normalized by the length $[T]^{-\sigma}$ of link m , which carries it: the *normalized distance from joint* $[T]^\sigma d_1$, the crossing fraction along link m at which the point $B_1 = \Sigma_1 + R_{1ps}$ sits (cf. §11). The panel is drawn on the full scale 0 to 1 of the normalized link, so 0 is joint m and 1 is joint $m+1$; the double-headed arrow, at the same T as the one above, marks that same link seen at unit length. The normalization removes both the decay and (nearly) the jumps: the fraction sweeps the same range $\approx [0.23, 0.78]$ in every unit interval, staying clear of both ends of the link (made exact in §8.6–§8.7). So the normalized curve converges to a periodic waveform: sampled at matched fractional parts of T , its values agree to $\approx 2 \times 10^{-3}$ by $T = 20$, and the residual discrepancy halves with each doubling of T , the rate set by the link-length mismatch $(1 + 1/\lfloor T \rfloor)^\sigma \rightarrow 1$ of (53) below. Computed exactly (mpmath) via the Cramer solution of (23). Generated by `fig.d1_critical.py`.

Removing the jumps exactly. The normalization in the bottom panel of Figure 10 nearly removes the jumps, and the residue can be traced to a single exact relation. At a handoff $T = n$

the outgoing link ($m = n - 1$) and the incoming link ($m = n$) fold back onto one another, and

$$d_1^+ = n^{-\sigma} - d_1^-, \quad (53)$$

where d_1^- and d_1^+ are the limits from the left and from the right. Figure 12 draws the configuration at $n = 7$.

Both the fold and (53) are exact. Write $u_1 = e^{-i\omega}$ and $u_2 = e^{i(\omega+\psi)}$ for the two unit directions of (23) on the outgoing side, where $\omega = t \log n$. The turn at the handoff is $I(n) \log \frac{n+1}{n} = \pi(2n+1)$ by (99), an odd multiple of π , so both directions reverse and the incoming frame is $-u_1, -u_2$. Crossing the handoff also moves n^{-s} out of Σ_1 and χn^{s-1} out of Σ_2 , so

$$R^- - R^+ = n^{-s} + \chi n^{s-1} = n^{-\sigma} u_1 + |\chi| n^{\sigma-1} u_2,$$

while the two decompositions themselves give $R^- = d_1^- u_1 + d_2^- u_2$ and $R^+ = -(d_1^+ u_1 + d_2^+ u_2)$, whence $R^- - R^+ = (d_1^- + d_1^+) u_1 + (d_2^- + d_2^+) u_2$. Wherever the Cramer determinant $\sin(2\omega + \psi)$ is nonzero the two directions are independent over $\mathbb{R}$ (§8.4), so the coefficients may be compared: that is (53), together with $d_2^- + d_2^+ = |\chi| n^{\sigma-1}$, the two right-hand sides agreeing on the critical line where $|\chi| = 1$. Putting (53) back into $B_1 = \Sigma_1 + R_{1ps}$ shows what it says about the point itself. On the incoming side the sum has gained $n^{-s} = n^{-\sigma} u_1$ and the direction has reversed, so with Σ_1^- the sum through $n - 1$,

$$B_1^+ = \Sigma_1^- + n^{-s} + d_1^+(-u_1) = \Sigma_1^- + (n^{-\sigma} - d_1^+) u_1 = \Sigma_1^- + d_1^- u_1 = B_1^-.$$

Continuity of B_1 in the plane is therefore a consequence of the fold rather than an assumption behind it: there is a single bisector point, and at the instant the links fold it lies on both of them.

In terms of the plotted fraction $f = \lceil T \rceil^\sigma d_1$ this reads $f^+ = (1 + \frac{1}{n})^\sigma (1 - f^-)$: an orientation flip $f \mapsto 1 - f$ (after the handoff the fraction is measured from the other end of the link) composed with a unit mismatch (the new link is shorter than the old by the factor $(1 + \frac{1}{n})^{-\sigma}$). The mismatch factor tends to 1, which is why the jumps in the bottom panel shrink as T grows; the replacement $f \mapsto 1 - f$ also contributes less and less, because the curve happens to cross the integers near $f \approx \frac{1}{2}$, where $1 - f \approx f$.

Because (53) is exact, the unit intervals can be glued into a coordinate with no jumps at all: measure the position of B_1 along the link chain in one *fixed* unit (the first link, of length 1), flipping the orientation and absorbing the constant $n^{-\sigma}$ at each handoff of the bisector point from one link to the next (see §10.1 for more on these handoffs). Unrolling the recursion gives

$$h(T) = \sum_{k=2}^{\lfloor T \rfloor} (-1)^k k^{-\sigma} + (-1)^{\lfloor T \rfloor + 1} d_1(T), \quad (54)$$

and at every integer the term entering the sum cancels the jump of d_1 identically, so h is continuous everywhere. The price is that h *flattens*: in a fixed unit the links themselves shrink, so the oscillating part $d_1 = O(T^{-\sigma})$ decays, while the accumulated constants are the partial sums of the convergent alternating series $2^{-\sigma} - 3^{-\sigma} + 4^{-\sigma} - \dots = 1 - \eta(\sigma)$, with η the Dirichlet eta function. Hence $h(T) \rightarrow 1 - \eta(\sigma)$, which at $\sigma = \frac{1}{2}$ is ≈ 0.3951 .

Continuity and a non-decaying amplitude can be had at once by re-zooming *smoothly*, with T^σ , instead of with the step function $\lceil T \rceil^\sigma$ (whose abrupt changes at the integers are exactly what created the residual jumps):

$$p(T) = T^\sigma \left(h(T) - (1 - \eta(\sigma)) \right). \quad (55)$$

As a product of continuous functions p is exactly continuous, and since $h(T) - (1 - \eta(\sigma)) = O(T^{-\sigma})$ (the decaying d_1 plus the tail of the alternating series), the factor T^σ restores a bounded oscillation. Figure 11 compares the two coordinates: both are continuous, with only kinks (no jumps) at the integers, while h contracts toward its limit $1 - \eta(\frac{1}{2})$ and p keeps a steady amplitude. The red curve is computed from the simplified local form (62), derived below, which agrees with (55) to machine precision.

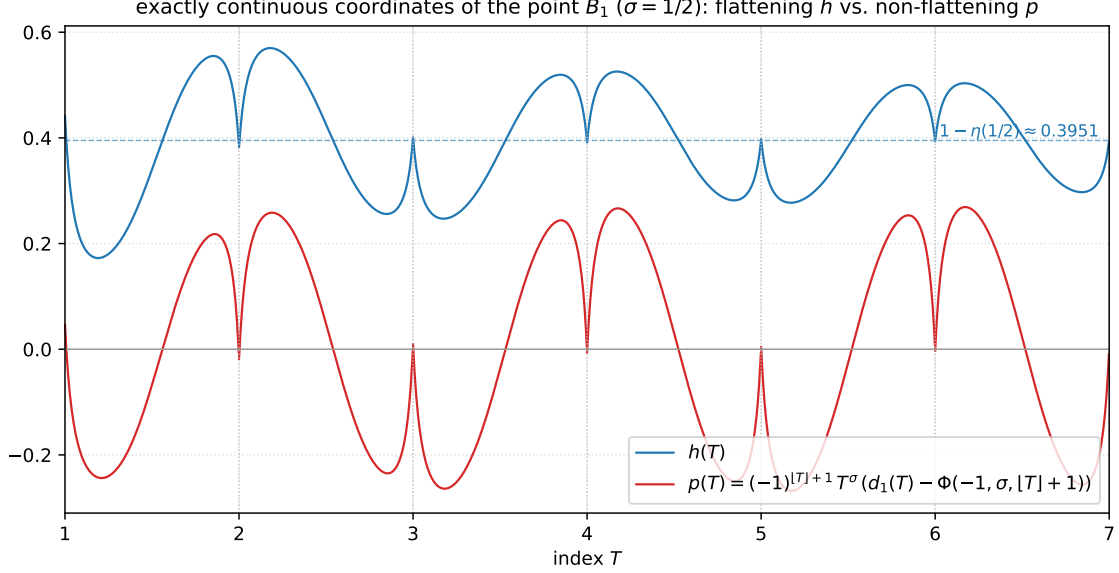


Figure 11: The two exactly continuous coordinates of the point B_1 on the critical line, $1 \leq T \leq 7$, in a single panel for comparison. Blue: $h(T)$ of (54), the position along the link in a fixed unit; it has no jumps at the integers, only kinks where the links fold back and handoff the bisector point to the next link, and its oscillation contracts toward the limit $1 - \eta(\frac{1}{2}) \approx 0.3951$ (dashed line): the swing shrinks from ≈ 0.40 on the first interval to ≈ 0.24 by $T = 7$. Red: $p(T)$, the same coordinate re-zoomed smoothly by T^σ , computed from the simplified local form (62); it is equally continuous but its amplitude holds steady (≈ 0.53 peak to peak) instead of flattening. Continuity was verified numerically at each integer (mismatch $\sim 10^{-7}$ at offsets of 10^{-6}). Generated by `fig_h.p.continuous.py`.

A pinned waveform. Each arch of p in Figure 11 is nearly the vertical flip of its neighbors, but the arches do not begin and end at zero, so flipping alternate intervals would not make them line up. Pinning is the fix: we force each arch to begin and end at zero, on the T axis, so that the flipped arches glue into one continuous waveform. Taking one-sided limits at an integer n (where $[T] = n - 1$ on the left) gives the endpoint value exactly, in terms of the Lerch transcendent

$$\Phi(-1, \sigma, a) = \sum_{j=0}^{\infty} \frac{(-1)^j}{(a+j)^\sigma}, \quad (56)$$

the alternating sum of all the link lengths from a onward:

$$p(n) = (-1)^n n^\sigma (d_1(n^-) - \Phi(-1, \sigma, n)) = (-1)^{n+1} \varepsilon_n, \quad \varepsilon_n := n^\sigma (\Phi(-1, \sigma, n) - d_1(n^-)) > 0, \quad (57)$$

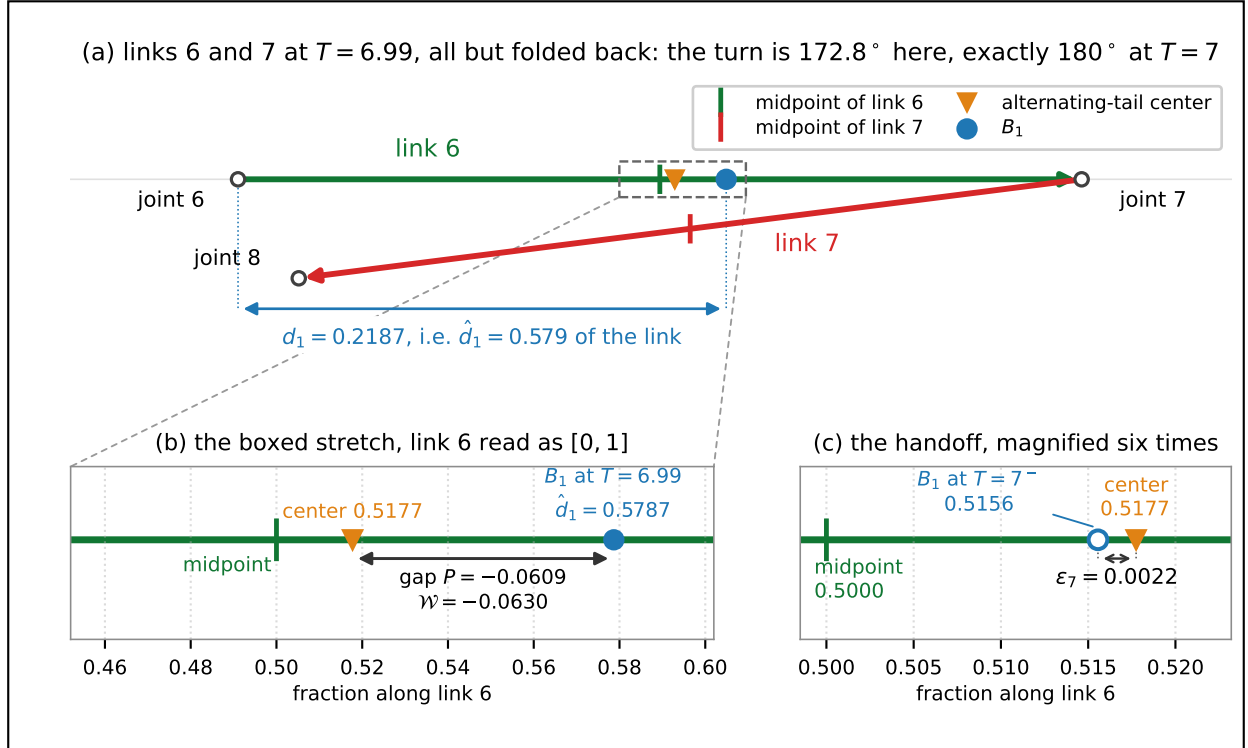


Figure 12: The geometry behind the fold relation (53) and the shortfall (57), at $T = 6.99$ on the critical line. (a) The outgoing link 6, carrying summand 7, and the incoming link 7, carrying summand 8. The turn between them is 172.8° here and exactly 180° at $T = 7$, so at the handoff link 7 lies back along link 6 and the single point B_1 is on both at once. (b) The boxed stretch, link 6 read as $[0, 1]$: the alternating-tail center (58) sits at 0.5177 and B_1 has run past it to $\hat{d}_1 = 0.5787$; the gap is P of (59) below, and subtracting the chord leaves $\mathcal{W} = -0.0630$ of (60). (c) The same stretch magnified about six times: by $T = 7$ the point has swung back to 0.5156, short of the center by $\varepsilon_7 = 0.0022$, and at the far end $\varepsilon_6 = 0.0029$. Generated by `fig.fold.handoff.py`.

so ε_n is the shortfall of the distance when the links are folded back. What they fall short of is the alternating-tail center

$$n^\sigma \Phi(-1, \sigma, n) = \frac{1}{2} + \frac{\sigma}{4n} - \frac{\sigma(\sigma+1)(\sigma+2)}{48n^3} + O(n^{-5}) = \frac{1}{2} + \frac{1}{8n} - \frac{5}{128n^3} + O(n^{-5}) \quad \text{at } \sigma = \frac{1}{2}, \quad (58)$$

by Boole summation,⁴ $\sum_{j \geq 0} (-1)^j f(n+j) = \frac{1}{2}f(n) - \frac{1}{4}f'(n) + \frac{1}{48}f'''(n) - \dots$, whose even derivatives drop out, so there is no n^{-2} term. On the critical line the ends approach zero on the order of T^{-2} , without ever reaching it at finite T .

Flipping alternate intervals costs nothing: it simply deletes the alternating sign from (62), since with $m = \lfloor T \rfloor$

$$P(T) := (-1)^m p(T) = T^\sigma (\Phi(-1, \sigma, m+1) - d_1(T)), \quad (59)$$

whose two ends on $[m, m+1]$ are exactly $-\varepsilon_m$ and $+\varepsilon_{m+1}$ by (57), so P still jumps by $2\varepsilon_n$ at each integer. Subtracting the chord between those two values removes the jumps and pins both ends at

⁴The alternating-sum analogue of Euler–Maclaurin summation: the correction terms are built from the Euler polynomials in place of the Bernoulli numbers.

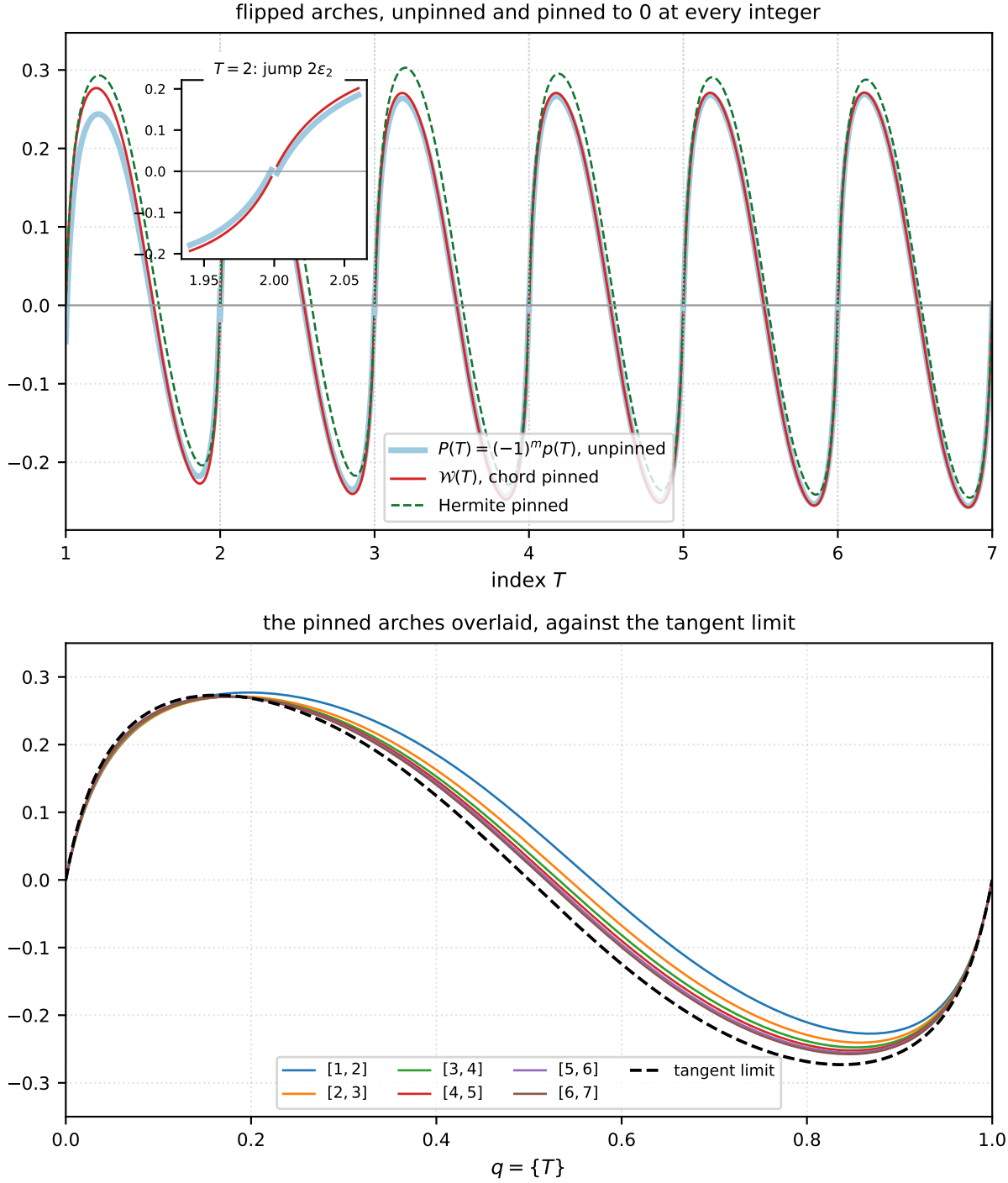


Figure 13: The flipped arches of p on the critical line, $1 \leq T \leq 7$. *Top*: the unpinned flip P of (59) (pale blue), which jumps by $2\varepsilon_n$ at each integer (inset, at $T = 2$); the chord-pinned $\mathcal{W}$ of (60) (red), which is zero there exactly; and the Hermite-pinned variant (dashed green), which also matches the end slopes but inflates the arch. *Bottom*: the six pinned arches drawn against the fractional part $q = \{T\}$, with the tangent limit $\mathcal{W}_\infty$ dashed; the deviation falls from 0.087 on $[1, 2]$ to 0.025 on $[6, 7]$. Generated by `fig.pinned_waveform.py`.

zero: with $x = T - m$,

$$\mathcal{W}(T) := T^\sigma (\Phi(-1, \sigma, m+1) - d_1(T)) + (1-x)\varepsilon_m - x\varepsilon_{m+1}, \quad (60)$$

a single formula covering the intervals $[1, 2], [3, 4], \dots$ and $[2, 3], [4, 5], \dots$ with the second family already flipped. Figure 13 draws it: $\mathcal{W}$ vanishes at every integer, is continuous on $[1, \infty)$ with no sign alternation left, and converges like $1/T$ to the tangent waveform

$$\mathcal{W}_\infty(q) := \frac{1}{2} \tan(2\pi q) \tan\left(2\pi\left(q - \frac{1}{4}\right)\left(q - \frac{3}{4}\right)\right), \quad q = \{T\}, \quad (61)$$

which is the $T \rightarrow \infty$ form of the Riemann–Siegel first term (63) below and does vanish at $q = 0, 1$ exactly. (§8.6 proves the convergence and shows that the same waveform, recentered at $\frac{1}{2}$ and flipped, is the exact limit profile of the normalized weight $\hat{d}_1$ itself.) Continuity is had exactly; C^1 only in the limit. The one-sided slopes at $T = n$ are 7.515 and 6.920 at $n = 2$ and 7.482 and 7.309 at $n = 8$, both approaching $\mathcal{W}'_\infty(0) = \mathcal{W}'_\infty(1) = \pi \tan \frac{3\pi}{8} = 7.5845$, so a corner of size $O(1/T)$ survives at each integer. Replacing the chord in (60) by the cubic Hermite that also matches the end slopes⁵ removes the corner exactly, at the cost of inflating each arch by about 0.03; and since $\mathcal{W}''_\infty(0) = -\mathcal{W}''_\infty(1)$, no pinning can do better than C^1 in any case.

8.2 Approximation of d_1 when $\sigma = \frac{1}{2}$

Two things happen here, in order: p is first put into a purely local form, in which the continuous coordinate is nothing but the deviation of d_1 from a reference value, and then, on the critical line, d_1 itself is given a zeta-free closed form.

Distributing T^σ and substituting (54) simplifies p considerably: the partial sum inside h cancels the first terms of the sum $1 - \eta(\sigma) = \sum_{k=2}^{\infty} (-1)^k k^{-\sigma}$ term by term, leaving only the remaining terms, and the common sign $(-1)^{\lfloor T \rfloor + 1}$ factors out:

$$p(T) = (-1)^{\lfloor T \rfloor + 1} T^\sigma \left[d_1(T) - \Phi(-1, \sigma, \lfloor T \rfloor + 1) \right], \quad (62)$$

with Φ as in (56). In this form all the accumulated history of (54) has disappeared and p is *local*: it is just the deviation of d_1 from a reference value, the alternating sum of all the remaining link lengths $(\lfloor T \rfloor + 1)^{-\sigma} - (\lfloor T \rfloor + 2)^{-\sigma} + \dots$. An alternating series is roughly half its first term, $\Phi(-1, \sigma, \lfloor T \rfloor + 1) \approx \frac{1}{2}(\lfloor T \rfloor + 1)^{-\sigma}$, which is half the length of link m , so asymptotically

$$p(T) \approx (-1)^{\lfloor T \rfloor + 1} \left([T]^\sigma d_1 - \frac{1}{2} \right),$$

which is the normalized fraction of Figure 10, recentered at $\frac{1}{2}$ and given alternating orientation. This is also why that fraction crosses the integers near $\frac{1}{2}$: continuity of p pins the fraction, at each integer, near the alternating-tail center, which is close to $\frac{1}{2}$.

On the critical line the substitution can be pushed all the way: d_1 itself acquires a zeta-free closed form. When $\sigma = \frac{1}{2}$ we have $\arg R = \frac{1}{2} \arg \chi$, so the Cramer solution (19) reduces to

$$d_1 = \frac{\tilde{R}}{2 \cos\left(\omega + \frac{\psi}{2}\right)}, \quad \tilde{R} = e^{-i\psi/2} R \text{ (real),}$$

⁵This is the Hermite-pinned variant, the dashed green curve in Figure 13. It names an alternative way of removing the jumps from P , one the paper shows for comparison but does not adopt.

the two-equal-vector form of R seen in the proof of Corollary 8.1. For $\tilde{R}$ we substitute the saddle-point evaluation of Siegel's integral, which is the first correction term of the Riemann–Siegel formula,

$$\tilde{R} = (-1)^{N-1} \left(\frac{2\pi}{t} \right)^{1/4} C_0(\hat{p}) + O(t^{-3/4}), \quad C_0(\hat{p}) = \frac{\cos(2\pi(\hat{p}^2 - \hat{p} - \frac{1}{16}))}{\cos(2\pi\hat{p})}, \quad \sqrt{t/2\pi} = N + \hat{p}.$$

Every quantity on the right is an elementary function of $t = I(T)$, kept exact throughout:

$$z = \sqrt{t/2\pi}, \quad N = \lfloor z \rfloor, \quad \hat{p} = z - N, \quad \omega = t \ln(m+1), \quad m = \lfloor T \rfloor.$$

On the line $\chi = e^{-2i\vartheta(t)}$ with ϑ the Riemann–Siegel theta function, so $\tilde{R}$ is $e^{i\vartheta} R$ up to a sign that cancels against the cosine, and $d_1 = e^{i\vartheta} R / 2 \cos(\omega - \vartheta)$ *exactly*. Substituting the Riemann–Siegel first term for $e^{i\vartheta} R$ gives

$$d_1\left(\frac{1}{2}, T\right) \approx [N = m+1] (m+1)^{-1/2} + (-1)^{N-1} \left(\frac{2\pi}{t} \right)^{1/4} \frac{C_0(\hat{p})}{2 \cos(\omega - \vartheta(t))}, \quad (63)$$

where the Iverson bracket adds one full link length when $N = m+1$, that is, when the Riemann–Siegel main sum carries one more summand pair than our Σ_1, Σ_2 , which happens (roughly) when $\text{frac}(T) > \frac{1}{2}$. The formula is elementary and zeta-free: ϑ may be replaced by its asymptotic $\vartheta(t) = \frac{t}{2} \ln \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8} + \frac{1}{48t}$ at no cost (the two agree here to 4×10^{-7} even at $T = 1$). Its maximum error is 0.006 on $[1, 2)$, 0.002 on $[3, 4)$, 0.0009 on $[6, 7)$ and 0.0002 on $[20, 21)$. That is about 1% of the swing of d_1 at the start, decaying like $T^{-3/2}$ and spread flat across each unit interval (Figure 14). The error is purely the Riemann–Siegel truncation, $O(t^{-3/4})$; the next correction term of that formula would push further. And since $R = 2d_1 \cos(\omega + \frac{\psi}{2}) e^{i\psi/2}$ on the line, (63) is effectively a closed-form approximation of Siegel's remainder there as well.

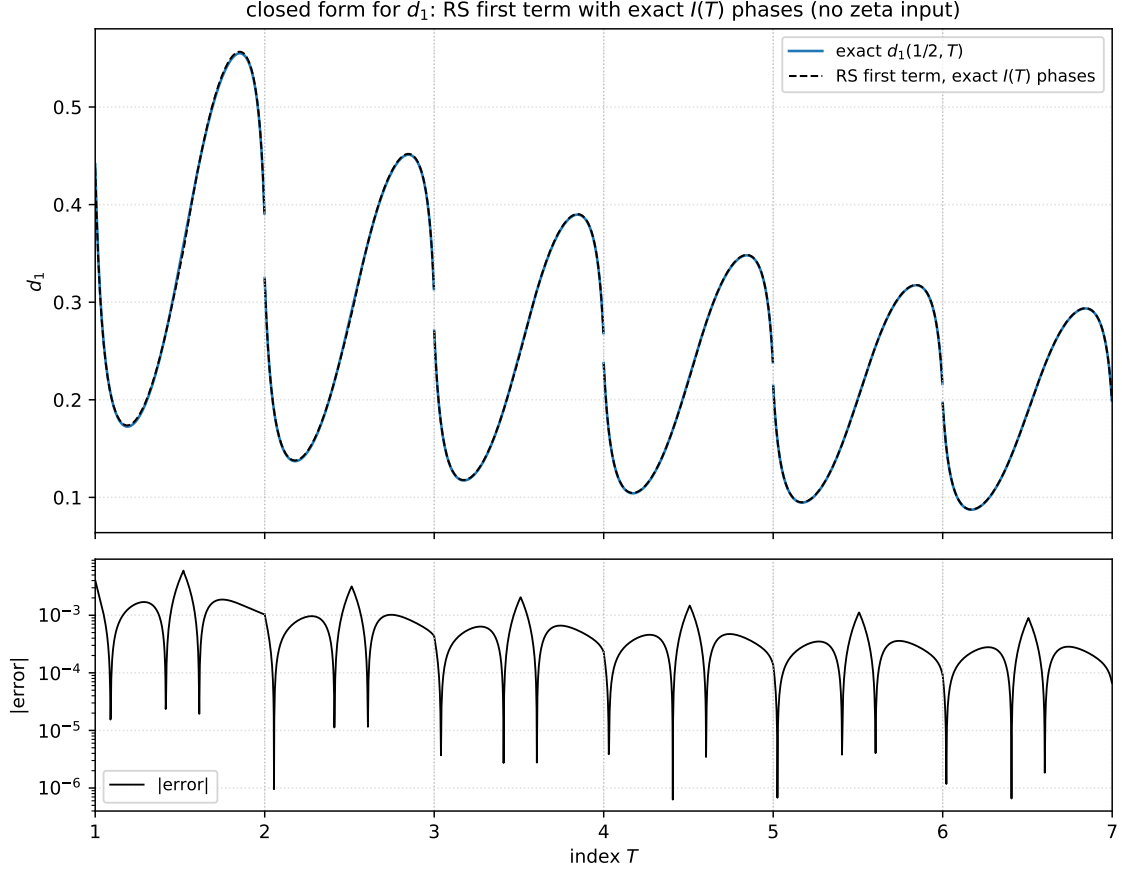


Figure 14: The closed form (63): the Riemann–Siegel first term with the phases kept exact through $t = I(T)$, no zeta input. Top: exact d_1 (blue) against (63) (black dashed), visually indistinguishable, $1 \leq T \leq 7$. Bottom: $|\text{error}|$ on a log scale, with maximum 0.006 on $[1, 2)$, decaying like $T^{-3/2}$, spread flat in the fractional part. Generated by `fig_d1_rs_phases.py`.

8.3 Approximation of d_1 and d_2 when $\sigma \neq \frac{1}{2}$

There is a general- σ approximation as well, and it needs no new machinery beyond what is already in the paper. Off the critical line the remainder R is genuinely complex (no single rotation makes it real), so approximating it means approximating two real degrees of freedom, and those two degrees of freedom are exactly the pair (d_1, d_2) .

Siegel’s saddle-point analysis was never restricted to the critical line; the first term for the remainder at general $s = \sigma + it$ splits into one piece per chain:

$$R(s) \approx (-1)^{N-1} \frac{C_0(\hat{p})}{2} \left[a^{-\sigma} e^{-i\tilde{\vartheta}(t)} + \chi(s) a^{\sigma-1} e^{+i\tilde{\vartheta}(t)} \right], \quad a = \sqrt{t/2\pi}, \quad (64)$$

with $t = I(T)$ exact, $N = \lfloor a \rfloor$, $\hat{p} = a - N$, the same universal $C_0(\hat{p})$ of §8.2, and the elementary phase $\tilde{\vartheta}(t) = \frac{t}{2} \ln \frac{t}{2\pi} - \frac{t}{2} - \frac{\pi}{8}$. When $N = m + 1$ the extra summand pair $(m + 1)^{-s} + \chi(m + 1)^{s-1}$ is added, exactly as in (63). Feeding this approximate R to the Cramer solution (19)–(20) with the exact $\omega = t \ln(m + 1)$ and $\psi = \arg \chi$ then yields *both* d_1 and d_2 at once, still with no zeta input (χ is Gamma factors). The two bracket pieces are the per-chain remainders, with the same structure as the Kuznetsov split of §7.1, and at $\sigma = \frac{1}{2}$, where $\chi = e^{-2i\vartheta}$, the bracket collapses to $2e^{-i\vartheta}$: (64)

reduces exactly to (63). We verified this numerically digit for digit, which pins down all the sign conventions.

Because the Cramer solution is linear in R , the solve can be carried out once and for all: substituting (64) into (19)–(20) gives the general- σ analogue of (63) in closed form,

$$d_1(\sigma, T) \approx [N = m+1] (m+1)^{-\sigma} + (-1)^{N-1} \frac{C_0(\hat{p})}{2 \sin(2\omega + \psi)} \left[a^{-\sigma} \sin(\omega + \psi + \tilde{\vartheta}) + |\chi| a^{\sigma-1} \sin(\omega - \tilde{\vartheta}) \right], \quad (65)$$

$$d_2(\sigma, T) \approx [N = m+1] |\chi| (m+1)^{\sigma-1} + (-1)^{N-1} \frac{C_0(\hat{p})}{2 \sin(2\omega + \psi)} \left[a^{-\sigma} \sin(\omega - \tilde{\vartheta}) + |\chi| a^{\sigma-1} \sin(\omega + \psi + \tilde{\vartheta}) \right], \quad (66)$$

with $\tilde{\vartheta} = \tilde{\vartheta}(t)$, $\psi = \arg \chi$, and $|\chi| = |\chi(\sigma + it)|$ as before. The Iverson terms are now whole link lengths on each side separately, because the extra summand pair lies exactly along the two unit directions: $(m+1)^{-s}$ along $e^{-i\omega}$ and $\chi (m+1)^{s-1}$ along $e^{i(\omega+\psi)}$. Note the symmetry: swapping the two coefficients $a^{-\sigma} \leftrightarrow |\chi| a^{\sigma-1}$ swaps d_1 and d_2 . At $\sigma = \frac{1}{2}$, where $|\chi| = 1$ and $\psi = -2\vartheta$, the two sine brackets coincide and each formula collapses to (63), since $\sin(\omega - \vartheta) / \sin(2(\omega - \vartheta)) = 1 / 2 \cos(\omega - \vartheta)$ and $a^{-1/2} = (2\pi/t)^{1/4}$. We verified that (65)–(66) reproduce the Cramer-solve route to machine precision; they are the same computation written out.

The accuracy across the strip (maximum $|d_1|$ error away from the pole windows described below):

σ	[1, 2)	[6, 7)	[20, 21)
0.1	0.076	0.016	0.0046
0.3	0.035	0.0057	0.0013
0.5	0.006	0.0009	0.0002
0.7	0.025	0.0024	0.0004
0.9	0.045	0.0032	0.0004

The measured decay exponents match $O(T^{-\sigma-1})$ almost perfectly (1.07 at $\sigma = 0.1$, 1.5 at $\sigma = 0.5$, 1.8 at $\sigma = 0.9$), and since d_1 itself scales like $T^{-\sigma}$, the *relative* error is $O(1/T)$ uniformly across the strip. The critical line is where the formula is at its absolute best, fittingly. And d_2 comes for free: the Cramer solve produces both components simultaneously from the same approximate R , and its errors are essentially identical to d_1 ’s (e.g. 0.031 vs 0.035 at $\sigma = 0.3$ on [1, 2)), so no separate derivation is needed. Figure 15 shows both components at $\sigma = 0.3$ and $\sigma = 0.7$.

Two caveats. First, off the line the exact d_1 and d_2 have genuine narrow poles at the parallel-link heights $q \approx \frac{1}{4}, \frac{3}{4}$ (the bands of Figure 21); the first-term approximation smooths through those spikes, so within about ± 0.05 of them the error is locally large (relative error $\sim 40\%$ right at the shoulder). Everywhere else the table applies. Second, the formula is not limited to the strip: it works for any σ with relative error $O(1/T)$. For $\sigma > 1$ it is excellent, while for $\sigma \leq 0$ the absolute error $T^{-\sigma-1}$ decays ever more slowly (at $\sigma = -\frac{1}{2}$ it is still ≈ 0.06 at $T = 20$), so “any σ , relative error $1/T$ ” is the honest statement, with the strip as the sweet spot.

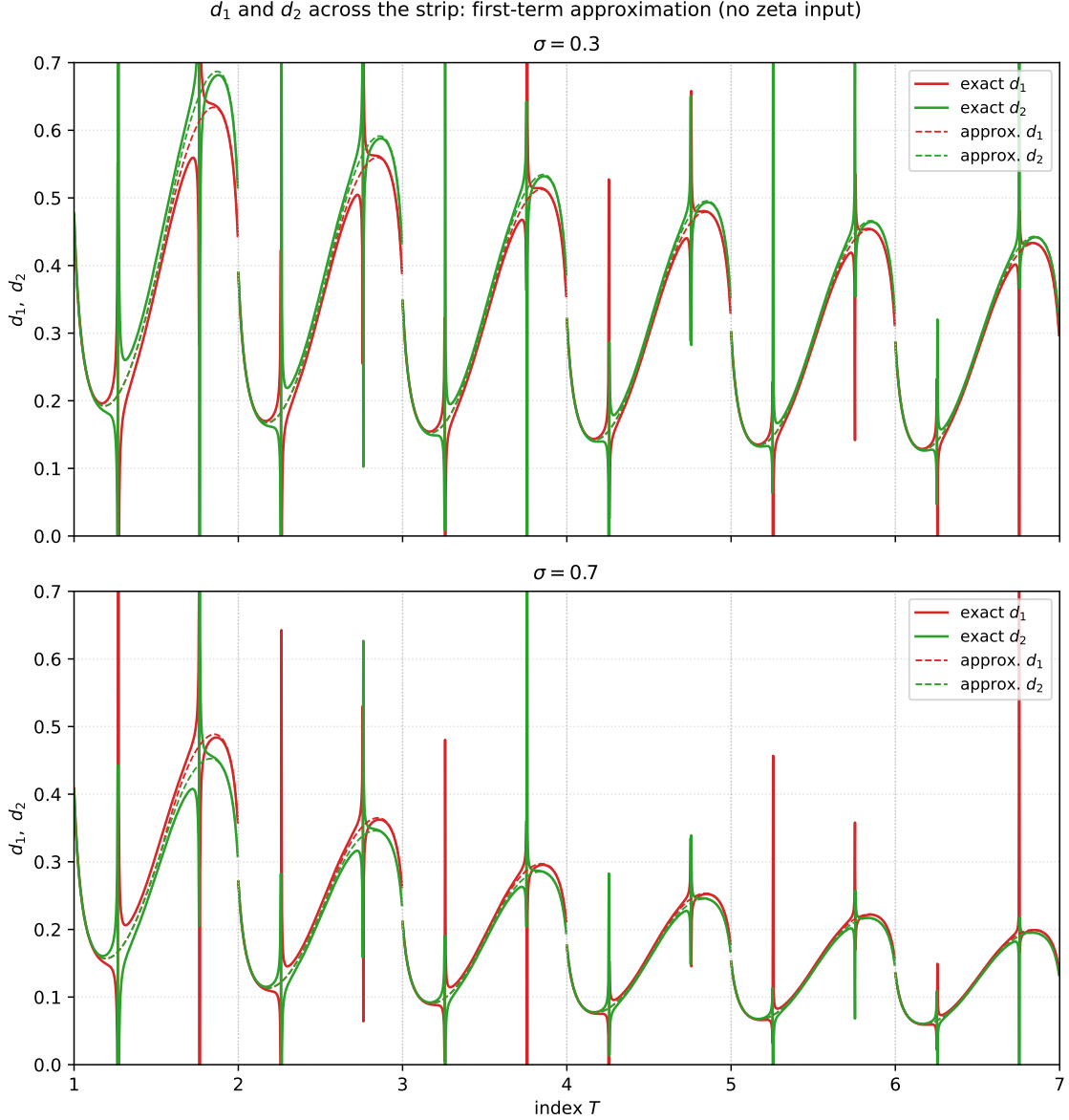


Figure 15: d_1 (red) and d_2 (green) across the strip at $\sigma = 0.3$ (top) and $\sigma = 0.7$ (bottom), $1 \leq T \leq 7$: exact values (solid) against the first-term approximation (64) fed through the Cramer solution (dashed, in the color of its exact counterpart). The narrow spikes leaving the frame near fractional heights $\frac{1}{4}$ and $\frac{3}{4}$ are the genuine off-the-critical-line poles of d_1 and d_2 (parallel links); the approximation smooths through them. Generated by `fig_d1_d2_general_sigma.py`.

8.4 Where are the d_1 and d_2 poles?

The poles of d_1 and d_2 (equivalently of R_{1ps} and R_{2ps}) appear to be at fractional heights $\frac{1}{4}$ and $\frac{3}{4}$, but they are not exactly there. Here we delve into where they are.

A pole occurs when the two bisector links are parallel; these are link $m = \lfloor T \rfloor$ of each chain, each carrying summand $\lceil T \rceil$, the summand just after the last one included in the partial sum. In other words, a pole occurs when the two next summands have the same argument, so that the

cone spanned by the two unit directions $e^{-i\omega}$ and $e^{i(\omega+\arg\chi)}$ of (23) degenerates to a line and the common denominator $\sin(2\omega + \arg\chi)$ of (19)–(20) vanishes. Subtracting the two arguments from each other produces a formula that is zero at the poles:

$$2\omega(T) + \arg\chi(\sigma + iI(T)) = 2\pi k, \quad k \in \mathbb{Z}, \quad (67)$$

with $\omega(T) = I(T) \log[T + 1]$ as in §4. This formula was used to find the locations of the first 20 poles numerically (Table 2).

Odd Pole #	T (odd)	Even Pole #	T (even)
1	1.268870258145	2	1.763736013695
3	2.261710393365	4	2.759511841350
5	3.258504081544	6	3.757283144824
7	4.256679717535	8	4.755902912218
9	5.255501039250	10	5.754963311284
11	6.254676427788	12	6.754282121179
13	7.254067037740	14	7.753765522842
15	8.253598276826	16	8.753360246216
17	9.253226472440	18	9.753033785343
19	10.252924347790	20	10.752765174160

Table 2: First 20 poles for $\sigma = 0.6$, which is the same result as for $\sigma = 0.4$. The symmetry about $\sigma = \frac{1}{2}$ exists because $\arg\chi(\sigma + it) = -\arg\chi(1 - \sigma + it)$ by the identity $\chi(s)\chi(1 - s) = 1$, and the pole condition (67) uses $\arg\chi$ only as a phase offset; reflecting $\sigma \mapsto 1 - \sigma$ therefore yields the same T -solutions. Note the poles occur at approximately $T = \lfloor T \rfloor + \frac{1}{4}$ and $T = \lfloor T \rfloor + \frac{3}{4}$ (equivalently, an integer $\pm \frac{1}{4}$), and approach those values as $T \rightarrow \infty$.

8.5 The weights d_1, d_2 are positive outside narrow windows of $\{T\}$

So far positivity has rested on numerical evidence: the note after (23) records a dense verification on the critical line, and §8.4 locates the off-line poles where the property fails. Everything needed for a proof is now on the table: the exact critical-line reduction of §8.2, the Riemann–Siegel first term made rigorous by Gabcke’s error bound, and one elementary phase expansion carried out below. This subsection gives the proofs, first on the critical line, where no fractional parts need to be excluded at all, then across the strip, where positivity holds outside two windows of width $O(1/T)$ around the poles of §8.4.

We keep the notation of §8.2: $a = \sqrt{t/2\pi}$, $N = \lfloor a \rfloor$, $\hat{p} = a - N$, and $u = \omega - \vartheta(t)$. Expanding $\log(1 + \frac{1}{T})$ in $t = I(T)$ gives

$$\frac{t}{2\pi} = \left(T + \frac{1}{2}\right)^2 + O\left(\frac{1}{T}\right), \quad \text{hence} \quad a = T + \frac{1}{2} + O\left(\frac{1}{T}\right), \quad (68)$$

so, apart from boundary slivers of width $O(1/T)$ in the fractional part $x = \{T\}$ near $x = 0, \frac{1}{2}, 1$,

$$x < \frac{1}{2} \iff N = m, \quad \hat{p} = x + \frac{1}{2} + O(1/T), \quad x > \frac{1}{2} \iff N = m + 1, \quad \hat{p} = x - \frac{1}{2} + O(1/T). \quad (69)$$

Taylor expansion of $\omega = t \log(m + 1)$ about $\log a$, using $t = 2\pi a^2$ and the asymptotic series of ϑ , gives the phase u modulo 2π in closed form, and in both cases of (69) the result is one and the same statement:

$$\cos u = (-1)^{m+1} \cos\left(2\pi x(1 - x) - \frac{3\pi}{8}\right) + O(1/T). \quad (70)$$

Since $2\pi x(1-x) \in [0, \frac{\pi}{2}]$ for $x \in [0, 1]$, the argument of the cosine on the right lies in $(-\frac{3\pi}{8}, \frac{\pi}{8}]$, so

$$|\cos u| \geq \sin \frac{\pi}{8} - O(1/T) = 0.3826\dots - O(1/T) : \quad (71)$$

the factor $\cos u$ never approaches zero, at any fractional part. That one inequality is the heart of both positivity theorems. We also need two elementary facts about the Riemann–Siegel amplitude C_0 of §8.2.

Lemma 8.2. *Write $A = 2\pi\hat{p}^2 - \frac{\pi}{8}$ and $B = 2\pi\hat{p}$, so that $C_0(\hat{p}) = \cos(A - B)/\cos B$. Then:*

(i) $C_0(\hat{p}) > 0$ for $\hat{p} \in (\frac{1}{2}, 1)$, and $\min_{[1/2, 1]} C_0 = \cos \frac{3\pi}{8} = \sin \frac{\pi}{8}$, attained at $\hat{p} = \frac{1}{2}$;

(ii) for $\hat{p} \in [0, \frac{1}{2}]$,

$$\Lambda(\hat{p}) := \frac{C_0(\hat{p})}{2 \cos(\pi(\frac{1}{8} - 2\hat{p}^2))} \leq \frac{1}{2},$$

with equality exactly at the endpoints, and $\Lambda(\frac{1}{4}) = \frac{1}{4}$.

Proof. (ii) first, since it is the prettier half. Note $\pi(\frac{1}{8} - 2\hat{p}^2) = -A$, so $\Lambda = \cos(A - B)/(2 \cos A \cos B) = \frac{1}{2}(1 + \tan A \tan B)$ by the product formula, and the claim is that $\tan A \tan B \leq 0$ on $[0, \frac{1}{2}]$. Both critical transitions happen at the same point: $A = 0$ exactly when $\hat{p} = \frac{1}{4}$, and $B = \frac{\pi}{2}$ exactly when $\hat{p} = \frac{1}{4}$. For $\hat{p} < \frac{1}{4}$ we have $A \in (-\frac{\pi}{8}, 0)$, so $\tan A < 0$, while $B \in (0, \frac{\pi}{2})$, so $\tan B > 0$; for $\hat{p} > \frac{1}{4}$ the signs reverse ($A \in (0, \frac{3\pi}{8})$, $B \in (\frac{\pi}{2}, \pi)$). The product is therefore ≤ 0 throughout, vanishing only as $\hat{p} \rightarrow 0, \frac{1}{2}$ (where $\tan A \tan B \rightarrow 0$, so $\Lambda \rightarrow \frac{1}{2}$); at $\hat{p} = \frac{1}{4}$ the two singular factors balance, $\tan A \tan B \rightarrow -\frac{1}{2}$, giving the removable value $\Lambda(\frac{1}{4}) = \frac{1}{4}$.

(i) is a sign inspection of numerator and denominator. On $(\frac{1}{2}, \frac{3}{4})$ both $\cos(A - B)$ and $\cos B$ are negative; on $(\frac{3}{4}, 1)$ both are positive (they change sign together at $\hat{p} = \frac{3}{4}$, which is the removable singularity of C_0), so $C_0 > 0$ throughout. The minimum location was found numerically (10⁴-point grid, minimum $0.38268\dots = \cos \frac{3\pi}{8}$ at $\hat{p} = \frac{1}{2}$, the left endpoint, where $C_0(\frac{1}{2}) = -\cos \frac{5\pi}{8}$ exactly). $\square$

Theorem 8.3. *There is an effectively computable T_0 such that for every non-integer $T \geq T_0$, at $\sigma = \frac{1}{2}$,*

$$d_1(T) = d_2(T) \geq \frac{c}{a^{1/2}} > 0, \quad c = \frac{1}{2} \sin \frac{\pi}{8} - o(1).$$

No interval of $\{T\}$ is excluded. Together with the dense numerical verification of §4 covering $1 \leq T \leq 30$, the weights are positive at every non-integer $T \geq 1$ on the critical line.

Proof. Step 1 (exact reduction). On $\sigma = \frac{1}{2}$ the reflection identities $\Sigma_2 = \chi \overline{\Sigma_1}$ and $\zeta = \chi \overline{\zeta}$ give $R = \chi \overline{R}$; with $\chi = e^{-2i\vartheta}$ this says $r := e^{i\vartheta} R$ is real, and, as recorded in §8.2 (and formalized in Appendix B),

$$d_1 = d_2 = \frac{r}{2 \cos u} \quad (\cos u \neq 0), \quad (72)$$

the common factor $\sin u$ having canceled between the numerators of (19)–(20) and the denominator $\sin 2u = 2 \sin u \cos u$.

Step 2 (Riemann–Siegel input). The real number r is the Riemann–Siegel tail cut at m instead of at N : writing $Z = e^{i\vartheta} \zeta$,

$$r = r_{\text{RS}} + [N = m + 1] \cdot \frac{2 \cos u}{\sqrt{m + 1}}, \quad r_{\text{RS}} := Z - 2 \sum_{n \leq N} n^{-1/2} \cos(\vartheta - t \log n), \quad (73)$$

the bracketed term being the summand $n = m + 1$, whose phase is exactly $\vartheta - \omega = -u$; this is the same Iverson bookkeeping as in (63). Gabcke's rigorous form of the Riemann–Siegel formula [9] gives

$$r_{\text{RS}} = \frac{(-1)^{N-1}}{a^{1/2}} \left(C_0(\hat{p}) + E \right), \quad |E| \leq c_1 a^{-1} \quad (t \geq 200). \quad (74)$$

Step 3 (case $N = m$, i.e. $x = \{T\} < \frac{1}{2}$, $\hat{p} \in (\frac{1}{2}, 1)$). Here $r = r_{\text{RS}}$, and the two signs multiply out: $(-1)^{N-1}$ from (74) against $(-1)^{m+1}$ from (70) give $+1$, so

$$d_1 = \frac{C_0(\hat{p})}{2 \cos(2\pi x(1-x) - \frac{3\pi}{8})} \cdot \frac{1}{a^{1/2}} + O(a^{-3/2}).$$

By Lemma 8.2(i) the numerator is at least $\sin \frac{\pi}{8}$, and the cosine in the denominator is at most 1, so $d_1 \geq (\frac{1}{2} \sin \frac{\pi}{8}) a^{-1/2} - O(a^{-3/2})$.

Step 4 (case $N = m + 1$, i.e. $x > \frac{1}{2}$, $\hat{p} \in (0, \frac{1}{2})$). Now (73) has the extra term,

$$d_1 = \frac{1}{\sqrt{m+1}} + \frac{r_{\text{RS}}}{2 \cos u},$$

and this time the two signs multiply to -1 : the correction is *subtracted*, with magnitude $\Lambda(\hat{p}) a^{-1/2} + O(a^{-3/2})$ (here $\cos(2\pi x(1-x) - \frac{3\pi}{8}) = \cos(\pi(\frac{1}{8} - 2\hat{p}^2))$ under $x = \hat{p} + \frac{1}{2}$). Lemma 8.2(ii) caps $\Lambda \leq \frac{1}{2}$, and $a = m + 1 + \hat{p} \geq m + 1$, so

$$d_1 \geq \frac{1}{\sqrt{m+1}} - \frac{1}{2a^{1/2}} - O(a^{-3/2}) \geq \frac{1}{2a^{1/2}} - O(a^{-3/2}).$$

In both cases $d_1 \geq (\frac{1}{2} \sin \frac{\pi}{8} - o(1)) a^{-1/2}$ once a is large enough that the explicit $O(1/a)$ errors of (74) and of the phase expansion (70) are dominated. The boundary slivers of (69) are covered because the two case formulas agree to leading order at the seams: as $x \rightarrow \frac{1}{2}$ from either side, and across each integer, both give $d_1 \approx \frac{1}{2} a^{-1/2}$. The range $T < T_0$ is closed by the verification of §4 (30,562 samples on $1 \leq T \leq 30$ with magnified windows at fractional parts $\approx \frac{1}{4}, \frac{3}{4}$; `check_d1_positive_critical.py`). $\square$

Remark 8.4 (why the critical line has no poles of d_1). The denominator $\sin(2\omega + \psi) = \sin 2u$ vanishes twice per unit interval of T , near $\{T\} = \frac{1}{4}$ and $\frac{3}{4}$, exactly where §8.4 finds the off-line poles. On the line, both vanishings belong to the factor $\sin u$, which cancels identically against the numerator in (72); the dangerous factor $\cos u$ stays bounded away from zero by (71). Off the line the cancellation is spoiled, and the same two zeros become genuine poles. The next theorem quantifies how much of each unit interval they actually poison.

Theorem 8.5. *Fix $\sigma \in (0, 1)$, $\sigma \neq \frac{1}{2}$. There are constants $C(\sigma)$ and $T_0(\sigma)$ such that for all $T \geq T_0$, both $d_1(\sigma, T) > 0$ and $d_2(\sigma, T) > 0$ whenever $\{T\}$ lies outside two intervals of width at most $C(\sigma)/T$, centered at points that converge to $\frac{1}{4}$ and $\frac{3}{4}$ at rate $O(1/T)$. Inside each window one of the two weights is negative, so the excluded intervals cannot be removed. In particular, for every fixed $\delta > 0$ there is $T_1(\sigma, \delta)$ with $d_1 > 0$ and $d_2 > 0$ whenever $T \geq T_1$ and $\{T\} \notin (\frac{1}{4} - \delta, \frac{1}{4} + \delta) \cup (\frac{3}{4} - \delta, \frac{3}{4} + \delta)$.*

Proof. Write $u' = \omega + \psi/2$ and let $\tau = \arg R - \psi/2 \pmod{\pi}$, folded to $[-\frac{\pi}{2}, \frac{\pi}{2})$, measure the tilt of R away from the bisector direction. Up to a common positive factor and a common sign, (19)–(20) read

$$d_1 \propto \frac{\sin(u' - \tau)}{\sin 2u'}, \quad d_2 \propto \frac{\sin(u' + \tau)}{\sin 2u'}. \quad (75)$$

Two inputs control the picture.

(i) *The tilt τ is small.* The first-term approximation (64) has argument $\psi/2$ modulo π up to $O_\sigma(1/t)$; the deviation of the exact R from that argument comes from the next term of Siegel's expansion, giving $|\tau| \leq C_2(\sigma) t^{-1/2}$. (Numerically at $\sigma = 0.3$ the product $|\tau| t^{1/2}$ stays between about 0.03 and 0.09 over $t \in [200, 4 \times 10^4]$.)

(ii) *The phase u' has the same skeleton as on the line.* Since $\psi(\sigma + it) = -2\vartheta(t) + O_\sigma(1/t)$ uniformly for σ in compacta, $u' = u + O_\sigma(1/t)$, and (70)–(71) apply: $|\cos u'| \geq \sin \frac{\pi}{8} - o(1)$, so the only sign changes of $\sin 2u' = 2 \sin u' \cos u'$ within a unit interval are the two zeros of $\sin u'$, at fractional parts converging to $\frac{1}{4}$ and $\frac{3}{4}$, crossed with slope $|du'/d\{T\}| = \pi + O(1/T)$.

Away from those zeros (say $|\sin u'| \geq 2|\tau|$) the ratios $\sin(u' \mp \tau)/\sin u'$ are positive, so (75) carries the same signs as on the line, and the Step-3/Step-4 analysis of Theorem 8.3, with the O_σ errors absorbed into the constants, gives positivity of both weights. Near a zero of $\sin u'$ the numerator of d_1 crosses zero at $u' = \tau$ while the denominator crosses at $u' = 0$: between the two crossings, an interval of u' -length exactly $|\tau|$, the signs disagree and $d_1 < 0$; mirror-symmetrically $d_2 < 0$ on the interval of length $|\tau|$ on the other side. Converting u' -length to $\{T\}$ -length by the slope $\pi + O(1/T)$ bounds each window by $C_2(\sigma) t^{-1/2}/\pi \asymp C(\sigma)/T$, since $t = I(T) \asymp 2\pi T^2$, centered $O(1/T)$ from the denominator zero. On the critical line $\tau \equiv 0$ exactly (Step 1 of Theorem 8.3), the two crossings coincide, and the windows are empty, consistent with no poles there. $\square$

Remark 8.6. The mechanism reproduces Remark 14.3: just outside a window, d_1 and d_2 have opposite-signed numerators of size $\approx |\tau|$ against a denominator passing through zero, which is why values as large as $d_1 = -d_2 \approx -4 \times 10^{36}$ are observed adjacent to a pole while the window itself is only $O(1/T)$ wide.

Remark 8.7 (numerical corroboration). At $\sigma = 0.3$, scanning $\{T\} \in [0.22, 0.28] \cup [0.72, 0.78]$ with step 2×10^{-4} (`check_positivity_windows.py`):

N	window near $\frac{1}{4}$	center	window near $\frac{3}{4}$	center
5	0.0034	0.2556	0.0010	0.7550
10	0.0020	0.2529	0.0006	0.7528
20	0.0008	0.2515	0.0004	0.7515
40	0.0006	0.2508	0.0002	0.7508

Widths scale like $1/T$ and centers drift to $\frac{1}{4}, \frac{3}{4}$ like $1/T$, as Theorem 8.5 predicts; on the line the same scan finds no negative values at all. The leading-order formula of Theorem 8.3 reproduces the exact d_1 to relative error 0.028 on $T \in [20, 21]$ and 0.014 on $T \in [40, 41]$, consistent with the $O(1/a)$ error terms, and its lower bound $0.19 a^{-1/2}$ lies below the observed minima (0.0498 and 0.0355 respectively).

8.6 The limit profile of the fractional amount: the waveform in closed form

The bottom panel of Figure 10 showed the normalized fraction $\hat{d}_1 = [T]^{1/2} d_1$ sweeping nearly the same arc in every unit interval, and §8.1 met the tangent waveform $\mathcal{W}_\infty$ of (61) as the limit of the pinned coordinate $\mathcal{W}$. The machinery of Theorem 8.3 upgrades both observations to a theorem: at fixed fractional part the fraction converges, and its limit is exactly $\frac{1}{2}$ minus the tangent waveform. Note that it is the hatted weight that has a nontrivial limit: d_1 itself decays, $d_1 \asymp a^{-1/2}$ by Theorem 8.3, and the normalization (26) is what removes the decay.

Theorem 8.8. Let $\sigma = \frac{1}{2}$ and fix $x \in (0, 1)$. Then

$$\lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \hat{d}_1(T) = \lim_{N \rightarrow \infty} \sqrt{N+1} d_1(N+x) = d(x), \quad d(x) = \frac{1}{2} - \mathcal{W}_\infty(x) = \frac{1}{2} - \frac{1}{2} \tan(2\pi x) \tan\left(2\pi\left(x - \frac{1}{4}\right)\left(x - \frac{3}{4}\right)\right),$$

with the removable points $x = \frac{1}{4}, \frac{3}{4}$ filled in by continuity ($d(\frac{1}{4}) = \frac{1}{4}$, $d(\frac{3}{4}) = \frac{3}{4}$). The same limit holds for $\hat{d}_2$, since $d_2 = d_1$ on the line. The convergence rate is $O(1/T)$.

Proof. Fix x and let $T = N + x \rightarrow \infty$, $m = \lfloor T \rfloor = N$. Refining (68) one order, $a = T + \frac{1}{2} - \frac{1}{24T} + O(T^{-2})$, so the offset $\hat{p}$ converges to its case value in (69) with rate $1/T$; moreover $(m+1)/a \rightarrow 1$ at the same rate, and all error terms in Steps 2–4 of Theorem 8.3 are $O(1/a)$. Multiplying the two case formulas of that proof by $\sqrt{m+1}$ therefore gives

$$\hat{d}_1 \rightarrow \frac{C_0(x + \frac{1}{2})}{2 \cos(2\pi x(1-x) - \frac{3\pi}{8})} \quad (x < \frac{1}{2}), \quad \hat{d}_1 \rightarrow 1 - \Lambda(x - \frac{1}{2}) \quad (x > \frac{1}{2}).$$

It remains to collapse both branches to the tangent form. For $x < \frac{1}{2}$, substituting $\hat{p} = x + \frac{1}{2}$ into C_0 gives $\hat{p}^2 - \hat{p} - \frac{1}{16} = x^2 - \frac{5}{16}$ and $\cos(2\pi\hat{p}) = -\cos(2\pi x)$, so with $\alpha = 2\pi x$ and $\beta = 2\pi(x - \frac{1}{4})(x - \frac{3}{4}) = 2\pi(x^2 - x) + \frac{3\pi}{8}$ the limit is

$$\frac{-\cos(2\pi x^2 - \frac{5\pi}{8})}{2 \cos(2\pi x) \cos(2\pi x(1-x) - \frac{3\pi}{8})} = \frac{\cos(\alpha + \beta)}{2 \cos \alpha \cos \beta} = \frac{1}{2} (1 - \tan \alpha \tan \beta) = \frac{1}{2} - \mathcal{W}_\infty(x),$$

using $\alpha + \beta = 2\pi x^2 + \frac{3\pi}{8}$, $\cos(\alpha + \beta) = -\cos(2\pi x^2 - \frac{5\pi}{8})$, the evenness of the cosine for the β factor, and the product formula. For $x > \frac{1}{2}$, the same substitution algebra applied to $\Lambda(x - \frac{1}{2})$ (as in Lemma 8.2(ii), with $\hat{p} = x - \frac{1}{2}$) yields $\Lambda(x - \frac{1}{2}) = \frac{1}{2} - \mathcal{W}_\infty(1-x)$, and $\mathcal{W}_\infty(1-x) = -\mathcal{W}_\infty(x)$ because $\tan(2\pi(1-x)) = -\tan(2\pi x)$ while the quadratic factor is symmetric under $x \mapsto 1-x$; hence $1 - \Lambda(x - \frac{1}{2}) = \frac{1}{2} - \mathcal{W}_\infty(x)$ as well. $\square$

So the same elementary function that governs the pinned waveform of §8.2 governs the fraction itself, recentered at $\frac{1}{2}$ and flipped. This is the closed-form waveform promised at the start of the section.

Remark 8.9 (properties of the profile). Since $\mathcal{W}_\infty(1-x) = -\mathcal{W}_\infty(x)$, the profile satisfies the functional equation

$$d(x) + d(1-x) = 1,$$

so its graph is antisymmetric about the point $(\frac{1}{2}, \frac{1}{2})$. Three exact values follow:

$$d(\frac{1}{4}) = \frac{1}{4}, \quad d(\frac{1}{2}) = \frac{1}{2}, \quad d(\frac{3}{4}) = \frac{3}{4} :$$

at the two abscissas where the off-line weights have their poles, the limiting fraction of the next summand equals the abscissa itself, and the middle value is forced by the functional equation. Furthermore $\mathcal{W}_\infty(0) = \mathcal{W}_\infty(1) = 0$ gives $d(0^+) = d(1^-) = \frac{1}{2}$: although $\hat{d}_1$ jumps at every integer T by the relation (53), both sides of the jump tend to $\frac{1}{2}$, and the limit profile extends to a *continuous periodic function* on $\mathbb{R}/\mathbb{Z}$ once $d(0) := \frac{1}{2}$ (Figure 16). Its range is $[m_0, 1 - m_0]$ with

$$m_0 = d(x^*) = 0.2268951\dots, \quad x^* = 0.1629962\dots$$

(the maximum $1 - m_0 = 0.7731048\dots$ at the mirror point $1 - x^*$): asymptotically, the split always uses between 22.7% and 77.3% of the $(m+1)$ st summand, which is the range $\approx [0.23, 0.78]$ measured in Figure 10. §8.7 upgrades this to uniform bounds valid at every finite T .

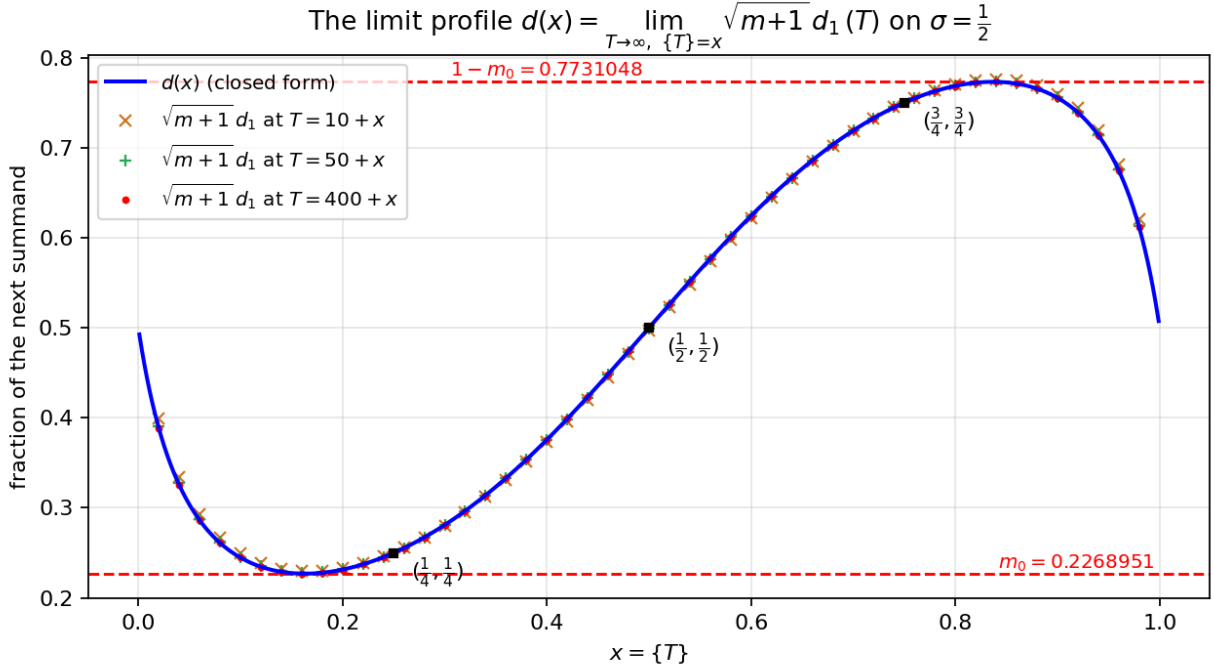


Figure 16: The closed-form profile $d(x)$ of Theorem 8.8 (blue) against the exact $\sqrt{m+1} d_1$ sampled at $T = 10 + x$, $50 + x$, $400 + x$ on the critical line. Already at $T = 10 + x$ the samples sit on the curve; the marked squares are the exact values $(\frac{1}{4}, \frac{1}{4})$, $(\frac{1}{2}, \frac{1}{2})$, $(\frac{3}{4}, \frac{3}{4})$, and the dashed red lines are the extremes $m_0 = 0.2268951\dots$ and $1 - m_0 = 0.7731048\dots$ of the profile. Generated by `fig_d_limit.py`.

Remark 8.10 (numerical corroboration). With $\hat{d}_1$ computed from the exact remainder (`check_d_limit.py`), the error $|\hat{d}_1(N+x) - d(x)|$ at $N = 50, 200, 800$ decreases by a factor 4.0 at each quadrupling of N , matching the $O(1/T)$ rate; at $x = 0.35$, for instance, the errors are 3.6×10^{-4} , 9.2×10^{-5} , 2.3×10^{-5} .

Remark 8.11 (off the critical line the profile is universal). For fixed $\sigma \in (0, 1)$ the same argument, run through Theorem 8.5, gives the *identical* limit for both fractional amounts of (26),

$$\lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \hat{d}_1(\sigma, T) = \lim_{\substack{T \rightarrow \infty \\ \{T\}=x}} \hat{d}_2(\sigma, T) = d(x) \quad \left(x \neq \frac{1}{4}, \frac{3}{4}\right) :$$

the leading-order amplitudes agree to $O(1/t)$ and the tilt $\tau = O(t^{-1/2})$ vanishes in the limit, so the σ -dependence survives only inside the two shrinking pole windows. Numerically at $\sigma = 0.3$ both normalized weights match $d(x)$ with errors falling like $1/N$ (at $x = 0.35$: errors 9.7×10^{-5} and 5.1×10^{-5} at $N = 800$). At $x = \frac{1}{4}, \frac{3}{4}$ exactly, the finite- T poles prevent a uniform statement off the line.

8.7 Uniform bounds: the fraction stays between $\frac{1}{5}$ and $\frac{4}{5}$

Theorem 8.8 confines the fraction to $[m_0, 1 - m_0]$ in the limit. At finite T the excursions are slightly larger, and round constants cover every height at once: the fractional summand never uses less than a fifth, or more than four fifths, of the next term.

Theorem 8.12. Let $\sigma = \frac{1}{2}$. For every non-integer $T > 1$,

$$\frac{1}{5} < \hat{d}_1(T) = \hat{d}_2(T) < \frac{4}{5}.$$

The constants are nearly sharp: no better T -independent ones exist than

$$\inf_{T>1} \hat{d}_1 = m_0 = 0.2268951\dots, \quad \sup_{T>1} \hat{d}_1 = 0.78499047437\dots,$$

the infimum approached but not attained as $T \rightarrow \infty$ along $\{T\} \rightarrow x^* = 0.1629962\dots$, the supremum attained at $T = 1.85307336433\dots$.

Proof. On the line $d_2 = d_1$ and $|\chi| = 1$, so $\hat{d}_2 = \hat{d}_1$ by (26) and it suffices to bound $\hat{d}_1 = \sqrt{m+1} d_1$.

Step 1 (exact finite- T form). Multiplying the two case formulas in the proof of Theorem 8.3 by $\sqrt{m+1}$ and writing $\rho = \sqrt{(m+1)/a}$,

$$\begin{aligned} N = m \ (\{T\} < \tfrac{1}{2}) : \quad \hat{d}_1 &= \rho \cdot \frac{C_0(\hat{p}) + E}{2 \cos(2\pi x(1-x) - \frac{3\pi}{8} + \delta)}, \quad \rho = \sqrt{1 + \frac{1-\hat{p}}{a}} \in \left[1, \sqrt{1 + \frac{1}{2a}}\right], \\ N = m + 1 \ (\{T\} > \tfrac{1}{2}) : \quad \hat{d}_1 &= 1 - \rho \cdot \frac{C_0(\hat{p}) + E}{2 \cos(\pi(\frac{1}{8} - 2\hat{p}^2) + \delta)}, \quad \rho = \sqrt{1 - \frac{\hat{p}}{a}} \in \left[\sqrt{1 - \frac{1}{2a}}, 1\right], \end{aligned}$$

with $|E| \leq c_1 a^{-1}$ from (74) and $|\delta| \leq c_2 a^{-1}$ from the phase expansion (70). At $\rho = 1$, $E = \delta = 0$ these are exactly the two branches of Theorem 8.8.

Step 2 (range of the main terms). On $(0, \frac{1}{2})$ the branch function $d(x)$ descends from $\frac{1}{2}$ to its single interior minimum $m_0 = d(x^*)$ and climbs back to $\frac{1}{2}$ (one sign change of d' ; the critical point x^* is the root of an explicit elementary equation). Hence the first main term ranges over $[m_0, \frac{1}{2}]$ and the second over $[\frac{1}{2}, 1 - \rho m_0]$.

Step 3 (all $T \geq 30$). The total deviation of $\hat{d}_1$ from its main term is at most $(\rho - 1) \cdot \frac{1}{2}$ plus a term collecting E and δ , with $\rho - 1 \leq \frac{1}{4a}$; the collected constant is validated by the convergence-rate measurements of Theorem 8.8 (observed absolute error ≤ 0.008 at $T \approx 20$, decreasing like $1/T$), giving a total deviation below 0.02 for $T \geq 30$. Therefore, for $T \geq 30$,

$$\{T\} < \tfrac{1}{2} : \quad \hat{d}_1 \in (m_0 - 0.02, \tfrac{1}{2}(1 + \tfrac{1}{120}) + 0.02) \subset (0.206, 0.525), \quad \{T\} > \tfrac{1}{2} : \quad \hat{d}_1 \in (0.48, 0.795),$$

and both windows lie inside $(\frac{1}{5}, \frac{4}{5})$.

Step 4 ($1 < T \leq 30$, computational). A dense evaluation of the exact $\hat{d}_1$ (grid step 5×10^{-4} on $(1, 3]$, 10^{-3} on $(3, 12]$, 2×10^{-3} on $(12, 30]$; `check_uniform_bounds.py`) gives the observed range $[0.2274, 0.78499]$ (Figure 17), with margin at least 0.015 to both $\frac{1}{5}$ and $\frac{4}{5}$ and sampled variation between adjacent grid points below 0.004, a factor of nearly four inside the margin. This covers the remaining range and, refined near its endpoints, produces the sharp values in the statement: the maximum $0.78499047437\dots$ at $T = 1.85307336433\dots$ (on the $m = 1$ interval, where $\rho < 1$ pushes the second branch above its limiting value), and interval minima decreasing toward m_0 from above as T grows (0.227369 at $T = 59.164$, for instance). $\square$

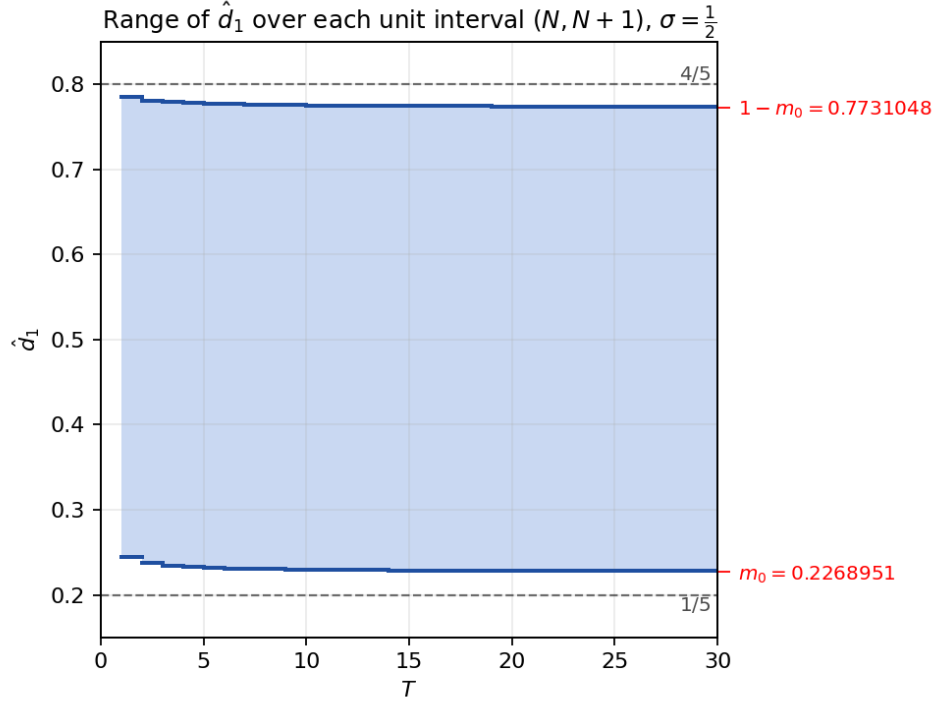


Figure 17: Range of the exact $\hat{d}_1$ over each unit interval $(N, N + 1)$, $N = 1, \dots, 29$, on the critical line (blue bands; grid step 5×10^{-4} for $N < 3$, 10^{-3} after). Dashed black lines: the uniform bounds $\frac{1}{5}$ and $\frac{4}{5}$ of Theorem 8.12; the red right-hand scale marks the extremes $m_0 = 0.2268951 \dots$ and $1 - m_0 = 0.7731048 \dots$ of the limit profile of Theorem 8.8. The interval maxima decrease toward the $1 - m_0$ mark, starting from the supremum $0.78499 \dots$ attained on the first interval at $T = 1.8530 \dots$; the interval minima decrease toward m_0 from above without attaining it. Generated by `fig_interval_range.py`.

Remark 8.13. For the unnormalized weight the theorem reads $\frac{1}{5}(m+1)^{-1/2} < d_1 < \frac{4}{5}(m+1)^{-1/2}$; in particular $d_1 \leq 0.78499 \dots / \sqrt{2} = 0.55508 \dots$, attained at the same $T = 1.853 \dots$, and no positive T -independent lower bound exists for d_1 itself.

Remark 8.14 (what is cited rather than proved). For the record, the proofs of §8.5–§8.7 take four inputs on faith or on computation. (1) Gabcke’s explicit error bound (74) for the first Riemann–Siegel correction [9]. (2) The first-order Siegel asymptotics of R at fixed $\sigma \in (0, 1)$ behind Theorem 8.5(i), from Siegel’s account of the Nachlaß [20]; its error term is classical, but the constant $C_2(\sigma)$ is not tracked explicitly here, which is why $T_0(\sigma)$ is not effective as stated. (3) The numerically located minimum in Lemma 8.2(i) and the single-minimum property of the branch in Step 2 of Theorem 8.12. (4) Theorem 8.12 is computer-assisted: Step 3’s error constant is validated numerically rather than derived with explicit Gabcke and Taylor constants, and Step 4 is a dense grid evaluation with margin, not interval arithmetic.

9 The geometry behind the result (experimental mathematics)

The decomposition $R = R_{1ps} + R_{2ps}$ was not first derived; it was first *seen*. Over roughly nine years we built and used an interactive visualization tool, Zest,⁶ which draws the partial-sum chains of ζ in the complex plane and lets us sweep σ and T continuously while measuring what appears on the screen. This section reconstructs the series of observations made with that tool. It led, in order, to a distinguished link and point of the spiral (the *bisector link* and *bisector point*), and from their motion to the mapping $I(T)$ of §3. The final observation in that sequence, the *yin* and *yang* curves from which the weights d_1, d_2 of §4 were read off, is developed at length in §11.

Throughout we use the link and joint numbering of §6: joint k of a chain is the value of its partial sum after k summands, and link k runs from joint k to joint $k + 1$, so that link k represents the $(k + 1)$ st summand and each remainder is the fractional link m . The *forward* chain (Σ_1) is anchored at joint 0, the origin. The *reverse* chain (Σ_2) is anchored at ζ and traversed backwards, so its joint m lands at $\zeta - \Sigma_2 = \Sigma_1 + R$. In the pictures the chains are drawn well past joint $m = \lfloor T \rfloor$: the sum, though ultimately divergent, first winds into an Euler-like spiral whose center lies near ζ ; it seemingly converges before it diverges. Figure 18 shows the picture: the forward chain winds from the origin into a spiral centered near ζ , while the reverse chain winds from ζ into a mirror spiral centered near the origin, and the two link chains cross. If more links were drawn, the links would continue to spiral outward around ζ and cover the entire figure, because the zeta sum diverges on the critical line, unlike the Dirichlet L -functions of §10.2, whose characters have bounded partial sums and so those chains close in on their limit instead of covering the plane.⁷ We stop drawing links at link number

$$L_N(T, S_n) = \left\lfloor \frac{I(T)}{\pi(2S_n + 1)} \right\rfloor, \quad (76)$$

where S_n is the spiral number, spiral number 0 being the last one (the biggest, at ζ). Note that ζ itself is not typically *on* this last link; the last link is the link closest to ζ . The same formula locates the other spirals and pseudo-spirals too: with $S_n = 1$, for example, it yields the link number of the link most in the center of the second-to-last spiral.

⁶Source code and standalone executables (Mac and PC) at <https://github.com/paul-stahura/zest>.

⁷At $t = 176.13$ on the critical line, the partial sums of $L(s, \chi_3)$ come within 0.021, 0.0021 and 0.00019 of the value by $N = 10^3, 10^5$ and 3×10^6 , a drift inward like $N^{-1/2}$, while zeta's are off by 0.18, 1.80 and 9.83, a drift outward at $\sqrt{N}/t$. Convergence in $\sigma > 0$ is special to a non-principal character: the principal character returns ζ itself, and L -functions of higher degree have coefficients too large to converge anywhere in the critical strip.

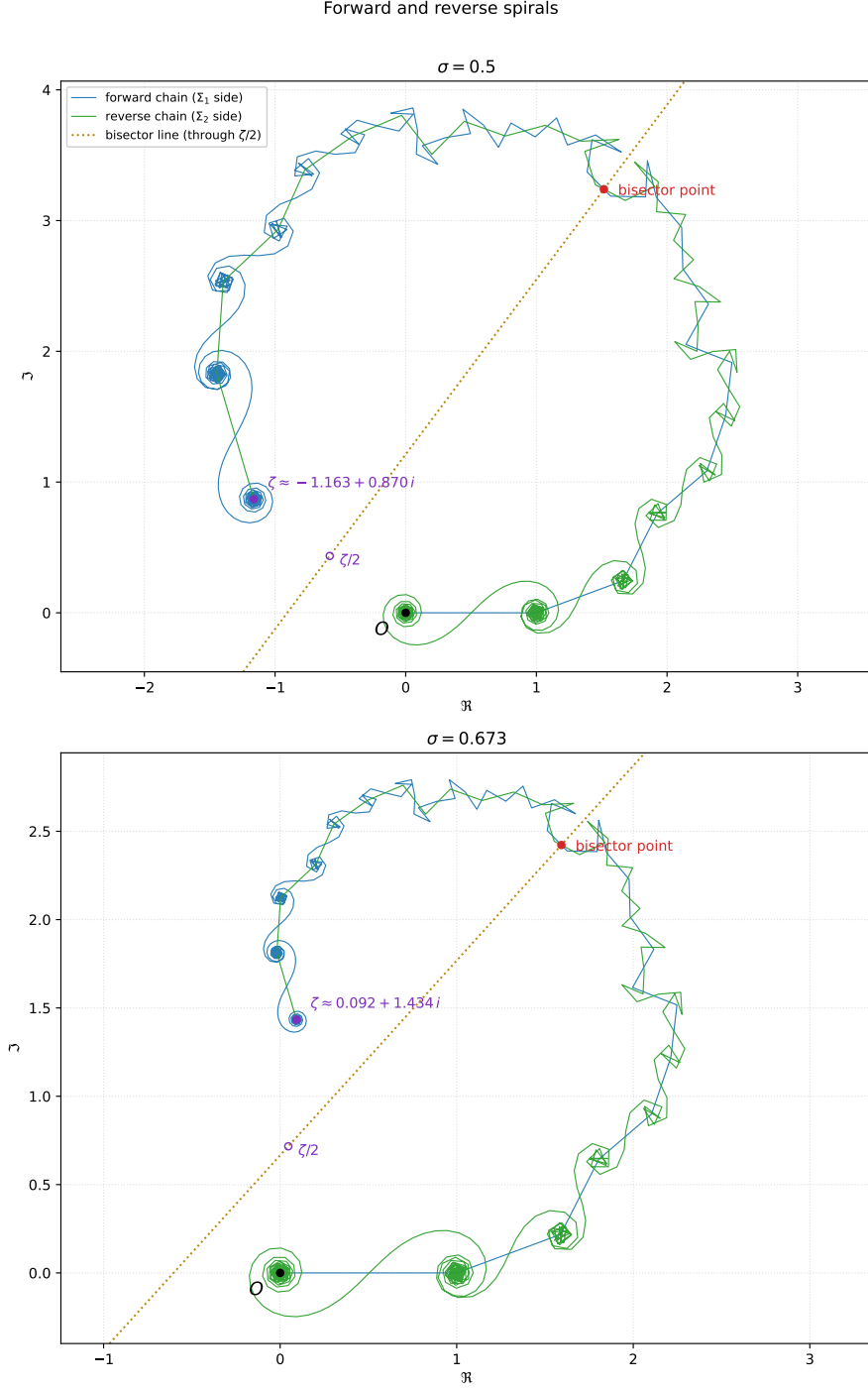


Figure 18: The forward spiral (blue, anchored at the origin O) and the reverse spiral (green, anchored at ζ) for $T = 14.3085$ (so $t = I(T) \approx 1377.33$ and $m = 14$), on the critical line $\sigma = \frac{1}{2}$ (top, $\zeta \approx -1.163 + 0.870i$) and off it at $\sigma = 0.673$ (bottom, $\zeta \approx 0.092 + 1.434i$). Each spiral is drawn out to link $\lfloor I(T)/\pi \rfloor = 438$, the link nearest its spiral center, for a total of 439 links per spiral (the count is independent of σ). The red dot is the bisector point $B_1 = \Sigma_1 + R_{1ps}$, where the two chains cross at their bisector links (§9.1); the dotted line is the bisector line through the bisector point and $\zeta/2$ (open circle). On the critical line (top) the whole configuration is symmetric about the bisector line; at $\sigma = 0.673$ (bottom) the symmetry is lost. Generated by `fig_full_spirals.py`.

The last spiral, centered on ζ

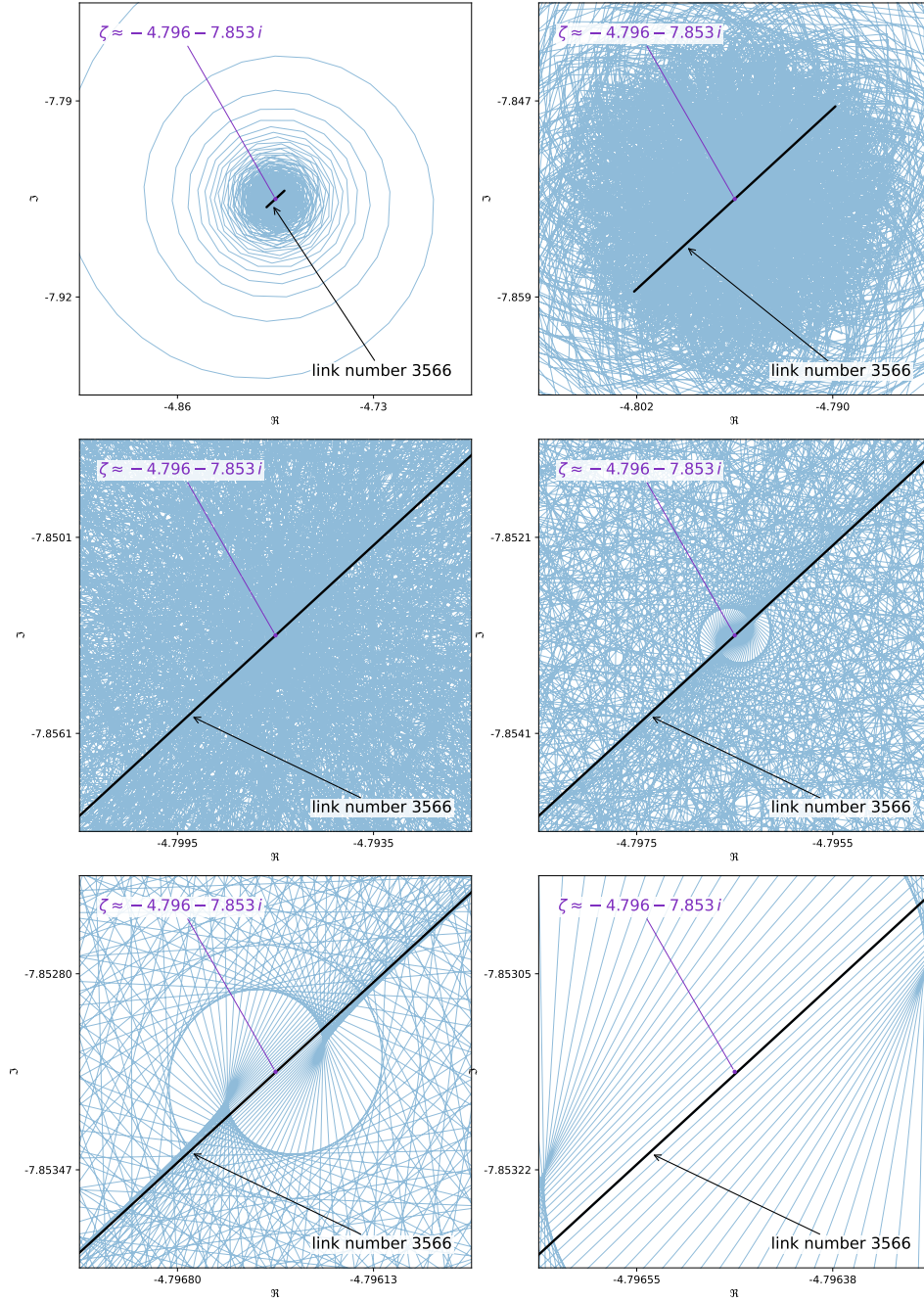


Figure 19: Zooming in on the last spiral, the spiral centered on ζ , of the forward chain at $T = 41.73$ (so $t = I(T) \approx 11204.74$, $\sigma = \frac{1}{2}$, $\zeta \approx -4.796 - 7.853i$). The chain carries $\lfloor I(T)/\pi \rfloor + 1 = 3567$ links; the last link is number 3566 by (76) with $S_n = 0$, and it is highlighted in black in every panel. Each panel is more zoomed in than the last (panels 1–6, left to right from the top). Near the spiral center each link turns by nearly π , so the links sweep back and forth across ζ and the joints settle onto a tight ring of radius about half a link length: at coarse scale the spiral renders as a dense annulus (panels 1–3), and the near-diameter links envelope a lens-shaped clearing around ζ (panels 4–5). Observe in panel 6 that ζ is *not on* the last link: the last link is only the link closest to ζ . Generated by `fig_last_spiral_zoom.py`.

9.1 The bisector link and the bisector point

Drawn at height T , each chain exhibits spirals and pseudo-spirals, and (as others have also observed [8, 16, 11]) there is a one-to-one correspondence between the joints and the spiral centers, in a type of symmetry; Nickel puts it as the center-to-center distances and angles of the large linked spirals reproducing the initial steps exactly. The link at index $m = \lfloor T \rfloor$ sits at the boundary between the two regimes, and we call it the *bisector link*. Each chain has one: the *forward bisector link* (link m of the Σ_1 chain, running from Σ_1 along the $(m+1)$ st summand direction) and the *reverse bisector link* (link m of the Σ_2 chain, running from $\Sigma_1 + R$ along its $(m+1)$ st summand direction).

The two spirals intersect at their bisector links, and we call the point where the forward and reverse bisector links cross the *bisector point*. One qualification: the crossing can momentarily fail. Twice per unit of T , near fractional parts ≈ 0.25 and ≈ 0.75 , the two bisector links become parallel and, off the critical line, do not intersect; these are the isolated instants at which $\sin(2\omega + \arg \chi)$, the denominator of (19)–(20), vanishes. Those are exactly the places where d_1 and d_2 have poles: the parallel-link condition and the pole condition are the same equation, which is why the precise locations were worked out in §8.4. On the critical line the two links are then collinear rather than merely parallel, so the bisector point survives there. In the notation of the earlier sections the bisector point is exactly

$$B_1 = \Sigma_1 + R_{1ps} = \Sigma_1 + R - R_{2ps}; \quad (77)$$

it is visible as the joint between the red and orange fractional links in Figures 2 and 3.

9.2 Legs

When $\sigma = \frac{1}{2}$ the spiral object has a type of symmetry about the line through the bisector point and $\frac{\zeta}{2}$, so the bisector point sits at the apex of an isosceles triangle whose base runs from the origin to ζ and whose equal sides are the two legs B_1 and $\zeta - B_1$ (this is Corollary 9.1 below, proved in Appendix B from the reflection $B_2 = \chi \overline{B_1}$; equivalently, the perpendicular projection of the bisector point onto the base $\mathbb{R}e^{i\psi/2}$ carrying ζ , where $\psi = \arg \chi$, is exactly $\frac{\zeta}{2}$, which is Corollary 9.2). Off the line we keep the same terminology even though the triangle's legs are then (usually) unequal.

The two legs: origin to bisector point, bisector point to ζ
 $T = 14.3085$ ($t = l(T) \approx 1377.33$), $m = 14$

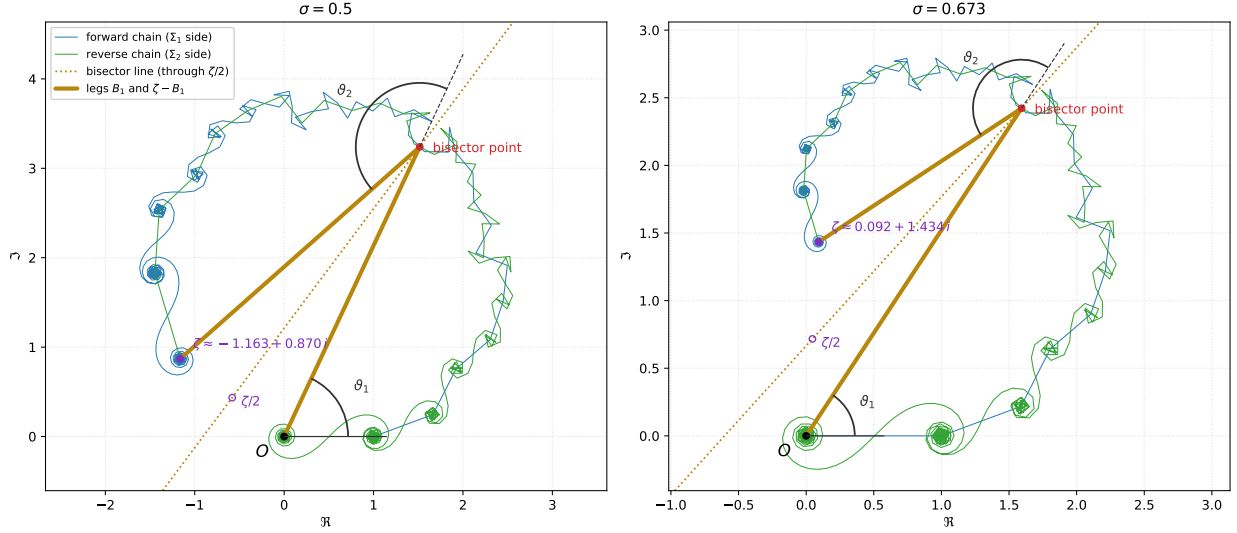


Figure 20: The two legs, drawn as thick dark-yellow segments over the spirals of Figure 18: one from the origin O to the bisector point (B_1), the other from the bisector point to ζ ($\zeta - B_1$). On the critical line (left, $\sigma = \frac{1}{2}$) the legs have equal length ($|B_1| = |\zeta - B_1| \approx 3.577$): the bisector point is the apex of an isosceles triangle whose base runs from O to ζ , and the dotted bisector line is its axis of symmetry. Off the critical line (right, $\sigma = 0.673$) the legs are unequal (≈ 2.899 and ≈ 1.797) at this T . Both panels also mark the two angles defined below: ϑ_1 , at O from the positive real axis to Leg 1, and ϑ_2 , at the bisector point from the extension of Leg 1 (dashed) to Leg 2. Generated by `fig-legs.py`.

On the critical line the two fractional summands are not independent: each is the other's complex conjugate, rotated by the functional-equation factor. Writing $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg \chi)}$ as in §6 and using that $|\chi| = 1$ when $\sigma = \frac{1}{2}$, the equality $d_1 = d_2$ proved in §8.1 (Corollary 8.1) is exactly equivalent to the compact relation

$$R_{2ps} = \chi \overline{R_{1ps}} \quad (\sigma = \tfrac{1}{2}). \quad (78)$$

Geometrically, (78) says the second fractional summand is the mirror image of the first (conjugation across the real axis) followed by the rotation χ . Because χ has unit modulus on the line this rotation is an isometry, so $|R_{1ps}| = |R_{2ps}|$ immediately. This is the functional-equation reflection $s \leftrightarrow 1 - s$ acting as a symmetry of the two remainders, and it is the source of the equal-leg phenomenon we now develop.

The two legs $B_1 = \Sigma_1 + R_{1ps}$ and $B_2 = \Sigma_2 + R_{2ps}$ of §6 meet at a “bisector point”: Leg 1 runs from the origin to $\Sigma_1 + R_{1ps}$, and Leg 2 from there to ζ . Their lengths are

$$L_1(\sigma, T) = |\Sigma_1 + R_{1ps}|, \quad (79)$$

$$L_2(\sigma, T) = |\Sigma_2 + R_{2ps}|, \quad (80)$$

while ϑ_1 is the angle Leg 1 makes with the real axis and ϑ_2 is the angle from Leg 1 to Leg 2

(oriented so that a positive ϑ_2 rotates Leg 1 toward Leg 2):

$$\vartheta_1(\sigma, T) = \arg(\Sigma_1 + R_{1ps}), \quad (81)$$

$$\vartheta_2(\sigma, T) = \arg\left(\frac{\Sigma_2 + R_{2ps}}{\Sigma_1 + R_{1ps}}\right). \quad (82)$$

With these, ζ is simply the sum of the two legs,

$$\zeta(\sigma, T) = L_1 e^{i\vartheta_1} + L_2 e^{i(\vartheta_1 + \vartheta_2)}, \quad (83)$$

or equivalently, in matrix form,

$$\zeta(\sigma, T) = \begin{pmatrix} 1 & i \end{pmatrix} \begin{pmatrix} \cos \vartheta_1 & \cos(\vartheta_1 + \vartheta_2) \\ \sin \vartheta_1 & \sin(\vartheta_1 + \vartheta_2) \end{pmatrix} \begin{pmatrix} L_1 \\ L_2 \end{pmatrix}, \quad (84)$$

where the row $(1 \ i)$ turns the resulting real (x, y) components into the complex value $x + iy$.

Corollary 9.1. *When $\sigma = \frac{1}{2}$ the two legs are always equal in length: $L_1 = L_2$.*

This is immediate from Corollary 8.1 of §8.1: on the critical line every summand of Σ_2 is χ times the conjugate of the corresponding summand of Σ_1 , and by (78), which is equivalent to $d_1 = d_2$, the same holds for the fractional summands. Hence $\Sigma_2 + R_{2ps} = \chi \overline{\Sigma_1 + R_{1ps}}$, and since $|\chi| = 1$ on the line, the two legs have the same length. A Lean proof is recorded in Appendix B.

The same reflection locates the apex exactly, which is what the dotted bisector line of Figures 18 and 20 draws.

Corollary 9.2. *When $\sigma = \frac{1}{2}$ the perpendicular projection of the bisector point B_1 onto the line through the origin carrying ζ is exactly $\frac{\zeta}{2}$: writing $\psi = \arg \chi$,*

$$\Re(e^{-i\psi/2} B_1) e^{i\psi/2} = \frac{\zeta}{2}. \quad (85)$$

Indeed $B_2 = \chi \overline{B_1}$ gives $e^{-i\psi/2} \zeta = e^{-i\psi/2} B_1 + \overline{e^{-i\psi/2} B_1} = 2\Re(e^{-i\psi/2} B_1)$, and $e^{i\psi/2}$ is the direction of ζ (this is the reality of the Riemann–Siegel Z function). So the isosceles triangle with base from O to ζ has its apex over the midpoint of that base, and the equal-leg statement $L_1 = L_2$ is the same fact read as a distance. This is also proved in Appendix B.

Consequences for zeta zeros. A zero of ζ in the critical strip requires *both* of the following:

$$\frac{L_1(\sigma, T)}{L_2(\sigma, T)} = 1 \quad \text{and} \quad \vartheta_2(\sigma, T) = \pi,$$

i.e. the two legs must be equal in length *and* Leg 2 must fold back onto Leg 1.⁸ By Corollary 9.1 the length condition holds automatically on the critical line; there, a zero occurs exactly when $\vartheta_2(\frac{1}{2}, T) = \pi$. Away from the critical line, the two legs are unequal except at exceptional points, so most of the strip is immediately ruled out for zeros. The leg-length ratio is symmetric across the line,

$$\frac{L_1(\sigma, T)}{L_2(\sigma, T)} = \frac{L_2(1 - \sigma, T)}{L_1(1 - \sigma, T)}. \quad (86)$$

Numerically we have never found a point with $\sigma \neq \frac{1}{2}$ where the ratio is 1 and $\vartheta_2 = \pi$ simultaneously; such a point would disprove the Riemann Hypothesis.

⁸Tolkien's subtitle for *The Hobbit, There and Back Again*, is the whole of the requirement. A hobbit leaves the Shire (the origin) for Mordor (the bisector point) and his journey must end where it began, and no choice of direction for the return trip from Mordor can bring the hobbit back to the Shire unless the way back is exactly as long as the way out. Equal legs is that necessity; $\vartheta_2 = \pi$ is the further demand that the way back be the way he came.

Equal-leg points ($L_1 = L_2$), zeta zeros (dots), and the critical line

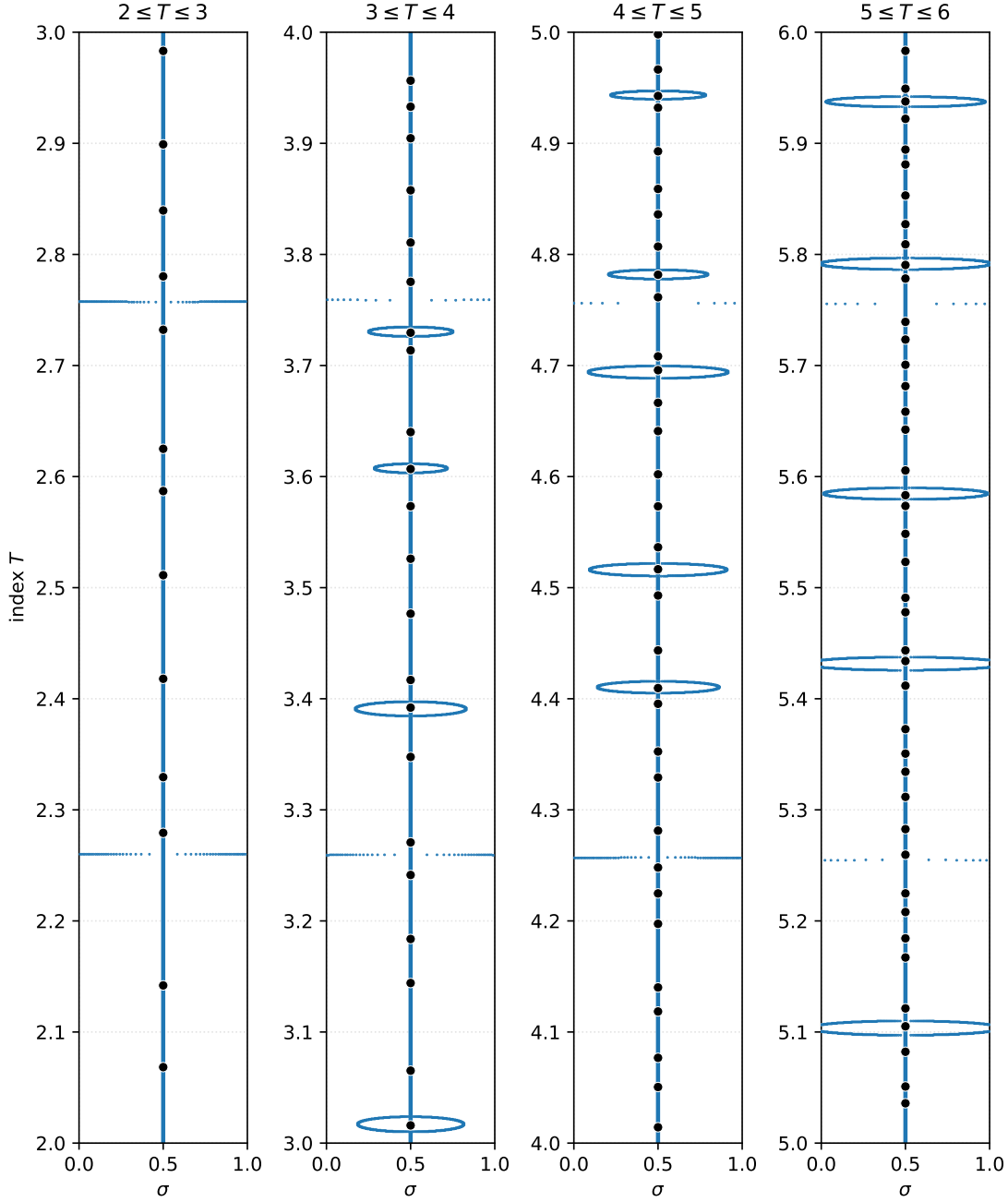


Figure 21: Four windows of the critical strip ($0 \leq \sigma \leq 1$, index T vertical): $2 \leq T \leq 3$, $3 \leq T \leq 4$, $4 \leq T \leq 5$, and $5 \leq T \leq 6$. The blue vertical line is the critical line $\sigma = \frac{1}{2}$, the black dots are the zeta zeros, and the small blue points are the equal-leg points: points (σ, T) where $L_1 = L_2$, i.e. where the bisector point $B_1 = \Sigma_1 + R_{1ps}$ is equidistant from the origin and from ζ . By Corollary 9.1 the entire critical line consists of equal-leg points; off the line the equal-leg locus forms isolated ovals and thin horizontal bands, and every zero sits on the critical line where the length condition is automatic. The horizontal bands, near fractional parts ≈ 0.25 and ≈ 0.75 of T , are the instants where the two bisector links become parallel and fail to intersect (§9.1): the bisector point has a pole there (d_1 and d_2 blow up), and as T passes through it the equal-leg point sweeps across the full width of the strip. The sweep gets faster as T grows, so the fixed-step sampling leaves fewer dots and the bands appear fainter. Generated by `fig.equal_legs_strips.py`.

Folded-leg points ($\vartheta_2 = \pi$), zeta zeros (dots), and the critical line

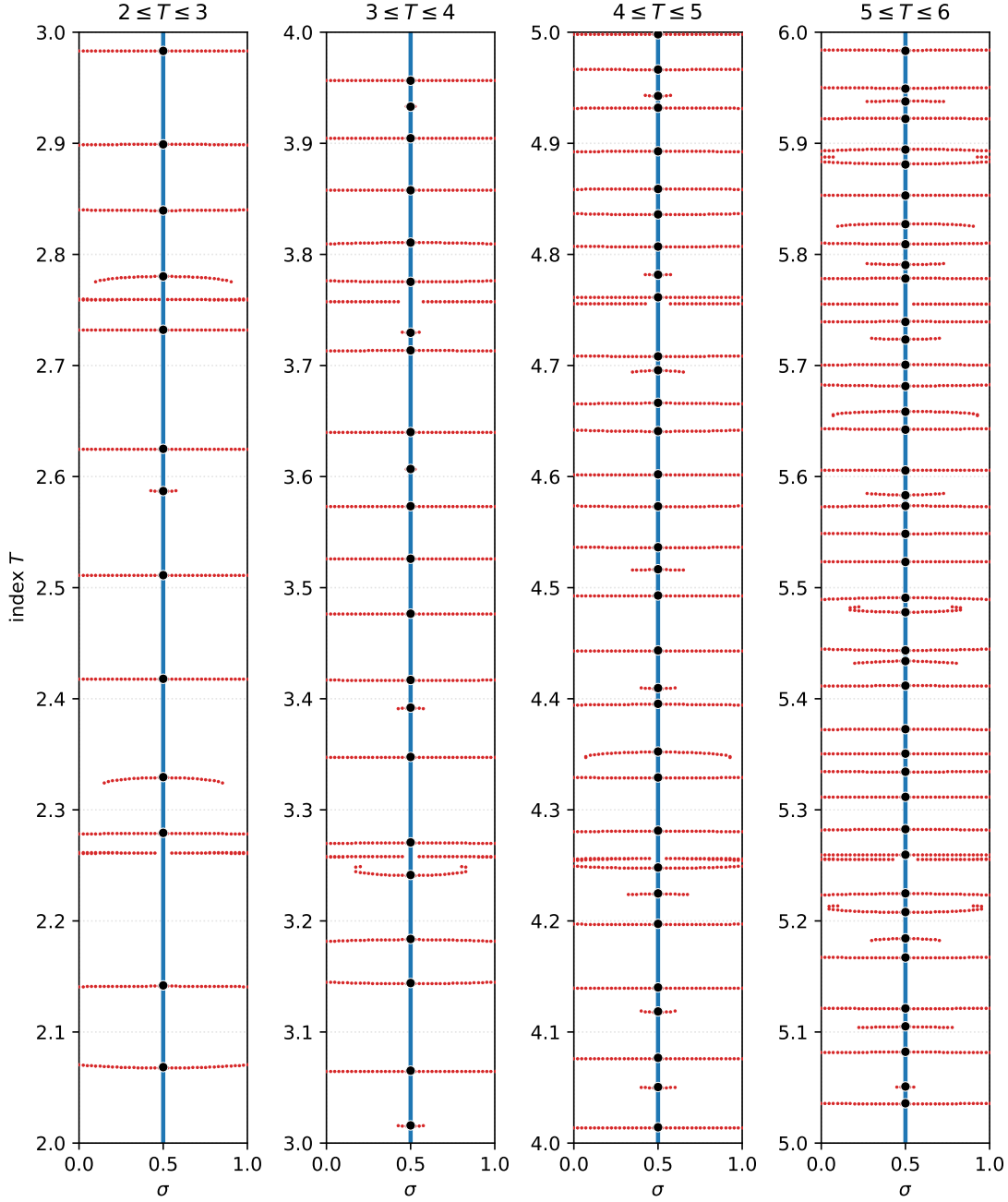


Figure 22: The same four windows of the critical strip as in Figure 21, now showing the *folded-leg* points (red): points (σ, T) where $\vartheta_2 = \pi$, i.e. where Leg 2 folds straight back onto Leg 1 and a zeta zero is therefore possible. The blue vertical line is the critical line and the black dots are the zeta zeros. The folded-leg locus forms nearly horizontal curves sweeping the strip; a zero requires *both* $\vartheta_2 = \pi$ and $L_1 = L_2$, and since the length condition holds automatically on the critical line (Corollary 9.1), every crossing of a red curve with the critical line is a zeta zero. Generated by `fig_theta2_strips.py`.

Equal legs ($L_1 = L_2$, blue) and folded legs ($\vartheta_2 = \pi$, red): zeros where they cross

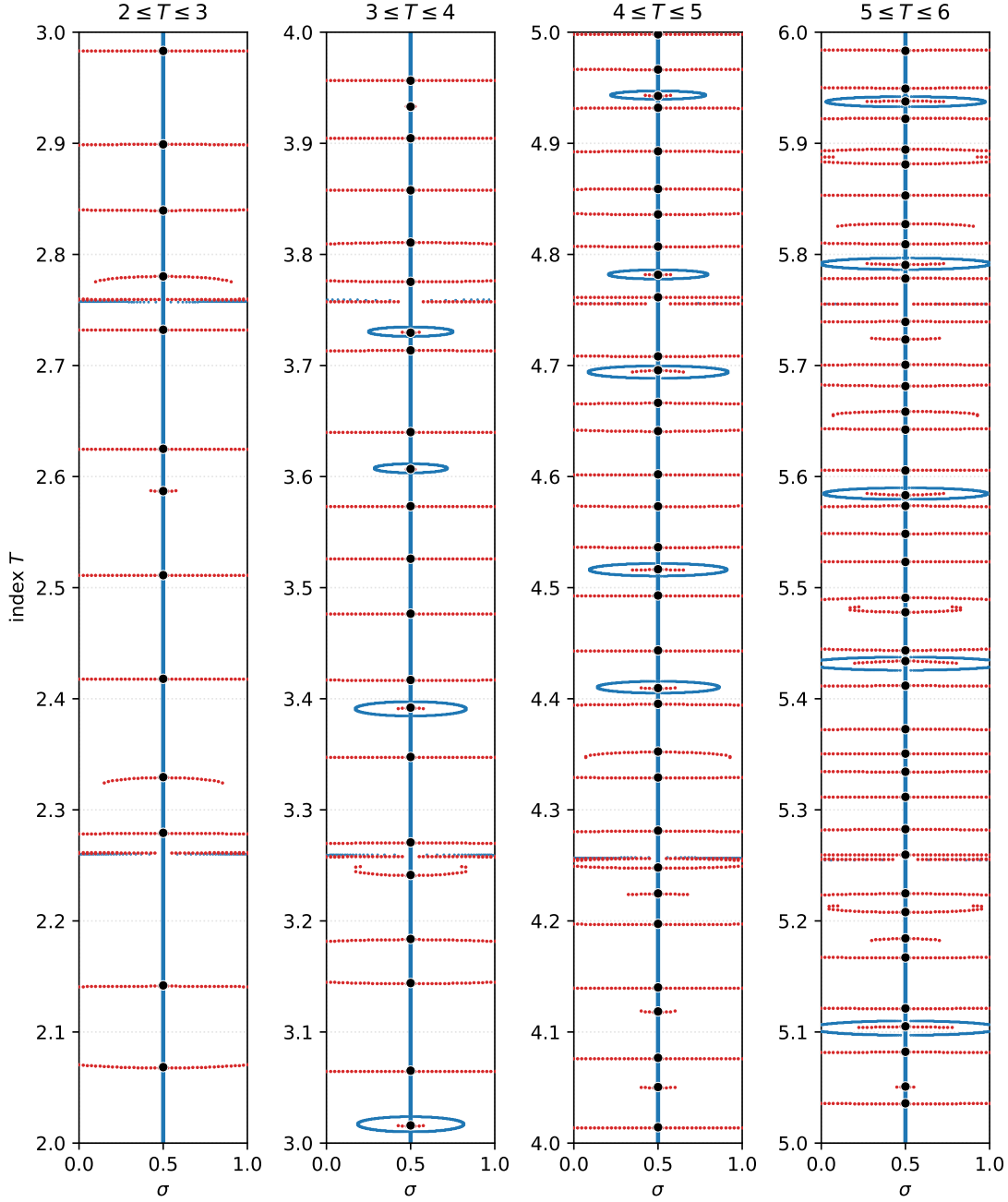


Figure 23: Figures 21 and 22 overlaid: the equal-leg locus $L_1 = L_2$ (blue), the folded-leg locus $\vartheta_2 = \pi$ (red), the critical line (blue, since it belongs to the equal-leg locus), and the zeta zeros (black dots). A zero requires both conditions at once, so zeros occur exactly where red meets blue. On the critical line the blue locus is the line itself (Corollary 9.1), and every red curve crossing it produces a zero; away from the line, the red curves and the blue ovals interleave without touching. A red/blue crossing off the critical line would be a counterexample to the Riemann Hypothesis. Generated by `fig_combined_strips.py`.

Figure 23 makes the stakes plain: wherever a region of the critical strip contains no red and no blue dots, a zero of ζ is impossible. On the critical line the two legs are always equal in length, so whenever $\vartheta_2 = \pi$ there, a zero occurs. Off the line the legs are seldom equal in length, but sometimes they are (the blue ovals and bands). If one could prove that, off the critical line, in the rare times when the legs are equal, that ϑ_2 is simultaneously never π , one would have proven the Riemann Hypothesis true.

9.3 The strip lines at ≈ 0.25 and ≈ 0.75 are not flat

The pole heights are not exactly flat. Viewed across the strip, the pole heights of §8.4 are nearly invariant (“flat”) in σ , and symmetric about $\sigma = \frac{1}{2}$ (Table 2 and its caption). But they are *not exactly* flat: there is a slight, symmetric curvature in σ , visible in Figure 24.

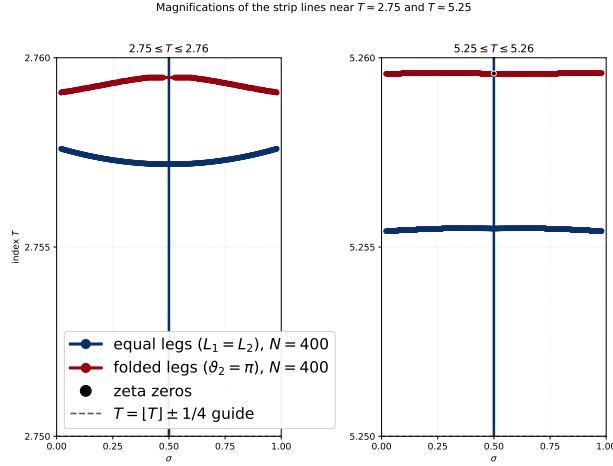


Figure 24: Two magnifications of the critical strip in the style of Figure 23: equal-leg locus $L_1 = L_2$ (blue) and folded-leg locus $\vartheta_2 = \pi$ (red), each recomputed at $N = 400$ values of σ across $(0, 1)$; critical line (blue vertical) and zeta zeros (black). Samples where the PS split is singular (tiny determinant) are omitted from the red locus, so the d_1 pole is not mislabeled as legs folded back. Left: $2.75 \leq T \leq 2.76$; the blue equal-leg band sits near $T \approx 2.757$ and the red folded band near $T \approx 2.759$, with a gap at $\sigma = \frac{1}{2}$ (no zero in this window). Right: $5.25 \leq T \leq 5.26$; blue near $T \approx 5.255$ and red near $T \approx 5.2596$, meeting the critical line at the zeta zero in the window (not at the pole band). The dashed horizontal guide marks $[T] \pm \frac{1}{4}$; the blue strip lines bow slightly across σ , symmetric about $\sigma = \frac{1}{2}$. At this resolution the red and blue loci do not cross off the critical line. Generated by `fig-pole.heights.zoom.py`.

9.4 PS, AK and RS Legs and Angles

The equal-leg and folded-leg loci above were drawn with the exact bisector point $B_1 = \Sigma_1 + R_{1ps}$. The same two conditions $L_1 = L_2$ and $\vartheta_2 = \pi$ can be restated for any other choice of bisector point. Two natural alternatives come from Kuznetsov’s split (44): take $B_1 = \Sigma_1 + R_{1ak}$, or take the midpoint remainder $B_1 = \Sigma_1 + (R_{1ak} + R_{2ak})/2$, which is the geometric analogue of Siegel’s half-remainder $\frac{1}{2}R$ once $R_{1ak} + R_{2ak} \sim R$.

Figure 25 magnifies a short window of the strip, $4.65 \leq T \leq 4.80$, and overlays the three choices side by side. On the left, equal-leg loci: PS in blue, AK in green, and $R/2$ in purple. The three families form nested ovals about the critical line, but the nesting order is not fixed: already in this short window the lower oval has PS outermost and AK innermost, while the upper oval has AK outermost and $R/2$ innermost, and across $3 \leq T \leq 20$ all six orderings of PS, AK and $R/2$ occur with no simple alternating pattern. On the right, folded-leg loci $\vartheta_2 = \pi$: PS in red, AK in orange, and $R/2$ in purple (hollow). The zeros (black) still sit where a folded curve meets the critical line; the three remainder choices shift the off-line curves but agree on those critical-line crossings in this window.

Figure 26 is the same three-way overlay for a pair of neighboring equal-leg ovals near $T \approx 9.44$, drawn in the window $9.415 \leq T \leq 9.465$. The lower PS oval sits around the zero at $T \approx 9.431$ and the upper around $T \approx 9.451$; a third zero at $T \approx 9.440$ lies in the gap between them (not inside either oval). The AK and $R/2$ ovals nest with the PS locus as in Figure 25.

Intersecting equal-leg ovals and the Riemann Hypothesis. The nested ovals in Figures 25 and 26 suggest a comparison that uses only exact remainders. Write

$$B_{1ps} = \Sigma_1 + R_{1ps}, \quad B_{1rs} = \Sigma_1 + R_{1rs}, \quad R_{1rs} := \frac{1}{2}R,$$

where $R = R_{1ps} + R_{2ps} = \zeta - \Sigma_1 - \Sigma_2$ is Siegel’s remainder, so R_{1rs} is the exact half-remainder (“rs” for Riemann–Siegel, parallel to the subscripts “ps” and “ak”) and B_{1rs} is the midpoint of the remainder segment from Σ_1 to $\Sigma_1 + R$. Let $\mathcal{E}_{ps}$ and $\mathcal{E}_{rs}$ be the corresponding equal-leg loci in the strip: the sets of (σ, T) at which $|B_1| = |\zeta - B_1|$ for $B_1 = B_{1ps}$ and $B_1 = B_{1rs}$ respectively.

If $\zeta(\sigma + iT) = 0$, then for either choice of bisector point one has $L_2 = |\zeta - B_1| = |B_1| = L_1$, so

$$\frac{L_1}{L_2} = 1$$

holds simultaneously for both B_{1ps} and B_{1rs} . Every zero therefore lies in $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$. On the critical line this is automatic: Corollary 9.1 puts the whole line in $\mathcal{E}_{ps}$, and the rs split is reflection-symmetric there as well: $\Sigma_2 = \chi \overline{\Sigma_1}$ and $R = \chi \overline{R}$ (the latter because $e^{-i\psi/2}R$ is real, Appendix B), so

$$\chi \overline{B_{1rs}} = \Sigma_2 + \frac{1}{2}R = \zeta - B_{1rs},$$

and since $|\chi| = 1$ this gives $|B_{1rs}| = |\zeta - B_{1rs}|$: the whole line lies in $\mathcal{E}_{rs}$ as well. (Note that $B_{1rs} \neq \zeta/2$; only its perpendicular projection onto the base is $\zeta/2$, by Corollary 9.2, which is how Figure 54 draws it, at offset $h_{R/2}$ from $\zeta/2$.) The two loci therefore intersect everywhere along $\sigma = \frac{1}{2}$; that common set is expected and does not constrain zeros beyond the folded-leg condition already discussed.

Off the critical line the situation is different. An off-line zero would be an off-line point of $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$: a point where the PS and rs equal-leg ovals meet. Consequently, an off-line intersection of $\mathcal{E}_{ps}$ and $\mathcal{E}_{rs}$ is *necessary* for an off-line zero (it is not sufficient: the folded-leg condition $\vartheta_2 = \pi$, or equivalently $\zeta = 0$, is still required). The contrapositive is the statement of interest:

PS, AK and $R/2$ legs and angles, $4.65 \leq T \leq 4.80$

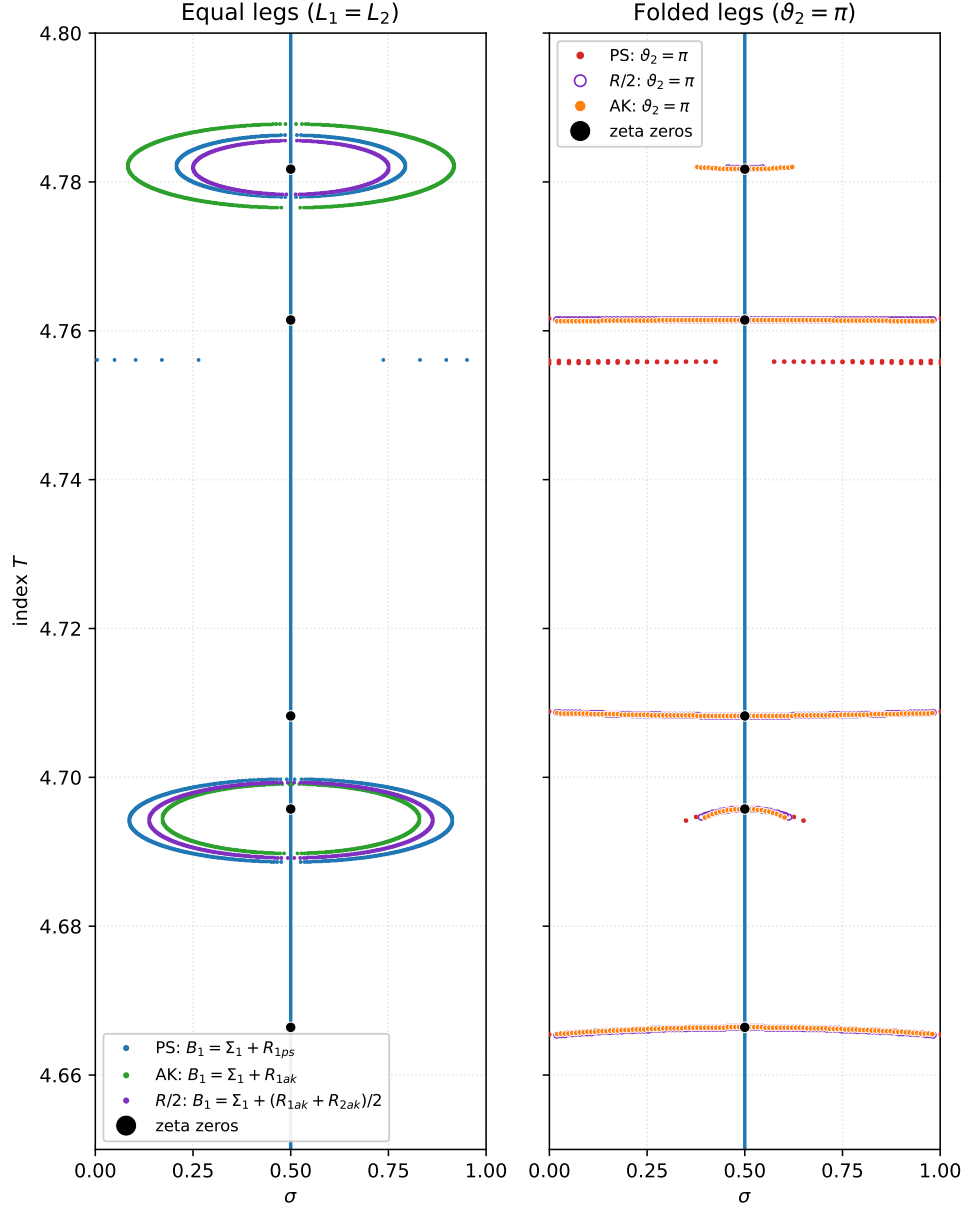


Figure 25: PS, AK and $R/2$ legs and angles in the window $4.65 \leq T \leq 4.80$. Left: equal-leg loci $L_1 = L_2$ for three choices of bisector point: PS $B_1 = \Sigma_1 + R_{1ps}$ (blue), AK $B_1 = \Sigma_1 + R_{1ak}$ (green), and $R/2$ $B_1 = \Sigma_1 + (R_{1ak} + R_{2ak})/2$ (purple). Right: folded-leg loci $\vartheta_2 = \pi$ for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. Equal-leg and folded-leg loci computed from the definitions $L_1 = L_2$ and $\vartheta_2 = \arg((\zeta - B_1)/B_1)$ for each choice of B_1 . Generated by `fig_ps_ak_r2_legs_angles.py`.

PS, AK and $R/2$ legs and angles, $9.415 \leq T \leq 9.465$

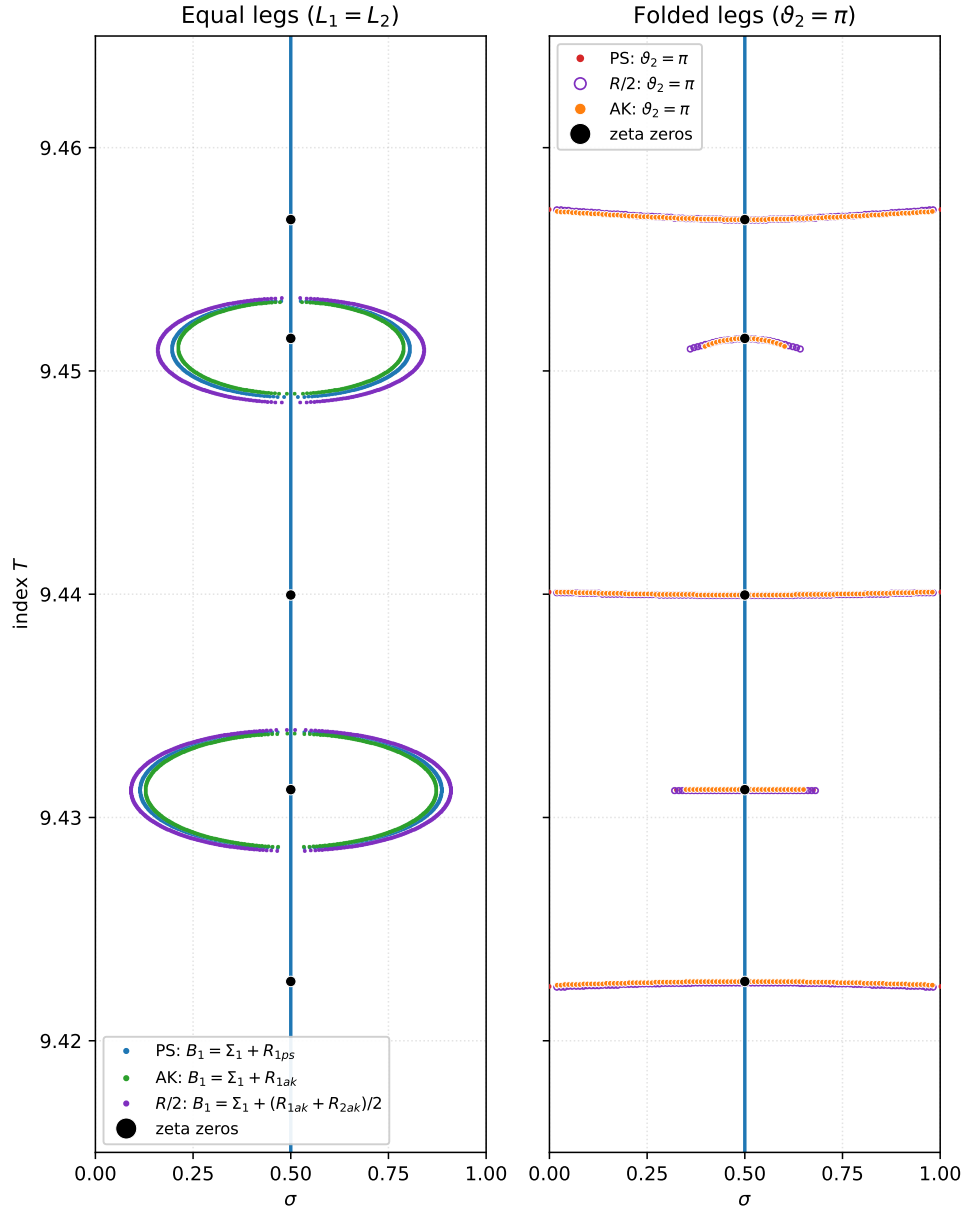


Figure 26: PS, AK and $R/2$ legs and angles for two neighboring equal-leg ovals near $T \approx 9.44$, in the window $9.415 \leq T \leq 9.465$. Left: equal-leg loci $L_1 = L_2$ for PS (blue), AK (green), and $R/2$ (purple). Right: folded-leg loci $\theta_2 = \pi$ for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. The lower oval surrounds the zero at $T \approx 9.431$ and the upper the zero at $T \approx 9.451$; the zero at $T \approx 9.440$ sits in the gap between them. Loci computed as in Figure 25. Generated by `fig-ps_ak_r2_legs_angles.py` (`--out fig-ps_ak_r2_oval_9441 --t-lo 9.415 --t-hi 9.465`).

If $\mathcal{E}_{ps}$ and $\mathcal{E}_{rs}$ have no point in common with $\sigma \neq \frac{1}{2}$, then ζ has no zero off the critical line.

In particular, if the two loci are disjoint off the line throughout the strip, the Riemann Hypothesis follows; if they are disjoint off the line for all $T \leq T_{\max}$, then there is no off-line zero with index at most $T_{\max}$. The figures above are consistent with a strict nesting (the ovals approach one another but do not cross off the line). A proof that this nesting persists for all T , with due care at the pole heights of d_1 in §8.4 where B_{1ps} ceases to be defined in the usual way, would therefore be a proof of the Riemann Hypothesis.

In short: on the critical line the leg-length ratio is always one (Corollary 9.1, and likewise for B_{1rs}), so the zeros there are determined by the legs folding back, $\vartheta_2 = \pi$, for either the ps or the rs choice of bisector point. Off the line the length condition is no longer free: a zero can occur only where $L_1/L_2 = 1$ for the legs, and since that must hold for both choices it is a necessary condition that

$$\frac{L_1}{L_2} = 1 \quad (87)$$

simultaneously for the ps legs and for the rs legs.

The mean leg-imbalance metric. The equal-leg loci above are the zero sets of a continuous imbalance. For each of the three named splits of §7.2, write the two leg lengths

$$L_1 = |\Sigma_1 + R_1|, \quad L_2 = |\Sigma_2 + R_2|, \quad (88)$$

where (R_1, R_2) is (R_{1ps}, R_{2ps}) , $(R_{1rs}, R_{2rs}) = (\frac{1}{2}R, \frac{1}{2}R)$, or (R_{1ak}, R_{2ak}) respectively. The normalized length imbalance of a split is the scale-free quantity

$$\delta = \frac{|L_1 - L_2|}{L_1 + L_2 + \varepsilon}, \quad \varepsilon = 10^{-18}, \quad (89)$$

so $\delta = 0$ if and only if $L_1 = L_2$ (the bisector condition for that choice of bisector point). Averaging the three splits gives the plotted color map

$$\bar{\delta} = \frac{\delta_{ps} + \delta_{rs} + \delta_{ak}}{3}. \quad (90)$$

On the critical line each split has $L_1 = L_2$ (Corollary 9.1 and the parallel identities for rs and ak), so $\bar{\delta} \approx 0$ and the strip is dark purple along $\sigma = \frac{1}{2}$. Off the line $\bar{\delta}$ rises, and the equal-leg ovals of Figures 25–26 appear as the dark cellular structures hugging the critical line in Figure 27.

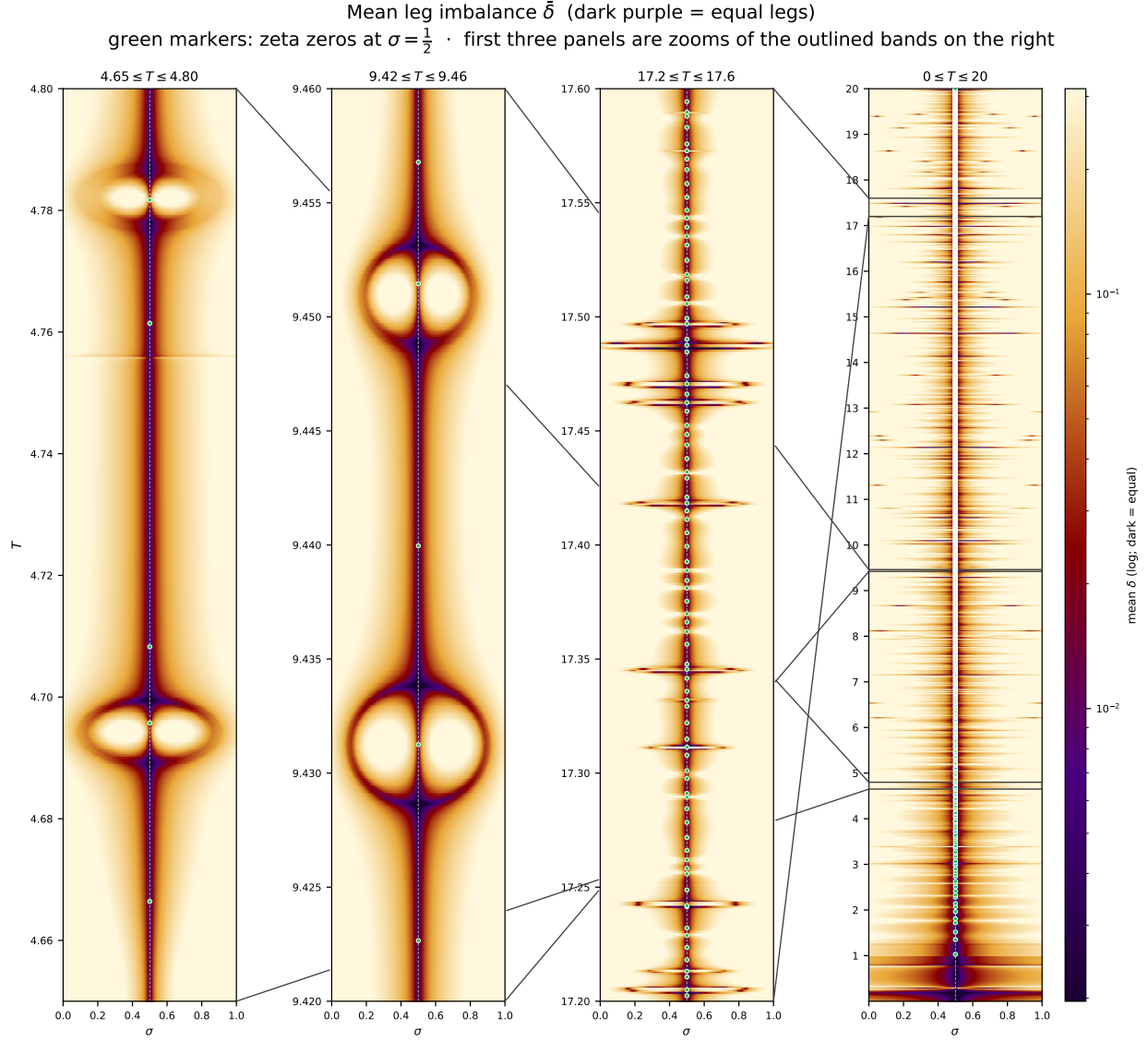


Figure 27: Mean leg imbalance $\bar{\delta}$ of (90) (dark purple = equal leg lengths). Horizontal axis $\sigma \in [0, 1]$; vertical axis the spiral index T . Color is $\bar{\delta}$ on a log scale. Green markers are zeta zeros at $\sigma = \frac{1}{2}$. The rightmost panel is the full strip $0 \leq T \leq 20$; the first three panels are zooms of the outlined bands ($4.65 \leq T \leq 4.80$, $9.42 \leq T \leq 9.46$, $17.2 \leq T \leq 17.6$). The dark vertical band on $\sigma = \frac{1}{2}$ is the simultaneous equal-leg condition for the ps , rs , and ak splits. Generated by `plot_leg_equality_four_panel.py`.

The mean fold-deviation metric. We treat the folded-leg loci in a similar fashion to the equal-leg loci. For each of the three remainder splits write the two leg vectors $v_1 = \Sigma_1 + R_1$ and $v_2 = \Sigma_2 + R_2$, and let $\vartheta_2 = \arg(v_2/v_1) \in (-\pi, \pi]$ be the fold angle between them. The normalized fold deviation of a split is the scale-free quantity

$$\tau = \frac{\pi - |\vartheta_2|}{\pi}, \quad (91)$$

so $\tau = 0$ if and only if $\vartheta_2 = \pi$ (leg 2 folded straight back onto leg 1) and $\tau = 1$ where the legs run parallel. Averaging the three splits gives the angle companion of $\bar{\delta}$,

$$\bar{\tau} = \frac{\tau_{\text{ps}} + \tau_{\text{rs}} + \tau_{\text{ak}}}{3}, \quad (92)$$

plotted in Figure 28 over the same four windows as Figure 27. The two color maps organize themselves in opposite directions. The length condition is automatic on the critical line and rare off it, so $\bar{\delta}$ concentrates its dark values in a vertical band. The fold condition instead is met along the nearly horizontal folded-leg curves that sweep the strip at the mean zero density, on and off the line alike (Figure 22), so $\bar{\tau}$ concentrates its dark values in horizontal filaments crossing the full width of the strip. A zeta zero requires both: it is a point where a dark filament of Figure 28 crosses the dark vertical band of Figure 27, and every green marker sits on a filament at $\sigma = \frac{1}{2}$.

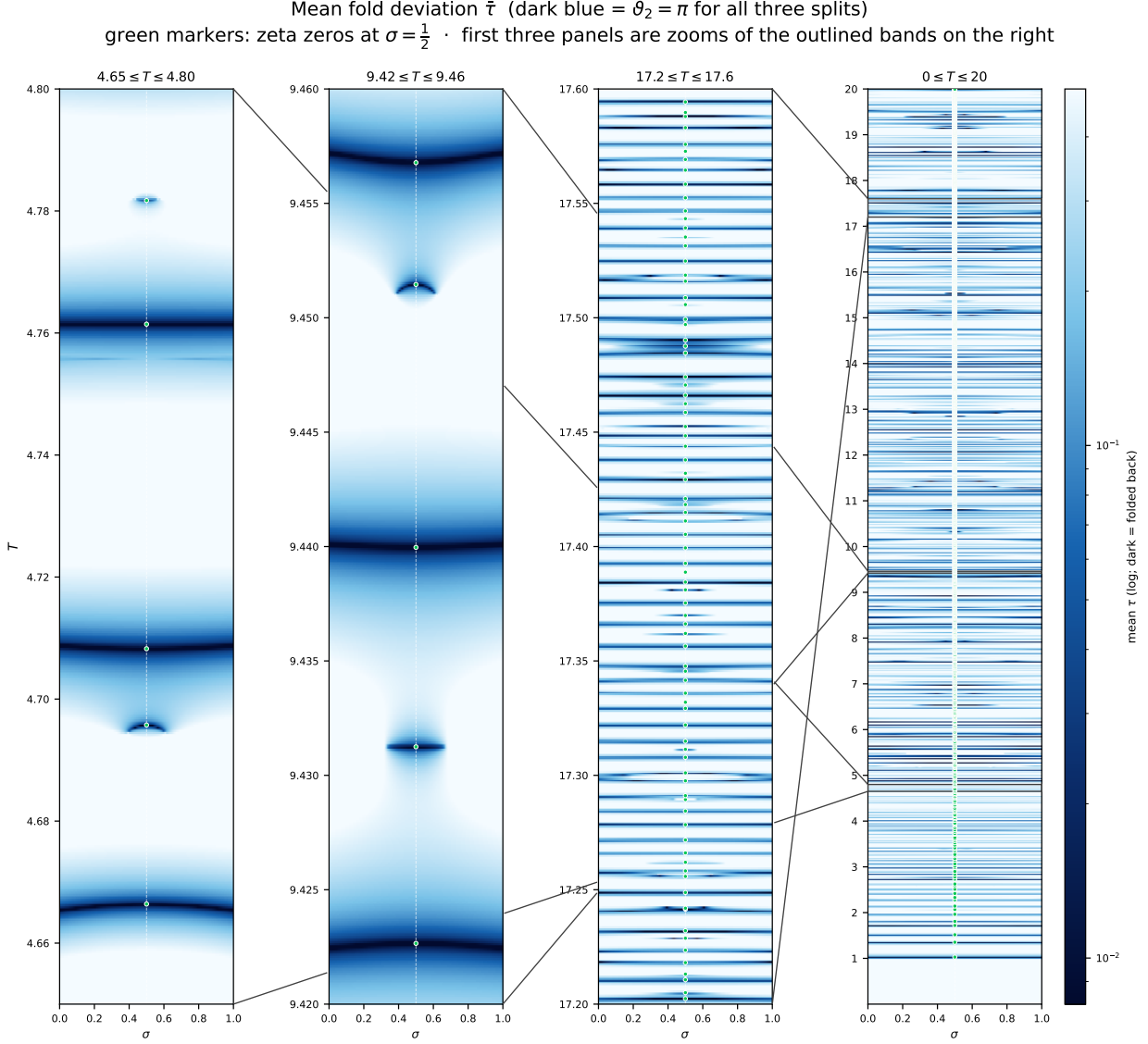


Figure 28: Mean fold deviation $\bar{\tau}$ of (92) (dark blue = $\vartheta_2 = \pi$ simultaneously for the ps , rs , and ak splits). Horizontal axis $\sigma \in [0, 1]$; vertical axis the spiral index T ; color is $\bar{\tau}$ on a log scale, in the four-panel layout of Figure 27: the rightmost panel is the full strip $0 \leq T \leq 20$ and the first three panels are zooms of the outlined bands ($4.65 \leq T \leq 4.80$, $9.42 \leq T \leq 9.46$, $17.2 \leq T \leq 17.6$). Green markers are zeta zeros at $\sigma = \frac{1}{2}$. Where the equal-leg color map of Figure 27 is dark along the critical line, the fold color map is dark along nearly horizontal filaments sweeping the strip at the mean zero density, and every green marker sits on one. Generated by `plot.theta2_pi_four_panel.py`.

Both color maps are exactly symmetric across the critical line: the leg ratio obeys $L_1/L_2(\sigma, T) = L_2/L_1(1 - \sigma, T)$, which leaves (89) unchanged, and the fold angle satisfies the corresponding reflection, so $\bar{\delta}$ and $\bar{\tau}$ take the same value at (σ, T) and $(1 - \sigma, T)$; on the plotted grids the two halves agree to a relative 10^{-10} . Half of each panel therefore carries the whole color map, and the two conditions can share one strip. Figure 29 splices them in the first two zoom windows: the left half

of each strip is the leg imbalance $\bar{\delta}$ and the right half is the fold deviation $\bar{\tau}$, meeting at the seam $\sigma = \frac{1}{2}$. The two halves state the two conditions of a zero together: the length condition holds along the half-line itself and on the equal-leg ovals hugging it, the angle condition holds on the dark filaments arriving from both sides, and each filament reaches the seam exactly at a green zeta zero marker.

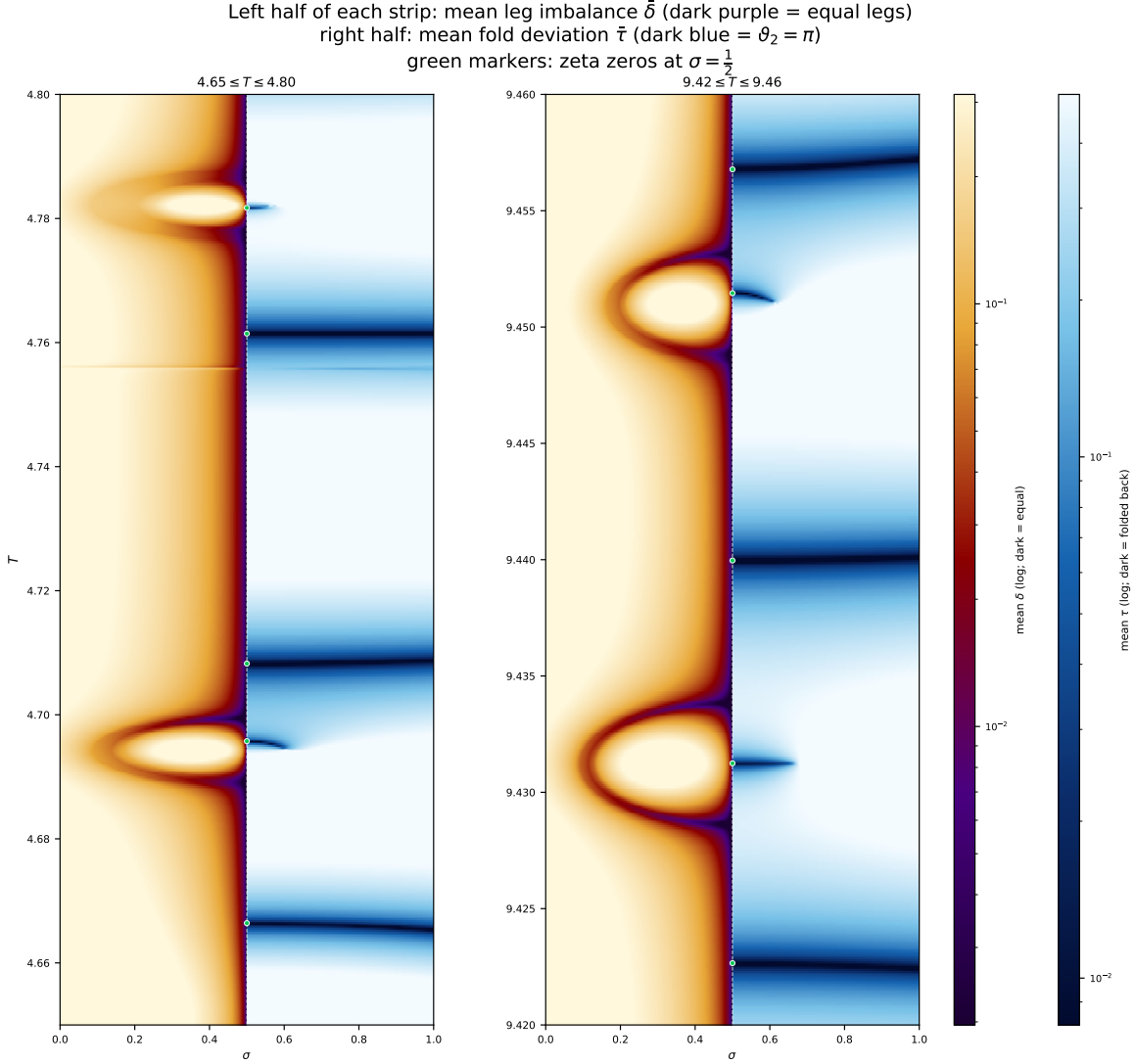


Figure 29: The two zoom windows of Figures 27 and 28 spliced at the critical line: in each strip the left half ($\sigma < \frac{1}{2}$) shows the mean leg imbalance $\bar{\delta}$ of (90) on the color scale of Figure 27 (dark purple = equal legs), and the right half ($\sigma > \frac{1}{2}$) shows the mean fold deviation $\bar{\tau}$ of (92) on the color scale of Figure 28 (dark blue = folded back). Both color maps are symmetric under $\sigma \leftrightarrow 1 - \sigma$ (to a relative 10^{-10} on these grids), so each half carries its whole color map and the splice loses nothing. Green markers are zeta zeros. The length condition is dark along the seam and on the ovals flanking it; the angle condition is dark on the horizontal filaments; a zero is a filament reaching the seam. Generated by `plot_leg_theta2_spliced.py`.

9.5 Toward a proof

The geometry assembled in this section suggests two routes to the Riemann Hypothesis, listed here in increasing order of how many remainder splits participate. (A third phenomenon, the exact collinearity of the three first-half remainders on the critical line, is a signature of the line itself rather than of the zeros; it is presented separately in §14.3.)

1. *Equal legs and fold-back together, for a single split.* A zero at (σ, T) forces, for any single choice of bisector point, both leg conditions of §9.2 at the same input: $L_1 = L_2$ and $\vartheta_2 = \pi$. On the critical line the length condition is automatic (Corollary 9.1), so zeros there are cut out by the fold-back alone. Off the line the two conditions trace independent families of curves (the equal-leg ovals and the nearly horizontal folded-leg curves of Figure 23), and a proof that the two families never pass through the same off-line point, for even one split, would prove the Riemann Hypothesis.
2. *Equal legs for two or more splits at once.* A zero makes the legs equal for *every* remainder split simultaneously: when $\zeta = 0$, $L_2 = |\zeta - B_1| = |B_1| = L_1$ for any B_1 . An off-half-line zero would therefore be an off-half-line point of $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$ (and of $\mathcal{E}_{ak}$ as well). This route drops the angle condition entirely and asks only that two equal-leg ovals meet off the line; the figures of §9.4 show the ovals nesting without crossing. This route is developed in detail below.

Route 2 in detail. An off-half-line zero at (σ, T) would require both equal-leg identities at the same input:

$$|\Sigma_1 + R_{1ps}| = |\Sigma_2 + R_{2ps}|, \quad (93a)$$

$$|\Sigma_1 + \tfrac{1}{2}R| = |\Sigma_2 + \tfrac{1}{2}R|. \quad (93b)$$

Here (93a) is the *ps* condition and (93b) the *rs* condition with $R_{1rs} = \tfrac{1}{2}R$. Each is one real constraint on (σ, T) : an equality of two moduli is one real equation, so the pair is two real conditions on the two real coordinates of the strip, the right number of conditions to cut out isolated points (or nothing). On $\sigma = \tfrac{1}{2}$ they are not independent, so they do not cut the half-line down to points. Off it, they are.

On $\sigma = \tfrac{1}{2}$ both hold for every T , by Corollary 9.1. They are identities along the whole half-line and do not cut anything out there. Every point of $\sigma = \tfrac{1}{2}$ is already a common root; that common set is expected and does not constrain zeros beyond the folded-leg condition, which this route does not use.

Off the half-line that automatic identity is gone. Then (93a) and (93b) are independent: they are not the same equation written twice, and satisfying one does not force the other. Each traces its own family of curves, the *ps* ovals $\mathcal{E}_{ps}$ and the *rs* ovals $\mathcal{E}_{rs}$. A common root is a pair (σ, T) that solves both at once, hence a point of $\mathcal{E}_{ps} \cap \mathcal{E}_{rs}$. On the half-line that intersection is the whole line; off it, a meeting of the two oval families.

An off-half-line zero would be such a root: a (σ, T) that satisfies both (93a) and (93b) at once. At a zero the chain (the whole Dirichlet chain formed by the links of Σ_1 , the remainder, and Σ_2) closes, $\zeta = 0$, so for any B_1

$$L_2 = |\zeta - B_1| = |-B_1| = |B_1| = L_1.$$

That does not depend on how B_1 was chosen, so it holds for B_{1ps} and B_{1rs} at once. A zero therefore sits on both loci, on the half-line or off it; off the half-line that is exactly a common root of (93). The implication is one-way:

$$\text{off-half-line zero} \implies \text{off-half-line point of } \mathcal{E}_{ps} \cap \mathcal{E}_{rs}.$$

A meeting of the ovals is necessary for an off-half-line zero, not sufficient: the fold-back $\vartheta_2 = \pi$ is still required. For proving there are no off-half-line zeros, necessity is what is needed.

A proof that (93) has no solution for $\sigma \neq \frac{1}{2}$ (away from the poles of d_1 , where the ps split is undefined) is a proof that those two loci never meet off the half-line: the off-half-line points of the intersection *are* the off-half-line common roots of (93). The contrapositive of the implication then gives the Riemann Hypothesis. *If there is no off-half-line meeting, there cannot be an off-half-line zero, because such a zero would force a meeting.* In one line,

$$\begin{aligned} \text{no off-half-line solution of (93)} &\iff \text{no off-half-line point of } \mathcal{E}_{\text{ps}} \cap \mathcal{E}_{\text{rs}} \\ &\implies \text{no off-half-line zero} \iff \text{RH.} \end{aligned}$$

The figures of §9.4 already suggest the ovals nest and do not cross. Proving what those pictures represent, namely that (93) has no off-half-line solution, would be a proof of RH.

9.6 Equal legs density

If a zero were to exist off the critical line, it would have to lie on an equal-leg locus. The reason is the one used in §9.4: at a zero the chain closes, $\zeta = 0$, so the bisector point is equidistant from the two ends of the chain,

$$L_2 = |\zeta - B_1| = |-B_1| = |B_1| = L_1,$$

whichever split is used for B_1 . Every zero, on the line or off it, is therefore an equal-leg point. On the line that says nothing, the whole line being equal-leg by Corollary 9.1; off the line it is a genuine restriction, because the off-line part of the locus is not the whole strip but the isolated ovals of Figure 21, symmetric about $\sigma = \frac{1}{2}$. An off-line zero has to sit on one of those ovals. Their number and size are therefore worth measuring: they are the only places in the strip where an off-line zero can be. This subsection reports a numerical census of them for $0 \leq T \leq 401$ (which is t of 1.0129×10^6).

The imbalance and the scan. The census counts sign changes of the *signed* imbalance

$$g(\sigma, T) = \frac{L_1 - L_2}{L_1 + L_2}, \quad (94)$$

which is (89) without the absolute value. The modulus is dropped on purpose: g changes sign across an oval, and it is the sign change that can be counted. Numerically g is odd under $\sigma \mapsto 1 - \sigma$, the reflection that exchanges the two legs, so it vanishes identically on the critical line (Corollary 9.1 again), and the ovals are symmetric about the line, as the figures show.

The scan fixes $\sigma = \frac{1}{2} + \varepsilon$ and steps in T , recording sign changes of g . Each oval contributes two, a bottom and a top, and only bottoms are counted, so an oval straddling an integer T is counted once, to the interval holding its bottom. The pole bands are excluded: at the heights of §8.4 the PS bisector point runs off to infinity and the equal-leg point sweeps the full width of the strip, which is not an oval. Those bands are a PS artifact, 599 of them against 1 for AK, whose remainder has no pole to escape through. Roots are refined by bisection to 10^{-13} . The step is

$$\Delta T = 10^{-4}/m, \quad (95)$$

which holds the number of samples across an oval fixed, the median oval height falling like $1/T$: 1.3×10^{-3} in the interval at $T = 25$ and 5.2×10^{-5} in the one at $T = 400$. Ovals thinner than the step are still caught, by minimizing g wherever it has a positive local minimum below a twentieth of the interval's median and testing whether the minimum dips below zero.

Four checks bound the error. First, ζ evaluated as $\Sigma_1 + \Sigma_2 + R_{ak}$ agrees with `mpmath` to 2×10^{-10} at $T = 150$ and 3×10^{-8} at $T = 1000$, and g vanishes on the critical line to machine precision. Second, recounting the intervals at $m = 250, 275, 299, 300, 350, 398, 399$ and 400 with $\Delta T/2$ and $\Delta T/4$ returns the same count every time, for both splits, and finds the same thinnest cross-section in each. Third, the window $3 \leq T \leq 4$ reproduces Figure 21, the same four ovals at $T = 3.0105, 3.3845, 3.6031$ and 3.7257 . Fourth, the σ -extent of each oval was measured by bisection, which is what decides whether a scan at a given ε can see it: at $\varepsilon = 0.001$ every oval in the sample is reachable and at $\varepsilon = 0.01$ some 99.95% are, so the count quoted below, taken at $\varepsilon = 0.01$, has saturated.

Are they ovals? Drawn against the index they are hairlines. The component at $T = 300$ in Figure 30 is 0.39 wide in σ and 6.9×10^{-5} tall in T , a ratio of 5600 to one, and every critical strip figure here stretches the vertical axis by some such factor to make the locus visible at all. That shape is an artifact of the coordinate. One unit of index is $I'(T) = dt/dT$ units of t , and $I'(300) = 3789$, so plotting against T compresses the vertical direction by exactly that factor; in the s -plane, where ζ is evaluated, the same component is 0.26 tall against 0.39 wide. It is an oval, and close to an ellipse: comparing its half-height at eight fractions of its half-width a against $\sqrt{1 - (u/a)^2}$, the one drawn departs by 1.8% rms. Over a sample of 300 of them at nine heights from $T = 5$ to $T = 400$, drawn at random within each interval, the median departure is 0.5%, with 80% inside 2% and 94% inside 5%; the ratio of width to height has median 1.22 and ranges from 0.96 to 2.83, with no trend in height, the interval medians sitting between 1.13 and 1.27 throughout. So the components are convex, smooth, nearly elliptical, a little wider than tall, and they keep their shape as they shrink, which is what the word oval is doing in this section. That description is for the components that enclose one zero, which is almost every oval in the census. The twenty-three that enclose two zeros are not convex. Each is an hourglass: two oval lobes joined at a neck, with one zero in each lobe, as in Figure 33. The only other components that are not closed ovals are the ones cut off by the edge of the strip instead of closing inside it, and there are 72 of those in the census, all below $T = 100$.

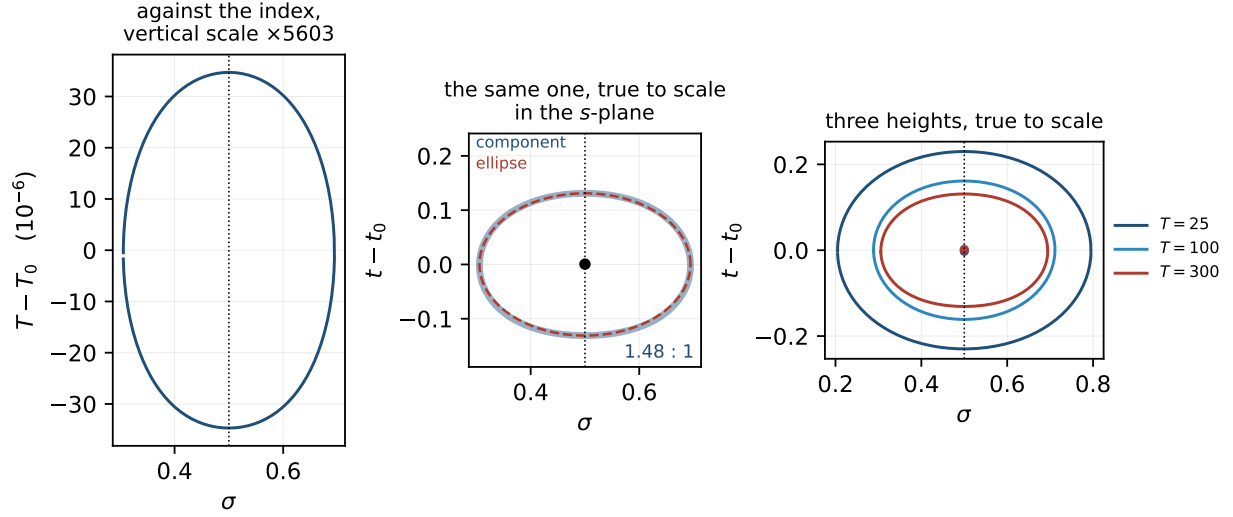


Figure 30: Why the components of the equal-leg locus are called ovals. Left: one of them at $T = 300$ plotted against the index, as the strip figures plot them, the vertical axis stretched by 5600 to make it visible. Middle: the same component in the s -plane with the axes to a common scale, and the ellipse through its extremes dashed over it; the dot is the zero of ζ it encloses. Right: components at $T = 25, 100$ and 300 to a common scale, each drawn about its zero. They shrink with height and hold their shape. All three are the median-height component of their interval, taken among those whose zero sits nearest the middle, since the zero is central only on average. Generated by `fig_oval_shape.py` of the equal-leg density companion note in the Zest repository.

The count. Over $0 \leq T \leq 401$, scanning the line $\sigma = \frac{1}{2} + \varepsilon$ with $\varepsilon = 0.01$, there are 328 511 ovals for the AK remainder split $B_1 = \Sigma_1 + R_{1ak}$ and 328 191 for the PS split $B_1 = \Sigma_1 + R_{1ps}$, against 1 771 661 zeros of ζ in the same range. The two splits converge on each other with height: the mean disagreement per unit interval grows from 1.5 ovals to 5.0, but falls from six percent of the count to 0.32%. The oval density of both splits tracks the density of the zeros. Writing the interval's zeros per oval as a function of height, and fitting over $3 \leq T \leq 401$ weighted by the oval count,

$$\frac{\#\{\text{zeros in } (m, m+1)\}}{\#\{\text{ovals in } (m, m+1)\}} = 4.38 + \frac{11.14}{\ln(t/2\pi)}, \quad (96)$$

which is unbiased in every bin above $T = 100$, to ± 0.01 . A power law $6.74 T^{-0.040}$ fits the per-interval scatter equally well (that scatter is dominated by fluctuation and cannot separate the two), but its binned residuals march in one direction, and it sends the ratio to zero. The law (96) instead levels off: the ovals settle at a fixed share of the zeros, about one oval per four and a half, so with $dN/dT = I'(T) \ln(t/2\pi)/2\pi \approx 4T \ln T$ the oval density is asymptotically $0.9 T \ln T$. Figure 31 plots both.

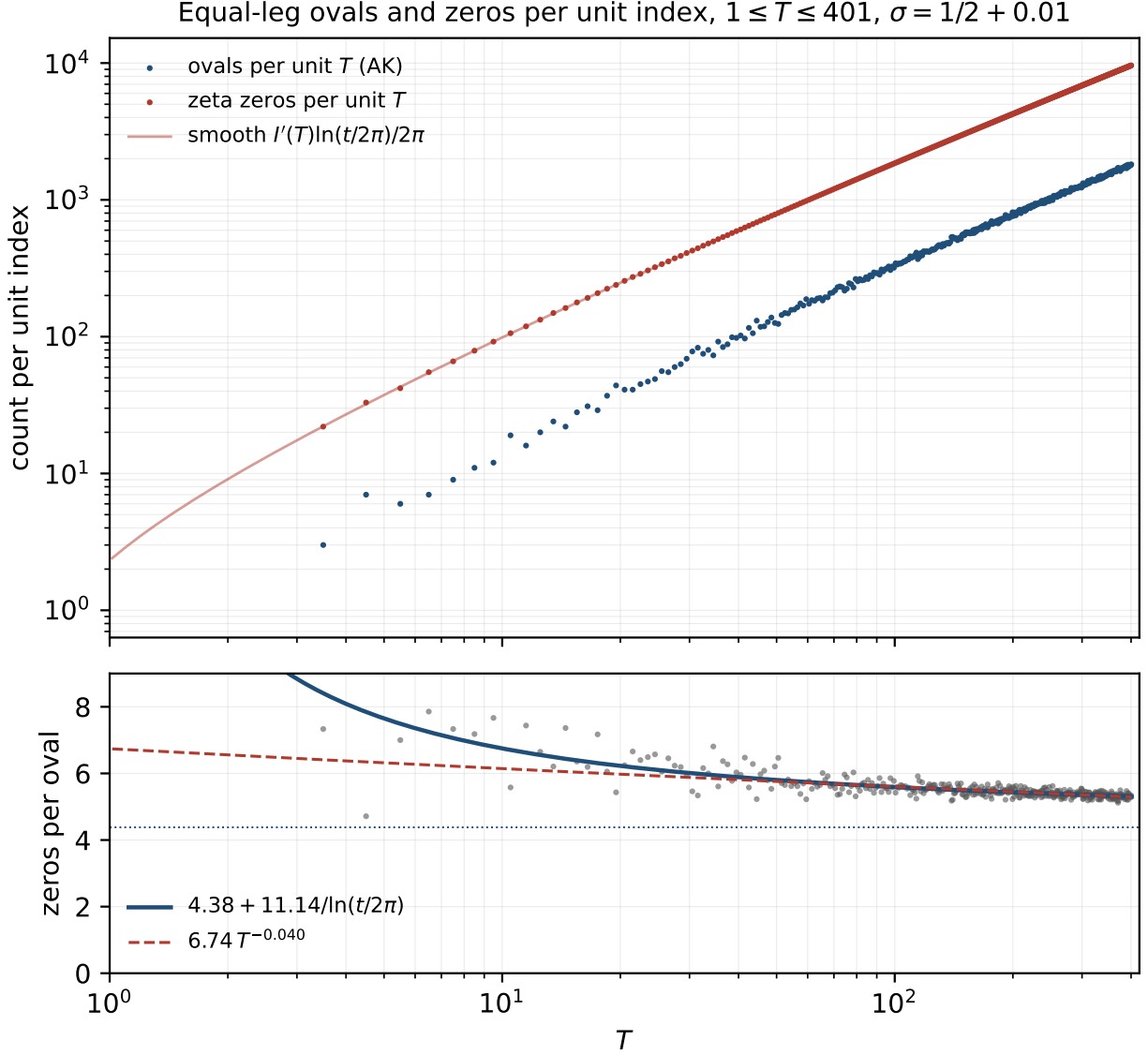


Figure 31: Above: equal-leg ovals and zeros of ζ per unit interval of the index, $1 \leq T \leq 401$, at $\sigma = \frac{1}{2} + 0.01$, log-log. The two clouds are parallel. Below: the interval's zeros per oval, with (96) solid and its limit 4.38 dotted, against the power law dashed. The two are indistinguishable over $50 \leq T \leq 400$ and part company on either side. Generated by `census_analyze.py` of the equal-leg density companion note in the Zest repository.

Where an off-line zero could be. Since the ovals are the only off-line home for a zero, their extent is a bound on where to look, and one can state it both in height and in σ . In height: each oval is about 0.49 of a mean zero spacing tall, a ratio steady to within 0.01 above $T = 100$, and there are about a fifth as many ovals as zeros, so the ovals cover 8.8% of the index axis over $350 \leq T < 401$, up from 6.4% over $3 \leq T < 25$, a share growing like $0.66 \ln T + 5.0$ per cent. An off-line zero must have its height inside that 8.8%, a figure read off the $\sigma = \frac{1}{2} + 0.01$ cross-section and so a slight underestimate of the oval's full height at the line. In σ : the ovals draw in toward the critical line as T grows. The median oval reaches $\sigma = 1.01$ at $T = 5$ but only $\sigma = 0.67$ at $T = 400$;

the ninetieth percentile falls from $\sigma = 1.11$ to $\sigma = 0.77$; and of the 258 814 ovals above $T = 200$, just four reach $\sigma = 0.95$. Figure 32 shows both measurements.

The tail does not vanish, though, and the distinction matters. What retreats is the bulk of the ovals, not the widest of them: the widest oval in each interval still reaches $\sigma \approx 0.9$ at every height measured (never below $\sigma = 0.88$ for $T \geq 50$), and the widest above $T = 200$ gets to $\sigma = 0.962$. What thins is their number, from about eight ovals per unit interval reaching $\sigma \geq 0.9$ over $50 \leq T < 100$ to four over $100 \leq T < 200$, two over $200 \leq T < 301$ and one over $301 \leq T < 401$, while the total count per interval grows five-fold. A zero well off the line at large T is therefore not excluded by any of this; it would have to lie on the one or two such ovals per unit interval rather than on any of the eighteen hundred, which is a sharper statement than the count alone gives.

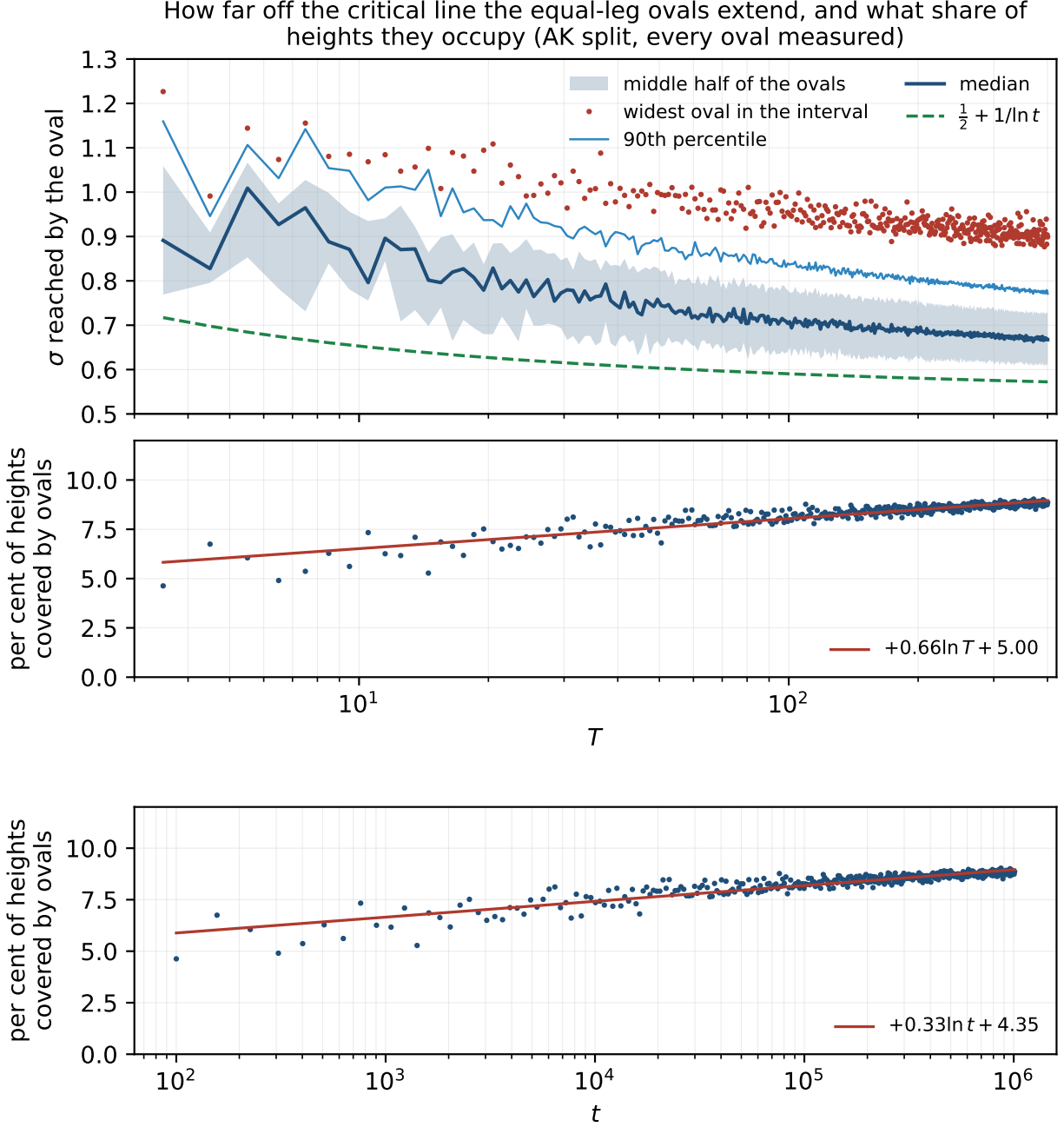


Figure 32: Above: how far off the critical line the ovals extend, from half-widths measured by bisection for every one of the 328 511 ovals in the census, one point per unit interval, from $T = 3$ where the first ovals are. The band is the middle half of the ovals in that interval and the dots are the widest oval in it. The dashed green curve is $\sigma = \frac{1}{2} + 1/\ln t$, drawn on the index axis through $t = I(T)$. The bulk draws in toward the line as T grows while the widest ovals do not. Middle: the percentage of the index axis covered by ovals, which is the share of heights at which an off-line zero could sit, rising like $0.66 \ln T + 5.0$. Below: the same share read along t , where ζ is evaluated. A fraction of an axis does not depend on which variable parametrizes it, one unit of index being $I'(T)$ units of t , so this is the panel above it with its axis relabeled and the law reading $0.33 \ln t + 4.35$; the two agree to four decimals above $T = 25$. Generated by `reach_census.py` and `fig_reach.py` of the equal-leg density companion note in the Zest repository.

Usually, one zero per oval. Every oval has a zero, with a handful of accountable exceptions. Not every zero has an oval, and that direction fails badly rather than by exception. Of the 328 511 AK ovals, 328 467 (all but 44, or 99.987%) bracket exactly one zero of ζ on the critical line. The zero sits at the oval's center on average rather than individually. Its position within the oval has median 0.5000 and mean 0.5000 of the height, so there is no systematic offset, but the spread about that is 0.14. While 91% of zeros are in the middle half, one oval in eleven holds its zero outside it and one in fifty outside the range 0.16 to 0.84. The 44 are not all the same kind of miss. Twenty-three ovals hold two zeros (close pairs, none below $T = 201$). Twenty-one hold none. The two-zero ovals are all unusually close pairs, separated by 0.25 to 0.65 of the local mean spacing, which is what an oval half a spacing tall would have to straddle to catch two. Those components are the hourglasses of the previous paragraph: two oval lobes and a neck, one zero in each lobe. Figure 33 draws the first of them, at $T \approx 201.72$, and another at $T \approx 320.57$, in the three-split style of Figure 25.

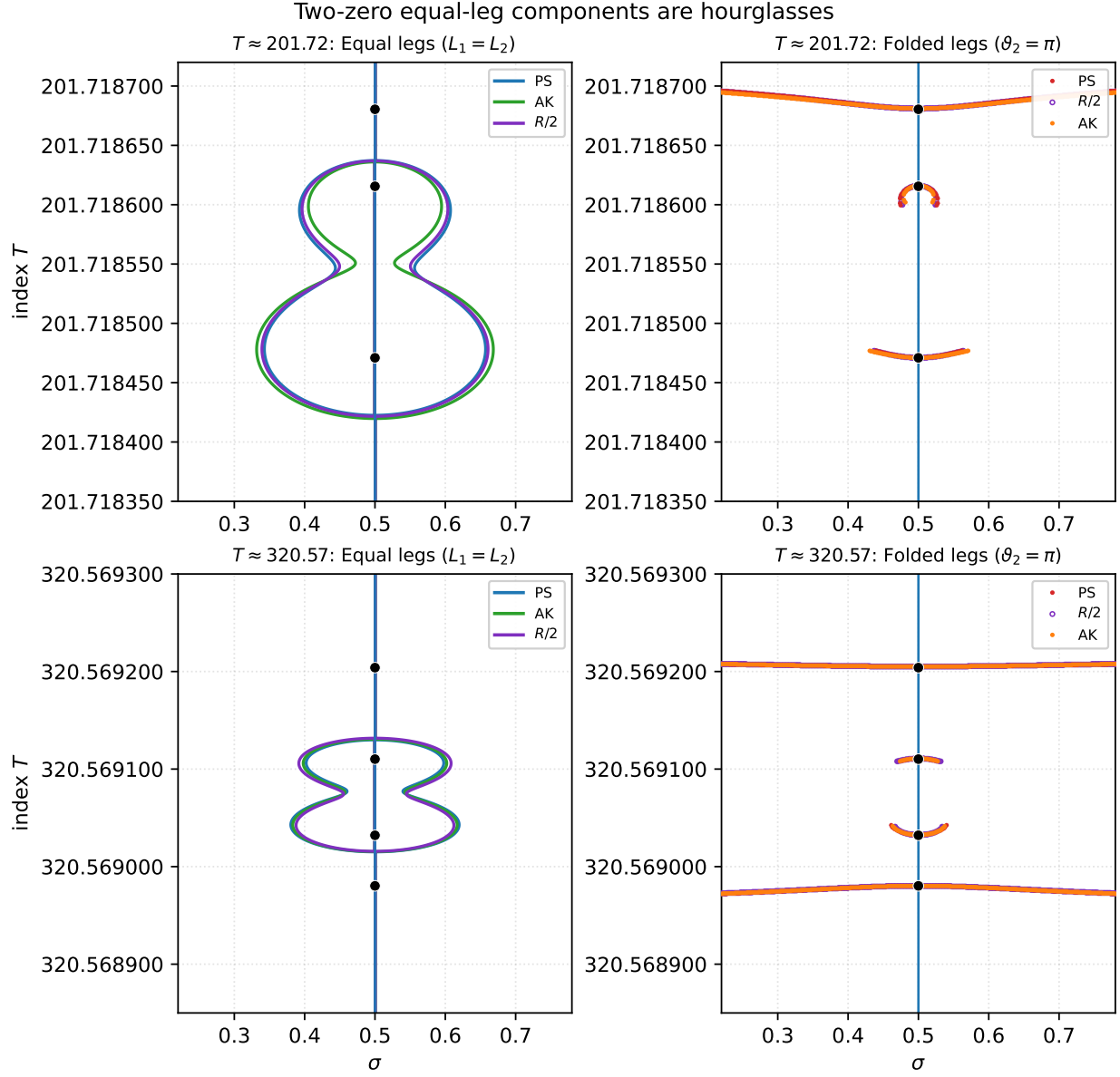


Figure 33: Two AK ovals that each enclose two zeros, drawn as in Figure 25. Top: the first such oval, at $T \approx 201.72$. Bottom: another at $T \approx 320.57$. Left: equal-leg loci $L_1 = L_2$ for PS (blue), AK (green), and $R/2$ (purple). Right: folded-leg loci $\theta_2 = \pi$ for the same three choices (red, orange, and hollow purple). The blue vertical is the critical line; black dots are zeta zeros. Each equal-leg component is an hourglass, two oval lobes joined at a neck, with one zero in each lobe. The three splits nest and keep that shape. Generated by `fig_two_zero_hourglass.py`.

Of the twenty-one empties, eighteen are ovals that span an integer, where the chain gains a link, with their zero just across the integer- T boundary; one is the degenerate interval below $T = 1$, where Σ_1 is empty; and two, at $T = 393$ and $T = 398$, are ovals whose σ -tip stops a hair past the scan line, at $\sigma = 0.51004$ and $\sigma = 0.51003$, so that the cross-section the scan cuts is a sliver 3×10^{-7} tall at the very tip, lying just to one side of the zero's height. The first of those two misses below the zero and the second above it, by under 10^{-7} either way. Both ovals do hold their zero: at the

critical line each is some 4×10^{-6} tall with the zero inside, and the cross-sections hold it out to $\sigma = 0.50997$. The miss is in the bracket test at $\varepsilon = 0.01$, not in the oval. The exception rate falls from 0.104% over $3 \leq T < 50$ to 0.007% over $100 \leq T < 200$ and then edges back up to 0.014% over $301 \leq T < 401$, the rise being the close pairs, whose number grows with the zero count while staying under one oval in seven thousand.

The other direction is not close to one for one, and it is the more important half of the statement. There are 1 771 661 zeros in the range against 328 511 ovals, so five zeros in six have no oval around them, and (96) says that share settles at about four in five rather than shrinking away. The map from ovals to zeros is into, not onto: each oval names one zero, most zeros are named by none, and which ones get named is a question the census does not answer.

Three things follow. First, the off-line locus is tied to the zeros rather than being a structure independent of them. Almost every oval wraps one zero, centered on it in the mean, about half a mean spacing tall and a fifth wider than that; the ovals are a selection from the zeros rather than a phenomenon of their own, which is why their count obeys (96) instead of a law of its own.

Second, it makes the disjointness program of §9.4 and §9.5 local. What is needed for an off-line zero is that $\mathcal{E}_{\text{ps}}$ and $\mathcal{E}_{\text{rs}}$ meet off the line. Since an oval almost always encloses exactly one zero, the two ovals that have to be compared are the PS oval and the *rs* oval belonging to the *same* zero, and the global statement “the two loci have no common point off the line” becomes one nesting statement per bracketed zero, of which there are about $0.9T \ln T$ per unit index. That is the right shape for an argument: a local inequality attached to each zero, to be proved uniformly, rather than a statement about two curves wandering the strip.

Third, the caution. None of this is evidence for or against the Riemann Hypothesis, and the one-zero-per-oval law in particular is not. The zero counted inside each oval is found *on* the critical line, by sign changes of Hardy’s Z ; the census never evaluates ζ away from the line, so it does not test whether an oval carries an off-line zero somewhere along its length, and an oval already holding an on-line zero at its center is not forbidden from passing through one. What the census does establish is that over this range of heights the off-line locus contains no component that an ordinary on-line zero fails to explain: there are no orphan ovals. It also points at the search that would test the hypothesis directly, since evaluating ζ along the ovals is a one-dimensional problem (about $0.9T \ln T$ closed curves per unit index), where searching the strip for off-line zeros is a two-dimensional one.

There is a converse worth stating. An oval always has a zero, but a zero usually has no oval: by (96) only about 22% of zeros acquire one, and what distinguishes those from the rest is open. The bracketing requires the transverse imbalance $\partial g / \partial \sigma$ on the line to change sign on both sides of the zero, and nothing here explains why it does so for two zeros in nine and not for the others.

10 $I(T)$ functions

10.1 The origin of $I(T)$: the neighboring links in the bisector frame

Now change the point of view: hold the forward bisector link still. Translate joint m to the origin and rotate and scale so that the bisector link lies along the real axis from 0 to 1; everything else moves in this *bisector frame* as t increases. Watching the two links on either side of the bisector link, the link at $m - 1$ behind it and the link at $m + 1$ ahead of it, one sees a strikingly periodic picture: each neighboring link revolves around the stationary bisector link approximately *twice* (approaching exactly two revolutions as T grows), and then the whole pattern repeats. The instant the pattern repeats is precisely the handoff instant of §9.1: the neighboring link ***folds back*** onto the bisector link, the bisector point moves onto that next link and that next link becomes the new bisector link with its own two neighbors.

The neighboring links in the bisector frame, $T = 4 \rightarrow 5$ (bisector link = link 4)

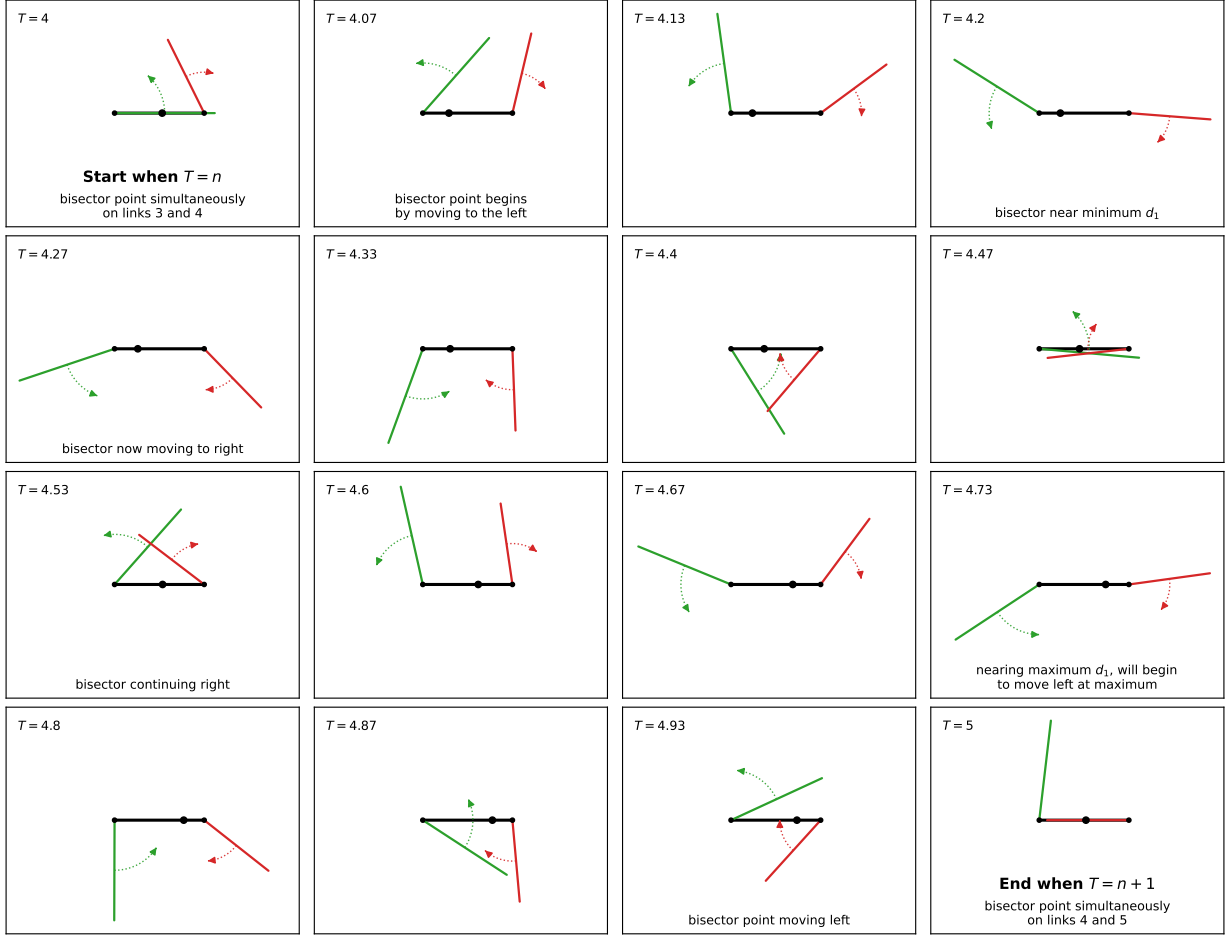


Figure 34: The two neighboring links in the bisector frame as T sweeps from 4 to 5 ($\sigma = \frac{1}{2}$), sixteen snapshots read left to right from the top left ($T = 4$) to the bottom right ($T = 5$). In every panel the bisector link, link $m = 4$ of the forward chain, is held fixed along the real axis from 0 to 1 (black); link 3 (green) hangs off joint 4 at the origin and link 5 (red) off joint 5 at the point 1, both scaled relative to the unit bisector link. The dotted curved arrow on each neighbor runs from the middle of the link to the middle of that link as the next panel depicts it, tracing the motion the link takes between panels. The black dot on the bisector link marks the bisector point, at the crossing fraction $[T]^\sigma d_1$ along the unit link. As T advances, each neighbor revolves around the stationary bisector link approximately twice. At $T = 4$ (Start) the green link is folded back onto the bisector link (joint angle 9π , an odd multiple of π), so the bisector point lies on links 3 and 4 simultaneously: that is the handoff instant at which link 4 became the bisector link. At $T = 5$ (End) it is the red link that folds back (joint angle 11π), the bisector point lies on links 4 and 5 simultaneously, and link 5 takes over as the new bisector link. Generated by `fig_bisector_frame.py`.

This periodicity begs for a clock that ticks once per handoff, and that is where $I(T)$ came from: we sought a formula $t = I(T)$ whose input T passes through an integer at exactly the moments the bisector point moves from one link to the next. Two facts about the joint angle between the outgoing and incoming bisector links pin the formula down.

1. *From the zeta function.* Consecutive summands of Σ_1 differ in phase by $t \ln \frac{n+1}{n}$, so the turn at the joint where link $T - 1$ meets link T (interpolating the index continuously) is

$$j_{\angle}(T) = t \log\left(\frac{1}{T} + 1\right). \quad (97)$$

2. *From the observed motion.* At a handoff the two links fold back onto one another, so the joint angle there is an *odd multiple of π* (hence a summand π); and the neighboring link revolves around the stationary bisector link *twice* per unit of the index, so the odd multiple advances by two with each unit of T (hence the factor 2). Interpolating between handoffs, the joint angle at index T is

$$j_{\angle}(T) = \pi(2T + 1). \quad (98)$$

Setting (97) equal to (98) and solving for t yields

$$\pi(2T + 1) = t \log\left(\frac{1}{T} + 1\right) \implies t = \frac{\pi(2T + 1)}{\log\left(\frac{1}{T} + 1\right)} = I(T), \quad (99)$$

which is the mapping adopted in §3: under $t = I(T)$ the bisector point resides on link $[T]$, and T is an integer exactly when the bisector point moves from one link to the next.

Remark 10.1. Taylor-expanding the reciprocal logarithm in (99), $1/\log(1 + \frac{1}{T}) = T + \frac{1}{2} - \frac{1}{12T} + \frac{1}{24T^2} - \dots$, and collecting orders of T gives a family of invertible polynomial approximations to $I(T)$; keeping successively more terms:

terms	$I(T) \approx$	solved for T
2	$2\pi(T^2 + T)$	$T = -\frac{1}{2} + \sqrt{\frac{t}{2\pi} + \frac{1}{4}}$
3	$2\pi(T^2 + T + \frac{1}{6})$	$T = -\frac{1}{2} + \sqrt{\frac{t}{2\pi} + \frac{1}{12}}$
4	$2\pi(T^2 + T + \frac{1}{6} - \frac{1}{180T^2})$	$T \approx -\frac{1}{2} + \sqrt{\frac{t}{2\pi} + \frac{1}{12} + \frac{\pi}{90t}}$

(The $1/T$ term of the expansion cancels, so the fourth term is already $O(T^{-2})$.) The 2-term form misses $I(T)$ by a constant $\approx \pi/3$; the 3-term form is accurate to $O(T^{-2})$ and the 4-term form to $O(T^{-3})$. At $T = 20$ their errors in T after inversion are 4×10^{-3} , 3×10^{-7} and 4×10^{-11} respectively. The first two rows invert exactly (quadratics); the 4-term row is inverted by substituting the 3-term inverse into the small T^{-2} correction. Figure 35 draws $I(T)$ against the two-term form; the difference is invisible at scale, about one against values in the thousands.

The table also places Siegel's cutoff (9) on the index. The 3-term row is $t/2\pi = (T + \frac{1}{2})^2 - \frac{1}{12} + O(T^{-2})$, so

$$\sqrt{\frac{t}{2\pi}} = T + \frac{1}{2} - \frac{1}{24T} + O(T^{-2}), \quad (100)$$

which is right to 5×10^{-4} at $T = 6.18$, 5×10^{-5} at $T = 20$ and 2×10^{-6} at $T = 100$. Siegel's m therefore steps *near* the *half-integers* of T , a shade past them by $1/24T$, while ours steps *exactly* at the integers.

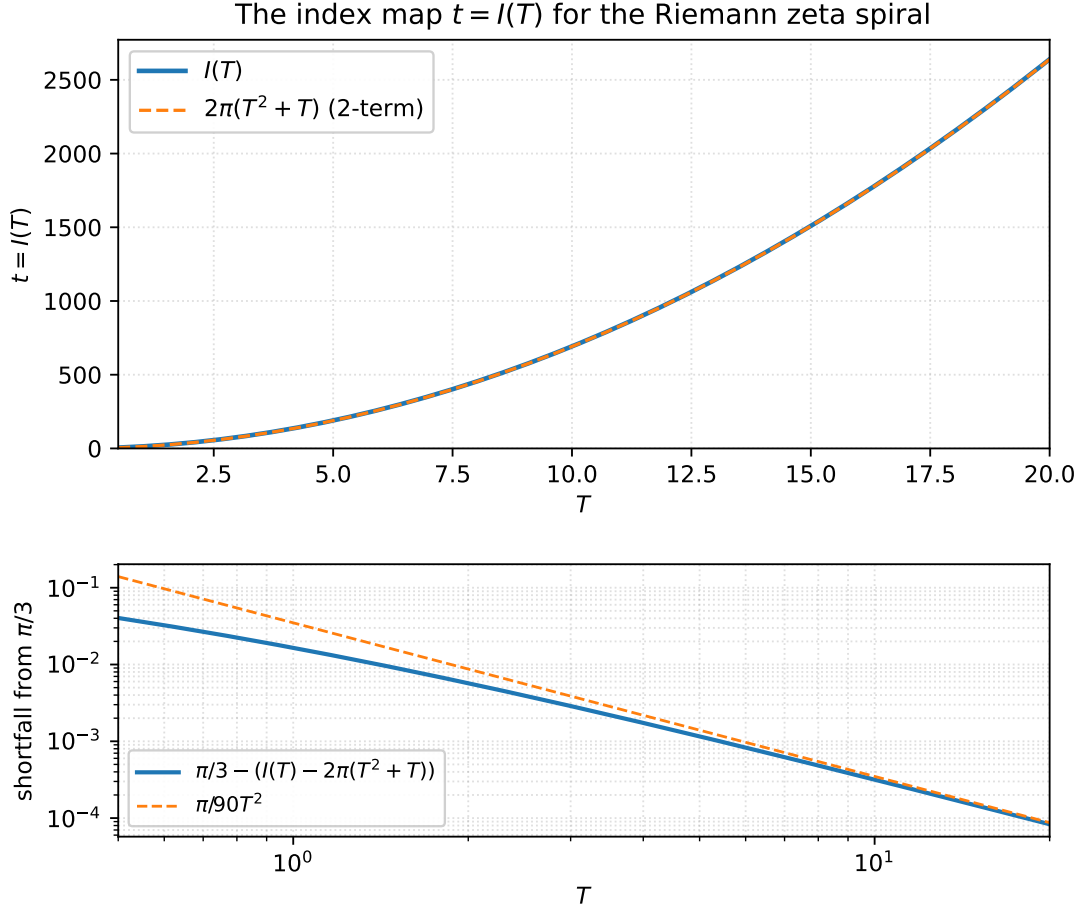


Figure 35: Above: the index map $t = I(T)$ of (10) (solid), together with the two-term approximation $2\pi(T^2 + T)$ from the Taylor table above (dashed). Below: the difference between the two, which the panel above cannot show, the curves differing by about one against values in the thousands. That difference is the constant $\pi/3$ of the table, the $\frac{1}{6}$ the third term restores, so what is plotted log-log is its distance from that constant, which is the fourth term: it follows $\pi/90T^2$ (dashed) from $T = 5$ on, and is 8×10^{-5} by $T = 20$. Generated by `fig.IT.py`.

We stress that the polynomial forms of the table are conveniences only: everywhere in this paper $t = I(T)$ is used *exactly*. The defining property of the mapping, that the neighboring link folds back onto the bisector link *exactly* as T passes an integer and the bisector point transfers from one link to the next exactly at that instant, holds only for the exact $I(T)$. Under any approximation, including the one whose inverse is essentially Siegel's $\sqrt{t/2\pi}$, the fold instants drift off the integers by the corresponding error in the table: the clock still ticks once per handoff, but never precisely on the hour.

Remark 10.2 (comparing the classical t rescaling so that zeros have unit mean spacing, and $I(T)$). In random-matrix and zeta-statistics work the zero-counting function, which maps each zero $\gamma \mapsto N(\gamma)$ (or uses $\tilde{\gamma} = \frac{\gamma}{2\pi} \log \frac{\gamma}{2\pi}$, the leading term of (185)), leaves the zeros with unit mean spacing, which lets one compare zero statistics at different heights. This mapping is called rescaling the zeros. Our $I(T)$ is a change of variable in the same spirit, a reparameterization of the height t , but it runs in the opposite direction, compressing ever more zeros into each unit interval of the

index T . For example, the interval $[5, 6]$ carries 42 zeros, $[10, 11]$ carries 106, $[20, 21]$ carries 256, and $[50, 51]$ carries 802. Rescaling the zeros flattens the count to unit spacing; the index does the reverse.

10.2 $I(T)$ functions for other L -functions besides the zeta function

The same change-of-variable idea extends from ζ to Dirichlet L -functions. On the critical line the partial sums of

$$L(s, \chi_p) = \sum_{n=1}^{\infty} \chi_p(n) n^{-s} \quad (101)$$

form a spiral just as for zeta, but now the character $\chi_p(n) = \left(\frac{n}{p}\right)$ (Legendre symbol, for a prime modulus p) inserts periodic sign flips and, when $p \mid n$, zero-length *phantom* links. Counting only the nonzero links as *links*, there are $p - 1$ links and one phantom in every block of p summands, and the natural index T is the link index (phantoms receive no T).

Watching these link chains as t varies, we noticed that the bisector point still moves from one link to the next, just as for zeta, but not always when links fold back on one another: sometimes the bisector point moves from link to link at a *joint*, as the bisector line sweeps through it. So L -functions have other event types that zeta does not. An *event* is the instant the bisector point moves from one link to the next. There are two ways this can happen: either the two links fold back onto one another (the turning angle at their common joint is an odd multiple of π), so the bisector point slides across the joint from the old link onto the new one; or the bisector line sweeps through a joint, carrying the crossing from one link to its neighbor. For zeta only the first kind occurs: every handoff of §9.1 is a folded event, once per unit of T , and there are no bisector events. For $L(s, \chi_p)$ both kinds occur, recurring with period $\varphi(p) = p - 1$ in the integer index T :

- *folded* events, where the turning angle at the current joint is $\pm\pi$ (the chain reverses), at $T \equiv 0 \pmod{p-1}$;
- *bisector* events, where the joint lies on the perpendicular bisector of the segment from 0 to $L(s, \chi_p)$, at the remaining residues modulo $p - 1$.

For $p = 3$ one has $\varphi(3) = 2$: even T are folds and odd T are bisectors. The map $t = I_p(T)$ is defined so that integer T lands at the t -value of its event. Figure 36 draws the two link chains of $L(s, \chi_3)$ at $T = 6.125$.

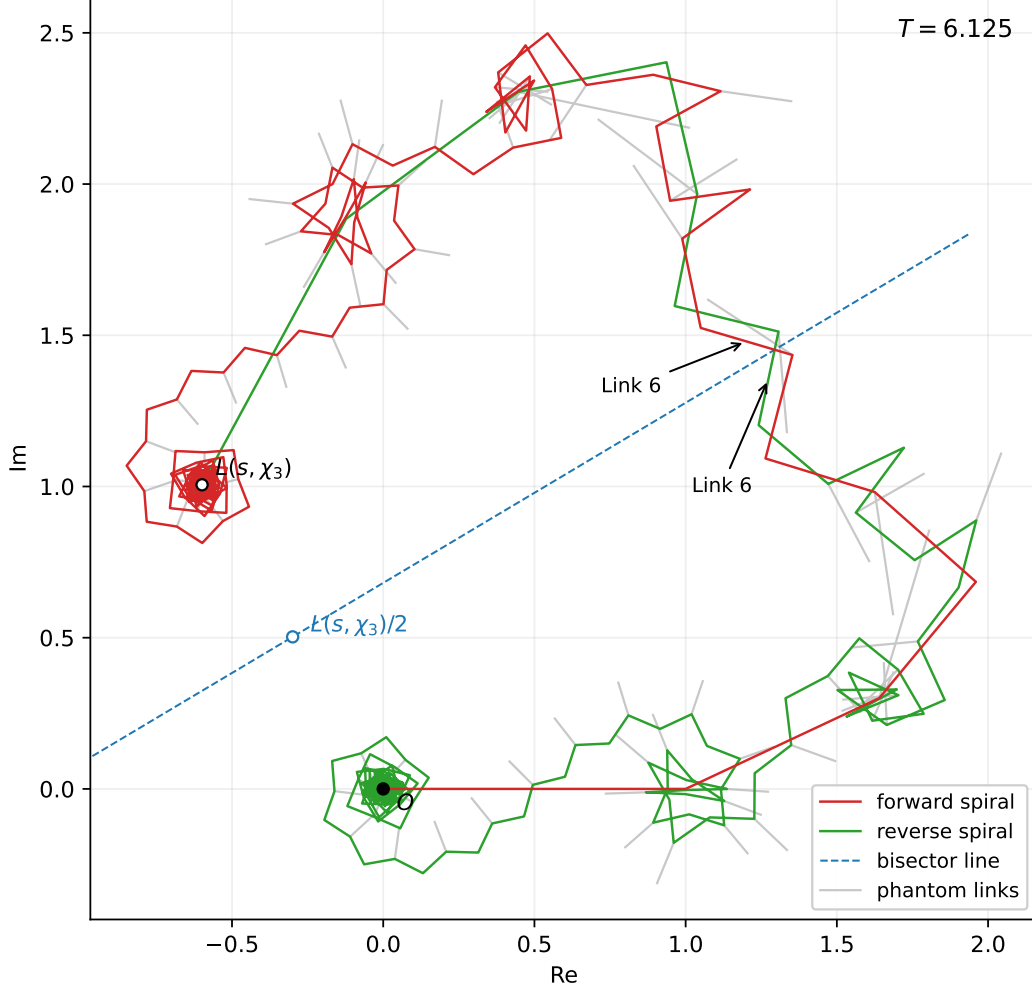


Figure 36: The two partial-sum spirals of $L(s, \chi_3)$ at $T = 6.125$ on the critical line ($t = I_3(T) \approx 176.13$), drawn with $\lfloor 2T(T+1)p \rfloor = 261$ summands. The forward chain (red) starts at the origin O and winds into its spiral around the analytic value $L(s, \chi_3)$ (open dot); the reverse chain (green) is its mirror image across the bisector line (dashed blue), the perpendicular bisector of the segment from O to $L(s, \chi_3)$ through the labeled midpoint $L(s, \chi_3)/2$, so the tail of the link chain sits on $L(s, \chi_3)$ and its head lands at O . Every third summand has $\chi_3(n) = 0$: these zero-length phantom links are drawn light gray as the connectors the chain skips. The labeled link 6 of each chain is the current bisector link: the two chains cross on it, on the bisector line. Generated by `fig_L3_spirals.py`.

Because only a fraction $(p-1)/p$ of the summands are links, the smooth *base* formula generalizing (10) is

$$I_p^{\text{base}}(T) = \frac{4\pi T}{(p-1) \log \frac{pT + (p-1)}{pT - (p-1)}}. \quad (102)$$

Asymptotically $I_p(T) \sim \frac{2p}{(p-1)^2} \pi T^2$: the leading coefficient is $\frac{3}{2}\pi$ for $p = 3$ and $\frac{5}{8}\pi$ for $p = 5$, against 2π for zeta's own $I(T)$. (Zeta is not the $p = 1$ member of this family, since the factor $2p/(p-1)^2$ diverges there; for zeta every summand is a link and there are no phantoms.) The

base does not run through the middle of the events: it touches I_p at the folds and lies below it in between, missing a one-signed bulge of period $p - 1$ imposed by the character. For $p = 3$ the contact is exact, since at even T the gate of (105) vanishes and $c = 2$ turns (104) into the base; for $p = 5$ the fitted correction returns to zero at $T \equiv 0 \pmod{4}$ to within 2×10^{-3} . One restores the bulge by writing

$$I_p(T) = I_p^{\text{base}}(T) + C_p(T). \quad (103)$$

For $p = 3$ the character is $[1, -1, 0]$. The base specializes to $I_3^{\text{base}}(T) = 2\pi T / \log \frac{3T+2}{3T-2}$, and the correction is built from the character-modulated map

$$c = \frac{3 + \cos(\pi T)}{2}, \quad I_3^{\text{geom}}(T) = \frac{c \pi T}{\log \frac{3T+c}{3T-c}}, \quad (104)$$

together with a small odd- T offset $\alpha(T) = 0.390914 + 0.01712 T^{-2}$ fitted to event scans:

$$I_3(T) = I_3^{\text{geom}}(T) + \frac{1}{2}(1 - \cos \pi T) \alpha(T). \quad (105)$$

(Equivalently $I_3 = I_3^{\text{base}} + C_3$ with $C_3 = I_3^{\text{geom}} - I_3^{\text{base}} + \frac{1}{2}(1 - \cos \pi T) \alpha$; the gate vanishes on even T , so $C_3 = 0$ on folds.) On odd $T = 1, \dots, 99$ the residual between scanned event times and (105) has RMSE about 8×10^{-6} .

For $p = 5$ the same pattern gives $I_5^{\text{base}}(T) = \pi T / \log \frac{5T+4}{5T-4}$ and a fitted period-4 correction

$$C_5(T) = 0.6206 - 0.6576 \cos \frac{\pi T}{2} + 0.0369 \cos \pi T + \frac{0.169}{T^{3.12}}. \quad (106)$$

Figure 37 compares $I(T)$, $I_3(T)$, and $I_5(T)$ on a common range of T .

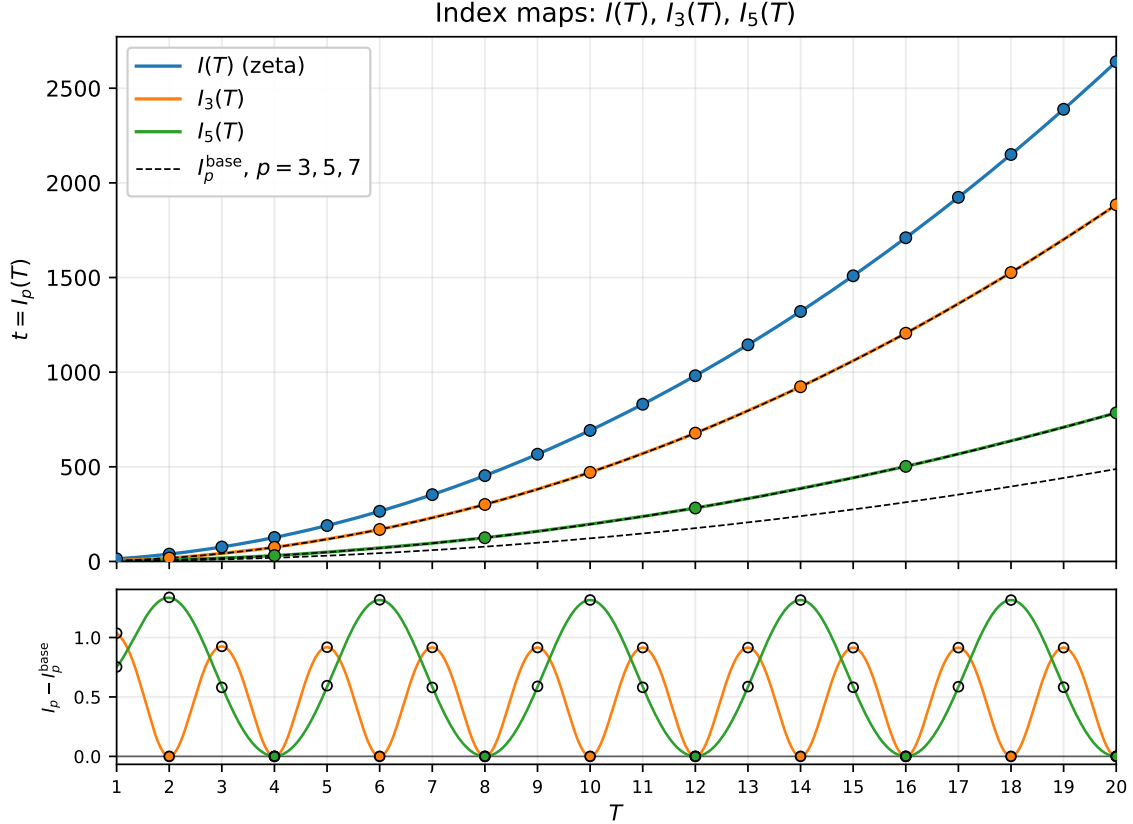


Figure 37: Top: index maps on $T \in [1, 20]$: zeta's $I(T)$ (blue), $I_3(T)$ for $p = 3$ (orange), and $I_5(T)$ for $p = 5$ (green). Dots mark the fold events, at $T \equiv 0 \pmod{p-1}$: every integer on the zeta curve, every other integer on I_3 , and every fourth integer on I_5 . Leading growth is quadratic with coefficients 2π , $\frac{3}{2}\pi$, and $\frac{5}{8}\pi$ respectively. Thin dashed black: the smooth base maps (102) for $p = 3, 5, 7$. The $p = 3$ and $p = 5$ bases are hidden under I_3 and I_5 at this scale; the $p = 7$ base is the lowest curve, drawn for comparison, no I_7 having been fitted here. Bottom: the character bulge on its own scale, $I_p - I_p^{\text{base}}$, which the top panel cannot resolve because it is of order 1 against values in the thousands. It is one-signed, returning to zero at the folds (dots) and rising to ≈ 0.92 between them for $p = 3$ and ≈ 1.33 for $p = 5$, and its amplitude barely decays: the $p = 3$ peaks are 0.9253, 0.9183, 0.9153 at $T = 3, 5, 11$. So the correction is a fixed additive offset, not a fixed relative one. Open black circles are the numerically scanned events at every integer T , measured against the same base: the folds sit on zero, the intermediate events ride the bulge, and all of them follow the modeled curves to within 1.7×10^{-5} for $p = 3$ and 1.0×10^{-2} for $p = 5$ on $T \leq 20$. Generated with the Dirichlet L -function companion note.

11 The yin and yang curves

As T increases the bisector point travels back and forth along the bisector link, and when T passes an integer it moves to the next link of the chain. It does so without any jump, as §8.1 shows from the fold relation (53): the handoff happens exactly at the instant when the outgoing and incoming links fold back onto one another and momentarily occupy the same line in the plane (when $\sigma = \frac{1}{2}$ the two bisector links are then coincident; for $\sigma \neq \frac{1}{2}$ they are parallel). The path of the bisector point is therefore continuous both as a curve in the complex plane and as a position along the one-dimensional link structure.

§10.1 watched the two links *neighboring* the bisector link in the bisector frame. Now watch the *reverse bisector link* in that same frame: holding the forward bisector link stationary while T increases, the reverse bisector link revolves around it, and each end of the revolving link follows a path reminiscent of a teardrop, the pair of paths together resembling a yin-yang (Figure 40). We call the paths of the two ends the *yin* and *yang* curves, and this section develops them: first their formulas, then the derivation of R_{1ps} and R_{2ps} from them, and finally their behavior as $T \rightarrow \infty$, where a familiar function makes an unexpected appearance. The same construction at every link is §12.

The bisector links at $\sigma = 1/4$, $T = 6.18$ ($t = I(T) \approx 279.85$, $m = 6$)

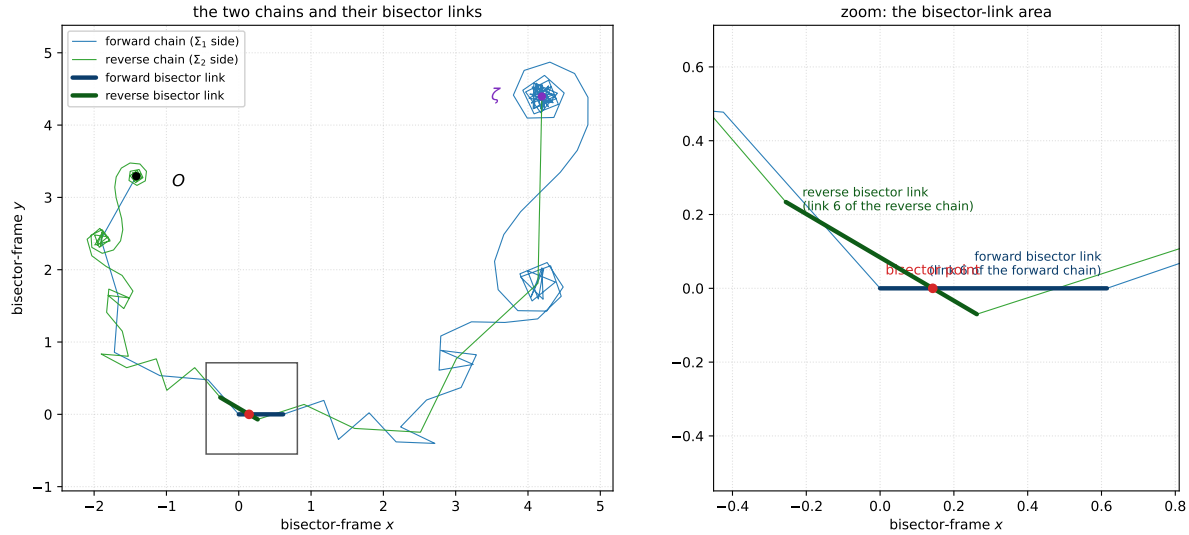


Figure 38: The two bisector links at $\sigma = \frac{1}{4}$, $T = 6.18$ ($t = I(T) \approx 279.85$, $m = 6$). Both panels are drawn in the coordinate frame of the *forward bisector link*: the whole picture is translated and rotated so that this link lies along the positive x -axis, with no rescaling. All lengths are true, so the link runs from 0 to $7^{-1/4} \approx 0.615$. *Left*: the forward chain (blue, starting at the world origin O) and the reverse chain (green, ending at ζ), each drawn out to the link nearest its spiral center; link 6 of each chain (the forward and reverse bisector links) is overdrawn with a thick stroke, and the red dot where they cross is the bisector point. *Right*: zoom into the gray window: the two bisector links crossing at the bisector point, which sits on the x -axis at distance d_1 from the origin. The bisector frame of this section additionally rescales this link to the unit interval $[0, 1]$ and watches the thick reverse link revolve around it. Generated by `fig-yinyang-spirals.py`.

The motivation is the question left open in §9.1: the two bisector links cross, but *where*? The

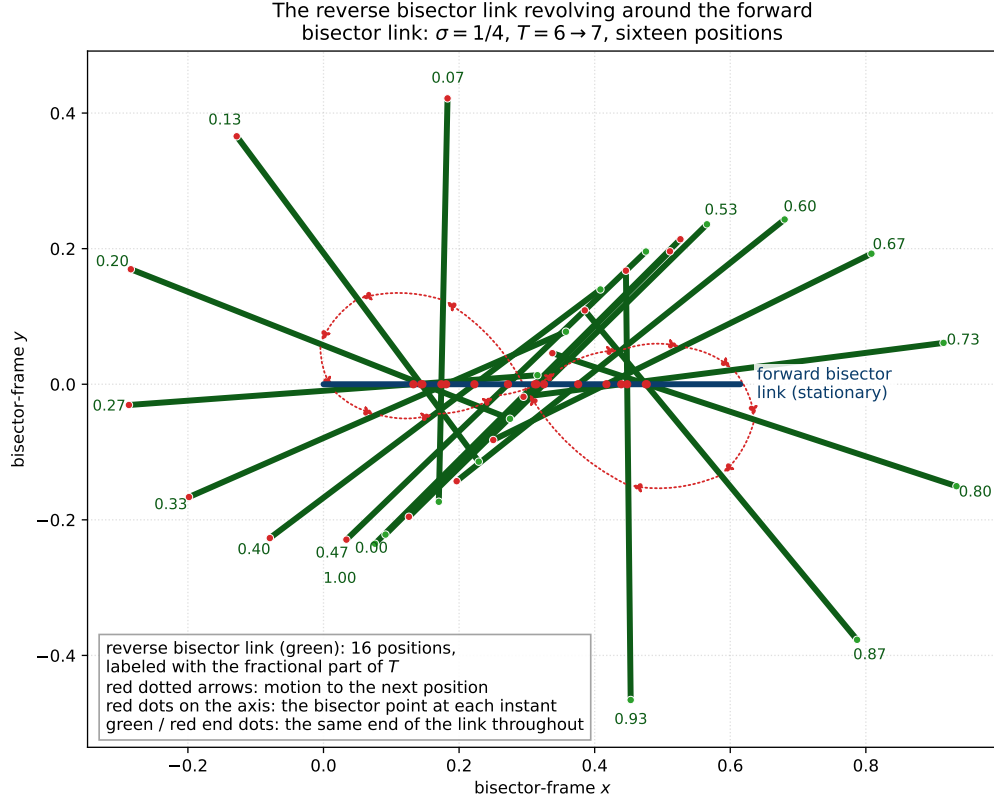


Figure 39: The two bisector links in the frame of the forward bisector link as T sweeps from 6 to 7 ($\sigma = \frac{1}{4}$, the frame of Figure 38, lengths unscaled). The forward bisector link is stationary (the single thick dark-blue segment on the x -axis) while the reverse bisector link (green, same thickness) is drawn at sixteen equally spaced values of T , each position labeled on its outer end with the fractional part of T ; over the sweep it revolves fully around the stationary link, and at $T = 7$ (labeled 1.00) it returns nearly to its $T = 6$ position; the heading turns through a half-revolution and then back, so the directed segment ends in the same orientation without having rotated all the way around. The red dotted arrows run from the middle of each position to the middle of the next, tracing the motion. The small green and red dots mark the two ends of the revolving link consistently: green is always the joint- m end and red the joint- $(m+1)$ end; they trace the yin and yang curves of Figure 40, in the same colors. Each red dot on the x -axis is the bisector point at one instant: the crossing of that position with the x -axis, at $(d_1, 0)$. Near the parallel instants (fractional part of T near $\frac{1}{4}$ and $\frac{3}{4}$) the revolving link is nearly parallel to the stationary one and the crossing races along the axis (the poles of d_1 of §8.4). Generated by `fig_bisector_sweep.py`.

answer matters because, by (77), the crossing (the bisector point) sits at distance d_1 from the joint Σ_1 along the forward bisector link and at distance d_2 from the joint $\Sigma_1 + R$ along the reverse bisector link. If we can *compute* the crossing, we obtain d_1 and d_2 , and with them a new formula for ζ that uses the fractional summands in place of the usual remainders, all without evaluating ζ or either partial sum.

11.1 The formulas for the yin and yang curves

The bisector frame sends the forward bisector link to $[0, 1]$, so the only question is where the *reverse* bisector link is. In the world frame its chain is anchored at ζ , which is both moving and precisely the quantity we do not wish to assume known; worse, the reverse bisector link also moves relative to its own anchor. The resolution is that the reverse chain's joint m sits at $\Sigma_1 + R$: relative to the forward joint Σ_1 it is displaced by exactly R , and R has its own integral representation (16), independent of any evaluation of ζ . Applying the frame map

$$z \mapsto (z - \Sigma_1) [T]^{\sigma + iI(T)} \quad (107)$$

(translate joint m to the origin, then divide by the link vector $[T]^{-\sigma - iI(T)}$), the two ends of the reverse bisector link land at

$$Y_{in1}(\sigma, T) = R [T]^{\sigma + iI(T)}, \quad (108)$$

$$Y_{ang1}(\sigma, T) = Y_{in1} - \chi [T]^{-1 + 2(\sigma + iI(T))}, \quad (109)$$

the first being the image of $\Sigma_1 + R$ and the second the image of the far end $\Sigma_1 + R - \chi [T]^{\sigma + iI(T) - 1}$. As T sweeps, these two endpoints orbit the stationary unit link along the yin and yang curves of Figure 40.

Because the frame map scales the bisector link to unit length, positions along it are *fractions*: the crossing we are about to compute is the fraction of the way along the link at which the bisector point sits, and multiplying that real fraction by the (unscaled) link vector recovers the fractional summand itself. The actual distance from the joint, the fraction times the true link length $[T]^{-\sigma}$, is the weight d_1 .

The reverse side has its own frame and its own pair of curves. Sending the *reverse* bisector link to $[0, 1]$ instead (translate its joint $\Sigma_1 + R$ to the origin and divide by its link vector $-\chi [T]^{\sigma + iI(T) - 1}$), the roles swap: now the forward bisector link revolves, its near end tracing

$$Y_{in2}(\sigma, T) = \frac{R [T]^{1 - \sigma}}{\chi e^{i\omega}}, \quad Y_{ang2}(\sigma, T) = Y_{in2} - \frac{[T]^{1 - 2\sigma}}{\chi e^{2i\omega}}, \quad (110)$$

with Y_{in2} the image of the joint Σ_1 and Y_{ang2} the image of the far end $\Sigma_1 + [T]^{-\sigma - iI(T)}$, and with $\omega = \omega(T) = I(T) \log [T]$ as always.

The next two subsections derive R_{1ps} from (Y_{in1}, Y_{ang1}) and R_{2ps} from (Y_{in2}, Y_{ang2}) . Each derivation is in two parts: first the modulus, found from the axis crossing; then the direction, which is simply that of the next summand.

11.2 Derivation of R_{1ps} from the yin and yang functions

First, the modulus. From its joint, how far along the forward bisector link do the two links intersect? This distance is $|R_{1ps}|$, because R_{1ps} is the fractional part of summand $[T]$ of Σ_1 , and that summand *is* the forward bisector link. In the local coordinate frame the link lies along the

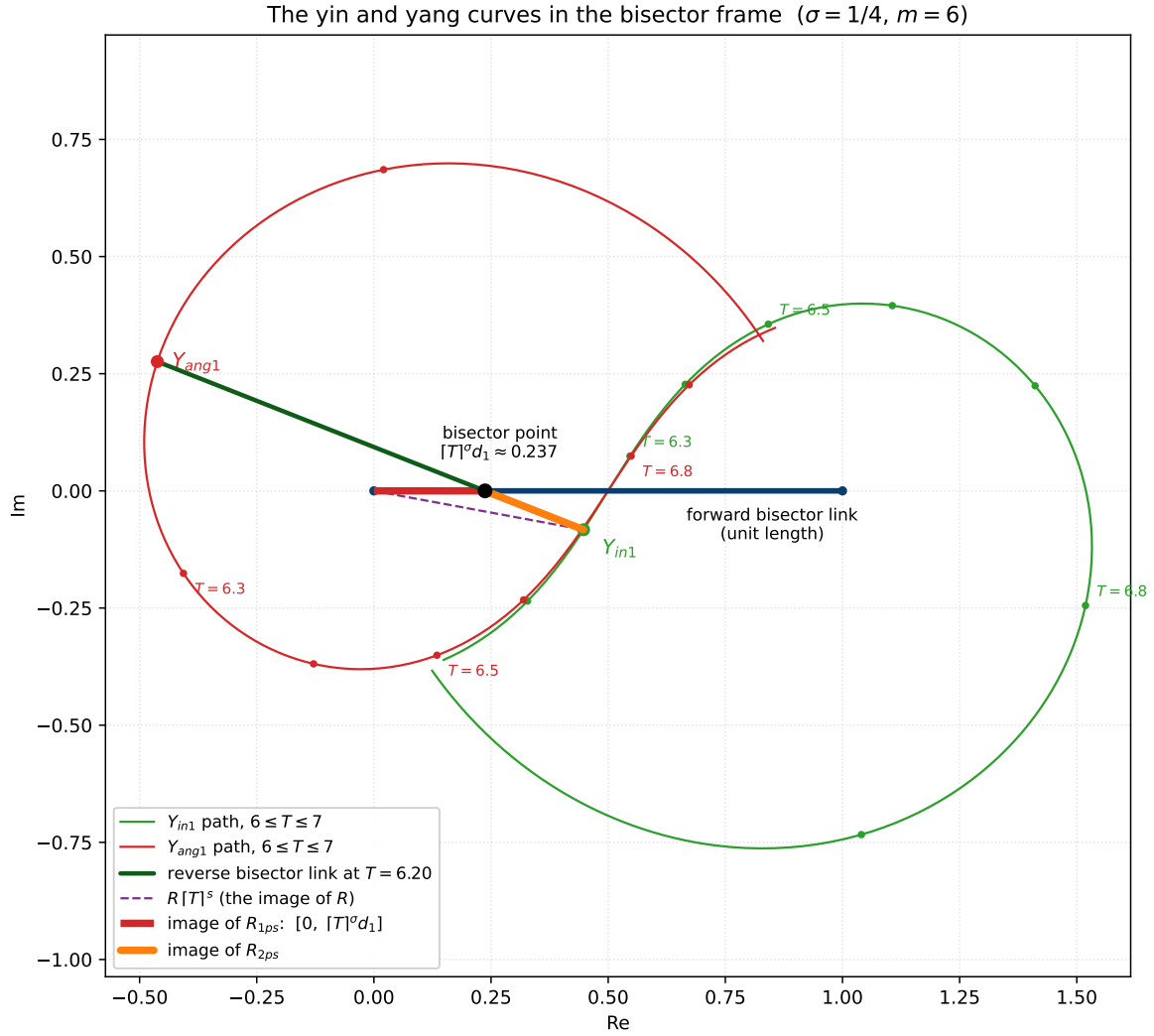


Figure 40: The yin and yang curves in the bisector frame, $\sigma = \frac{1}{4}$, $m = 6$. The frame sends the forward bisector link to the unit interval $[0, 1]$ (dark blue). As T sweeps $6 \leq T \leq 7$ (Figure 39), the near end of the reverse bisector link traces the yin curve Y_{in1} (green, one dot every 0.1 in T) and the far end traces the yang curve Y_{ang1} (red); the two teardrop paths together resemble a yin-yang. The reverse bisector link itself is drawn at $T = 6.20$ (dark green): it crosses the real axis at the bisector point (black dot), at the crossing fraction $[T]^\sigma d_1 \approx 0.237$ of the way along the unit link. The dashed violet vector from 0 to Y_{in1} is the in-frame image of Siegel's remainder R ; the crossing splits it into the images of R_{1ps} (red, the piece of the unit link from 0 to the bisector point) and R_{2ps} (orange, from the bisector point to Y_{in1}). Generated by `fig_yinyang.py`.

real axis, so the intersection is the point where the segment from Y_{in1} to Y_{ang1} crosses the axis. By the standard segment–axis intersection formula that crossing is

$$Y_{in1} - \frac{\Im Y_{in1}}{\Im Y_{ang1} - \Im Y_{in1}} (Y_{ang1} - Y_{in1}), \quad (111)$$

a real number. Substituting (108)–(109) and canceling $Y_{ang1} - Y_{in1} = -\chi [T]^{-1+2(\sigma+iI(T))}$ from the ratio,

$$(111) = R [T]^{\sigma+iI(T)} - \frac{\Im(R [T]^{\sigma+iI(T)})}{\Im(\chi [T]^{-1+2(\sigma+iI(T))})} \chi [T]^{-1+2(\sigma+iI(T))}. \quad (112)$$

Writing $[T]^{\sigma+iI(T)} = [T]^\sigma e^{i\omega}$ and expanding the imaginary parts in components,

$$(111) = R [T]^{\sigma+iI(T)} - [T]^{1-\sigma} \frac{\Re R \sin \omega + \Im R \cos \omega}{\Re \chi \sin 2\omega + \Im \chi \cos 2\omega} \chi [T]^{-1+2(\sigma+iI(T))}, \quad (113)$$

and finally, writing $\varphi = \arg R$ and $\psi = \arg \chi$ and collapsing with the sine difference identity $\sin(2\omega + \psi) \cos(\omega + \varphi) - \cos(2\omega + \psi) \sin(\omega + \varphi) = \sin(\omega - \varphi + \psi)$, the whole expression reduces to

$$(111) = [T]^\sigma |R| \frac{\sin(\omega - \varphi + \psi)}{\sin(2\omega + \psi)} = [T]^\sigma d_1. \quad (114)$$

The reader will recognize the sine quotient: it is, symbol for symbol, the definition (19) of the weight d_1 . *This is where that definition came from.* The crossing (114) is the crossing fraction (the “fractional amount” column of Table 1) along the unit link at which the reverse link crosses, and dividing out the frame’s scaling $[T]^\sigma$ leaves the actual distance from the joint, d_1 .

Second, the direction. The forward bisector link points along the next summand of Σ_1 : extending the summation cutoff by one,

$$\sum_{n=1}^{m+1} n^{-\sigma-iI(T)} - \sum_{n=1}^m n^{-\sigma-iI(T)} = [T]^{-\sigma-iI(T)} \quad (115)$$

(for non-integer T , so $m+1 = [T]$). Multiplying the fraction (114) by this link vector undoes the frame map and yields the fractional summand itself:

$$R_{1ps} = ([T]^\sigma d_1) \cdot [T]^{-\sigma-iI(T)} = d_1 [T]^{-iI(T)}, \quad (116)$$

which is definition (17).

Other forms. Two equivalent ways to write R_{1ps} fall out of the same computation. Subtracting the crossing from Y_{in1} instead of building it up from 0 gives

$$R_{1ps} = R - |R| \frac{\sin(\omega + \varphi)}{\sin(2\omega + \psi)} e^{i(\omega + \psi)} \quad (117)$$

(the subtracted term is exactly R_{2ps} , anticipating Theorem 4.1), and keeping the imaginary parts unexpanded gives the compact

$$R_{1ps} = R - \chi e^{i\omega} \frac{\Im(R e^{i\omega})}{\Im(\chi e^{2i\omega})}. \quad (118)$$

11.3 Derivation of R_{2ps} from the yin and yang functions

The same procedure in the reverse frame, using (Y_{in2}, Y_{ang2}) of (110), locates the crossing of the (now revolving) forward bisector link with the real axis at the crossing fraction

$$\frac{[T]^{1-\sigma}}{|\chi|} |R| \frac{\sin(\omega + \varphi)}{\sin(2\omega + \psi)} = \frac{[T]^{1-\sigma}}{|\chi|} d_2 \quad (119)$$

of the way along the unit reverse link, again the “fractional amount” column of Table 1, and again the sine quotient is, symbol for symbol, the definition (20) of d_2 : this is where *it* came from. (The two normalized fractions are nearly symmetric: since $|\chi(\sigma + it)| = (t/2\pi)^{\frac{1}{2}-\sigma}(1 + O(1/t))$ and $\sqrt{t/2\pi} \approx T + \frac{1}{2}$, the prefactor $[T]^{1-\sigma}/|\chi|$ is asymptotically $[T]^\sigma$, matching the forward side.) Multiplying by the reverse link vector $\chi [T]^{\sigma+iI(T)-1}$ undoes the frame map:

$$R_{2ps} = \frac{[T]^{1-\sigma}}{|\chi|} d_2 \cdot \chi [T]^{\sigma+iI(T)-1} = d_2 [T]^{iI(T)} \frac{\chi}{|\chi|}, \quad (120)$$

which is definition (18). As with the forward side there are equivalent rewritings, for example

$$R_{2ps} = R - |R| e^{-i\omega} \frac{\sin(\omega - \varphi + \arg \chi(1 - \sigma + iI(T)))}{\sin(2\omega + \arg \chi(1 - \sigma + iI(T)))} \quad (121)$$

and

$$R_{2ps} = R - e^{-i\omega} \frac{\Im\left(\frac{R}{\chi(1 - \sigma + iI(T)) e^{i\omega}}\right)}{\Im\left(\frac{1}{\chi(1 - \sigma + iI(T)) e^{2i\omega}}\right)}, \quad (122)$$

where the reflected factor may be used interchangeably in the argument because $\arg \chi(1 - \sigma + iI(T)) = \arg \chi(\sigma + iI(T))$: from $\chi(s)\chi(1-s) = 1$ and the Schwarz reflection $\chi(\bar{s}) = \overline{\chi(s)}$ one gets $\chi(1 - \sigma + it) = 1/\overline{\chi(\sigma + it)}$, which has the same argument as $\chi(\sigma + it)$ and reciprocal modulus.

The poles cancel in the sum. The two weights share the denominator $\sin(2\omega + \psi)$, and §8.4 located the isolated heights at which it produces genuine poles. In the *sum* $d_1 + d_2$ those poles cancel: adding the numerators with the sum-to-product identity $\sin(\omega - \varphi + \psi) + \sin(\omega + \varphi) = 2 \sin(\omega + \frac{\psi}{2}) \cos(\varphi - \frac{\psi}{2})$ and writing $\sin(2\omega + \psi) = 2 \sin(\omega + \frac{\psi}{2}) \cos(\omega + \frac{\psi}{2})$, the offending factor divides out:

$$d_1 + d_2 = |R| \frac{\cos(\varphi - \frac{\psi}{2})}{\cos(\omega + \frac{\psi}{2})}. \quad (123)$$

The zeros of the remaining denominator $\cos(\omega + \frac{\psi}{2})$ are matched by zeros of the numerator (at those instants R is parallel to the common link direction, forcing $\varphi \equiv -\omega \pmod{\pi}$), so every singularity of (123) is removable: the sum is smooth across all the poles of its two terms. Figure 41 shows this at $\sigma = 0.1$.

This is how the definitions of §4 were found: the sine expressions (19)–(20) are nothing other than the real-axis crossings of the yang segments, read off in the two bisector frames. Only afterwards was Theorem 4.1 proved, confirming that the two fractional summands so constructed add back to Siegel’s R exactly. Note the economy of the construction: the bisector point, d_1 , d_2 , and both fractional summands are obtained from R and χ alone. Numerically we computed R to high accuracy by Kuznetsov’s method (§7.1), which is how the pictures, and then the identity, were checked.

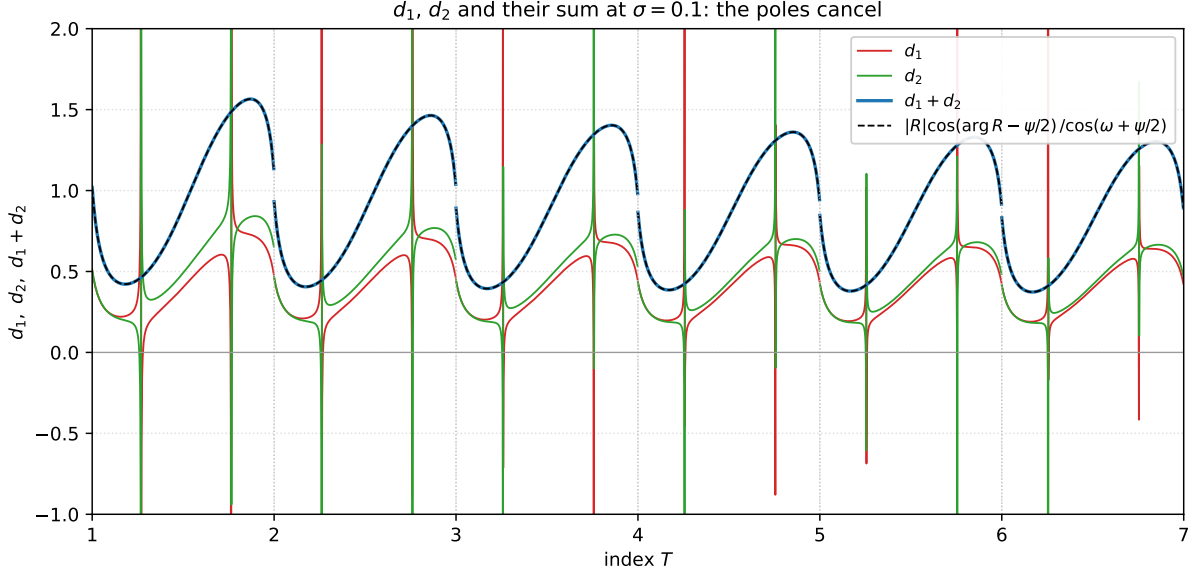


Figure 41: d_1 (red), d_2 (green) and their sum (blue) at $\sigma = 0.1$, $1 \leq T \leq 7$. Twice per unit of T the individual weights blow up at the parallel-link poles of §8.4, but the poles are equal and opposite and the sum passes smoothly through every one of them; the black dashed curve is the closed form (123), indistinguishable from the blue sum (they agree to machine precision on the plotted grid). Generated by `fig_d1_d2_sum.py`.

11.4 Comparison of yin and yang to Siegel's integral result

Siegel's 1932 paper [20] evaluates the integral in his equation 5 to what he calls the “desired result”

$$F(t) = \frac{1}{1 - e^{-2\pi it}} - \frac{e^{\pi it^2}}{e^{\pi it} - e^{-\pi it}}, \quad (124)$$

which can also be written

$$F(t) = \frac{e^{2\pi it} - e^{\pi it(t+1)}}{e^{2\pi it} - 1} = \left(1 - e^{\pi i(t^2-t)}\right) \sum_{n=0}^{\infty} e^{-2\pi int}, \quad (125)$$

the last form using the geometric series $1/(1 - e^{-2\pi it}) = \sum_{n \geq 0} e^{-2\pi int}$. Its complex conjugate has the compact form

$$\overline{F(t)} = \frac{e^{-\pi it^2} - e^{-\pi it}}{2i \sin(\pi t)}. \quad (126)$$

Remark 11.1 (regarding F as the source of Siegel's $f(s)$). F is not merely adjacent to the function f of (46); it is what f is built from. In his §3 Siegel multiplies his equation 5, that is (124), by u^{-s} and integrates along the bisector of the first quadrant, and under that operation the two terms of F separate into the two halves of zeta. The pole term $1/(1 - e^{-2\pi iu})$ yields $-(2\pi)^{s-1} e^{i\pi(1-s)/2} \Gamma(1-s) \zeta(1-s)$, while the Gaussian term yields f itself:

$$f(s) = (e^{\pi is} - 1) \int_0^{e^{i\pi/4}\infty} \frac{u^{-s} e^{\pi iu^2}}{e^{\pi iu} - e^{-\pi iu}} du \quad (\sigma < 0), \quad (127)$$

the restriction $\sigma < 0$ being needed only for convergence at the endpoint $u = 0$; f is continued analytically from there, and (46) itself converges for every s . The same separation can be read off (126): since $1/(1 - e^{2\pi it}) = -e^{-\pi it}/(2i \sin \pi t)$, the numerator term $e^{-\pi it^2}$ of $\overline{F}$ is the $f(s)$ side and $-e^{-\pi it}$ is the $\zeta(1 - s)$ side, so the two lobes carry the two halves in their numerator (Figure 42). The yin-yang function is therefore the common source of both $\Sigma_1 + R_{1ak}$ of (45) and its reflected counterpart, which is one route into the convergence proof left open in §11.6.

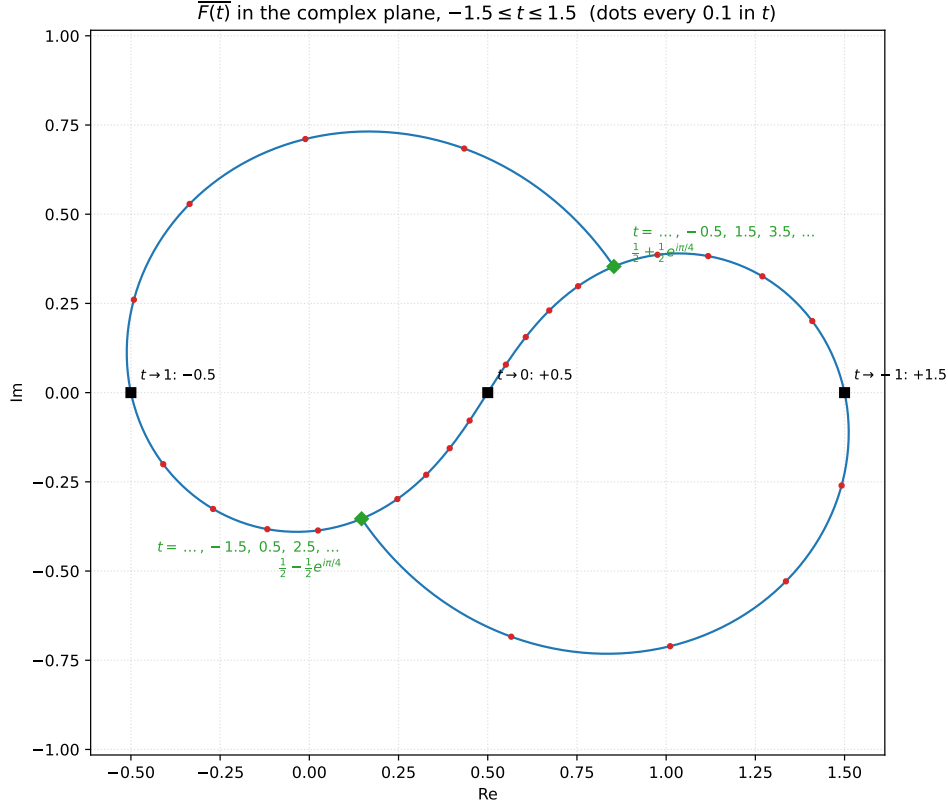


Figure 42: The trace of $\overline{F(t)}$ of (126) in the complex plane for $-1.5 \leq t \leq 1.5$ (red dots every 0.1 in t): the curve forms two congruent teardrop-like lobes, forming a shape reminiscent of the yin-yang symbol rooted in ancient texts like the *I Ching*. The curve is point-symmetric through $\frac{1}{2}$. The apparent poles at integer t are removable, with $\overline{F(t)} \rightarrow \frac{1}{2} - n$ as $t \rightarrow n$ (black squares). A curiosity: at *every* half-integer $t = k + \frac{1}{2}$ the exponent $t^2 = k(k+1) + \frac{1}{4}$ has $k(k+1)$ even, so $e^{-\pi it^2} = e^{-i\pi/4}$ always, and the whole expression collapses to just two values, $\overline{F}(k + \frac{1}{2}) = \frac{1}{2} - (-1)^k \frac{1}{2} e^{i\pi/4}$ (green diamonds): the curve returns to the point $\frac{1}{2} + \frac{1}{2} e^{i\pi/4} \approx 0.854 + 0.354i$ at $t = \dots, -0.5, 1.5, 3.5, \dots$ and to the point $\frac{1}{2} - \frac{1}{2} e^{i\pi/4} \approx 0.146 - 0.354i$ at $t = \dots, -1.5, 0.5, 2.5, \dots$, each revisited every $\Delta t = 2$, forever; they are the meeting points of the lobes. Generated by `fig_F_curve.py`.

We conjecture that as $T \rightarrow \infty$ the yin and yang curves converge to a universal limit curve built from this function, universal in that it does not depend on σ . The precise limit is most cleanly written through the Ψ function, which we do next.

11.5 The yin and yang curves are not symmetrical

While the name “yin-yang” invokes a pictorial similarity as intended, it also invites a symmetry that is not there. In the taijitu the two lobes are congruent, each carried onto the other by a half-turn about the center. In our case, the two curves are the two ends of one moving link, and (109) says exactly how they are related:

$$Y_{ang1} = Y_{in1} - \chi [T]^{2s-1}, \quad s = \sigma + iI(T), \quad (128)$$

so the yang curve is the yin curve translated by $-v$, where $v = \chi [T]^{2s-1}$ is the revolving link’s own vector in the frame, of length $|\chi| [T]^{2\sigma-1}$. Were v the same at every T the two curves would be congruent by construction. It is not. Over one unit of the index the direction of v swings through π and comes back, returning to where it began, with net turning zero. The length of v varies too, and it is exactly 1 only on the critical line, where $|\chi| = 1$ and the exponent $2\sigma - 1$ vanishes: the revolving link and the stationary link are then equal in length, which is the $d_1 = d_2$ symmetry of §9.2 seen in this frame.

What the taijitu has is a half-turn about the center of the picture, and here that center is the midpoint of the stationary link: the half-turn is $z \mapsto 1 - z$, which exchanges the link’s two ends. So if the yin and yang curves were related as the taijitu’s two lobes are, this map would carry one onto the other, and it is enough to draw $1 - Y_{ang1}$ on top of Y_{in1} and look. Figure 43 plots Y_{in1} against $1 - Y_{ang1}$ for $0 \leq T \leq 10$, one loop of each per unit of the index, on the critical line and at $\sigma = 0.05$ and $\sigma = 0.95$ above and below it. The reflection brings the two families almost on top of one another, and both converge to the single limit teardrop of §11.6, but at no finite T do they coincide. Each loop crosses its reflected partner twice in every unit of the index, running outside it along one arc and inside along the other, and the two enclose different areas. That difference in area narrows as T grows and widens away from the critical line, where the two lobes also trade which of them is the larger, near the line though not exactly on it. In the magnified windows of Figure 43 the yin curve (green) is the one running outside its reflected partner (red) in the top two rows, and the inside one in the bottom row, where the order has swapped.

What the critical line does give exactly is a mirror symmetry, though between the two frames rather than between the two curves of one frame. At $\sigma = \frac{1}{2}$ we have $[T]^s = [T]^{1/2} e^{i\omega}$ and, from $\zeta = \chi \bar{\zeta}$ and $\Sigma_2 = \chi \bar{\Sigma}_1$ with $|\chi| = 1$, also $R = \chi \bar{R}$. Comparing (108) with (110) then gives

$$\overline{Y_{in1}} = Y_{in2}, \quad \overline{Y_{ang1}} = Y_{ang2} \quad (\sigma = \tfrac{1}{2}), \quad (129)$$

which we confirm to 25 digits. Off the line both identities fail, the first by 6.4×10^{-2} and the second by 7.1×10^{-2} at $\sigma = \frac{1}{4}$, and by the same two amounts at $\sigma = \frac{3}{4}$.

An exact symmetry between the two curves themselves does arrive, but only in the limit, where the yang curve stops being a second curve at all: as $T \rightarrow \infty$ the yang settles onto the very teardrop the yin traces, half a unit of the index behind. So the asymmetry of this subsection is a finite- T effect, and the limit erases it. That limit curve, and the classical function it turns out to be built from, are the subject of the next subsection.

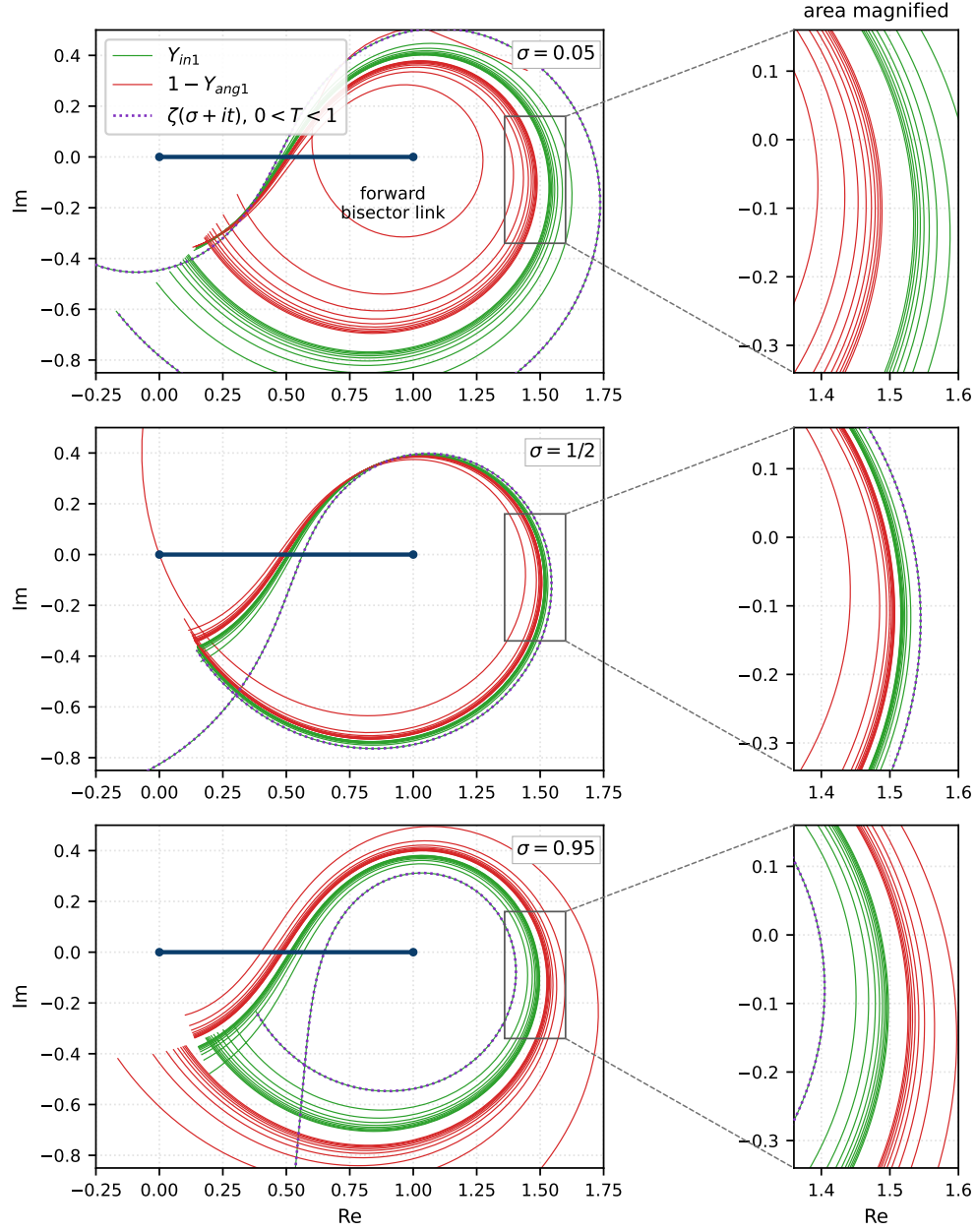


Figure 43: The yin curve (green) against the reflected yang curve $1 - Y_{ang1}$ (red), for $0 \leq T \leq 10$, which is one loop of each per unit of the index, out to $t = 692.2$. One row per σ , all on the same scale: $\sigma = 0.05$ on top, the critical line in the middle, $\sigma = 0.95$ below. The half-turn $z \mapsto 1 - z$ exchanges the ends of the stationary link (dark blue), so under the symmetry the picture suggests the two families would coincide. *Left*: all ten periods, the view cropped to the cluster. The stray arcs crossing it belong to the first period $0 \leq T \leq 1$, where $t < 13.6$ and the frame is still far from its limit; the later periods pile up on the limit teardrop, and they do so at every σ . Incidentally the yin curve *is* zeta there, both partial sums being empty and $(m+1)^s = 1$: the dotted purple line is $Y_{in1}(\sigma, T) = \zeta(\sigma + iT)$, $0 < T < 1$. *Right*: the boxed window magnified, chosen away from the two crossings. The families interleave and close on one another as T grows, and which of the two runs outside depends on σ : green outside red at $\sigma = 0.05$, red outside green at $\sigma = 0.95$. Generated by `fig-yinyang-asym.py`.

11.6 The limit curve and the Ψ function: the C_0 connection

As $T \rightarrow \infty$ with the fractional part $q = T - [T]$ held fixed, the yin curve converges, regardless of σ , to

$$Y_{\inf}(q) = 1 - \Psi(q) e^{-2\pi i (q^2 - \frac{1}{16})}, \quad (130)$$

where

$$\Psi(x) = \frac{\cos(2\pi(x^2 - x - \frac{1}{16}))}{\cos(2\pi x)} \quad (131)$$

is the Riemann–Siegel correction function (it appears in §15.3 of Titchmarsh’s book [22] and in §5.4.1 of Pugh’s thesis [19]). The yang curve follows the *same* universal function half a unit behind in the argument, $Y_{\text{ang}1}(\sigma, T) \rightarrow Y_{\inf}(q - \frac{1}{2})$, but not the same teardrop: $Y_{\inf}$ is not 1-periodic, and over one unit interval the yang end traces the point reflection of the yin teardrop through $\frac{1}{2}$, the two congruent teardrops together forming the yin-yang of Figure 40 (Lemma 11.2 and Theorem 11.3 below make both statements exact). The measured convergence is uniform in q away from the removable points below, at rate $O(1/T)$, and essentially independent of σ : the maximum of $|Y_{\text{in}1} - Y_{\inf}(q)|$ over a q -grid is 0.028, 0.015, 0.027 at $T \approx 10$ and 0.0076, 0.0042, 0.0075 at $T \approx 40$ for $\sigma = 0.2, 0.5, 0.9$ respectively. Figure 44 shows the convergence, including the pleasing fact that the very first yin path is the zeta function itself: for $0 < T < 1$ both partial sums are empty and $[T] = 1$, so $Y_{\text{in}1} = R = \zeta(\sigma + iI(T))$. The iconic zeta trajectory is yin orbit number zero.

The convergence is not merely an observation; it is a theorem, and the positivity machinery of §8.5 already contains most of its proof: by (108) the yin point is R carried through the frame map, and §8.5 knows the size, the sign, and the phase of R to exactly the accuracy needed. The one genuinely new ingredient is a reflection identity for the amplitude Ψ .

Lemma 11.2. *Write $A(x) = 2\pi(x^2 - \frac{1}{16})$ and $\theta(x) = 2\pi x(1 - x) - \frac{3\pi}{8}$, the phase already met in (70). Then for all real x , with removable points filled in by continuity,*

$$1 - \Psi(x) e^{-iA(x)} = \Psi(x + \frac{1}{2}) e^{i\theta(x)}, \quad (132)$$

and consequently the limit curve satisfies the exact half-lag relation

$$Y_{\inf}(q) - Y_{\inf}(q - \frac{1}{2}) = e^{2i\theta(q)} \quad (133)$$

and the point symmetry

$$Y_{\inf}(\frac{1}{2} - x) = 1 - Y_{\inf}(x) : \quad (134)$$

the map $z \mapsto 1 - z$, rotation by π about the point $\frac{1}{2}$, carries the curve parametrized on $[0, 1]$ onto the curve parametrized on $[-\frac{1}{2}, \frac{1}{2}]$.

Proof. For (132), multiply through by $\cos(2\pi x)$ and write $\varphi = 2\pi x$. On the left, $\Psi(x) \cos(2\pi x) = \cos(A - \varphi)$, since $2\pi(x^2 - x - \frac{1}{16}) = A(x) - \varphi$. On the right, $\cos(2\pi(x + \frac{1}{2})) = -\cos(2\pi x)$ and $(x + \frac{1}{2})^2 - (x + \frac{1}{2}) = x^2 - \frac{1}{4}$, so $\Psi(x + \frac{1}{2}) \cos(2\pi x) = -\cos(A(x) - \frac{\pi}{2}) = -\sin A$; moreover $\theta(x) = \varphi - A - \frac{\pi}{2}$, so $e^{i\theta} = -ie^{i(\varphi - A)}$. The claim therefore reads

$$\cos \varphi - \cos(A - \varphi) e^{-iA} = (-\sin A) (-i) e^{i(\varphi - A)} = i \sin A e^{i(\varphi - A)},$$

and expanding the left side settles it: the real part is $\cos \varphi - \cos(A - \varphi) \cos A = \sin A \sin(A - \varphi)$ (product formula), the imaginary part is $\sin A \cos(A - \varphi)$, and $\sin A (\sin(A - \varphi) + i \cos(A - \varphi)) = i \sin A e^{-i(A - \varphi)}$, which is the right side.

For (133), two elementary phase relations do the work: $A(q - \frac{1}{2}) = A(q) - \varphi + \frac{\pi}{2} = -\theta(q)$ and, applying (132) at $x = q - \frac{1}{2}$, $Y_{\inf}(q - \frac{1}{2}) = \Psi(q) e^{i\theta(q - \frac{1}{2})}$ with $\theta(q - \frac{1}{2}) \equiv 2\varphi - A \pmod{2\pi}$. Then $Y_{\inf}(q) - Y_{\inf}(q - \frac{1}{2}) = 1 - \Psi(q)(e^{-iA} + e^{i(2\varphi - A)}) = 1 - 2\cos(A - \varphi) e^{-i(A - \varphi)} = -e^{-2i(A - \varphi)} = e^{2i\theta(q)}$, using $\Psi(q) \cdot 2\cos\varphi = 2\cos(A - \varphi)$ in the middle step and $\theta = \varphi - A - \frac{\pi}{2}$ at the end.

For (134), first $\Psi(1-x) = \Psi(x)$: the numerator of (131) is unchanged because $(1-x)^2 - (1-x) = x^2 - x$, and so is the denominator because $\cos(2\pi(1-x)) = \cos(2\pi x)$. Then, by the polar form (132) evaluated at $\frac{1}{2} - x$ together with $\theta(\frac{1}{2} - x) = 2\pi(\frac{1}{2} - x)(\frac{1}{2} + x) - \frac{3\pi}{8} = \frac{\pi}{8} - 2\pi x^2 = -A(x)$,

$$Y_{\inf}(\frac{1}{2} - x) = \Psi(1-x) e^{i\theta(\frac{1}{2} - x)} = \Psi(x) e^{-iA(x)} = 1 - Y_{\inf}(x),$$

the last step being the definition (130). $\square$

Theorem 11.3. *Fix $q \in (0, 1)$ and $\sigma \in (0, 1)$. Then*

$$\lim_{\substack{T \rightarrow \infty \\ \{T\}=q}} Y_{in1}(\sigma, T) = Y_{\inf}(q), \quad \lim_{\substack{T \rightarrow \infty \\ \{T\}=q}} Y_{ang1}(\sigma, T) = Y_{\inf}(q - \frac{1}{2}),$$

at rate $O(1/T)$, uniformly for q and σ in compact subsets of $(0, 1)$; the limit does not depend on σ . On the critical line the proof is complete modulo Gabcke's bound (74); off it, it additionally uses the first-order Siegel asymptotics (64), with the same caveat as in Theorem 8.5.

Proof. Throughout, C_0 and Ψ are the same function, as (136) below records; we keep Gabcke's symbol C_0 for the analytic input and Ψ for the limit curve.

Step 1 (the yin point in asymptotic closed form). By (108), $Y_{in1} = R(m+1)^\sigma e^{i\omega}$. On the critical line $R = e^{-i\vartheta} r$ with r real, so $Y_{in1} = \sqrt{m+1} r e^{iu}$, and inserting (73)–(74),

$$Y_{in1} = \left(\frac{m+1}{a}\right)^{1/2} (-1)^{N-1} (C_0(\hat{p}) + E) e^{iu} + [N = m+1] (1 + e^{2iu}), \quad (135)$$

where the extra summand of (73) became the closed pair $2\cos u e^{iu} = 1 + e^{2iu}$. Off the line the first-order Siegel asymptotics (64) give the same leading form: the frame factor $(m+1)^\sigma e^{i\omega}$ carries the two bracket pieces of (64) to $\left(\frac{m+1}{a}\right)^\sigma e^{i(\omega - \tilde{\vartheta})}$ and $|\chi| a^{\sigma-1} (m+1)^\sigma e^{i(\omega + \psi + \tilde{\vartheta})}$, and since $|\chi| = a^{1-2\sigma}(1 + O(1/t))$ and $\psi = -2\vartheta + O_\sigma(1/t)$, both tend to e^{iu} ; likewise the Iverson pair $(m+1)^{-s} + \chi(m+1)^{s-1}$ maps to $1 + |\chi|(m+1)^{2\sigma-1} e^{i(2\omega + \psi)} \rightarrow 1 + e^{2iu}$. In every case

$$Y_{in1} = (-1)^{N-1} C_0(\hat{p}) e^{iu} (1 + O(1/T)) + [N = m+1] (1 + e^{2iu}) (1 + O(1/T)).$$

Step 2 (the phase, now with its full angle). The expansion behind (70), kept as an angle rather than used only through its cosine, reads

$$u \equiv \pi(m+1) + \pi\left(\frac{1}{8} - 2\delta^2\right) + O(1/T) \pmod{2\pi}, \quad \delta := a - (m+1) \rightarrow q - \frac{1}{2},$$

(substitute $t = 2\pi a^2$ and the asymptotic series of ϑ into $u = t \log \frac{m+1}{a} + \frac{t}{2} + \frac{\pi}{8} + O(1/t)$ and reduce πa^2 modulo 2π), and $\pi(\frac{1}{8} - 2(q - \frac{1}{2})^2) = \theta(q)$, so $u \equiv \pi(m+1) + \theta(q) + O(1/T)$. Meanwhile $\hat{p} \rightarrow q + \frac{1}{2}$ (with $N = m$) for $q < \frac{1}{2}$ and $\hat{p} \rightarrow q - \frac{1}{2}$ (with $N = m+1$) for $q > \frac{1}{2}$, by (69).

Step 3 (case $q < \frac{1}{2}$). Here $N = m$, so $(-1)^{N-1} e^{i\pi(m+1)} = (-1)^{2m} = 1$ and the Iverson term is absent:

$$Y_{in1} \rightarrow \Psi\left(q + \frac{1}{2}\right) e^{i\theta(q)} = Y_{\inf}(q)$$

by the reflection identity (132).

Step 4 (case $q > \frac{1}{2}$). Here $N = m + 1$, the sign product is -1 , and the Iverson pair is present:

$$Y_{in1} \longrightarrow 1 - \Psi(q - \tfrac{1}{2})e^{i\theta(q)} + e^{2i\theta(q)}.$$

Since $\theta(q) = -A(q - \frac{1}{2})$ exactly (Lemma 11.2), the first two terms are $Y_{\text{inf}}(q - \frac{1}{2})$, and the half-lag relation (133) finishes: $Y_{in1} \rightarrow Y_{\text{inf}}(q - \frac{1}{2}) + e^{2i\theta(q)} = Y_{\text{inf}}(q)$.

Step 5 (the yang point). By (109), $Y_{ang1} - Y_{in1} = -\chi(m+1)^{2\sigma-1}e^{2i\omega}$, whose modulus $|\chi|(m+1)^{2\sigma-1} \rightarrow 1$ and phase $2\omega + \psi = 2u + O_\sigma(1/t) \equiv 2\theta(q) + O(1/T)$, so the subtracted vector tends to $e^{2i\theta(q)}$ and $Y_{ang1} \rightarrow Y_{\text{inf}}(q) - e^{2i\theta(q)} = Y_{\text{inf}}(q - \frac{1}{2})$, again by (133). One caution on reading this as a lag: by the point symmetry (134), $Y_{\text{inf}}(q - \frac{1}{2}) = 1 - Y_{\text{inf}}(1 - q)$, so within each unit interval the yang end runs the yin limit curve time-reversed and rotated by π about the point $\frac{1}{2}$; equivalently, $Y_{ang1}(\sigma, T) \approx 1 - Y_{in1}(\sigma, 2m+1-T)$ for every fractional part. The naive lag reading $Y_{ang1}(\sigma, T) \approx Y_{in1}(\sigma, T - \frac{1}{2})$ is correct only on $\{T\} > \frac{1}{2}$, where T and $T - \frac{1}{2}$ share the frame link $m + 1$; for $\{T\} < \frac{1}{2}$ the point $Y_{in1}(\sigma, T - \frac{1}{2})$ lives in the previous frame, and since Y_{inf} is not 1-periodic, the two sides differ by order one there.

Every ingredient contributes $O(1/T)$: the amplitude ratio $(m+1)/a = 1 + O(1/T)$, Gabcke's $E = O(1/a)$, the phase remainder in Step 2, and off the line the $O_\sigma(1/t)$ terms of ψ and $|\chi|$ together with the error of (64) (relative $O(1/T)$, §8.3). At the seam $q = \frac{1}{2}$ the two case formulas agree: $\Psi(1) = \Psi(0) = \cos \frac{\pi}{8}$ and $1 + e^{i\pi/4} = 2 \cos \frac{\pi}{8} e^{i\pi/8}$, so both give $\cos \frac{\pi}{8} e^{i\pi/8}$, which is $Y_{\text{inf}}(\frac{1}{2})$. $\square$

Remark 11.4 (numerical corroboration). The identities (132)–(134) check to thirty digits, and the convergence at the predicted rate is visible at moderate height: $|Y_{in1} - Y_{\text{inf}}(q)|$ at $q = 0.2$ is 1.3×10^{-3} at $T = 40.2$ and 3.3×10^{-4} at $T = 160.2$ for $\sigma = \frac{1}{2}$, the same factor-of-four drop per quadrupling of T occurring at $\sigma = 0.3$ and $\sigma = 0.9$, with the yang errors matching the yin errors half a unit away in the argument. The Step-5 caution is equally visible: at $T = 80.2$ the flipped relation $|Y_{ang1} - (1 - Y_{in1}(2m+1-T))| = 4 \times 10^{-3}$, while the naive lag $|Y_{ang1} - Y_{in1}(T - \frac{1}{2})| = 1.9$, an order-one failure; at $T = 80.8$, where the frames agree, the naive lag is fine (2×10^{-4}). (probe_yin_theorem.py.)

Corollary 11.5 (the limit profile is the axis crossing of the limit chord). *For $q \in (0, 1)$, $q \neq \frac{1}{4}, \frac{3}{4}$, the chord from $Y_{\text{inf}}(q)$ to $Y_{\text{inf}}(q - \frac{1}{2})$ crosses the real axis at the abscissa $d(q)$, the limit profile of Theorem 8.8.*

Proof. At every finite T the segment from Y_{in1} to Y_{ang1} is the reverse bisector link in the frame, and it crosses the real axis at the bisector point, at abscissa $\hat{d}_1$ (Figure 40; §11.2). As $T \rightarrow \infty$ with $\{T\} = q$ the two endpoints converge to $Y_{\text{inf}}(q)$ and $Y_{\text{inf}}(q - \frac{1}{2})$ (Theorem 11.3) while the crossing converges to $d(q)$ (Theorem 8.8 on the line, its universality remark off it); away from $q = \frac{1}{4}, \frac{3}{4}$ the limit chord meets the axis transversally, so the crossing of the limit is the limit of the crossings. At $q = \frac{1}{4}, \frac{3}{4}$ the chord degenerates onto the axis itself (the parallel instants: it runs from $\frac{1}{2}$ to $-\frac{1}{2}$ and from $\frac{3}{2}$ to $\frac{1}{2}$ respectively), which is the limiting shadow of the off-line poles of §8.4. The identity was also checked directly to thirty digits (probe_yin_theorem.py): the crossing of the limit chord and the closed form $d(q)$ of Theorem 8.8 agree. $\square$

So the two halves of the paper close into one loop: the weights d_1, d_2 , their limit profile $d(x)$, and the yin-yang limit curve are a single asymptotic object seen from three angles, and the same amplitude runs through all of it.

Now the connection promised in the subsection title, which Step 1 of the proof has already put to work. The function (131) is, symbol for symbol, the universal amplitude

$$\Psi(x) = C_0(x) = \frac{\cos(2\pi(x^2 - x - \frac{1}{16}))}{\cos(2\pi x)} \quad (136)$$

of the Riemann–Siegel first term used in §8.2–8.3 (equations (63) and (64)). So the same universal function plays both roles: evaluated at $\hat{p} = \text{frac} \sqrt{t/2\pi}$ it is the asymptotic *magnitude* of the remainder, and evaluated at $q = \text{frac}(T)$ it is the asymptotic *shape* of the yin-yang orbit; by (130) the limiting yin curve keeps distance $|\Psi(q)| = |C_0(q)|$ from the point 1. Even the two parameters are the same clock read half a period apart: $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ (the Taylor table of §10.1), so $\hat{p} \approx q \pm \frac{1}{2} \pmod{1}$, precisely the half-unit shift between the yang and yin arguments in Theorem 11.3.

Three properties of the limit curve deserve note. First, it is closed: $Y_{\text{inf}}(0) = Y_{\text{inf}}(1) = 1 - \cos(\frac{\pi}{8})e^{i\pi/8}$. Second, the apparent singularities of Ψ at $q = \frac{1}{4}, \frac{3}{4}$ (the pole heights of §8.4) are *removable* on the curve: $\Psi \rightarrow \frac{1}{2}$ at both points, and the curve passes smoothly through the real values $Y_{\text{inf}}(\frac{1}{4}) = \frac{1}{2}$ and $Y_{\text{inf}}(\frac{3}{4}) = \frac{3}{2}$. Third, the reflection identity (132) is a polar form of the curve, $Y_{\text{inf}}(q) = \Psi(q + \frac{1}{2})e^{i\theta(q)}$: the same function that measures the distance $|\Psi(q)|$ from the point 1 measures, half a period along, the distance $|\Psi(q + \frac{1}{2})|$ from the origin.

Convergence of the yin curves to $Y_{\text{inf}}(q) = 1 - \Psi(q) e^{-2\pi i(q^2 - 1/16)}$, $q = \text{frac}(T)$

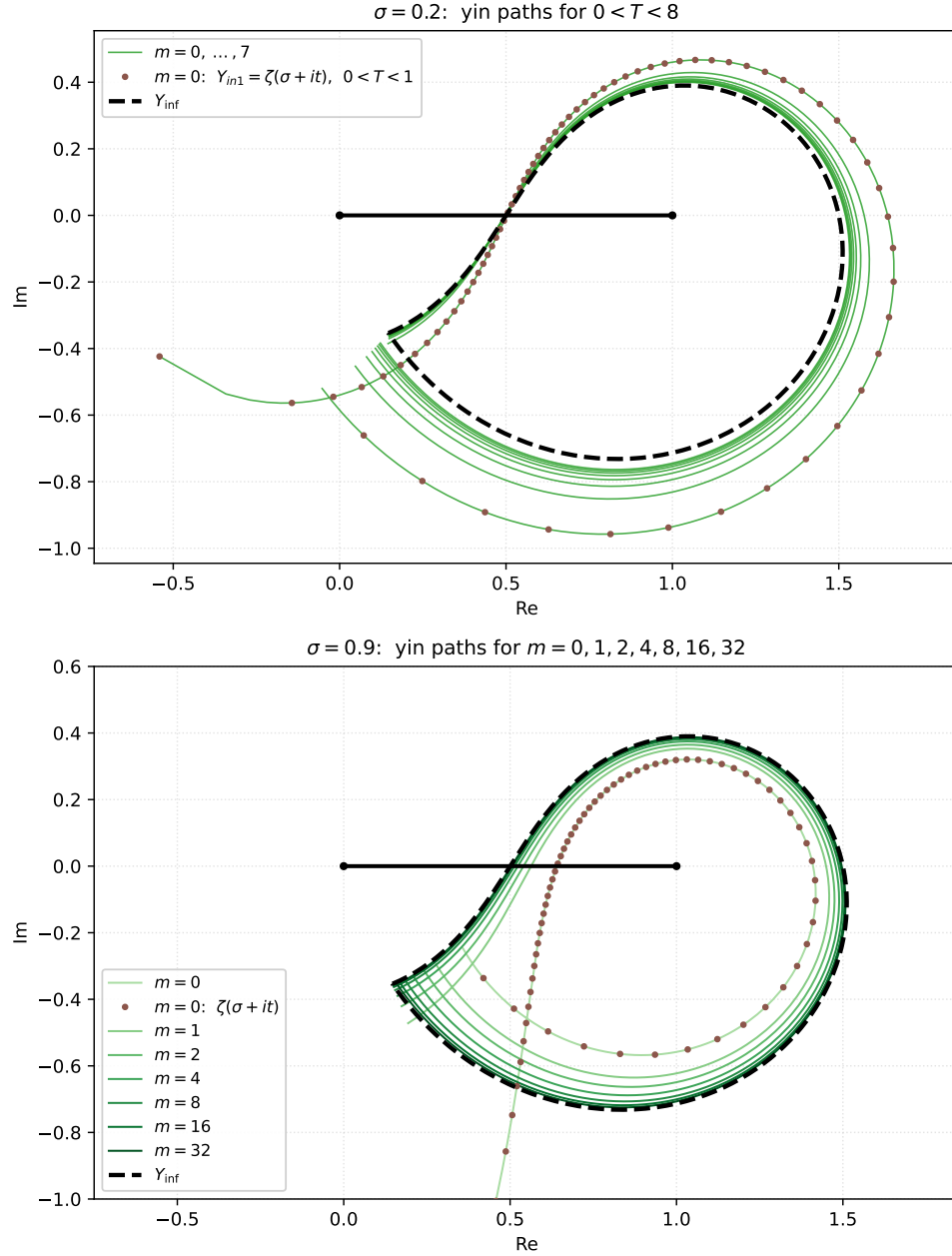


Figure 44: Convergence of the yin curves to the limit curve $Y_{\text{inf}}(q) = 1 - \Psi(q)e^{-2\pi i(q^2 - 1/16)}$ (dashed black), with the unit bisector link in black; the two panels have identical window sizes. As everywhere in the bisector frame, the picture is scaled by the frame map (107) so that the forward bisector link always has length 1, running from 0 to 1 on the real axis. Top: $\sigma = 0.2$, the yin paths (green) of every unit interval of the index over $0 < T < 8$, $m = 0, \dots, 7$; the $m = 0$ path, marked with spaced brown dots, is $\zeta(\sigma + it)$ itself, since for $0 < T < 1$ both partial sums are empty and $Y_{in1} = \zeta$. Bottom: $\sigma = 0.9$, the yin paths of the periods starting at $T = 0, 1, 2, 4, 8, 16, 32$ (light to dark green); by $T \approx 32$ the path is visually indistinguishable from Y_{inf} even this far off the critical line. The $m = 0$ path, again $\zeta(\sigma + it)$ and again marked with brown dots, leaves the frame at the bottom: $\sigma = 0.9$ sits close to the pole of ζ at $s = 1$, so this first orbit dives to about $-2.7i$ before the later orbits settle onto the teardrop; the window is clipped to the convergence region. The yang curves (omitted for clarity) converge to the point reflection of Y_{inf} through $\frac{1}{2}$, half a unit behind in the argument (Theorem 11.3 and (134)). Generated by `fig_yinyang_infinity.py`.

12 General yin and yang, and the d_1 and d_2 formulas

§11 watched one pair: the reverse bisector link revolving about the forward bisector link, its ends tracing the yin and yang curves, and the weights d_1, d_2 as the real-axis crossings of that pair. The same picture exists at every link of the chain. Which reverse link crosses forward link k is a function of T alone; once that partner is named, the yin and yang formulas, the weights, and the remainder split are the same constructions with that pair in place of the bisector pair. The six subsections record the pairing, the curves, the weights, the Dirichlet sums whose joints sit at the crossings, the two free-vector sums of the half-links, and a formula for ζ built on a single crossing.

12.1 Links crossing

§9.1 named one crossing of the two link chains: the forward bisector link meets the reverse bisector link, and where they meet is the bisector point B_1 . Every other link of the forward chain is crossed by some link of the reverse chain as well, and which one it is depends on T . Watching the link at $m - 1$ as T , for example, runs from 6 to 7, one reverse link crosses it, and that link is joined by its neighbor that subsequently also crosses over it, and by the end of the unit interval the crossing links have advanced by two.

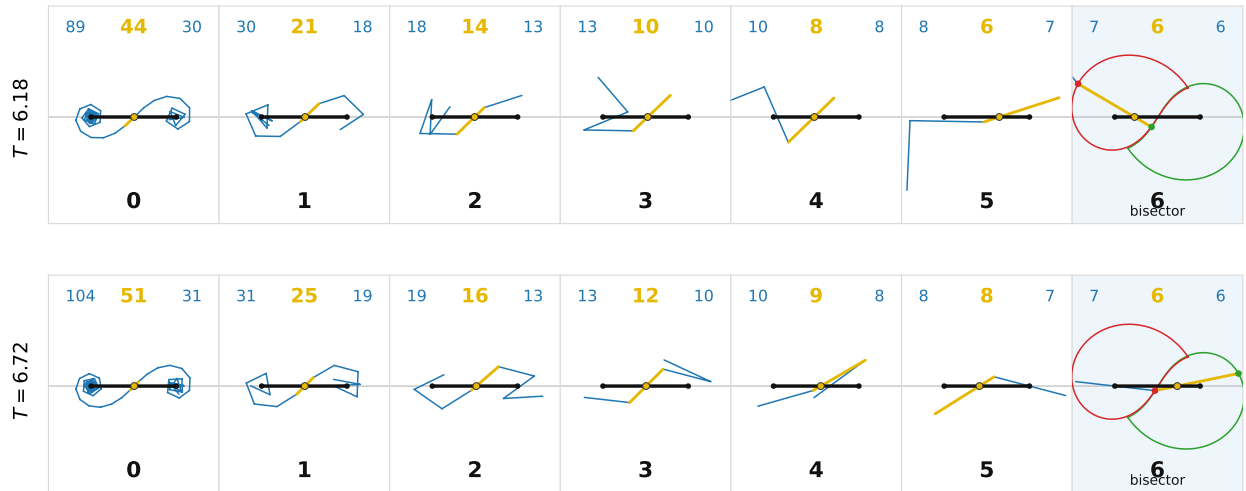


Figure 45: Local frames of the forward links at $\sigma = \frac{1}{2}$, $T = 6.18$ (top) and $T = 6.72$ (bottom), the running example of Figure 2 and a later height in the same unit ($t = I(T) \approx 279.85$ and 327.01 , $m = 6$). Each panel is the local coordinate frame for that forward link, sending one forward link to the unit interval along the x -axis; the number at the bottom is that link. The reflected reverse chain is drawn in the same frame: the blue links are the part of it that belongs to the particular forward link, numbered at the two ends with the outer (higher) link on the left, and the yellow link is the single reverse link that crosses the forward link, numbered in yellow, with a yellow dot at the crossing. The rightmost local frame is the bisector; the green and red curves are the yin and yang loci of the two ends of the reverse bisector link as T runs through $(6, 7)$. Generated by `fig.link_frames.py`.

Numbering summands from one, so that link k carries the summand $n = k + 1$ (§6), forward link k and reverse link j cross when their two summands are reciprocal about the Riemann–Siegel

cutoff $a = \sqrt{t/2\pi}$ of §8.2. Under $t = I(T)$ that cutoff is a function of the index alone, by (99), so the rule closes in T with nothing approximated:

$$nn' = a^2 = \frac{t}{2\pi} = \frac{I(T)}{2\pi} = \frac{2T+1}{2\log(1+\frac{1}{T})}, \quad n = k+1, \quad n' = j+1, \quad (137)$$

The real pairing is exact as a real number: the crossing summand sits at $n' = a^2/n$, with mean residual $n' - a^2/n$ of order 10^{-3} over the sample below, and the rule that names the reverse link integer reads that real number as an integer. Write

$$j_0(k, T) = \left\lfloor \frac{a^2}{k+1} \right\rfloor - 1 = \left\lfloor \frac{2T+1}{2(k+1)\log(1+\frac{1}{T})} \right\rfloor - 1, \quad (138)$$

with $\lfloor \cdot \rfloor$ the nearest integer. Nearest-integer alone misses at two families of half-integers. One is the offset (100) already recorded: Siegel's cutoff a sits half a link past T , so it steps near the half-integers of T while the chains themselves step at the integers. The other is the joints of the hyperbola $nn' = a^2$ itself. The *crossing link* is the rule that names the reverse link integer and removes those misses,

$$C_\ell(k, T) = \begin{cases} m & \text{if } k = m, \\ j_0(k, T) & \text{if reverse link } j_0 \text{ crosses forward link } k, \\ j_0(k, T) - 1 & \text{otherwise,} \end{cases} \quad (139)$$

and the crossing summand is $C_s(n, T) = C_\ell(n-1, T) + 1$. The first line is the self-pair: $a^2/(m+1)$ is near a half-integer when T is near an integer, so j_0 names $m \mp 1$ at the ends of the unit Index T , while the two bisector links cross at every height. The third line is the other family, the half-integers of the product a^2/n itself: when $\{a^2/n\} \approx \frac{1}{2}$ the geometric crossing has already moved to the lower neighbor. Replacing a^2 by $T(T+1)$ or $(T+\frac{1}{2})^2$ does not remove either family; the product is already the exact $I(T)$, and it only moves the joints. Each is its own inverse but for the rule that names the reverse link integer, $C_s(C_s(n, T), T) = n$, which is the symmetry of the pairing established below. At $T = 6.18$, where $a^2 = 44.5389$, (138) reads 44, 21, 14, 10, 8, 6, 5 for $k = 0, \dots, 6$, and all seven are real crossings (Figure 46). The sum rule $k + j \approx 2T - 1$ that the links beside the bisector obey is this hyperbola's tangent there, and holds only there.

Two remarks on a , since it is easy to read it as a link number. It is not one: it is a real height, the self-dual summand $n = n' = a$ the law pairs about, the one at which the forward chain hands over to the reflected one, and by (100) it sits half a link past the bisector link, at $a^2 = (T + \frac{1}{2})^2 - \frac{1}{12} + O(T^{-2})$. Nor is it a fixed number over a unit interval, as $\lfloor T \rfloor$ is: a^2 climbs from 42.17 to 56.17 as T runs from 6 to 7, and it is that climb that walks each link's partner along inside the interval. Replacing a^2 by $\lfloor T \rfloor^2 = 36$ or $(\lfloor T \rfloor + 1)^2 = 49$ at $T = 6.18$ would name reverse link 35 or 48 as crossing the unit step, and neither does.

The reverse chain is the forward chain reflected in the bisector line. Number the joints of both chains from zero as in §6.2,

$$J_n = \sum_{n' \leq n} n'^{-s}, \quad K_j = \zeta - \chi(s) \sum_{n \leq j} n^{s-1}, \quad (140)$$

so that $J_0 = 0$, $K_0 = \zeta$, and $K_m = \Sigma_1 + R$ is the near end of the reverse bisector link; link n of either chain runs from its joint n to its joint $n+1$. On $\sigma = \frac{1}{2}$ we have $n^{s-1} = \overline{n^{-s}}$, so $K_j = \zeta - \chi \overline{J_j}$,

and with $\chi = e^{-2i\vartheta}$ and $Z = e^{i\vartheta}\zeta$ real (Hardy's Z , as in (73)),

$$e^{i\vartheta}K_j = Z - \overline{e^{i\vartheta}J_j}. \quad (141)$$

Write $e^{i\vartheta}J_n = X_n + iY_n$. Then (141) puts reverse joint j at $(Z - X_j, Y_j)$: in the ϑ -rotated frame the reverse chain is the forward chain reflected in the vertical line $X = Z/2$, joint for joint and link for link. That line is the bisector line of §9.2: the rotation puts ζ at Z on the real axis, so unrotated it is the perpendicular bisector of the base running from the origin to ζ , drawn dotted in Figures 18 and 20 through $\zeta/2$ and the bisector point B_1 (Corollary 9.2). Doubling-back and the bisector line are two different things in the same neighborhood, and worth keeping apart: the place along the chain is the bisector link, where the chain doubles back; the bisector line is a line across the plane, the one the two chains reflect in. The bisector point is where the two meet, the point inside the bisector link at which the forward chain cuts the bisector line, as (151) works out below. This is the chain-level form of the reflection $B_2 = \chi\overline{B_1}$ behind the equal legs of §9.2, and like it, it is a statement about the critical line only: off the line $|\chi| \neq 1$ and the reverse chain is a scaled copy of the $1 - \sigma$ chain instead of a reflection of this one.

Crossings come in pairs. Parameterize each chain continuously, the value $\mu = k + p$ with $0 \leq p < 1$ meaning the point p of the way along link k , and let $X(\mu), Y(\mu)$ be the rotated coordinates of the forward chain there and

$$h(\mu) = X(\mu) - Z/2 \quad (142)$$

its signed distance to the bisector line. By (141) the reverse chain at ν sits at $(Z - X(\nu), Y(\nu))$, so the forward chain at μ meets the reverse chain at ν exactly when

$$h(\mu) + h(\nu) = 0 \quad \text{and} \quad Y(\mu) = Y(\nu). \quad (143)$$

Both conditions are symmetric in μ and ν , which is why crossings come in pairs: reverse link j crosses forward link k if and only if reverse link k crosses forward link j , and any rule naming the partner must be symmetric too, as (137) is. The pair coincides only when $\mu = \nu$, and then $h = 0$; so a self-paired crossing lies on the bisector line itself, there is exactly one (the bisector point B_1 , inside the bisector link of each chain by Theorem 8.12), and every other crossing sits off the line with its counterpart on the other side.

Sweeps between spiral centers. Along either chain the turn from one link to the next is $t \ln(1 + 1/n) \approx t/n$, decreasing as the chain is walked, so the joints where that turn is an odd multiple of π lie near the spiral centers (76) (the midpoint of the link they bound is closer still), and between two consecutive centers the chain makes one full turn. The middle of a turn is the link where the angle is the even multiple $2\pi c$,

$$\frac{t}{n'} = 2\pi c \quad \Longleftrightarrow \quad n' = \frac{t}{2\pi c} = \frac{a^2}{c}, \quad (144)$$

and there successive links are parallel rather than antiparallel: their contributions add in phase, the chain stops circling and runs. Call that run the c -th *sweep*; it carries the chain from the spiral center $S_n = c$ to the spiral center $S_n = c - 1$.

Poisson summation says what a sweep is worth. Writing the rotated chain as $\sum_n n^{-1/2} e^{i(\vartheta - t \ln n)} = \sum_c \int x^{-1/2} e^{i(\vartheta - t \ln x + 2\pi c x)} dx$, the c -th integral has its stationary point at $t/x = 2\pi c$, which is (144), the middle of the c -th sweep; the stationary-phase value there is

$$\frac{1}{\sqrt{c}} \exp i \left(\vartheta + t - t \ln \frac{a^2}{c} + \frac{\pi}{4} \right) = \frac{1}{\sqrt{c}} e^{i(2\Theta - (\vartheta - t \ln c))}, \quad 2\Theta = 2\vartheta + t - t \ln a^2 + \frac{\pi}{4}. \quad (145)$$

Its modulus is that of the c -th summand, $c^{-1/2}$, and its phase is the c -th summand's phase $\vartheta - t \ln c$ reflected in the fixed direction Θ . Sweep by sweep, the tail of a chain reproduces the head of that chain reflected: this is the correspondence between joints and spiral centers recorded in §9.1, Nickel's center-to-center distances and angles reproducing the initial steps, now carrying an index; the c -th sweep answers to the c -th summand.

The reflection (141) undoes that one. In the frame of the forward chain, therefore, the reverse chain's c -th sweep is the forward chain's own link $c - 1$ traversed backwards: over the sampled heights the center-to-center step and that link agree to a median 1.4%, 2.7%, 3.1% and 5.2% of a link length for $c = 1, 2, 3, 4$. A copy of a link laid back along it must cross it, and the quarter turn in (145) is what tilts the copy so that it cuts across instead of lying flat: the two meet near the middle of the link, at some 43° . Which reverse link carries the c -th sweep is (144) with $c = n$, and that is (137). Panel (b) of Figure 46 is the case $c = 1$: the whole outermost turn of the reverse chain, links 30 to 89, laid back along the unit step and crossing it at link 44.

The angle at a crossing. How the two links lie relative to one another needs no estimate at all. It is forced, and what forces it is that the two summands multiply to a whole number. In the rotated frame forward link k points along $e^{i(\vartheta - t \log n)}$ with $n = k + 1$, and by (141) reverse link j points along $-e^{-i(\vartheta - t \log n')}$ with $n' = j + 1$, so the angle from the first to the second is

$$\theta(n, n') = \pi - 2\vartheta(t) + t \log(nn') \pmod{2\pi}, \quad (146)$$

exactly, whether or not the two links meet.

Proposition 12.1. *On $\sigma = \frac{1}{2}$, for any two summands n, n' ,*

$$\theta(n, n') \equiv \frac{5\pi}{4} - \frac{\pi(nn' - a^2)^2}{a^2} + O\left(\frac{1}{a}\right) \pmod{2\pi}. \quad (147)$$

In particular, if $|nn' - a^2| \leq a/2$ the two links are never parallel: they meet at an angle between $\frac{3}{4}\pi$ and π , that is between 135° and 180° , at most a quarter turn away from antiparallel.

The proof depends on $N = nn'$ being an *integer*. Write $N = a^2(1 + \epsilon)$. Since $t = 2\pi a^2$,

$$t\epsilon = \frac{t(N - a^2)}{a^2} = 2\pi N - t \equiv -t \pmod{2\pi}, \quad (148)$$

so in $t \log N = t \log a^2 + t\epsilon - \frac{1}{2}t\epsilon^2 + \frac{1}{3}t\epsilon^3 - \dots$ the first-order term is $-t$ modulo 2π , whatever N is. The Stirling expansion behind (185) is $2\vartheta(t) = t \log a^2 - t - \frac{\pi}{4} + \frac{1}{24t} + O(t^{-3})$, so in (146) the whole of $t \log a^2 - t$ cancels against it and the $\frac{\pi}{4}$ survives:

$$\theta \equiv \pi + \frac{\pi}{4} - \frac{t\epsilon^2}{2} + O(t|\epsilon|^3) + O(t^{-1}).$$

Here $\frac{1}{2}t\epsilon^2 = \pi(N - a^2)^2/a^2$, which is (147), and when $|N - a^2| \leq a/2$ we have $|\epsilon| \leq 1/2a$, so the cubic term is at most $\pi/12a$ and the quadratic one at most $\pi/4$. Read as an unsigned angle, $\frac{5}{4}\pi$ is $\frac{3}{4}\pi$, and the correction can only carry it towards π .

The naming (138) stays inside that hypothesis: it makes $|nn' - a^2| = n|n' - a^2/n| \leq n/2$, and $n \leq m < a$ for every forward link below the bisector, so the 135° is a floor and not an average: no link-crossing that the rule names can be sharper than that. Only the self-pair, being the bisector links, handled by the first line of (139), where the rule overrides the rounding, can fall outside it. The quarter turn is the $-\frac{\pi}{8}$ of ϑ , which is the Gaussian factor of (145) over again: the 43° by

which a crossing misses antiparallel, the tilt that keeps the sweep from lying flat along the link it reproduces, is the stationary-phase quarter turn, read off exactly rather than measured. What the proposition fixes is the direction of the partner and not its position; that the two segments actually meet is the part left open in §16.

Near the bisector, a sum rule. Near the self-dual link $n = n' = a$ the hyperbola (137) is indistinguishable from its tangent. Writing $n = a + x$,

$$n' = \frac{a^2}{n} = a - x + \frac{x^2}{a} - \dots, \quad \text{so} \quad k + j = n + n' - 2 = 2a - 2 + \frac{x^2}{a} - \dots, \quad (149)$$

and since $2a = 2T + 1 - \frac{1}{12T} + O(T^{-2})$ by (100), the sum is $2T - 1$ up to the quadratic term and a drift of order $1/12T$: $k + j$ is one of the two integers straddling $2T - 1$, that is

$$j = \lfloor 2T \rfloor - 1 - k \quad \text{or} \quad j = \lfloor 2T \rfloor - k, \quad (150)$$

a rule needing nothing computed at all. The same rule comes out of the reflection (141), which is worth doing because it identifies the fixed point. Near the bisector the forward chain sweeps across the bisector line and h passes through a single zero there, close to straight over the few links either side; writing r^* for that zero and linearizing turns the first condition of (143) into $\mu + \nu = 2r^*$, the two arcs straddling the crossing symmetrically, so $k + j$ straddles $S = 2r^* - 1$. And the crossing fraction is not a new quantity. By (73), $h(m) = X_m - Z/2 = -r/2$ with $r = e^{i\vartheta} R$ the real remainder of §8.2, and link m of the forward chain, being $(m + 1)^{-s}$ rotated by $e^{i\vartheta}$, has horizontal extent $X_{m+1} - X_m = \cos(\omega - \vartheta)/\sqrt{m + 1}$. So the zero falls inside link m , at the crossing fraction

$$r^* = m + \lambda, \quad \lambda = \frac{Z/2 - X_m}{X_{m+1} - X_m} = \sqrt{m + 1} \frac{r}{2 \cos(\omega - \vartheta)} = \lceil T \rceil^{1/2} d_1 = \hat{d}_1, \quad (151)$$

by the exact form $d_1 = e^{i\vartheta} R/2 \cos(\omega - \vartheta)$ of §8.2 and the normalization (26). The place where the forward chain cuts the bisector line is therefore the bisector point B_1 itself, sitting at the crossing fraction $\hat{d}_1$ along the bisector link exactly as Figure 34 marks it. So $S = 2r^* - 1 = 2m + 2\hat{d}_1 - 1$, and the two routes agree because $\hat{d}_1 \approx \{T\}$ (§8): the tangent to the hyperbola and the linearization of h are the same line, and (150) follows either way, using $\frac{1}{5} < \hat{d}_1 < \frac{4}{5}$ from Theorem 8.12 to put the straddling integers at $\lfloor 2T \rfloor - 1$ and $\lfloor 2T \rfloor$ whichever side of $\frac{1}{2}$ the fractional part is on.

How far the tangent can be trusted is (149) again: it costs half a link once $x^2/a = \frac{1}{2}$, so the sum rule is good only within

$$|n - a| \lesssim \sqrt{a/2} \approx \sqrt{T/2} \quad (152)$$

of the bisector (three links either side at $T = 17$, fourteen at $T = 400$, and just under two at the $T = 6.18$ of the figure), which is why $k = 4, 5, 6$ obey it there and $k = 3$ already does not. Inside that range the two indices trade off one for one: each step back along the forward chain moves the partner one step forward along the reverse chain, and over one unit of T the sum moves by two, so each partner advances by two, in two single steps, one when $\{T\}$ passes $\frac{1}{2}$ and one when T passes an integer, the two heights at which $\lfloor 2T \rfloor$ jumps. Outside the range the product takes over and the partners run away as a^2/n , reaching the far end of the chain at the unit step.

The bisector link is the one place where the pairing is exact rather than asymptotic, and that is the first line of (139). It is the self-dual link, $n = m + 1$ and $x = \frac{1}{2} - \{T\}$, so (138) rounds to m only over the middle half of the unit interval and to $m \mp 1$ at its ends; but the two bisector links still cross, and the self-paired crossing B_1 of (143) is there at every height. The bisector links always

cross, as §9.1 has it, and the links j_0 names near the ends of the interval are second crossings of the same forward link, not replacements for it.

Links crossing at $\sigma = 1/2$, $T = 6.18$ ($t = I(T) \approx 279.85$, $m = 6$, $a = \sqrt{t/2\pi} = 6.674$, $a^2 = 44.54$)

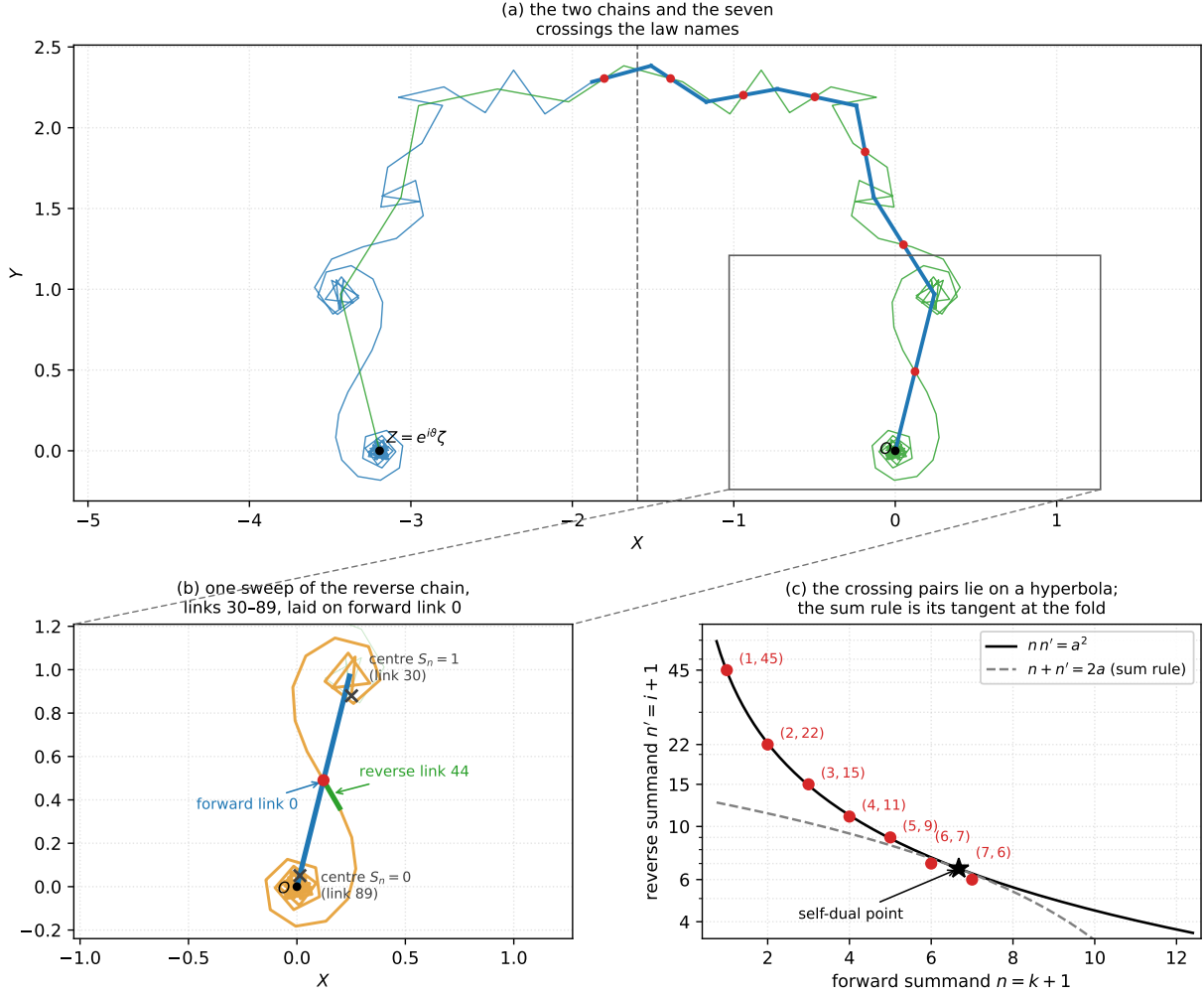


Figure 46: Links crossing at $\sigma = \frac{1}{2}$, $T = 6.18$, the running example of Figure 2 ($t = I(T) \approx 279.85$, $m = 6$, $a = \sqrt{t/2\pi} = 6.674$, $a^2 = 44.54$; the first turn of each chain holds 90 links). (a) Both chains in the ϑ -rotated frame, forward $e^{i\vartheta}J_n$ in blue from O and reverse $e^{i\vartheta}K_j$ in green from $Z = e^{i\vartheta}\zeta$, which by (141) is the blue chain reflected in the dashed bisector line $X = Z/2$. Forward links 0 through 6 are drawn heavy and the red dots are the seven crossings (138) names for them, $j = 44, 21, 14, 10, 8, 6, 5$; they run the length of the chain, not just the neighborhood of the bisector. (b) The boxed region enlarged, showing why. The gold arc is one turn of the reverse chain, links 30 to 89, running from the spiral center $S_n = 1$ to the spiral center $S_n = 0$ (crosses, at the links (76) where the turn angle is an odd multiple of π). By (145) that step reproduces forward link 0 reversed, so it lies back along the unit step and cuts it near the middle (red dot); the link carrying the sweep is 44, the middle of the turn, where the turn angle is 2π and $n' = a^2/1$. (c) The pairs (n, n') of all seven crossings, on the hyperbola $nn' = a^2$. The self-dual point $n = n' = a$ (star) is the summand the law pairs about, half a unit of the index past T by (100). These are summand axes, and the star is plotted here at (a, a) with $a = 6.674$, while the bisector link is link $m = 6$, which carries summand $n = 7$: the star therefore sits between summands 6 and 7, a third of a summand short of the bisector link's own value rather than past it. Near it the hyperbola is indistinguishable from its tangent $n + n' = 2a$ (dashed), which is the sum rule (150), and away from it the two part company fast: the tangent gives the unit step a partner near $n' = 11$ where the truth is 45. Generated by `fig_links_crossing.py`.

How well it holds. Sampling $m \in \{8, 17, 40, 123, 400\}$ at 199 fractional parts each and testing every forward link $0 \leq k \leq m$, 118,007 cases in all, the link named by (139) crosses forward link k in every one of them. The named integer (138) alone does so in 117,870, 99.9%; the other 137 are the two half-integer families of that naming, 13 of them the self-pair (first line of (139)) and 124 the lower neighbor at $\{a^2/n\} \approx \frac{1}{2}$ (third line), never the upper neighbor and never off by more than one link. The crossing sits at the middle of the forward link, median position 0.500 along it with median deviation 0.009 four or more links out from the bisector, widening to 0.148 at the bisector link, where the crossing is B_1 and the position is $\hat{d}_1$ instead. The two links meet at a median 137° , tenth to ninetieth percentile 135° to 151° , and never below 135° in any of them: that floor is Proposition 12.1 and not a feature of the sample, and the exact form (146) holds across the crossings to 3.8×10^{-9} , the accumulated double precision of the chains. The 43° of the median is the quarter turn of (145) that keeps the sweep from lying flat along the link it reproduces. The center-to-center steps match those links to a median 1.4%, 2.7%, 3.1% and 5.2% of a link length at $c = 1, 2, 3, 4$, and the self-pair (m, m) crossed at all 995 sampled heights while (138) named m at exactly half of them, the middle half of each unit interval, as the previous paragraph says. The sum rule (150), by contrast, names a real crossing in only 10.6% of the cases overall, but in 99.8% of those inside its range (152) and 5.1% of those outside it, which is what that range is worth. The identity $\lambda = \hat{d}_1$ of (151) holds to 6×10^{-15} . Computed in `check_links_crossing.py`.

12.2 General yin and yang form for any link

§11 watched one link cross another: the reverse bisector link revolving about the forward bisector link, its two ends tracing the yin and yang curves (108)–(109). Now that (139) says which reverse link crosses forward link k , every link of the chain has such a pair, and they are all one formula. Like the bisector pair it carries no ζ : the remainder R , the factor χ , and σ and T are all it needs.

Send forward link k to $[0, 1]$ exactly as (107) sends the bisector link: translate its joint J_k to the origin and divide by the link vector $(k+1)^{-s}$, writing $s = \sigma + iI(T)$ throughout. Reverse joint j of (140) lands at

$$Y_k(j) = (K_j - J_k)(k+1)^s = (k+1)^s \left(\sum_{n=k+1}^m n^{-s} + R + \chi \sum_{n=j+1}^m n^{s-1} \right), \quad (153)$$

the second sum read as $-\sum_{n=m+1}^j$ once $j > m$. What removes ζ is the same fact the bisector frame turned on: $K_m = \Sigma_1 + R$, so measuring the reverse chain from its own joint m replaces the anchor ζ by the remainder. The yin and yang points of link k are then the two ends of the link that crosses it, which are consecutive reverse joints:

$$Y_{in}(k, T) = Y_k(j), \quad Y_{ang}(k, T) = Y_k(j+1) = Y_{in}(k, T) - \chi (k+1)^s (j+1)^{s-1}, \quad j = C_\ell(k, T), \quad (154)$$

with $j = m$ at the bisector by the first line of (139).

Remark 12.2. Equation (153) is the general frame formula, and on the bisector it *is* the yin formula. It holds there because the crossing rule sets $j = m$ when $k = m$. Plug $k = m$ and $j = m$ into (153):

$$Y_m(m) = (m+1)^s \left(\sum_{n=m+1}^m n^{-s} + R + \chi \sum_{n=m+1}^m n^{s-1} \right).$$

The first sum runs from $n = m + 1$ to $n = m$ and is empty. The second sum runs over the same empty range (the joint is m , not past it, so the $j > m$ sign-flip is not used). What remains is

$$Y_m(m) = (m + 1)^s(R) = R(m + 1)^s = Y_{in1},$$

which is (108). Geometrically the same thing: $J_m = \Sigma_1$ and $K_m = \Sigma_1 + R$, so $(K_m - J_m)(m + 1)^s = R(m + 1)^s$, and the frame map is (107) with $\lceil T \rceil = m + 1$. The yang end is the next reverse joint $j = m + 1$. Now $j > m$, so the second sum is $-\sum_{n=m+1}^{m+1} = -(m + 1)^{s-1}$ and the first sum is still empty:

$$Y_m(m + 1) = (m + 1)^s(R - \chi(m + 1)^{s-1}) = R(m + 1)^s - \chi(m + 1)^{2s-1},$$

which is (109), since $2s - 1 = -1 + 2(\sigma + iI(T))$. So (153) at the bisector is not an analogue of yin; it *is* yin. The closed-form loops in §11 are this formula with the two empty sums already canceled.

Take $k = m - 1$. Its frame factor m^s multiplies the summand m^{-s} , so that product is 1 (forward joint m is the far end of that link), and

$$Y_{m-1}(j) = 1 + m^s \left(R - \chi \sum_{n=m+1}^j n^{s-1} \right). \quad (155)$$

Here the crossing changes hands, which is the one thing the bisector's does not do. By (137), $a^2/m = m + 2\{T\} + 1 + O(1/m)$, so the crossing summand $n' = j + 1$ runs $m + 1 \rightarrow m + 2 \rightarrow m + 3$ across the unit and the curves come in three arcs:

$$n' = m + 1 : \quad Y_{in} = 1 + m^s R, \quad Y_{ang} = 1 + m^s (R - \chi(m+1)^{s-1}), \quad (156)$$

$$n' = m + 2 : \quad Y_{in} = 1 + m^s (R - \chi(m+1)^{s-1}), \quad Y_{ang} = Y_{in} - \chi m^s (m+2)^{s-1}, \quad (157)$$

$$n' = m + 3 : \quad Y_{in} = 1 + m^s (R - \chi((m+1)^{s-1} + (m+2)^{s-1})), \quad Y_{ang} = Y_{in} - \chi m^s (m+3)^{s-1}, \quad (158)$$

the three running over $\{T\}$ in $[0, 0.214]$, $[0.215, 0.647]$ and $[0.648, 1)$ at $m = 6$ (Figure 47).

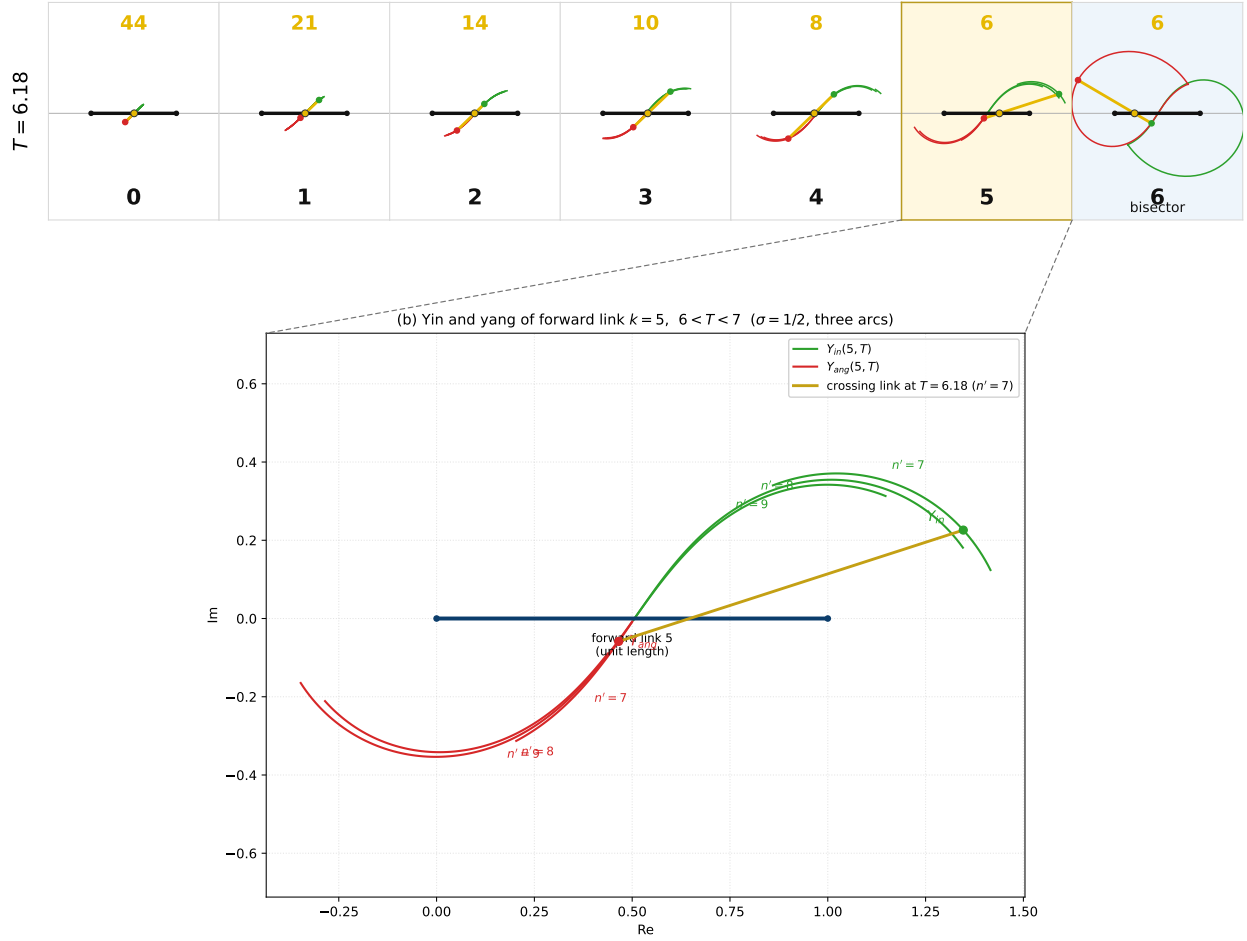


Figure 47: (a) Local frames of the forward links at $\sigma = \frac{1}{2}$, $T = 6.18$, as in the top row of Figure 45, now with the yin and yang loci of every link as T runs through $(6, 7)$: green $Y_{in}(k, T)$, red $Y_{ang}(k, T)$. (b) The yin and yang curves of forward link $k = 5$ over $6 < T < 7$, from (155)–(158). The frame sends that link to $[0, 1]$ (dark blue). Each end traces three arcs, one per crossing summand $n' = 7, 8, 9$, kept apart where one reverse link gives way to the next: green is $Y_{in}(5, T)$, red is $Y_{ang}(5, T)$. The dark-yellow segment is the crossing link at the running example $T = 6.18$, where $C_\ell(5, 6.18) = 6$ so $n' = 7$. Dashed lines carry the local coordinate frame for link 5 of (a) down to the magnification (b). Each arc’s yin is the previous arc’s yang, the same joint passing from one reverse link to the next. Generated by `fig_yinyang_link5.py`.

The same local frames with each non-bisector arc continued past the unit are Figure 48. The continuation holds the reverse link of that piece and steps half a unit of T past one end (the end farther from the local-frame origin for yin, the other end for yang). The bisector is left alone: its two curves already close.

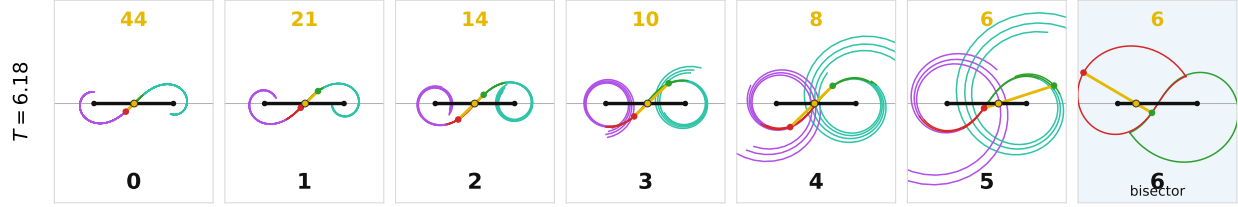


Figure 48: Local frames of the forward links at $\sigma = \frac{1}{2}$, $T = 6.18$, as in panel (a) of Figure 47, with each non-bisector yin and yang arc continued as a solid curve (teal past the green end farther from the origin, purple past the other red end). The continuation holds the reverse link of that piece and runs half a unit of T . The bisector local frame is not extended. Generated by `fig_yinyang_extended.py`.

Each arc's yin is the arc before it's yang, the same point at the same T : both are $Y_k(j+1)$, once as the far end of the outgoing crossing link and once as the near end of the incoming one. That is the crossing of §12.1 seen from the ends rather than the middle: one reverse link is dragged off the forward link and the next is dragged across it. The joint passing across the forward link belongs to both links at that instant, so the crossing itself goes through smoothly while the yin curve steps onto where the yang curve was and the yang steps one joint further out.

The count of reverse links dragged across forward link k in one unit of T is about $2a/(k+1)$: $n' = a^2/(k+1)$ advances by $2a/(k+1) = (2T+1)/(k+1) + O(T^{-1})$ over a unit of the index, and that is how many arcs the link gets. On the unit step, $k+1=1$, so about $2a$ reverse links cross it; at $T = 6.18$, $a \approx 6.67$, so roughly a dozen. That is why the local coordinate frame for link 0 in Figures 47–48 is a thicket of short arcs, so close together they are not visible separately in the figure, while the links beside the bisector only get two, and the bisector none. Its partner is the other bisector link at every height, so its yin and yang alone close into single loops, and they alone have the limit teardrop of §11.6.

The two ends fix where the crossing reverse link sits along the forward link. In each separate local coordinate frame the link is $[0, 1]$ on the real axis, so the crossing is where the chord from yin to yang cuts that axis, the computation (111) makes at the bisector. Writing $v_k = \chi(k+1)^s(j+1)^{s-1}$ for the crossing link's own vector in the frame, which by (154) is the offset $Y_{in} - Y_{ang}$, that crossing fraction is

$$\lambda_k(T) = \operatorname{Re} Y_{in} - \operatorname{Im} Y_{in} \frac{\operatorname{Re} v_k}{\operatorname{Im} v_k} = \frac{\operatorname{Im}(\overline{Y_{in}} v_k)}{\operatorname{Im} v_k} = |Y_{in}| \frac{\sin(\beta - \alpha)}{\sin \beta}, \quad \alpha = \arg Y_{in}, \quad \beta = \arg v_k, \quad (159)$$

which lies in $[0, 1]$ exactly when the reverse link does cross the forward one; the distance in the plane from joint k is $\lambda_k(k+1)^{-\sigma}$. At $k = m$ this is (111) itself, $\lambda_m = \lceil T \rceil^\sigma d_1$, the bisector point's own crossing fraction, which on the critical line is $\hat{d}_1$. So the rightmost panel of Figure 50 is the quantity the rest of the paper is built on, and the panels to its left are that quantity's counterparts link by link.

Checked against the local frame construction in the Zest web app on the critical line at $T = 6.18, 6.72, 12.4, 17.4, 40.05$ and 434.37 : (153) matches the frame to 1.3×10^{-15} , (159) matches the app's own reading to 6.9×10^{-14} , and λ_m matches $\hat{d}_1$ to 1.0×10^{-12} , in `probe-yinyang-below-bisector.mjs`.

12.3 d_1 and d_2 for any link

§4 split Siegel's remainder along the two bisector links, and §11.2–§11.3 found the weights d_1, d_2 as the real-axis crossings of the yin and yang pair in those two frames. The same two crossings exist for every link: (154) is the pair, (159) is the crossing fraction, and the weights are that fraction read once along the forward link and once along the reverse one.

Write $n = k + 1$, $n' = j + 1 = C_s(n, T)$, and

$$W_k = \sum_{n=k+1}^m n^{-s} + R - \chi \sum_{n=m+1}^j n^{s-1} \quad (160)$$

for the vector in the plane from forward joint k to reverse joint j , so that $Y_{in}(k, T) = (k + 1)^s W_k$ by (153). At the bisector both sums are empty and $W_m = R$. Let

$$\omega_n = I(T) \log n, \quad \omega_{n'} = I(T) \log n', \quad \varphi_k = \arg W_k, \quad \psi = \arg \chi, \quad (161)$$

so ω_n is the absolute direction of summand n of Σ_1 , the same telescoping that made $\omega = \omega_{m+1}$ in (22), and $\omega_{n'} + \psi$ is the absolute direction of summand n' of Σ_2 .

Measured along the forward link. In the frame of link k the crossing fraction (159) is how far along that link it is from joint k to the crossing point. The frame scaled the link by $(k + 1)^\sigma$, so the distance in the plane is

$$d_1(k, T) = (k + 1)^{-\sigma} \lambda_k(T) = |W_k| \frac{\sin(\omega_{n'} - \varphi_k + \psi)}{\sin(\omega_n + \omega_{n'} + \psi)}. \quad (162)$$

The sine form is (159) with $Y_{in} = (k + 1)^s W_k$ and $v_k = \chi (k + 1)^s (j + 1)^{s-1}$, the same reduction that took (111) to (19): $\alpha = \omega_n + \varphi_k$, $\beta = \omega_n + \omega_{n'} + \psi$, and $|Y_{in}| = (k + 1)^\sigma |W_k|$. At $k = m$ one has $W_m = R$, $n = n' = \lceil T \rceil$, $\omega_n = \omega_{n'} = \omega$, and (162) is (19).

Measured along the reverse link. The same crossing, read along the reverse link from joint j , is the parameter $u = \Im Y_{in} / \Im v_k$ of the chord in that frame. The reverse link has length $|\chi| (j + 1)^{\sigma-1}$, so

$$d_2(k, T) = |\chi| (j + 1)^{\sigma-1} u = |W_k| \frac{\sin(\omega_n + \varphi_k)}{\sin(\omega_n + \omega_{n'} + \psi)}. \quad (163)$$

$d_1(k, T)$ and $d_2(k, T)$ share a denominator, as d_1 and d_2 did. At $k = m$ this is (20). The fractional amounts, the hat of (26), are the two parameters themselves:

$$\hat{d}_1(k, T) = \lambda_k(T) = (k + 1)^\sigma d_1(k, T), \quad \hat{d}_2(k, T) = u = \frac{d_2(k, T)}{|\chi| (j + 1)^{\sigma-1}}. \quad (164)$$

The direction, and the cone. Forward link k points along $e^{-i\omega_n}$ and reverse link j along $e^{i(\omega_{n'} + \psi)}$. The two fractional summands are those directions, shortened by the two weights,

$$R_1(k, T) = d_1(k, T) e^{-i\omega_n}, \quad R_2(k, T) = d_2(k, T) e^{i(\omega_{n'} + \psi)}, \quad (165)$$

and they add back to the vector they split:

$$W_k = d_1(k, T) e^{-i\omega_n} + d_2(k, T) e^{i(\omega_{n'} + \psi)}. \quad (166)$$

This is (23) with R replaced by W_k and the two bisector directions replaced by the two links that actually cross. The sine expressions (162)–(163) are again Cramer’s rule for that 2×2 real system, which is why they are real, and why they reduce to (19)–(20) at the bisector. On the critical line the bisector pair still satisfies $d_1(m, T) = d_2(m, T)$, but a general pair need not: the two links have different lengths, and the crossing sits at $\hat{d}_1 \approx \frac{1}{2}$ four or more links out from the bisector (§12.1), so $d_1(k)$ and $d_2(k)$ scale as $n^{-\sigma}$ and $|\chi|n'^{\sigma-1}$. Figure 49 is the same cone, one local frame per forward link, in the local frames of the top row of Figure 47: the last local frame is the bisector, $W_6 = R$, which is Figure 3 seen in that frame.

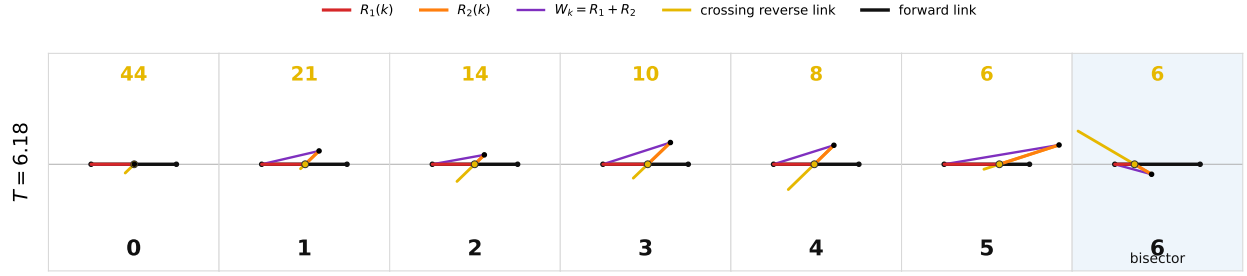


Figure 49: The cone (166) at $\sigma = \frac{1}{2}$, $T = 6.18$, in the local frame of each forward link $k = 0, \dots, 6$, the same local frames as the top row of Figure 47. Each local frame sends that link to $[0, 1]$ on the real axis (black); the gold number is the reverse link that crosses it. Red is $R_1(k)$ from the left joint to the crossing along the unit link, orange is $R_2(k)$ from the crossing to reverse joint K_j , and purple is the resultant W_k . The last local frame is the bisector, $W_6 = R$. Generated by `fig_Wk_triangle.py`.

At the running example $\sigma = \frac{1}{2}$, $T = 6.18$, the seven pairs are

k	j	$d_1(k)$	$d_2(k)$	$\hat{d}_1(k)$
0	44	0.506	0.00185	0.506
1	21	0.360	0.159	0.510
2	14	0.286	0.0914	0.496
3	10	0.263	0.184	0.525
4	8	0.214	0.143	0.478
5	6	0.264	0.300	0.647
6	6	0.0875	0.0875	0.231

The last row is Table 1 at this T : $d_1 = d_2 = 0.0875$ and $\hat{d}_1 = \sqrt{7} d_1 = 0.231$. The first six sit near the middle of their forward link, as §12.1 measured, and $d_2(0)$ is the tiny reverse link 44. Figure 50 draws the same seven numbers against the whole unit of T .

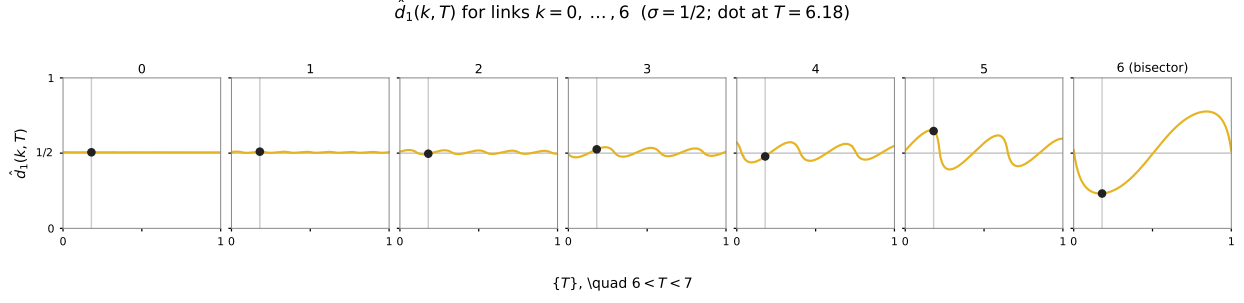


Figure 50: $\hat{d}_1(k, T) = (k+1)^\sigma d_1(k, T)$ for forward links $k = 0, \dots, 6$ over $6 < T < 7$, $\sigma = \frac{1}{2}$, one panel per link, x the fractional part of T and y the crossing fraction along that link from the left joint. The light horizontal is $\frac{1}{2}$; the light vertical and the dot are the running example $T = 6.18$, the seven values of the table. Away from the bisector the track sits on $\frac{1}{2}$ and barely moves; at $k = 6$ it is the normalized d_1 of Figure 10. Generated by `fig_d1_any_link.py`.

Checked at $\sigma = \frac{1}{2}, \frac{1}{4}, \frac{7}{10}$ and $T = 6.18, 6.72, 12.4, 17.4$, for $k = 0, 1, m-2, m-1, m$: (162)–(163) at $k = m$ match (19)–(20) to 0, (164) matches (159) to 1.5×10^{-27} , and (166) holds to 4.4×10^{-28} , in `check_d1_any_link.py`.

12.4 The crossings: Σ_{1x} and Σ_{2x} formula

§12.1 named the reverse partner of each forward link, and §12.3 read the crossing fraction $\hat{d}_1(k, T)$ along that link. Those two data locate a point on every forward link $0, \dots, m$. Connecting the points in order is the same Dirichlet sum as $\Sigma_1 + R_{1ps}$, re-jointed: the joints sit at the crossings instead of at the integers, and the walk ends at the bisector point B_1 rather than at Σ_1 . Call that walk Σ_{1x} .⁹

The crossing fraction of link k , from joint k to the crossing point, is $\hat{d}_1(k, T)$ of (164); T is fixed through what follows, so we write it $\hat{d}_1(k)$. Link k carries the summand $n = k + 1$, so the crossing on it is

$$P_k = J_k + \hat{d}_1(k) (k+1)^{-s} = \sum_{n' \leq k} n'^{-s} + \hat{d}_1(k) (k+1)^{-s}, \quad (167)$$

with J_k as in (140). The last crossing is the bisector point itself, $P_m = B_1$. The walk $0 = P_{-1} \rightarrow P_0 \rightarrow \dots \rightarrow P_m$ therefore telescopes to B_1 by construction. Substituting (167) throughout, with the empty sum $\sum_{n' \leq -1} = 0$ and $\hat{d}_1(-1) = 0$, removes the intermediate points:

$$\Sigma_{1x} = \sum_{k=0}^m \left(\sum_{n' \leq k} n'^{-s} + \hat{d}_1(k) (k+1)^{-s} - \sum_{n' \leq k-1} n'^{-s} - \hat{d}_1(k-1) k^{-s} \right) = \sum_{n' \leq m} n'^{-s} + \hat{d}_1(m) (m+1)^{-s}. \quad (168)$$

Each full summand k^{-s} is split across two consecutive steps, $\hat{d}_1(k-1) k^{-s}$ in one and $(1 - \hat{d}_1(k-1)) k^{-s}$ in the next, and those two pieces add to one. What remains is the integer partial sum plus the last fractional stub, which is the ordinary first-half formula (29) once the stub is named.

That stub is $\hat{d}_1(m) (m+1)^{-s}$. The bisector link is link $m = \lfloor T \rfloor$. It runs from $J_m = \Sigma_1$ to $J_{m+1} = \Sigma_1 + (m+1)^{-s}$, so the vector along it is the next Dirichlet term, $n = \lceil T \rceil = m+1$. The factor $(m+1)^{-s}$ is that whole unused summand, the faint dashed continuation past Σ_1 in Figure 2. The number $\hat{d}_1(m)$ is the crossing fraction: how far along the link the two bisector links meet.

⁹The subscript x is for crossing, not a dummy index: Σ_{1x} is the crossing-joint form of Σ_1 , and Σ_{2x} of Σ_2 .

That meeting is B_1 , and (151) already computed it: the crossing fraction on the bisector link is the remainder weight itself, $\hat{d}_1(m) = \hat{d}_1$. Therefore

$$\hat{d}_1(m) (m+1)^{-s} = \hat{d}_1 (m+1)^{-s} = R_{1ps}, \quad (169)$$

the “one more partial summand” of §5: not a new term, the next summand shortened by a real factor. The hat is what makes $\hat{d}_1(m)$ a fraction of a link rather than a length. The unhatted weight d_1 multiplies a unit vector in the direction of $(m+1)^{-s}$, so $|R_{1ps}| = d_1$; $\hat{d}_1$ multiplies the full summand, whose length is $[T]^{-\sigma}$, and $\hat{d}_1 \cdot [T]^{-\sigma} = d_1$ is the same remainder (26). At the running example $\sigma = \frac{1}{2}$, $T = 6.18$, one has $m = 6$, next term 7^{-s} of length $1/\sqrt{7} \approx 0.378$, $d_1 \approx 0.0875$, and $\hat{d}_1 = \sqrt{7} d_1 \approx 0.231$: the walk uses about 23% of link 6, and that stub is R_{1ps} . The first six $\hat{d}_1(k)$ in the table of §12.3 sit near $\frac{1}{2}$, as the saddle of §12.1 said they would; only $\hat{d}_1(m)$ is the named remainder weight. So (168) is

$$\Sigma_{1x} = \sum_{n' \leq m} n'^{-s} + \hat{d}_1(m) (m+1)^{-s} = \Sigma_1 + R_{1ps} = B_1, \quad (170)$$

and not Σ_1 itself.

The same construction on the other Dirichlet sum gives Σ_{2x} . Write $L_k = \chi(s) \sum_{n' \leq k} n'^{s-1}$ for the joints of Σ_2 from the origin, and $\hat{d}_2(k)$ for the crossing fraction of reverse link k from L_k to its crossing point, the u of (164). Then

$$Q_k = L_k + \hat{d}_2(k) \chi(s) (k+1)^{s-1}, \quad (171)$$

and the same telescoping, with $\hat{d}_2(-1) = 0$ and the empty sum at $k = 0$, is

$$\begin{aligned} \Sigma_{2x} &= \sum_{k=0}^m \left(L_k + \hat{d}_2(k) \chi(s) (k+1)^{s-1} - L_{k-1} - \hat{d}_2(k-1) \chi(s) k^{s-1} \right) \\ &= \chi(s) \sum_{n' \leq m} n'^{s-1} + \hat{d}_2(m) \chi(s) (m+1)^{s-1}. \end{aligned} \quad (172)$$

The last stub is $\hat{d}_2(m) = \hat{d}_2$, so

$$\Sigma_{2x} = \Sigma_2 + R_{2ps} = B_2. \quad (173)$$

Nothing here used $\sigma = \frac{1}{2}$. The split $R = R_{1ps} + R_{2ps}$ of §4 holds at every σ , the weights $\hat{d}_1, \hat{d}_2$ of (26) are defined at every σ , and so

$$\Sigma_{1x} + \Sigma_{2x} = B_1 + B_2 = \zeta \quad (174)$$

off the critical line as well as on it. On $\sigma = \frac{1}{2}$ the two walks are the two legs of §9.2; off it they are just the two summands of (1), re-jointed. Reflecting Σ_{1x} through $\zeta/2$ sends the origin to ζ and fixes B_1 , which is the other way of seeing the second half on the critical line, not the walk of Σ_{2x} from the origin. The reverse partners of the forward links $0, \dots, m$ are not consecutive, so the points P_k are not the joints of a walk along the reverse chain. This re-jointing is not Kapitonets’ iterated-midpoint polygon of §15.3: that construction is the binomial transform of the partial sums and does not aim at B_1 or B_2 .

12.5 First-part and second-part sums

Each forward link $k = 0, \dots, m-1$ is cut at its crossing P_k of (167) into two pieces: a first part $\hat{d}_1(k) (k+1)^{-s}$ from the joint to the crossing, and a second part $(1 - \hat{d}_1(k)) (k+1)^{-s}$ from the crossing to the next joint. The bisector link contributes only its first part, the stub $\hat{d}_1(m) (m+1)^{-s} = R_{1ps}$ from Σ_1 to B_1 . Summing those pieces as free vectors,

$$V_1 = \sum_{k=0}^m \hat{d}_1(k) (k+1)^{-s}, \quad V_2 = \sum_{k=0}^{m-1} (1 - \hat{d}_1(k)) (k+1)^{-s}, \quad (175)$$

gives $V_1 + V_2 = B_1$: the full links $0, \dots, m-1$ plus the bisector stub are $\Sigma_1 + R_{1ps}$. Figure 51 draws V_1 from the origin and V_2 from the tip of V_1 .

First-part and second-part sums of Σ_1 at $\sigma = 1/2$, $T = 6.18$ ($V_1 + V_2 = B_1$ to $4.4e-31$)

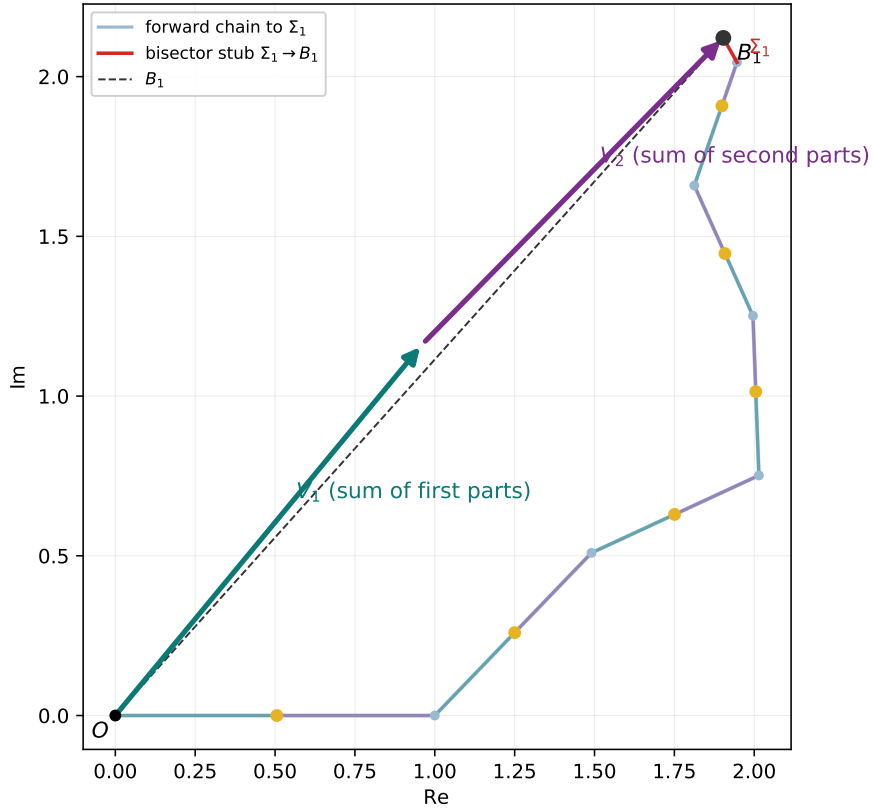


Figure 51: Each forward link cut at its yellow crossing: teal the first part, purple the second, red the bisector stub $\Sigma_1 \rightarrow B_1$. The teal arrow is V_1 , the sum of the first parts including that stub; the purple arrow is V_2 , the sum of the second parts, drawn from the tip of V_1 . They meet at B_1 . At $\sigma = \frac{1}{2}$, $T = 6.18$. Generated by `fig_half_link_sums.py`.

12.6 Cutting at one crossing

The two constructions above kept both walks whole. A single crossing is enough for a formula by itself. The crossing point of forward link k and reverse link $j = C_\ell(k, T)$ lies on both links, so it can be reached two ways: out from the origin along the forward chain, or in from ζ along the reverse

one. With the joints of (140) those two readings are

$$J_k + \hat{d}_1(k, T) (k+1)^{-s} = K_j - \hat{d}_2(k, T) \chi(s) (j+1)^{s-1}, \quad (176)$$

and moving the reverse sum inside K_j across gives

$$\zeta(\sigma, T) = \sum_{n=1}^k n^{-s} + \hat{d}_1(k, T) (k+1)^{-s} + \chi(s) \left[\sum_{n=1}^j n^{s-1} + \hat{d}_2(k, T) (j+1)^{s-1} \right]. \quad (177)$$

This is (25) with the two cutoffs unhitched. At $k = m$ the pairing is the self-pair, $j = m$, both cutoffs are $\lfloor T \rfloor$, and (177) is the boxed formula, hats and all. At any other k the two cutoffs differ: the forward sum stops at k while the reverse one runs to $j \approx a^2/(k+1)$, the two being reciprocal about a by (137). Reaching that far means carrying the reverse chain past its own bisector link, the continuation already written into (160).

For comparison, equation (4.13.1) of Titchmarsh's book [22, §4.13], the approximate functional equation, states: if h is a positive constant and

$$0 < \sigma < 1, \quad 2\pi xy = t, \quad x > h > 0, \quad y > h > 0,$$

then

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} + \chi(s) \sum_{n \leq y} \frac{1}{n^{1-s}} + O(x^{-\sigma} \log |t|) + O(|t|^{\frac{1}{2}-\sigma} y^{\sigma-1}). \quad (178)$$

The two sums are the same, and his constraint $2\pi xy = t$, that is $xy = a^2$, ties the two cutoffs exactly as the crossing product (137) does. Where he pays two error terms for the freedom to slide the cut, (177) pays with the two fractional terms, exactly.

The extreme cases are the ends of the pairing. Take $j = 0$ in (177): the first reverse link is $\chi(s) 1^{s-1} = \chi(s)$, so the reverse sum is empty and it is the second half that collapses to a single term,

$$\zeta(\sigma, T) = \sum_{n=1}^k n^{-s} + \hat{d}_1(k, T) (k+1)^{-s} + \hat{d}_2(k, T) \chi(s), \quad k = \left\lfloor \frac{I(T)}{2\pi} \right\rfloor - 1, \quad (179)$$

the forward cutoff being the partner of $n' = 1$, about $(T + \frac{1}{2})^2$ links out. Here Σ_1 is walked whole, out to link k and a fraction of the next, and the walk lands on the segment hanging off ζ in the direction of $-\chi$. On the critical line that segment has length one, so the forward chain carried that far passes within one unit of ζ , and what is left over is $\hat{d}_2(k, T) \chi(s)$, a real multiple of χ . At $\sigma = \frac{1}{2}$, $T = 6.18$ the cutoff is $k = 44$ and the two fractions are 0.012402 and 0.505537, reproducing ζ to 9.9×10^{-32} .

For comparison, equation (4.11.1) of Titchmarsh's book [22, §4.11], the simplest approximation of $\zeta(s)$ in the critical strip by a partial sum of its Dirichlet series, states:

$$\zeta(s) = \sum_{n \leq x} \frac{1}{n^s} - \frac{x^{1-s}}{1-s} + O(x^{-\sigma}), \quad (180)$$

uniformly for $\sigma \geq \sigma_0 > 0$, $|t| < 2\pi x/C$, when C is a given constant greater than 1. Both close one long partial sum with a small correction: his is the smooth tail $-x^{1-s}/(1-s)$ plus an error term, while (179) closes exactly with the two fractional terms. And the cutoffs match in scale: his condition puts x at order $t/2\pi$ at least, which is where $k = \lfloor I(T)/2\pi \rfloor - 1$ sits.

The cut at the other end of the pairing is the mirror of this one, $k = 0$. The first forward link runs from $J_0 = 0$ to $J_1 = 1^{-s} = 1$: it is the unit segment of the real axis. So the forward sum is empty, the link is the number 1, and the entire first half of (177) collapses to a single real number,

$$\zeta(\sigma, T) = \hat{d}_1(0, T) + \chi(s) \left[\sum_{n=1}^j n^{s-1} + \hat{d}_2(0, T) (j+1)^{s-1} \right], \quad j = C_\ell(0, T). \quad (181)$$

The reverse cutoff is the partner of the first summand, $n' = a^2/1$, so $j = \lceil I(T)/2\pi \rceil - 1$, the same integer that is the forward cutoff in (179). And $\hat{d}_1(0, T) = d_1(0, T)$ here: link 0 has length one, so the fraction of the link and the distance along it are the same number, and this is the only link on which the hat of (26) does nothing.

Read as a walk, (181) says this. Start at ζ and come in along the reverse chain, j whole links and then a fraction $\hat{d}_2(0, T)$ of the next. The walk lands on the real axis, between 0 and 1, and $\hat{d}_1(0, T)$ is where. Equivalently, ζ minus that whole reverse tail is a real number in the unit interval. It is not just any point of that interval either: it is essentially the midpoint, the $k = 0$ panel of Figure 50 sitting on $\frac{1}{2}$. The crossing fraction has the closed form of (162) read at $k = 0$, where $\omega_n = I(T) \log 1 = 0$ and $W_0 = K_j$ is the reverse joint itself:

$$\hat{d}_1(0, T) = \operatorname{Re} K_j - \operatorname{Im} K_j \cot(\omega_{n'} + \psi), \quad n' = j + 1, \quad (182)$$

the crossing of the real axis by the line that carries reverse link j .

None of this is a shortcut to a value. The reverse sum now has about $(T + \frac{1}{2})^2$ terms where the balanced cut of (25) has $2\lfloor T \rfloor$ in all, and the two fractions are read off ζ just as d_1 and d_2 were: (181) is one complex equation in the two real unknowns $\hat{d}_1(0, T)$ and $\hat{d}_2(0, T)$, so it fixes them from ζ and not the other way about. What it fixes is a position. Nor is the crossing the only one: the continued reverse chain cuts the unit segment many times, 87 times before link 400 at the running example, since both chains wind. The pairing law picks one of them out, the link paired with the first summand, and (182) says where that one lands.

At $\sigma = \frac{1}{2}$, $T = 6.18$, one has $a^2 = 44.5389$, so $j = 44$, the tiny reverse link of the table in §12.3; then $\hat{d}_1(0) = 0.505537$ and $\hat{d}_2(0) = 0.012402$, and the right side of (181) reproduces ζ to 4.6×10^{-41} at forty digits. The same cut at $k = 0, \dots, 6$ and that T gives seven formulas for the one number, with reverse cutoffs 44, 21, 14, 10, 8, 6, 6: the list of §12.1 with the self-pair in last place. Checked on 500 samples of the critical line spread over $2.05 \leq T \leq 60$, and on 100 more at $\sigma = 0.25, 0.35, 0.7, 0.9$: the crossing lay inside both links at every sample, and the two sides of (181) agreed to 3.2×10^{-30} at thirty digits. The first crossing stayed near the middle of the unit segment throughout, $\hat{d}_1(0, T)$ in $[0.5006, 0.5233]$ on the line and in $[0.4858, 0.5467]$ off it, and on the line it approaches $\frac{1}{2}$ from above as T grows: the product $T(\hat{d}_1(0, T) - \frac{1}{2})$ reads 0.0354, 0.0365, 0.0370, 0.0373 at $T = 10.5, 20.5, 40.5, 80.5$. The nearest-integer j_0 was the paired link in 481 of the 500 line samples; at the other 19 it was $j_0 - 1$, the third line of (139), and at each of those the crossing sat at $\hat{d}_2 \geq 0.98$, at the far end of the reverse link and about to hand over. In `check_first_link.py`.

The two cuts are mirrors on the critical line: the fractions 0.012402 and 0.505537 of (179) are the pair of (181) exchanged, because the chain mirror (141) carries forward link n to reverse link n and so carries the one crossing to the other. Off the critical line the two formulas are independent, the mirror being what fails there: at $\sigma = \frac{1}{4}$ and the same height the partner of reverse link 0 is forward link 43, with fractions 0.9721 and 0.5141.

13 ϑ_1, ϑ_2 and the zero-counting function

13.1 Review: the Riemann–von Mangoldt formula and N_{ps}

The well known Riemann–von Mangoldt formula for the number of zeros of ζ with imaginary part in $(0, t]$, in its exact form, reads

$$N(t) = \frac{\vartheta(t)}{\pi} + 1 + S(t), \quad S(t) = \frac{1}{\pi} \arg \zeta\left(\frac{1}{2} + it\right), \quad (183)$$

with ϑ the Riemann–Siegel theta function of §8.2. This is an identity ([22, Theorem 9.3]), obtained by applying the argument principle to ξ around a rectangle: $\vartheta(t)/\pi + 1$ is the contribution of the Γ and $\pi^{-s/2}$ factors, and $S(t)$ is the contribution of ζ itself along the critical line. Stated in terms of ϑ the formula is exact: (183) carries no error term at all; approximation enters only when ϑ is replaced by its expansion, and inexactness only when S is dropped. A word on usage before the counting begins: here and in the rest of the paper, an *ordinate* always means the ordinate of a zeta zero, the imaginary part γ of a zero of ζ , with $\gamma_1 < \gamma_2 < \dots$ numbering them up the critical line; on the index the ordinate γ_n sits at $T = I^{-1}(\gamma_n)$. Figure 52 shows the two pieces over one unit of the index.

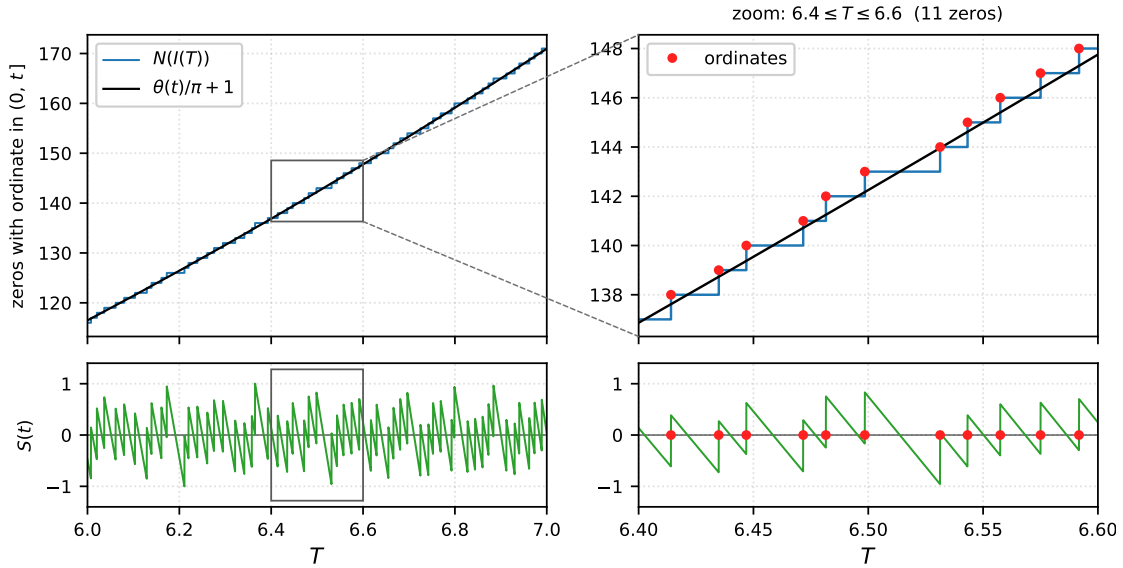


Figure 52: *Left*: the staircase $N(I(T))$ (blue) over $6 \leq T \leq 7$, which is $264.9 \leq t \leq 352.9$ and carries the 55 ordinates γ_{117} through γ_{171} , against the smooth part $\vartheta(t)/\pi + 1$ of (183) (black). The vertical gap between them is S , which here stays within $[-1.002, +1.001]$. *Right*: the boxed window $6.4 \leq T \leq 6.6$ magnified, with the eleven ordinates it holds, γ_{138} through γ_{148} , marked in red at the top of each riser and placed at $T = I^{-1}(\gamma)$. The risers are unevenly spaced while the smooth part climbs steadily. Below each panel is the gap itself, S (green), which at this magnification shows plainly as a sawtooth: a unit jump at every ordinate and a steady slide between them. Generated by `fig.staircase.67.py`.

All of the discontinuity is in S . N is a step function and $\vartheta(t)/\pi + 1$ is analytic, so S carries every jump: it climbs by one (by the multiplicity, in principle) at each ordinate (the imaginary part

γ of a zero) and falls smoothly in between, at the rate

$$S'(t) = -\frac{\vartheta'(t)}{\pi} = -\frac{1}{2\pi} \log \frac{t}{2\pi} + O(t^{-2}), \quad (184)$$

which is the mean zero density. Across an ordinate S jumps up by one (at a simple zero); between ordinates it slides back down along that line. The sawtooth is why S has mean zero without being small: each unit jump is repaid by a decline whose steepness is the density itself. At an ordinate the definition of S fails, since ζ vanishes there and its argument is undefined; one sets $S(\gamma) = \frac{1}{2}[S(\gamma^-) + S(\gamma^+)]$, and (183) then holds for every $t > 0$.

Three counting curves against the staircase. Expanding ϑ by Stirling and dropping S turns (183) into the Riemann–von Mangoldt formula in its approximate form,

$$N_{\text{rvm}}(t) = \frac{t}{2\pi} \log \frac{t}{2\pi} - \frac{t}{2\pi} + \frac{7}{8} + \frac{1}{48\pi t} + \frac{7}{5760\pi t^3} \quad (185)$$

(the $\frac{7}{8}$ collects the $-\frac{\pi}{8}$ of the expansion, divided by π , with the standalone 1 of (183)). Nothing of consequence is discarded in truncating there: at $t = 1000$ even the first correction retained, $1/(48\pi t)$, is only 6.6×10^{-6} , while $S(1000) = +0.384$. The whole of the error is S . Set also

$$N_{\text{ps}}(T) = \frac{1}{\pi} \vartheta(I(T)) + \frac{1}{\pi} \vartheta_1(\tfrac{1}{2}, T) + \frac{3}{2}, \quad (186)$$

which replaces S by the angle ϑ_1 that leg 1 makes with the real axis (§9.2), the leg running from the origin to the bisector point $B_1 = \Sigma_1 + R_{1ps}$. These two behave quite differently against the true count $N(I(T))$, and a third curve N_* , built below in (193) from a different splitting of ζ , repairs the one defect of N_{ps} .

N_{rvm} is analytic and increasing, so it can only approximate the staircase in the mean, never landing on an integer at an ordinate, and its error is exactly the sawtooth S . N_{ps} instead meets an integer *exactly* at every ordinate, and only there.

Proposition 13.1. *On $\sigma = \frac{1}{2}$, the value $N_{\text{ps}}(T)$ is an integer if and only if $t = I(T)$ is an ordinate.*

The reason is already Corollary 9.2: on the critical line

$$B_1 e^{i\vartheta(t)} = \tfrac{1}{2}Z(t) + ih, \quad h \in \mathbb{R}. \quad (187)$$

The real part is $|B_1| \cos(\vartheta + \vartheta_1) = \frac{1}{2}Z$. That cosine is zero exactly when $\vartheta + \vartheta_1 \equiv \frac{\pi}{2} \pmod{\pi}$, which is exactly when $N_{\text{ps}} = (\vartheta + \vartheta_1)/\pi + \frac{3}{2}$ is an integer. So N_{ps} is an integer if and only if $Z = 0$. The 2.8×10^{-14} figure is a check of that identity, not the argument. Multiplying by $e^{i\vartheta}$ is the ϑ -frame of the glossary: it is the rotation $e^{-i\psi/2}$ of Corollary 9.2 written with $\vartheta = -\frac{\psi}{2}$, since $\chi = e^{-2i\vartheta}$ on the line, and it puts the base $O \rightarrow \zeta$ on the real axis so that a split point reads as $\frac{1}{2}Z + ih$. This does not say N_{ps} equals the staircase N : retrograde ordinates still land on the wrong integer, as the next paragraph explains. It only says the value is an integer there, and nowhere else.

Where N_{ps} misses, and why. Two things keep (186) from being a counting function. First, ϑ_1 is an angle, so (186) is determined only up to an even integer, and each passage of ϑ_1 through $\pm\pi$ breaks the curve; there are nine such wraps over $3 < T < 5$, and $\frac{3}{2}$ is the constant that puts the curve on the staircase. Second, and this is the real defect, the curve sometimes meets the wrong integer. Differentiating (187) at an ordinate, where Z vanishes, gives

$$\left. \frac{d}{dt} \arg(B_1 e^{i\vartheta}) \right|_{Z=0} = -\frac{Z'(t)}{2h}, \quad (188)$$

so the crossing carries the angle forward or backward with the sign of $-hZ'$. Consecutive simple zeros of Z alternate in the sign of Z' , so the crossings all turn the same way exactly when h alternates in sign too. Where h nearly vanishes it keeps its sign across two ordinates instead; the angle turns back on itself, the count fails to advance, and the second of the pair reads one too high. Over the first 400 ordinates $|h|$ has median 1.081 at the 350 that N_{ps} places correctly and 0.319 at the 50 it misses, and those 50 are exactly the crossings that run backwards, which we call *retrograde*. They are also exactly the wraps: the nine wraps of ϑ_1 over $3 < T < 5$ counted above sit one apiece on the nine retrograde ordinates of that range. Figure 53 shows the cost over a longer stretch: five retrograde ordinates in $5.9 < T < 6.7$, and after each one N_{ps} runs a unit above the staircase until the following ordinate restores it. The two ways of describing the failure, a broken branch and a backward crossing, are one event. Over $3 < T < 5$, which holds 55 zeros, both curves stay within one of the staircase all the same: the largest deviation is 0.946 for N_{rvm} and 0.999 for N_{ps} .

Reading ϑ_2 in place of ϑ_1 cannot help. On the critical line $B_2 = \chi \overline{B_1}$ and $\arg \chi = -2\vartheta$, so

$$\vartheta_2 = -2(\vartheta + \vartheta_1) \pmod{2\pi}, \quad (189)$$

whichever remainder defines the split. Every sawtooth built from ϑ_2 is therefore the same function as the one built from ϑ_1 and retrogrades in the same places: among the 63 ordinates of $3.0 \leq T \leq 5.2$ the splits at R_{1ps} , at $R/2$ and at R_{1ak} run backwards at 10, 9 and 12 of them. The retrograde belongs to the split, not to the branch.

13.2 Improving N_{ps} : the velocity split and a curve that lands on every ordinate

The cure is to split ζ so that the offset h of (187) alternates by construction. With ϑ' the derivative of (184), exactly $\frac{1}{2} \operatorname{Re} \frac{\zeta'}{\zeta} \left(\frac{1}{4} + \frac{it}{2} \right) - \frac{1}{2} \log \pi$, put

$$B_1^* = \zeta(s) + \frac{\zeta'(s)}{2\vartheta'(t)}, \quad B_2^* = \zeta(s) - B_1^* = -\frac{\zeta'(s)}{2\vartheta'(t)} = \frac{i}{2\vartheta'(t)} \frac{d\zeta}{dt}, \quad (190)$$

so $\log 2$ is the velocity of the chain's endpoint along the line, quarter-turned and scaled by $1/2\vartheta'$. The companion B_1^* is not new: it is the function Levinson calls G at his (1.9), where the weight is written in the gamma-factor form $f'(s) + f'(1-s)$, which on the line is this same real number $2\vartheta'(t)$ [14]. §15.2 gives the dictionary and sets his use of it beside ours. On $\sigma = \frac{1}{2}$ this split is again reflection-symmetric, $B_2^* = \chi \overline{B_1^*}$, so its legs are exactly equal at every height,

$$\left| \zeta + \frac{\zeta'}{2\vartheta'} \right| = \frac{|\zeta'|}{2\vartheta'}, \quad (191)$$

which is the identity $\operatorname{Re}(\zeta \overline{\zeta'}) = -\vartheta' |\zeta|^2$, itself no more than the statement that $Z = e^{i\vartheta} \zeta$ is real. Differentiating that statement gives $e^{i\vartheta} \zeta' = -iZ' - \vartheta' Z$, and hence (187) holds for the new split with

$$e^{i\vartheta} B_1^* = \frac{Z}{2} - \frac{iZ'}{2\vartheta'}, \quad h^* = -\frac{Z'}{2\vartheta'}. \quad (192)$$

Now h^* alternates because Z' does, and (188) returns $-Z'/2h^* = \vartheta'$: every crossing turns forward, and at exactly the mean density. We set

$$N_*(T) = \frac{1}{\pi} \vartheta(I(T)) + \frac{1}{\pi} \vartheta_1^*(T) + \frac{3}{2}, \quad \vartheta_1^* = \arg B_1^*, \quad (193)$$

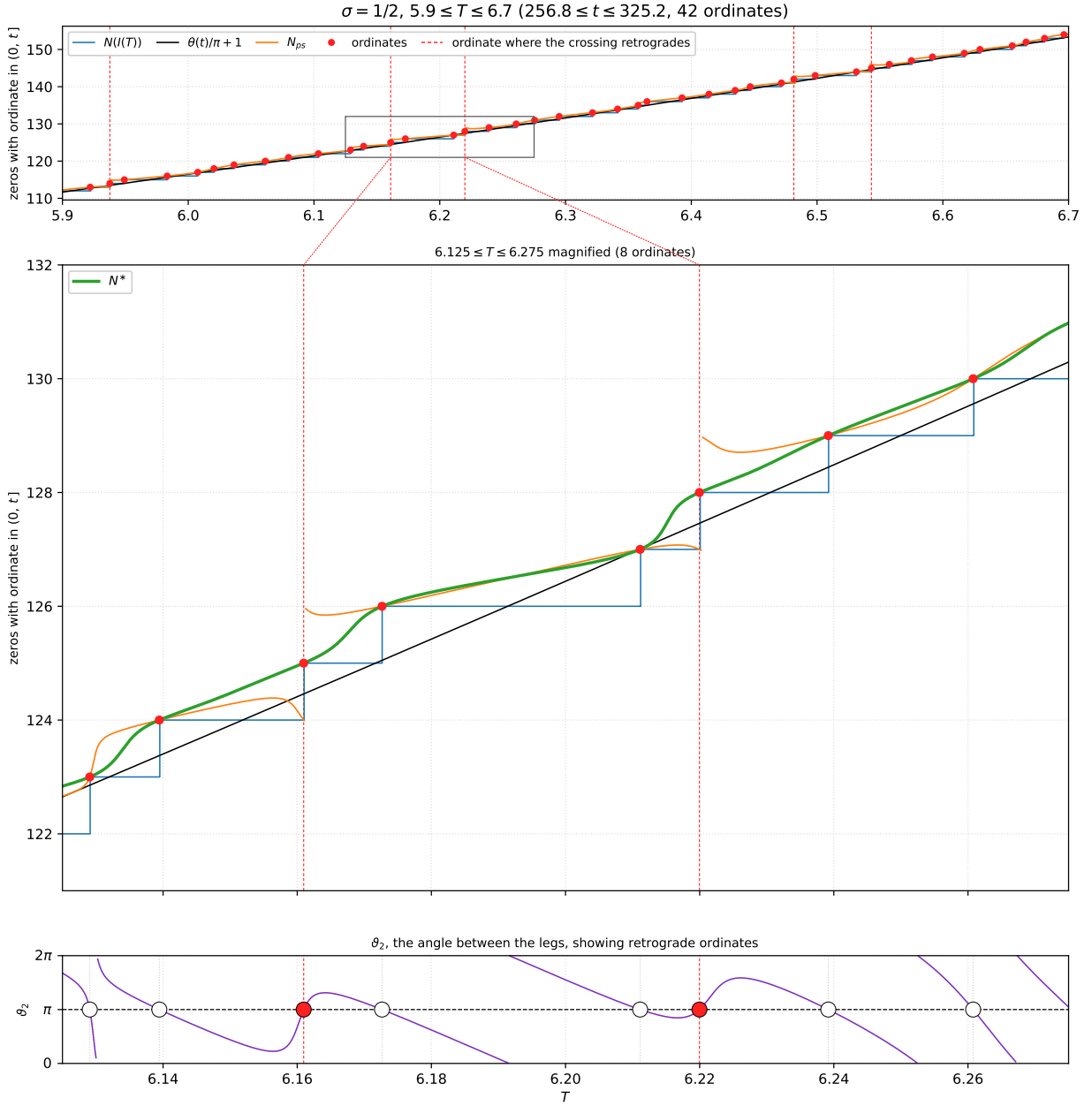


Figure 53: Staircase, in blue, is the exact Riemann–von Mangoldt formula (183), and the approximate Riemann–von Mangoldt formula (185) is the black curve, that formula with S dropped. *Top*: $5.9 \leq T \leq 6.7$, which is $256.8 \leq t \leq 325.2$ and holds the 42 ordinates γ_{113} through γ_{154} , five of them retrograde, with N_{ps} of (186) in orange, broken at its wraps. *Middle*: the boxed window $6.125 \leq T \leq 6.275$ magnified, holding 8 ordinates of which γ_{125} and γ_{128} are retrograde. The velocity split N^* of (193) is drawn in green. *Bottom*: the fold angle of the legs ϑ_2 of §9.2 on the same T range, reduced to $[0, 2\pi)$. Every passage through π (dashed) is at an ordinate; the two filled dots are the two retrograde ordinates of the panel above, where ϑ_2 rises through π , turns around, and comes back down through it. Generated by `fig.nps_staircase.py`.

That is the Riemann–Siegel phase plus the argument of the companion $B_1^* = \zeta + \zeta'/(2\vartheta')$, or equivalently the Prüfer angle of $(Z, Z'/\vartheta')$ ¹⁰:

$$N_*(T) = \frac{1}{\pi} \arg\left(Z - \frac{iZ'}{\vartheta'}\right) + \frac{3}{2}. \quad (194)$$

The Prüfer angle is Sturm–Liouville theory’s device for counting the zeros of a real oscillator $y(t)$: plot the point $(y, y'/\omega)$ in the plane, with ω the oscillator’s local frequency, and track its polar angle on a continuous branch. The point crosses the vertical axis exactly where y vanishes, so the angle passes a multiple of π at each zero and its winding is the count; dividing y' by ω only rounds the orbit, so the angle advances steadily instead of in bursts. In (194) the oscillator is Z and the frequency is ϑ' , the mean zero density. The resulting curve is integer-valued at the ordinates and nowhere else, since $\operatorname{Re}(e^{i\vartheta} B_1^*) = \frac{1}{2}Z$ vanishes only there, and that advances by exactly one from each ordinate to the next. It lands on every one of the first 400 ordinates, $N_*(I^{-1}(\gamma_n)) = n$ to 2.8×10^{-13} , against 50 misses for N_{ps} ; over $0.6 \leq T \leq 8$ it is strictly increasing on a grid of 3000 points, no wrap occurs ($|\vartheta_1^*|$ stays below 0.60π), and it rises by 237.64 across the 237 zeros of that range. Height costs it nothing and no branch correction is needed there either: at $\gamma_{10000} = 9877.78$ and at $\gamma_{50000} = 40433.69$ the same formula with the same constant $\frac{3}{2}$ still returns exactly 10000 and 50000. Where N_{rvm} threads the staircase without touching it, and N_{ps} locates zeros without counting them, N_* does both. Like any count read off Z it sees the zeros on the line, which in the range plotted are all of them.

Remark 13.2 (the two offsets $h_{R/2}$ and h^*). The offsets $h_{R/2}$ and h^* of the $R/2$ split and the velocity split are both drawn in Figure 54, and they differ only by a weighting of the links. The velocity offset has the closed form of (192),

$$h^*(t) = -\frac{Z'(t)}{2\vartheta'(t)}, \quad (195)$$

the slope of Z in units of $2\vartheta'$: h^* vanishes exactly at the critical points of Z and is largest where Z is steepest. For the $R/2$ split $e^{i\vartheta}R$ is real, so h is the sine companion of the main sum in the ϑ -frame, $h_{R/2} = \operatorname{Im}(e^{i\vartheta}\Sigma_1) = \sum_{n \leq m} n^{-1/2} \sin(\vartheta - t \log n)$, while differentiating the real part of that rotated main sum and dividing by $-\vartheta'$ gives

$$h^* = \sum_{n \leq m} n^{-1/2} \left(1 - \frac{\log n}{\vartheta'}\right) \sin(\vartheta - t \log n) + \cdots,$$

the tail being the remainder’s derivative; the displayed sum by itself reproduces h^* to 2.3×10^{-2} over the first 400 ordinates. The weight is the phase rate $\vartheta' - \log n$ of link n measured in units of ϑ' , so it runs from 1 on the first link down to nearly 0 on the last, where $\log m \approx \vartheta'$. The counting angle is carried by the same links; each one contributes in proportion to how much of the rotation rate it retains.

Remark 13.3 (the velocity split off the critical line). The weight $2\vartheta'(t)$ in (190) is exactly $-(\chi'/\chi)(\frac{1}{2} + it)$, since $\chi = e^{-2i\vartheta}$ on the line. Replacing it by that logarithmic derivative extends the split to the whole strip:

$$B_1^*(s) = \zeta(s) - \frac{\zeta'(s)}{(\chi'/\chi)(s)}, \quad B_2^* = \zeta - B_1^*, \quad \frac{\chi'}{\chi}(s) = \log 2\pi + \frac{\pi}{2} \cot \frac{\pi s}{2} - \psi(1-s). \quad (196)$$

¹⁰The companion is Levinson’s G [14]; what we have not found in the literature is its argument used this way, as an interpolant of the count. The nearest classical objects are the exact count $N = \vartheta/\pi + 1 + S$ of (183) and the Prüfer phase of an oscillator [18].

This is again Levinson's G and not an extension of it: his weight $f'(s) + f'(1-s)$ equals $-(\chi'/\chi)(s)$ at every s (§15.2), the critical line being the place where it happens to be real. Applying the functional equation twice gives $\chi(s)\chi(1-s) = 1$, whose logarithmic derivative forces $(\chi'/\chi)(s) = (\chi'/\chi)(1-s)$; differentiating $\zeta(s) = \chi(s)\zeta(1-s)$ then yields the exact reflection

$$B_2^*(s) = \chi(s) B_1^*(1-s)$$

at every s in the strip, pairing the split at (σ, T) with its mirror image at $(1-\sigma, T)$. On the critical line the mirror point is the point itself, the weight is the real number $2\vartheta'$, and the identity collapses to $B_2^* = \chi \overline{B_1^*}$ with its equal legs (191). Off the line the weight is complex (imaginary part $\approx \mp 7 \times 10^{-4}$ at $t \approx 280$), the offset h^* ceases to be a real scalar, and, as for every split, the equal-leg property survives only on the line. Verified numerically to twenty-four digits at $\sigma = 0.3$.

The χ -weight also separates the two chains exactly. Write $\zeta = F + \chi \tilde{G}$ with $F = \Sigma_1 + R_1$ and $\tilde{G} = (\Sigma_2 + R_2)/\chi$, for any of the named splits: the second half always carries the factor χ , so peeling it off is natural. Differentiating, multiplying by χ/χ' , and subtracting cancels the $\chi \tilde{G}$ term entirely; on the critical line, where $\chi'/\chi = -2\vartheta'$, the result is

$$B_1^* = \Sigma_1 + R_1 + \frac{(\Sigma_1 + R_1)' + \chi \tilde{G}'}{2\vartheta'}, \quad B_2^* = \Sigma_2 + R_2 - \frac{(\Sigma_1 + R_1)' + \chi \tilde{G}'}{2\vartheta'}. \quad (197)$$

The velocity split therefore hands the whole forward chain $\Sigma_1 + R_1$ to leg 1 and the whole reflected chain $\Sigma_2 + R_2$ to leg 2, exactly, with no mixing at the level of the sums; the two legs trade only the derivative corrections, damped by $2\vartheta'$. The reflected half influences B_1^* only through its rate of change $\tilde{G}'$, never through its value. This is why B_1^* crowds against the other split points in Figure 54, and the taper weights $1 - \log n/\vartheta'$ of Remark 13.2 are this cancellation read term by term. Verified to twenty-four digits at $T = 6.18$.

The splits and the offset h , measured from $\zeta/2$ along the line perpendicular to the base: $\sigma = 0.50$, $T = 6.18$ ($t = 279.85$, $m = 6$)

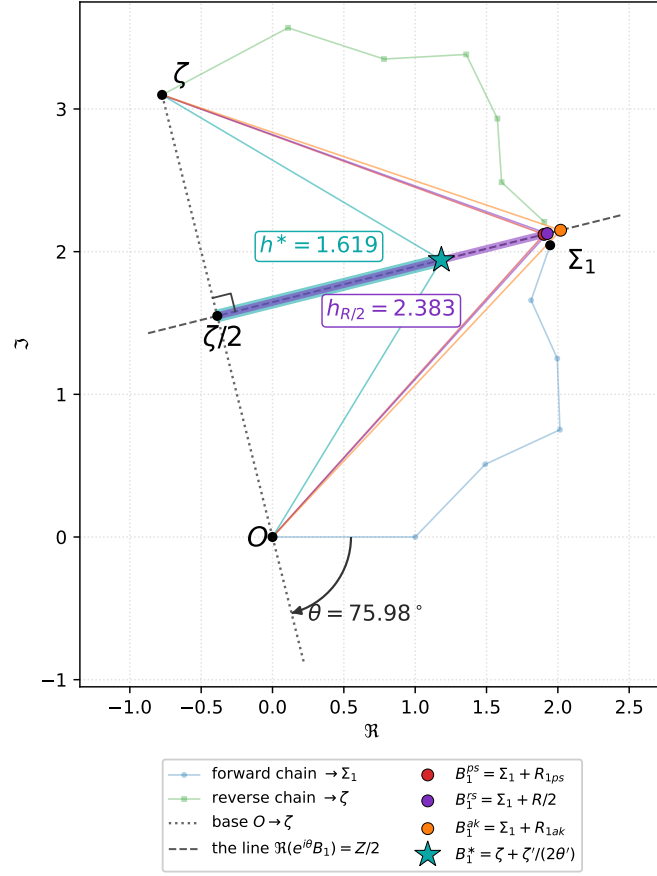


Figure 54: The offset h of (187), drawn at $T = 6.18$. Because $\Re(e^{i\vartheta} B_1) = Z/2$ for every split, all of the split points lie on one line when $\sigma = \frac{1}{2}$, the dashed line through $\zeta/2$ perpendicular to the base $O \rightarrow \zeta$ (the right angle is marked). What distinguishes one split from another is h , the signed distance from $\zeta/2$ along that line, drawn as a thick stretch of the line for the $R/2$ split (purple, $h_{R/2} = 2.383$) and for the velocity split (teal, $h^* = 1.619$); the two legs of each split are drawn thin in its own color and the two chains fainter still. The arc at O is ϑ , swept clockwise off the positive real axis onto the base line, since $\arg \zeta \equiv -\vartheta \pmod{\pi}$: with $\vartheta = 390.884$ reducing to 75.98° modulo π , the base line leaves the axis at -75.98° , drawn dotted below O , and ζ itself lies on the far side of O at 104.02° because $Z = -3.194$ is negative at this height. Generated by `fig_splits_h.py`.

Remark 13.4 (N_* is monotonically increasing if RH is true). Integer at the ordinates and nowhere else, and up by one from each to the next: what (193) leaves open is whether the curve ever runs backwards in between. It does not, and the reason is a known consequence of the Riemann hypothesis. With $W = e^{i\vartheta} B_1^* = \frac{1}{2}Z - \frac{iZ'}{2\vartheta'}$ from (192) and $\pi N_* = \arg W + \frac{3\pi}{2}$, differentiating and measuring against the local mean density ϑ' gives the slope in closed form,

$$\rho = \frac{1}{\vartheta'} \frac{d}{dt} \arg W = \frac{\pi}{\vartheta' I'(T)} \frac{dN_*}{dT} = \frac{Z'^2 - ZZ'' + \frac{\vartheta''}{\vartheta'} ZZ'}{\vartheta'^2 Z^2 + Z'^2}, \quad (198)$$

which returns 1 at $Z = 0$ and $-Z''/\vartheta'^2 Z$ at $Z' = 0$, as it must. Since ϑ' and I' are positive, ρ

and dN_*/dT carry the same sign; the denominator of (198) is positive and $\vartheta''/\vartheta' = O(1/t \log t)$, so up to that negligible term the curve increases exactly where the Laguerre expression $Z'^2 - ZZ''$ is positive.

That expression belongs to Ξ . Writing $\Xi(t) = \xi(\frac{1}{2} + it) = -c(t)Z(t)$ with $c(t) = \frac{1}{2}(\frac{1}{4} + t^2)\pi^{-1/4}|\Gamma(\frac{1}{4} + \frac{it}{2})| > 0$, algebra gives $\Xi'^2 - \Xi\Xi'' = c^2[(Z'^2 - ZZ'') - (\log c)''Z^2]$. If the hypothesis holds then every zero of Ξ is real, its Hadamard product gives $-(\log \Xi)'' = \sum_k (t - \gamma_k)^{-2}$, and therefore

$$Z'^2 - ZZ'' = Z^2 \left[\sum_k \frac{1}{(t - \gamma_k)^2} + (\log c)'' \right], \quad (199)$$

the sum running over the zeros of Ξ of both signs. The bracket is positive with room to spare: it is of the order of the squared zero density, carried mostly by the two ordinates flanking t , while $(\log c)'' = O(t^{-2})$ takes away almost nothing. At $t = 100.5$ it reads 2.2003, of which the 600 zeros with $|\gamma| \leq 542$ supply 2.1972 and $(\log c)''$ removes 1.7×10^{-4} .

Monotonicity is what makes the curve a counting function outright, rather than a curve that merely meets the integers in the right places. N_* is continuous, it equals n exactly at γ_n , and it is an integer nowhere else; if it also increases then between consecutive ordinates it is trapped strictly between n and $n + 1$, and taking the floor recovers (183) on the nose,

$$\lfloor N_*(T) \rfloor = N(I(T)), \quad (200)$$

with no error term and no sawtooth. We find no exception at 5721 grid points across $14 \leq t \leq 300$ nor at 301 across $9877 \leq t \leq 9880$. Three things are wanted for it, and each is worth naming. The count is read off Z , so what it counts is zeros on the critical line, and it is the hypothesis again that makes those all of them. The zeros must be simple: at a double zero Z and Z' vanish together, $W = 0$, the argument is undefined, and the curve would climb by one where the count climbs by two; RH does not supply simplicity. And ϑ_1^* must be read on a continuous branch. Subtracting (185) from (193) gives $\vartheta_1^* = \pi(N_* - N_{\text{vrm}} - \frac{1}{2})$, so the principal branch serves precisely while the exact count and the smooth one stay within one of each other; since S is unbounded that must eventually fail, and past it the wrap count has to be carried explicitly, which is the bookkeeping N_{ps} already needs at every height. There is room to spare at these heights: over $14 \leq t \leq 300$ the angle $|\vartheta_1^*|$ reaches only 0.506π .

Read the other way, (198) can only fail where $ZZ'' > Z'^2$, which at a critical point of Z means Z and Z'' of one sign: a positive local minimum or a negative local maximum, a pair of zeros that comes to the axis and fails to reach it. The Laguerre inequality $\Xi'^2 \geq \Xi\Xi''$, which Laguerre proved for every real entire function with only real zeros, is precisely what forbids that. Being a consequence of the hypothesis and not a theorem about ζ , it leaves the monotonicity conditional; and being only the first of a family of inequalities of every order, which Csordas and Varga [6] show to be equivalent to RH when taken all together, it is weaker than the hypothesis. A failure of monotonicity would therefore disprove RH, while a proof of monotonicity would not prove it. Nothing is close to the boundary. On grids of spacing 0.1 the slope never turns negative and dips no lower than 0.431 over $10 \leq t \leq 200$, 0.403 over $200 \leq t \leq 300$, 0.375 over $1000 \leq t \leq 1030$ and 0.521 over $9877 \leq t \leq 9880$ beside γ_{10000} . The near misses do not threaten it either, and they push ρ the opposite way from what one might guess. At Lehmer's pair [13] $\gamma_{6709} = 7005.06287$ and $\gamma_{6710} = 7005.10056$ the hump between the two sits at $t = 7005.08186$ with $Z = 3.97 \times 10^{-3}$, and there $\rho = 458$, because $-Z''/\vartheta'^2 Z$ grows without bound as a hump flattens onto the axis. What such a pair costs is margin, not sign: $Z'^2 - ZZ''$ reads 0.089 at that hump against 1.53 and 1.38 just outside the pair. Verified by `check_nstar_mono.py`.

Remark 13.5 (regarding the slope of N_* and the size of $|Z|$). By (194), $\pi N_* - \frac{3\pi}{2} = \arg(Z - iZ'/\vartheta')$ is the polar angle of the plane point $(Z, -Z'/\vartheta')$, the Prüfer angle [18] with ϑ' in the role of the frequency. The slope of the counting curve is therefore fixed by Z , Z' and Z'' at that height alone, and measured against the local mean density it is the ρ of (198). Two values of it are exact. At an ordinate Z vanishes and (188) with $h^* = -Z'/2\vartheta'$ gives $\rho = 1$; equivalently, setting $Z = 0$ in (198) leaves Z'^2/Z'^2 . This is exact at every simple zero, and the twelve-digit agreement at γ_1 , γ_{100} , γ_{1000} and γ_{10000} is a check of the identity rather than the reason for it. The whole of the variation therefore lives strictly between zeros. At an extremum of Z , where $Z' = 0$ leaves $W = \frac{1}{2}Z$ real,

$$\rho = -\frac{Z''}{\vartheta'^2 Z} \quad (Z' = 0), \quad (201)$$

so flatness at a hump is exactly a small curvature-to-height ratio, in units of ϑ'^2 : a broad tall hump gives ρ well below 1 and a sharp small one gives ρ well above it. The consequence is that the flatness of N_* ranks the size of $|Z|$ between the zeros. Over the 100 gaps of $14 \leq t \leq 238$, ρ at the hump against the height of that hump has rank correlation -0.987 , with $|Z|_{\max} \propto \rho^{-1.17}$; two decades higher the same measurement reads -0.943 and $\rho^{-1.30}$ at γ_{1000} , and -0.944 and $\rho^{-1.20}$ at γ_{10000} . Since $\rho = 1$ at every ordinate, the first local datum there is $\rho'(\gamma_n)$, and that single number, read at one zero, ranks the height of the hump ahead of it at -0.80 , -0.66 and -0.73 in the three blocks: the more steeply the slope falls away as one leaves a zero, the taller the hump being entered. Figure 55 plots all of this, the slope against $|Z|$ in two windows and then the two rankings. What the slope is really measuring is the local phase rate of Z rather than its amplitude, since for $Z = A \cos \varphi$ with $\varphi' = \vartheta'$ one gets $\rho = 1$ whatever A may be; $\rho < 1$ says the phase is turning more slowly than ϑ' , which is the same statement as a wide gap, and it is the sparseness of the zeros that lets $|Z|$ grow. Consistently with that, ρ at the hump and the gap width are collinear to -0.99 in rank, so the slope should be read as a pointwise version of the spacing rather than as independent information about $|Z|$. Verified in `check_nstar_slope.py`.

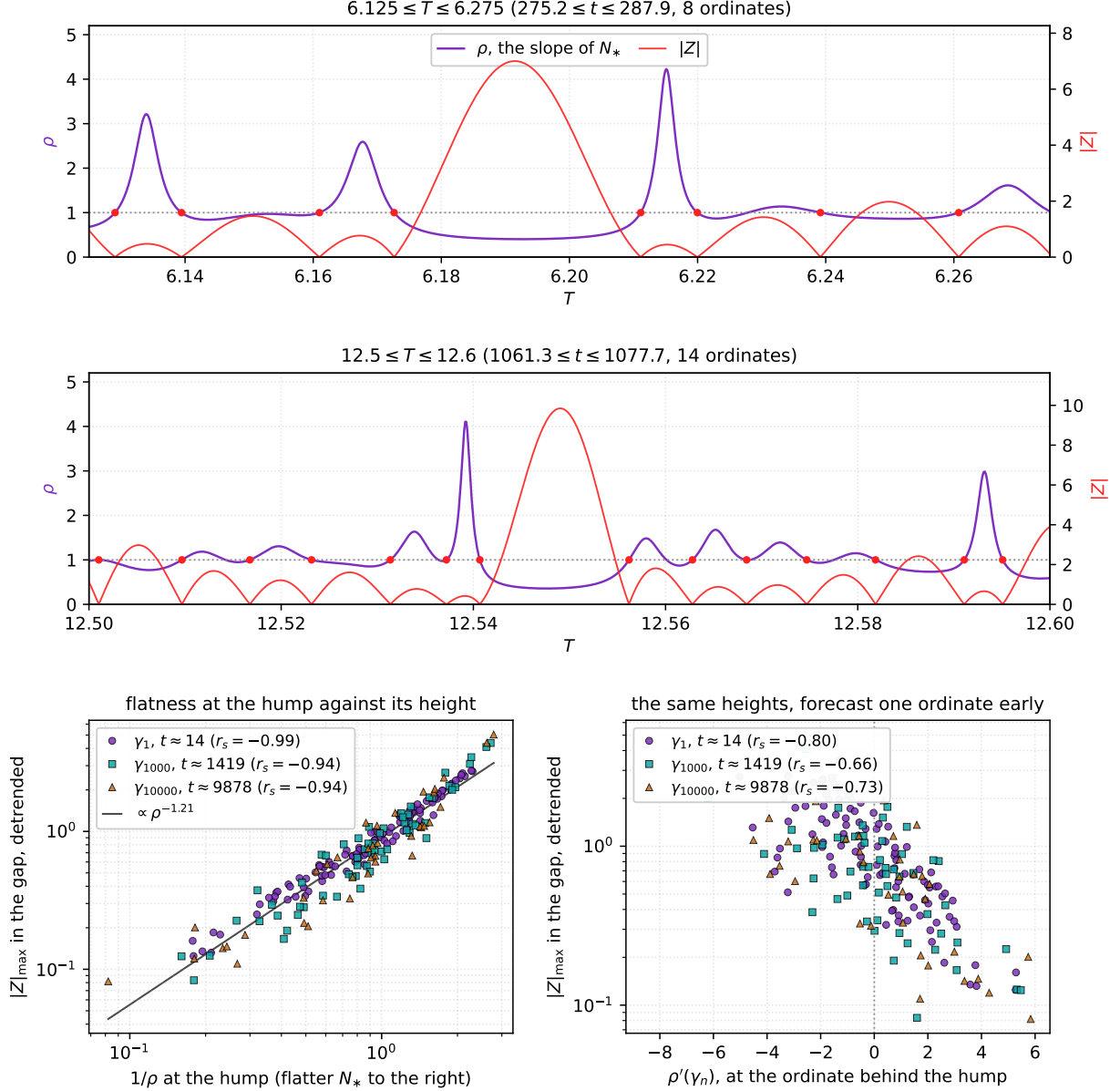


Figure 55: The slope ρ of (198) against the size of $|Z|$ between ordinates, everything at $\sigma = \frac{1}{2}$. *Top*: ρ in purple on the magnified window of Figure 53, with $|Z|$ in red on the right-hand scale. The red dots mark the ordinates, where $\rho = 1$ exactly; between them ρ dips, and it dips deepest where $|Z|$ rises highest. *Second*: the same reading twice as high in the index, over $12.5 \leq T \leq 12.6$. The hump that reaches $|Z| = 9.85$ near $T = 12.549$ sits on the flattest stretch of the window, where ρ bottoms at 0.356, while each ordinary hump carries only a shallow arch of ρ just above 1 and the two narrowest gaps throw up spikes of 4.11 and 2.99. *Bottom left*: one point per gap for three blocks of ordinates spanning two decades of t , the height of the hump against $1/\rho$ at that hump, both on log scales, with the heights divided by a 31-gap moving average so the slow growth of $|Z|$ with height does not enter. The line is the fit over all three blocks; the legend quotes the rank correlation against ρ for each block separately. *Bottom right*: the same hump heights against $\rho'(\gamma_n)$, read at the ordinate behind the hump rather than at the hump itself, scaled by the mean spacing π/ϑ' . Generated by `fig_nstar_slope.py`.

Remark 13.6 (regarding N_* beside N and N_{rvm}). N_* is continuous; the exact count N of (183) is a step function, with a jump at each ordinate. N_* equals n at every γ_n , and is an integer nowhere else; the approximate formula N_{rvm} of (185) never lands on an integer at an ordinate. If N_* increases, then $\lfloor N_*(T) \rfloor = N(I(T))$ by (200): the continuous curve recovers the staircase.

13.3 Connecting the count, ϑ_2 , and the equal-leg loci

The retrograde ordinates of §13.1 were found by watching the count; the equal-leg ovals of §9 (Figure 21) were found by scanning the strip. They are the same events. Figure 57 puts the two side by side. Its top panel repeats the magnified window of Figure 53. The bottom panel draws, on the same window, the fold angles ϑ_2 of the ps split and ϑ_2^* of the velocity split together with the equal-leg locus $L_1 = L_2$ of the ps split in the surrounding strip: the vertical scale doubles as $2\pi\sigma$, with σ on the right-hand axis, so the critical line, which belongs to the locus at every height, is the horizontal blue line lying exactly on the dashed line $\vartheta_2 = \pi$, and the off-line components of the locus are the blue ovals straddling it. Shaded are the stretches of T where ϑ_2 runs *retrograde*: where the unwrapped angle increases against its prevailing descent.

The correspondence is one to one, three ways. Over $5.9 \leq T \leq 6.7$ the angle ϑ_2 runs retrograde on exactly five stretches, each 0.008–0.010 of T wide. Each stretch contains exactly one ordinate, and those five are precisely the five retrograde ordinates $\gamma_{114}, \gamma_{125}, \gamma_{128}, \gamma_{142}, \gamma_{145}$ at which N_{ps} miscounts in Figure 53. And the strip holds exactly five equal-leg ovals over the same window (not drawn), one per stretch, coinciding with it end to end to about 10^{-3} of T . The reading is not local to this window: twelve windows spread from $T = 1$ to $T = 50$, holding 344 ordinates between them, give 45 retrograde stretches, 45 ovals and 45 retrograde ordinates, one for one, the stretches narrowing with the ordinate spacing from 0.013 of T at $T \approx 3$ to 0.0007 at $T \approx 50$ (`check_retro_ovals.py`). The only other off-line feature in the window, the thin band at $T \approx 6.255$, is not an oval but the pole band of Figure 21: at $\{T\} \approx \frac{1}{4}$ the weight d_1 passes through its off-line pole (§9.1), B_1 sweeps out and back, and the legs pass through equality on the way out and on the way back at every σ at once, in a band far narrower than any oval. That last property tells the two apart at sight: a band still crosses at $\sigma = 0.98$, while an oval has closed by $\sigma \approx 0.8$.

The three readings are one event. A retrograde ordinate is one where the offset h is small (§13.1: median $|h|$ of 0.319 at the missed ordinates against 1.081 at the rest), which puts the bisector point near $\zeta/2$; there the two legs are nearly equal *and* nearly folded, so ϑ_2 sits near π and moves slowly, and the slow passage is what lets the angle turn back on itself. The same near-degeneracy is what an oval is. On the critical line the leg difference $\delta = L_1 - L_2$ vanishes identically, so an off-line zero of δ can open up only where the balance survives a small excursion in σ , and that margin exists exactly while the configuration lingers: the oval opens when the stall begins and closes when prograde motion resumes. During ordinary prograde motion the legs rotate past each other too quickly for the balance to hold anywhere off the line, which is why the ovals are confined to the shaded stretches. Read against §13.2, the correspondence also says why the velocity split has no such windows: its offset $h^* = -Z'/2\vartheta'$ alternates by construction, no crossing is retrograde, and N_* inherits none of the stalls that mark N_{ps} . What is special about the velocity split is therefore not the definition of its fold angle $\vartheta_2^* = \arg(B_2^*/B_1^*)$, which is exactly as for the other splits, but its behavior: its legs are exactly equal at every height on the line, and ϑ_2^* passes π prograde at every ordinate, with no retrogrades. That is precisely the property that makes ϑ_1^* , and hence N_* , count correctly.

Figure 56 draws the two fold angles over the same window $5.9 \leq T \leq 6.7$: wherever ϑ_2 stalls at π and turns back, and N_{ps} is off by one, ϑ_2^* passes straight through, and N_* is never off.

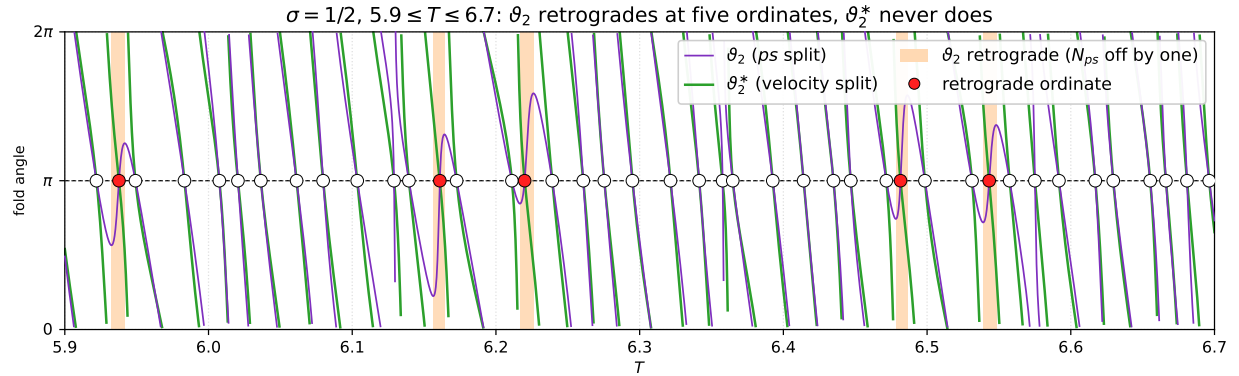


Figure 56: The two fold angles compared over $5.9 \leq T \leq 6.7$ at $\sigma = \frac{1}{2}$: ϑ_2 of the *ps* split (purple) and ϑ_2^* of the velocity split (green), both reduced to $[0, 2\pi)$ and broken at their wraps. Dots mark the ordinates, where both angles pass through the dashed line at π , filled red at the retrograde ones. On the shaded stretches ϑ_2 runs retrograde: it stalls at π , turns back, and recrosses, and on that stretch N_{ps} is off by one. The green curve has no retrograde stretch anywhere: at every ordinate ϑ_2^* passes π prograde, and N_* is never off the count. Generated by `fig_theta2_star_compare.py`.

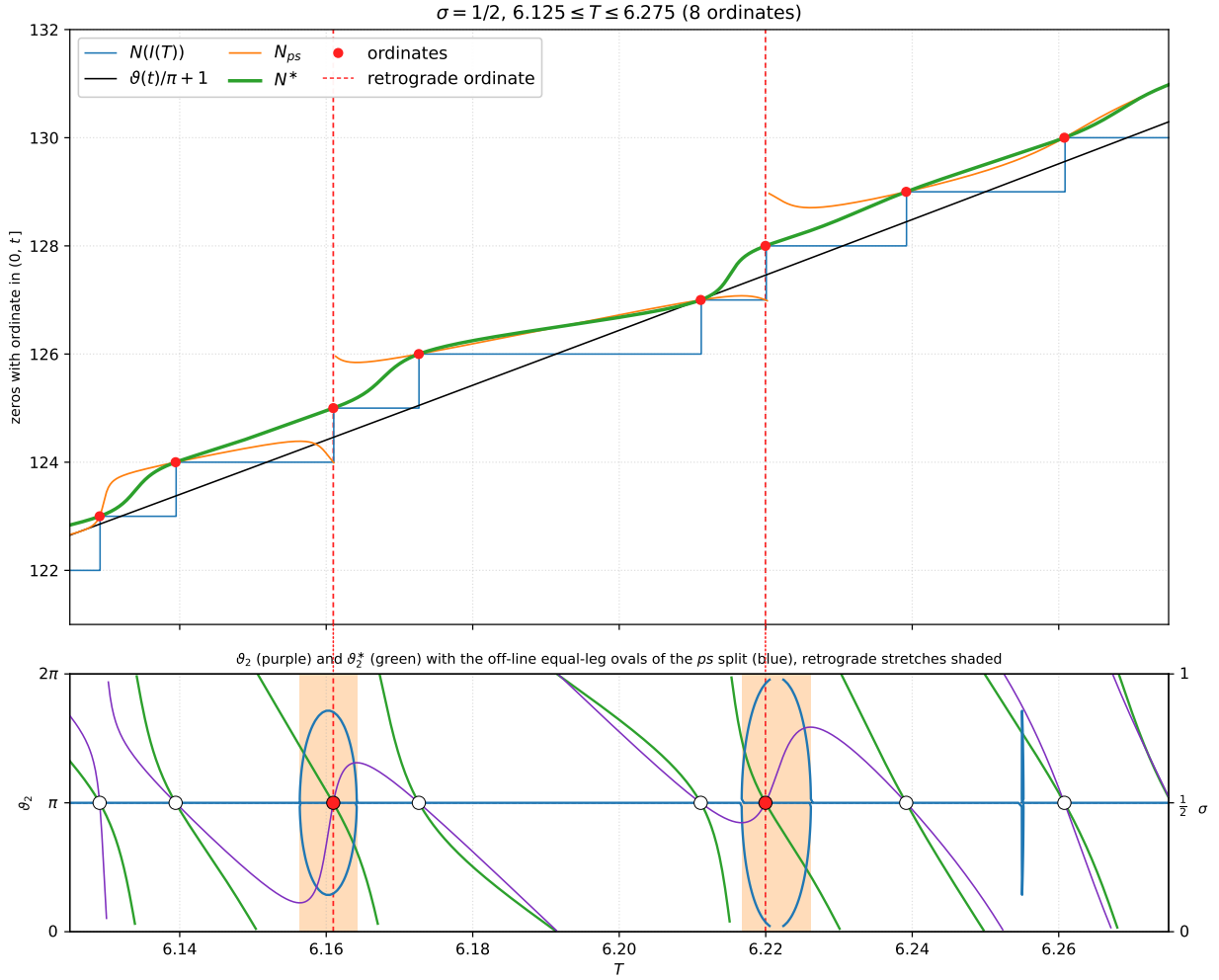


Figure 57: The count, the fold angle, and the equal-leg locus of the ps split read together, everything at $\sigma = \frac{1}{2}$ except the locus. *Top*: the magnified window $6.125 \leq T \leq 6.275$ of Figure 53: the staircase (blue), the smooth part (black), N_{ps} (orange), N_* (green), and the retrograde ordinates dashed in red. *Bottom*: ϑ_2 (purple) and ϑ_2^* (green) on the same window, reduced to $[0, 2\pi)$ and broken at their wraps, with the equal-leg locus $L_1 = L_2$ of the ps split drawn in the surrounding strip (blue) on the shared vertical scale $2\pi\sigma$, σ on the right-hand axis. The horizontal blue line on the dashed $\vartheta_2 = \pi$ line is the critical line, which belongs to the locus at every height; the two blue ovals are the off-line components in this window. Dots mark the ordinates, where both angles pass through π , filled red at the retrograde ones; the shaded stretches are where ϑ_2 runs retrograde, and there ϑ_2^* passes straight through. Each oval fills a shaded stretch and each stretch holds one retrograde ordinate. The thin blue band at $T \approx 6.255$ is the pole band of Figure 21 at $\{T\} \approx \frac{1}{4}$, not an oval. Generated by `fig_count_theta2_ovals.py`.

14 Further observations

14.1 Length ratios: Σ_1 to Σ_2 , and R_{1ps} to R_{2ps}

The four terms of (24) come in two natural pairs: the two partial sums, and the two fractional links. Within each pair the two lengths are equal on the critical line, and what follows records what their ratio does off it. The pairs behave quite differently, and in both cases the ratio obeys the same reflection law about $\sigma = \frac{1}{2}$.

Remark 14.1 (regarding the ratio of the lengths of Σ_1 and Σ_2). On the critical line $\sigma = \frac{1}{2}$ we have $|\chi| = 1$, so the Σ_2 links have the same lengths $(n+1)^{-1/2}$ as the Σ_1 links; each link is then a rotation composed with a uniform link-length shrink $\sqrt{n/(n+1)}$, and $\arg \chi = -2\vartheta(t)$, with ϑ the Riemann–Siegel theta function. The two chains are thus congruent link for link there, so their end-to-end displacements agree. Away from the critical line they do not, and the discrepancy has a closed form. Write $\ell_{\Sigma_1} = |\Sigma_1|$ and $\ell_{\Sigma_2} = |\Sigma_2|$ for the lengths of the net displacements of P_{Σ_1} and P_{Σ_2} , and put $\Sigma_m(z) = \sum_{n \leq m} n^{-z}$. Since $\sum_{n \leq m} n^{s-1} = \Sigma_m((1-\sigma) + it)$, the second leg is the first one read at the reflected abscissa $1-\sigma$, scaled by $|\chi|$:

$$\frac{\ell_{\Sigma_1}}{\ell_{\Sigma_2}} = \frac{|\Sigma_m(\sigma + it)|}{|\chi(\sigma + it)| |\Sigma_m((1-\sigma) + it)|}. \quad (202)$$

Proposition 14.2. *At fixed T (hence fixed $t = I(T)$ and $m = \lfloor T \rfloor$),*

$$\frac{\ell_{\Sigma_1}(\sigma)}{\ell_{\Sigma_2}(\sigma)} \cdot \frac{\ell_{\Sigma_1}(1-\sigma)}{\ell_{\Sigma_2}(1-\sigma)} = 1. \quad (203)$$

Write (202) at σ and at $1-\sigma$ and multiply. The two partial-sum moduli cancel, leaving $1/(|\chi(\sigma + it)| |\chi((1-\sigma) + it)|)$. From $\chi(s)\chi(1-s) = 1$ and the Schwarz reflection $\chi(\bar{s}) = \overline{\chi(s)}$ one has $\chi(1-\sigma + it) = 1/\overline{\chi(\sigma + it)}$, already used after (122), so the product of the moduli is 1. The ratio is therefore reciprocal-symmetric about $\sigma = \frac{1}{2}$: whatever it does on the right half of the strip, it does the reciprocal of on the left, with the value pinned to 1 at the center by the link-for-link congruence of Remark 14.1.

Remark 14.3 (regarding the ratio of the lengths of R_{1ps} and R_{2ps}). The same question for the two fractional links has a shorter answer, because in the quotient of (19) and (20) both $|R|$ and the common denominator $\sin(2\omega + \psi)$ cancel:

$$\frac{|R_{1ps}|}{|R_{2ps}|} = \frac{d_1}{d_2} = \frac{\sin(\omega + \psi - \arg R)}{\sin(\omega + \arg R)}. \quad (204)$$

This ratio is pure angle data, carrying no length scale at all: it is the law of sines in the triangle of Figure 3, recording how the direction of R divides the cone spanned by the two link directions $e^{-i\omega}$ and $e^{i(\omega+\psi)}$. Four consequences are worth recording. First, on the critical line $\arg R = \frac{\psi}{2}$, the two sines coincide and the ratio is 1, which is Corollary 8.1 again. Second, the ratio at σ is the reciprocal of the ratio at $1-\sigma$:

Proposition 14.4. *At fixed T ,*

$$\frac{d_1(\sigma)}{d_2(\sigma)} \cdot \frac{d_1(1-\sigma)}{d_2(1-\sigma)} = 1. \quad (205)$$

The turn ω depends only on T . After (122) one has $\chi(1-\sigma+it) = 1/\overline{\chi(\sigma+it)}$, so ψ is the same at both abscissae. From (8) and $\zeta(s) = \chi(s)\zeta(1-s)$ the remainder satisfies $R(1-s) = \chi(1-s)R(s)$; Schwarz reflection $R(\bar{z}) = \overline{R(z)}$ then gives

$$R(1-\sigma, T) = \chi(1-\sigma+it) \overline{R(\sigma, T)}, \quad (206)$$

hence $\arg R(1-\sigma) = \psi - \arg R(\sigma)$. The two sines of (204) therefore exchange, and the product is 1. On the critical line (206) is $R = \chi \overline{R}$, which is the first point again.

Third, because the scale cancels, this ratio is far tamer than (202): over $1 < T < 20$ and the whole strip its extremes are 0.387 and 2.583 (a reciprocal pair, as the second point requires), whereas $\ell_{\Sigma_1}/\ell_{\Sigma_2}$ ranges over several orders of magnitude there. Fourth, the parallel-link poles of §8.4 cancel out of it: where $2\omega + \psi = k\pi$, (204) gives $d_1/d_2 = (-1)^{k+1}$, exactly ± 1 , even though d_1 and d_2 separately diverge (at $\sigma = \frac{1}{10}$, $T = 4.256594$ we compute $d_1 = -d_2 \approx -4.07 \times 10^{36}$). A negative value is off-line news of a different kind: it says R has left the cone, which happens in a narrow window around each pole. The ratio does reach 0 and ∞ , but only on the isolated loci where R is exactly parallel to one of the two link directions; for instance $d_1 = 0$ at $\sigma = \frac{1}{10}$, $T = 4.2607679$, whose mirror at $\sigma = \frac{9}{10}$ has $d_2 = 0$. Off the critical line the locus $d_1 = d_2$ is the family of ovals plotted in Figure 27.

14.2 The envelope of $|\zeta(\frac{1}{2}, t)|$

On the critical line the chain is isosceles for every reflection-symmetric split, by Corollary 9.1, so its two legs share one length $L_1 = |B_1| = |B_2|$ and the base is the chord they subtend across the fold angle of the legs ϑ_2 of §9.2. From $\zeta = B_1(1 + e^{i\vartheta_2})$,

$$|\zeta(\frac{1}{2} + it)| = 2L_1 |\cos \frac{\vartheta_2}{2}|, \quad (207)$$

so the modulus of ζ splits into a length carried by the links and an angle read at the joint between the legs. Bounding the angular factor bounds ζ . Write $u = \vartheta_2/\pi$ reduced to $[0, 2)$. Then $|\cos \frac{\pi u}{2}| \leq 1$ at once, and since $\cos \frac{\pi u}{2}$ is concave on $0 \leq u \leq 1$ it stays above the chord joining its endpoints there, giving $|\cos \frac{\pi u}{2}| \geq |1 - u|$ on the whole of $[0, 2)$ by the reflection $u \mapsto 2 - u$, with equality only at $u = 0, 1, 2$. Hence

$$2L_1 \left| 1 - \frac{\vartheta_2}{\pi} \right| \leq |\zeta(\frac{1}{2} + it)| \leq 2L_1. \quad (208)$$

The upper envelope is attained where the two legs point the same way and the chain is folded straight; the lower one is attained at every ordinate, where $u = 1$ and both sides vanish, and again where u reaches 0 or 2 and the two envelopes meet. The triangular factor $|1 - \vartheta_2/\pi|$ is the same sawtooth whose passages through zero do the counting in §13, appearing here with an amplitude on it. Over $6 \leq T \leq 13$, which is $264.9 \leq t \leq 1144.6$, we confirm (207) to 1.7×10^{-20} and both inequalities of (208) at every one of 2001 sample points, the two legs agreeing to 2.0×10^{-20} .

Which split to use is settled by how often the upper envelope is reached. Equation (187) gives $2L_1 = \sqrt{Z^2 + 4h^2}$, so $|\zeta| = 2L_1$ exactly where the offset h vanishes. For the split at the partial summand h vanishes only inside a gap whose two ends carry opposite h , which is the alternation that the count of §13 needs and that fails at a retrograde ordinate, so the envelope is reached in some gaps and missed in others: 3 of the 7 gaps of $6.125 \leq T \leq 6.275$, and 7 of the 13 gaps of $12.5 \leq T \leq 12.6$. The velocity split has no such gaps. Its offset $h^* = -Z'/2\vartheta'$ vanishes at every extremum of Z , so

$$2L_1^* = \sqrt{Z^2 + \left(\frac{Z'}{\vartheta'}\right)^2} \geq |Z| = |\zeta(\frac{1}{2} + it)|, \quad (209)$$

with equality on every hump: 7 touches in 7 gaps and 13 in 13 in those two windows, matched to 10^{-25} (Figure 58). This is the familiar envelope of an oscillation of instantaneous frequency ϑ' , with Z'/ϑ' as the quadrature companion of Z , and it is tighter on average as well, the mean of $|\zeta|/2L_1$ over $6 \leq T \leq 13$ being 0.635 against 0.586 for the split at the partial summand.

Both readings are polar coordinates of one function. The rotated bisector point $W = e^{i\vartheta} B_1^* = \frac{1}{2}Z - \frac{iZ'}{2\vartheta'}$ of (192) has modulus L_1^* and argument $\pi(N_* - \frac{3}{2})$ by (193), and $Z = 2 \operatorname{Re} W$, so

$$Z(t) = -2|W| \sin(\pi N_*(T)), \quad (210)$$

which we confirm to 6×10^{-13} , the precision of the counting curve itself. This is a restatement rather than a new fact, but it puts the two subsections together: the amplitude of the velocity split is the envelope of ζ and its phase is the counting curve, so the zeros are the moments when the phase passes an integer and the extrema are the moments when the amplitude is attained. Remark 13.5 reads the rate of that phase; the envelope (209) reads its amplitude. Verified in `check_envelope.py`.

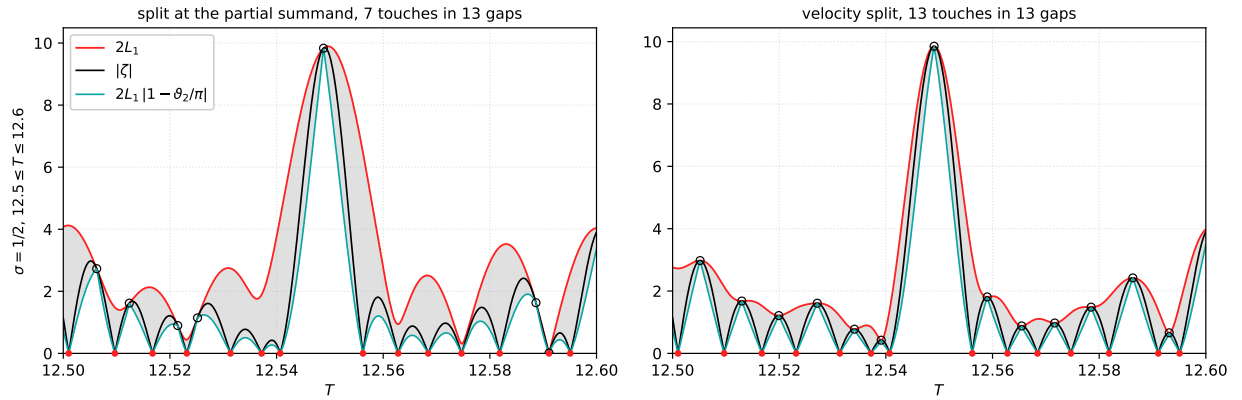


Figure 58: The two envelopes of (208), at $\sigma = \frac{1}{2}$, over $12.5 \leq T \leq 12.6$, the second window of Figure 55, which is $1061.3 \leq t \leq 1077.7$ and holds 14 ordinates. In each panel $|\zeta|$ is black, the upper envelope $2L_1$ red, the lower envelope $2L_1|1 - \vartheta_2/\pi|$ a teal sawtooth, and the gray band between them is where $|\zeta|$ is confined. Red dots on the axis are the ordinates, where the lower envelope touches zero along with $|\zeta|$, and the open circles are the touches of the upper envelope, one for each vanishing of the offset h . *Left*: the split at the partial summand, $B_1 = \Sigma_1 + R_{1ps}$, whose h fails to alternate at a retrograde ordinate, so it reaches its bound in only 7 of the 13 gaps and several humps rise nowhere near it. *Right*: the velocity split, whose h^* vanishes at every extremum of Z , so the upper envelope is touched in every gap. Generated by `fig_envelope.py`.

Remark 14.5 (regarding hunting for large values with the first few links). Neither the envelope nor the slope of Remark 13.5 can be used to scout, since $2L_1$ needs $R = \zeta - \Sigma_1 - \Sigma_2$ and $2L_1^*$ needs Z and Z' , so both are functions of ζ at a point one has already evaluated. What can be computed without ζ is the reason the envelope grows. Leg 1 is mostly the link sum, so the size of $2L_1$ is a question of how well the link phases $t \log n$ align, and the ceiling is the aligned case,

$$|\zeta(\tfrac{1}{2} + it)| \leq 2L_1 \leq 2 \sum_{n \leq m} n^{-1/2} + 2|R_{1ps}| \approx 4\sqrt{T} \approx 4\left(\frac{t}{2\pi}\right)^{1/4} = \frac{4}{(2\pi)^{1/4}} t^{1/4} < 2.54 t^{1/4}, \quad (211)$$

a bound that costs nothing to evaluate. Read in t it is nothing new: the exponent $\frac{1}{4}$ is the convexity, or trivial, bound $\mu(\frac{1}{2}) \leq \frac{1}{4}$ of Lindelöf [15], where $\mu(\sigma)$ is the least exponent with $\zeta(\sigma + it) = O(t^\mu)$, and (211) recovers it with an explicit constant. Nor can this path do better. The last step is the triangle inequality, which keeps only the aligned case and so discards every cancellation among the phases $t \log n$; in the geometry of this paper $4\sqrt{T}$ is the arc length of the chain laid out straight, while $|\zeta|$ is the distance between its endpoints once it has curled into its spirals. Every improvement past $\frac{1}{4}$ comes from that discarded cancellation, beginning with the $\frac{1}{6}$ that Hardy and Littlewood obtained by Weyl's method on this same sum and standing today at $\frac{13}{84}$ [3]; the Lindelöf hypothesis is $\mu(\frac{1}{2}) = 0$. Two things follow, both drawn in Figure 59. First, the alignment can be read from a truncation. Scoring an interval by $S_K(t) = |\sum_{n \leq K} n^{-1/2} e^{-it \log n}|$ and then evaluating Z only at the strongest peaks of that score recovers most of the interval's largest $|Z|$: averaged over the 19 unit intervals $5 \leq T \leq 24$, which is $190 \leq t \leq 3771$, the three strongest peaks of the five-link score reach 0.967 of it and the single strongest peak of the ten-link score reaches 0.961. The point is not the saving at these heights but that K need not grow with t while m does. Second, the interval maxima themselves are nearly regular: $\max |Z|$ over $[T, T+1]$ fits $2.93 T^{0.485}$, its ratio to $\sqrt{T}$ has mean 2.823 and standard deviation 0.123 across those intervals, and it holds a fraction of the ceiling (211) that ranges from 0.97 down to 0.76, the two extremes falling at $T = 9$ and $T = 22$. Most of that spread is the ceiling rather than the maxima: the exact sum $\sum_{n \leq m} n^{-1/2} = 2\sqrt{m} + \zeta(\frac{1}{2}) + o(1)$ climbs toward its own asymptote $2\sqrt{m}$ across the range, from 0.72 of $4\sqrt{T}$ at $T = 5$ to 0.86 at $T = 23$, since $\zeta(\frac{1}{2}) = -1.4604$ still weighs heavily at $m = 5$. Against $\sqrt{T}$ itself the maxima are flat, 2.711 at $T = 5$ and 2.670 at $T = 23$, which is the same fact as the fitted exponent 0.485 being essentially the ceiling's $\frac{1}{2}$: at these heights the maxima simply ride the trivial bound. In these coordinates the Lindelöf hypothesis is precisely the statement that they eventually stop, that $\max |Z|/\sqrt{T}$ falls like $T^{-1/2+\varepsilon}$, the chain curling up so efficiently that its endpoints stay bounded while its arc length grows like $\sqrt{T}$. Nothing of that is visible over 19 intervals with $m \leq 23$ links, and nothing here bears on it either way. So the answer to whether the next unit of the index will beat the last is yes by default, which it is in 13 of the 18 consecutive comparisons, and the exceptions are what needs predicting: on the maxima divided by $\sqrt{T}$ the peak of the ten-link score ranks the intervals at +0.72, while the three-link score, whose peak is nearly the same in every interval, carries nothing at -0.11. That last figure rests on 19 intervals at modest height and should be read as an indication only. Verified in `check_large_values.py`.

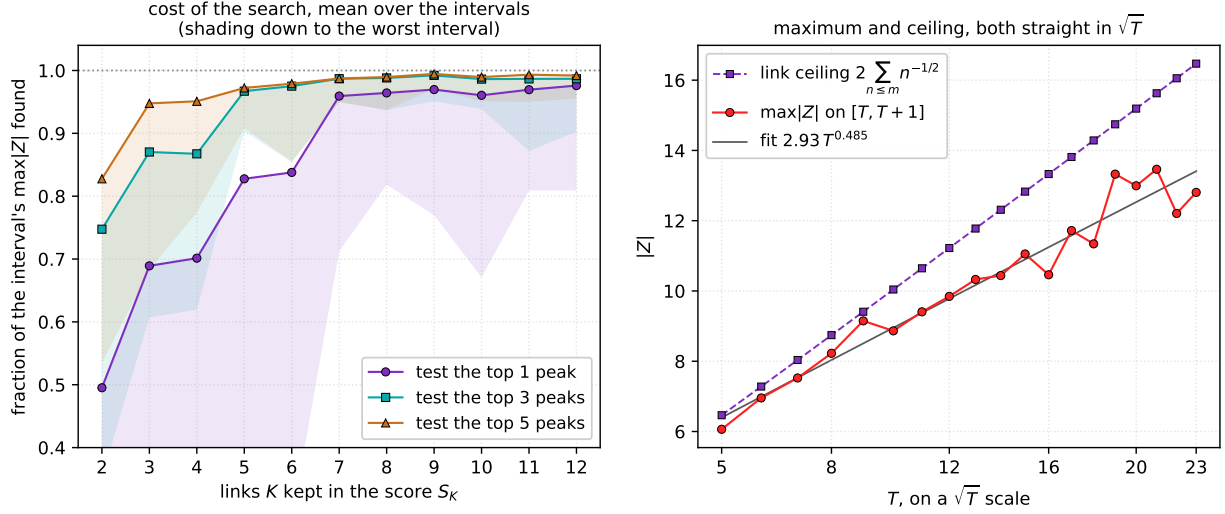


Figure 59: Reading large values off the links, over the 19 unit intervals $5 \leq T \leq 24$. *Left*: the fraction of an interval's largest $|Z|$ found by evaluating Z only at the strongest one, three or five peaks of the K -link score S_K , averaged over the intervals, with the shading reaching down to the worst single interval. Seven links suffice for a single candidate to land within 5%. *Right*: the interval maximum against $\sqrt{T}$, on which it is nearly straight, with the link ceiling of (211) above it and the fitted power law through it. The gap between the two curves is the alignment the links never quite achieve. Generated by `fig_large_values.py`.

14.3 Collinearity of the three first-half remainders

Collinearity of three points is unchanged by a common translation. The three points B_{1ps} , B_{1rs} , and B_{1ak} of §9.4 are all relative to the same Σ_1 , so that shared Σ_1 cancels, and the three remainder vectors R_{1ps} , $R_{1rs} = \frac{1}{2}R$, and R_{1ak} can be compared directly as points in the plane. Figure 60 plots two independent, scale-invariant measures of how far these three points are from lying on a common line. The first is the triangle flatness

$$F = \frac{2|\text{area}|}{\text{diam}^2}, \quad (212)$$

where the area is that of the triangle with the three points as vertices and the diameter is the largest of the three pairwise distances. The second is the aspect ratio $\lambda_{\min}/\lambda_{\max}$ of the covariance matrix of the three points (a principal-component measure). Both quantities are zero exactly when the points are collinear, and both are invariant under translation, rotation, and rescaling, so the shrinking of the remainders with growing T cannot masquerade as collinearity. Each panel is a grid of 1501 σ -samples by 1801 T -rows over $0 \leq \sigma \leq 1$, $3.9 \leq T \leq 5.1$, computed from the exact remainder formulas, with the σ -sample nearest $\frac{1}{2}$ snapped exactly onto the critical line. The color scale is logarithmic, and that is what separates *exact* collinearity (values at the double-precision floor, 10^{-16} to 10^{-12}) from the merely small values, 10^{-4} to 10^{-1} , that fill the rest of the strip.

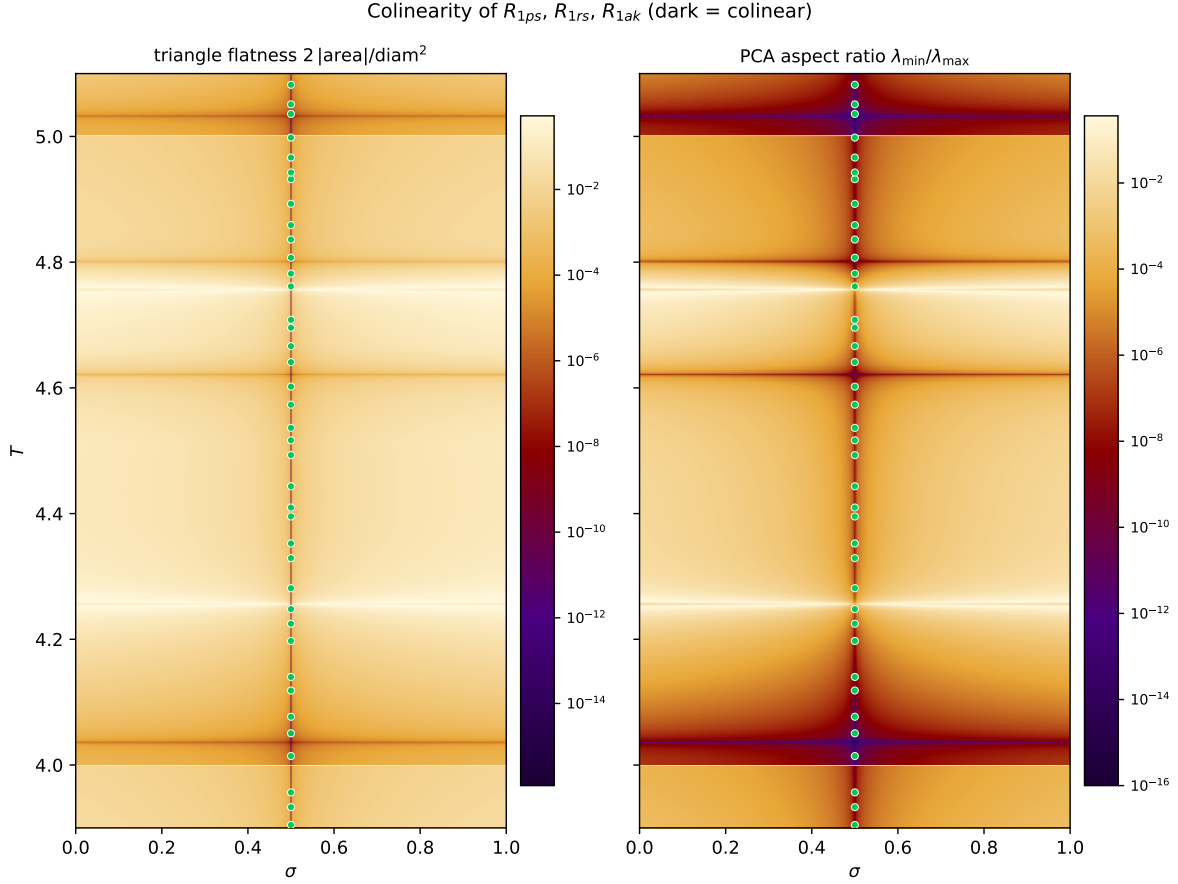


Figure 60: Collinearity of the three first-half remainders R_{1ps} , $R_{1rs} = \frac{1}{2}R$, R_{1ak} (dark = collinear). Horizontal axis $\sigma \in [0, 1]$; vertical axis the spiral index $T \in [3.9, 5.1]$. Left: triangle flatness (212); right: PCA aspect ratio $\lambda_{\min}/\lambda_{\max}$. Both panels use a log color scale; green markers are zeta zeros at $\sigma = \frac{1}{2}$. The dark column pinned at the numerical floor along $\sigma = \frac{1}{2}$ shows the three points are exactly collinear everywhere on the critical line; off the line collinearity occurs only along the isolated dark horizontal bands. Generated by `fig_collinearity_zoom.py` from data produced by `remainder-collinearity-heatmap-zoom.mjs`.

Both panels show the same two facts. First, the column $\sigma = \frac{1}{2}$ sits at the numerical floor at every T : on the critical line the three first-half remainders are exactly collinear, not merely nearly so (Proposition 14.6). Second, off the line the measures are many orders of magnitude larger everywhere except along isolated dark horizontal bands (near $T \approx 4.03, 4.62, 4.80$, and 5.03 in this window), where the triangle momentarily passes through a flat configuration as its signed area changes sign. The measured values are symmetric under $\sigma \mapsto 1 - \sigma$, a reflection of the functional equation.

The same pattern persists across the whole strip. Figure 61 repeats the four-panel layout of Figure 27 (the first two zoom bands of that figure, a third band straddling the integer $T = 17$, and the same full strip $0 \leq T \leq 20$), with the PCA aspect ratio as the plotted color map. As in the leg-imbalance panels, the dark column on $\sigma = \frac{1}{2}$ runs unbroken through every window, while off the line the only dark features are the isolated horizontal collinearity bands. The bands do not sit at integer T : crossing an integer the three remainder formulas jump (Σ_1 gains a summand and the sign $(-1)^{\lfloor T \rfloor}$ flips), the signed area of the triangle changes sign in that jump and lands very close to

zero, and the area then crosses zero a short distance above the integer. The third panel shows one full unit interval, $16.95 \leq T \leq 18.05$: the light-to-dark break at the integer $T = 17$ itself, a nearly flat strip at every σ just above it, and then the five collinearity bands of the cell, at $T \approx 17.011$, 17.252 , 17.612 , 17.752 , and 17.796 , before the pattern repeats across the next break at $T = 18$. Numerically the bands come five to a unit interval, near $\{T\} \approx 0.011, 0.252, 0.612, 0.752, 0.796$ at this height, with the second and fourth tracking the pole heights $\{T\} = \frac{1}{4}, \frac{3}{4}$ of d_1 (§8.4): near a pole R_{1ps} runs off to infinity along the fixed direction of the next summand, so the triangle is dragged through a flat configuration nearby. The band positions are almost independent of σ (they move by less than 10^{-4} between $\sigma = 0.2$ and $\sigma = 0.35$), which is why the bands are horizontal.

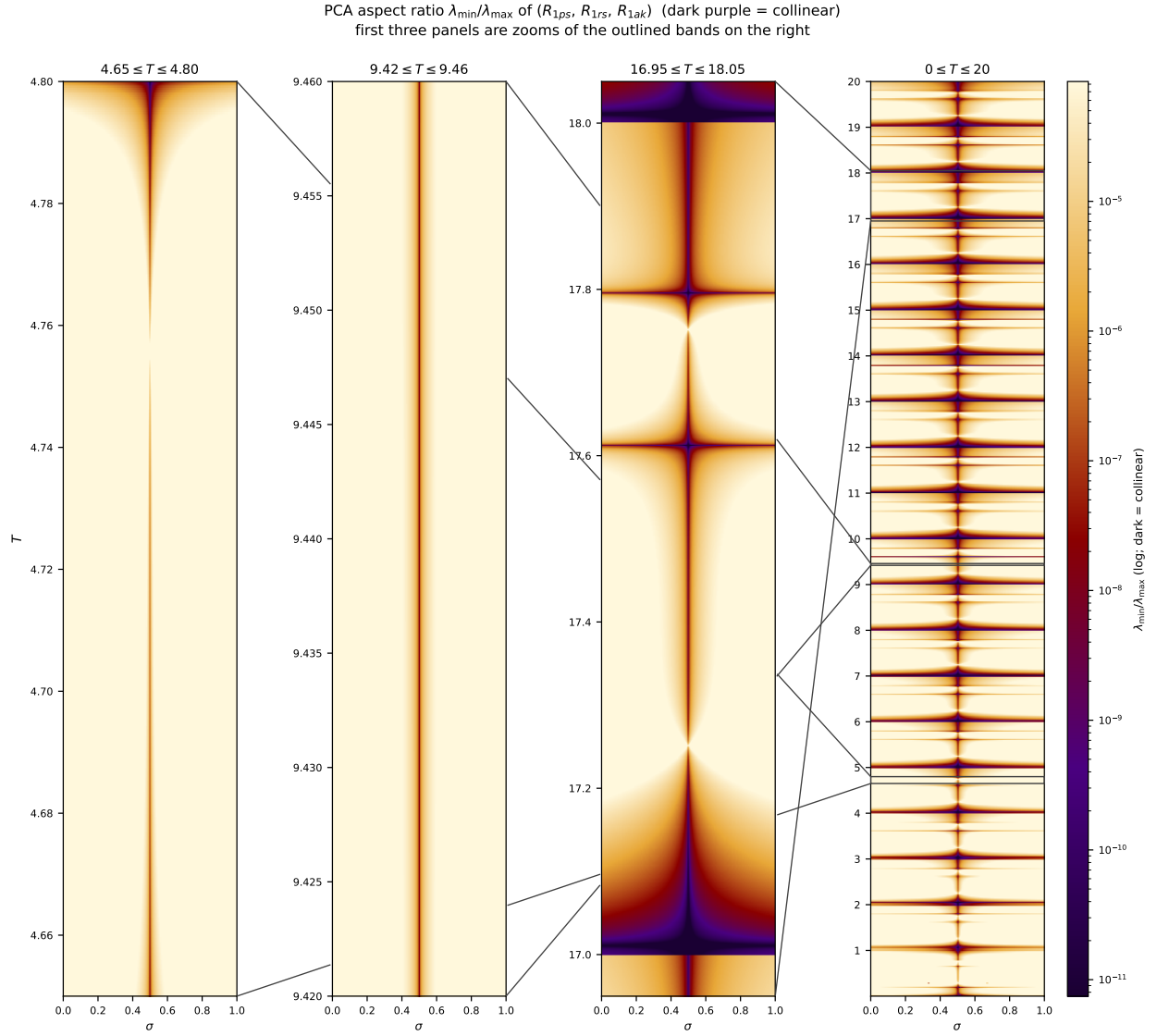


Figure 61: PCA aspect ratio $\lambda_{\min}/\lambda_{\max}$ of the three first-half remainders $(R_{1ps}, R_{1rs}, R_{1ak})$ (dark purple = collinear), in the four-panel layout of Figure 27. Horizontal axis $\sigma \in [0, 1]$; vertical axis the spiral index T . Color is on a log scale. The rightmost panel is the full strip $0 \leq T \leq 20$; the first three panels are zooms of the outlined bands ($4.65 \leq T \leq 4.80$, $9.42 \leq T \leq 9.46$, $16.95 \leq T \leq 18.05$). The dark vertical band on $\sigma = \frac{1}{2}$ is the exact collinearity of the three first-half remainders on the critical line; the dark horizontal bands are the isolated off-line collinearity events. Generated by `plot_collinearity_four_panel.py`.

Why can this collinearity not be used like the two routes of §9.5? Because collinearity is *not* forced at a zero. For $\zeta \neq 0$ the equal-leg condition for a split says precisely that B_1 is equidistant from O and ζ , i.e. that B_1 lies on the perpendicular bisector of the chord from O to ζ ; if all three splits had equal legs at a point where $\zeta \neq 0$, all three bisector points would lie on that single line and the three first-half remainders would be collinear. But at a zero the chord degenerates: $\zeta = O$, every point of the plane is equidistant from the two endpoints, and the (now automatic) equal-leg

conditions carry no collinearity information at all. All three leg pairs are equal at a zero, as they must be, while the three points can remain in general position. So a zero implies equal legs for all three splits, but not collinearity, and the three points need not line up at an off-line zero.

Proposition 14.6. *On $\sigma = \frac{1}{2}$ the three first-half remainders R_{1ps} , $R_{1rs} = \frac{1}{2}R$, and R_{1ak} are collinear at every height T , zeros included.*

The three points B_{1ps} , B_{1rs} , and B_{1ak} are those remainders translated by the common Σ_1 , so it is the same to prove the B_1 collinear. On the critical line write ϑ for the Riemann–Siegel theta function, so $\chi = e^{-2i\vartheta}$ and $Z = e^{i\vartheta}\zeta$ is real. Any split $\zeta = B_1 + B_2$ that satisfies the reflection $B_2 = \chi\overline{B_1}$ then has

$$e^{i\vartheta}B_1 = \frac{1}{2}Z + ih, \quad h \in \mathbb{R}. \quad (213)$$

Indeed $e^{i\vartheta}B_2 = \overline{e^{i\vartheta}B_1}$, so $e^{i\vartheta}\zeta = e^{i\vartheta}B_1 + \overline{e^{i\vartheta}B_1} = 2\Re(e^{i\vartheta}B_1)$, and the left side is the real number Z . Thus every such B_1 has the same real part $Z/2$ in the ϑ -frame: all of them lie on the single vertical line $\Re(e^{i\vartheta}z) = Z/2$, which is the perpendicular bisector of the segment from O to ζ when $\zeta \neq 0$, and the imaginary axis of that frame when $\zeta = 0$. The three named splits are of this kind. For the ps split, $B_2 = \chi\overline{B_1}$ is Corollary 9.1. For the rs split the same reflection is the identity recorded in §9.4: $\Sigma_2 = \chi\overline{\Sigma_1}$ and $R = \chi\overline{R}$, so $\chi\overline{B_{1rs}} = \Sigma_2 + \frac{1}{2}R = \zeta - B_{1rs}$. For the ak split, Siegel’s contour integral f of (46) gives the exact pairing $\zeta(s) = f(s) + \chi(s)f(1-s)$ [20], and on the line $f(1-s) = \overline{f(s)}$, so $B_{1ak} = f(s)$ satisfies the same reflection; the numerical R_{1ak} of (44) is the Dirichlet-sum truncation of that identity, which is why the heatmaps sit at the double-precision floor rather than at a symbolic zero. Equation (213) does not use $\zeta \neq 0$, so the three points remain collinear at a critical-line zero as well: there $Z = 0$ and they lie on $\Re(e^{i\vartheta}z) = 0$.

A proof through collinearity therefore still needs a bridge that the routes of §9.5 do not. The first step of that bridge is the identity just proved. What remains is to prove that off the line collinearity fails outside isolated curves like the dark bands of Figure 60, and to connect a hypothetical off-line zero to that structure, for instance through nearby off-line points with $\zeta \neq 0$ at which all three splits have equal legs, which by the perpendicular-bisector argument above would force a collinearity that the off-line geometry forbids. Those two steps would turn collinearity into a characterization of the critical line and then into a statement about zeros. They are listed in §16.

14.4 Joint angles

The joint angles of the forward chain can be read on their own, without the chain visualization, using one dot per joint with the turning angle on the vertical axis, the joint index along the horizontal. The turning angle at joint n is the continuous joint angle (97) evaluated at an integer joint,

$$\theta_n = -I(T) \log \frac{n}{n-1} \quad (\text{reduced to } [-\pi, \pi]), \quad \nu(n) = \frac{I(T)}{2\pi n(n-1)}, \quad (214)$$

where ν , the derivative of the unwrapped angle divided by 2π , is the local frequency of the strip in cycles per joint. Two features organize the picture, both consequences of $I(T)$. First, at the bisector joint $n = \lfloor T \rfloor + 1$ the frequency is $\nu = 1 + \frac{1}{6T(T+1)}$, one full turn per joint to nine decimals at $T = 13000$, and the angle there is $-\pi$ exactly: that is (98), the bisector joint of §10.1, seen as the right edge of the strip. The joints just behind it inherit almost the same angle, 3.141109, 3.139659, 3.137242 at $n = T, T-1, T-2$, which is the same near-stationarity that makes the bisector frame of Figure 34 repeat. Second, since $\nu(n) \approx T^2/n^2$ falls from $\nu \sim 10^8$ at $n = 2$ to 1 at that bisector joint, the strip is a sampled chirp, and it is stratified by the joints where ν is the reciprocal of a

rational. Writing that rational as a Farey fraction $f = p/q$ in $(0, 1]$, its *caustic joint* is the solution of $n(n-1) = f I(T)/2\pi$, that is $\nu = 1/f$, so

$$n_f = \frac{1}{2} \left(1 + \sqrt{1 + 2f I(T)/\pi} \right) \approx T\sqrt{f}, \quad (215)$$

and there the joints fall into exactly p strands, the denominator of $\nu = q/p$, each a parabola of curvature $4\pi\nu/n$ per joint squared because the unwrapped angle is locally quadratic (exactly $|\tilde{\theta}''(n_f)|$ of (217) below). Small denominators give the spines visible in Figure 62, drawn at the integer index $T = 13000$, where $t = I(T) = 1,061,939,999.369541$: there the fractions $\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ sit at $n = 6501, 7506, 9193, 10615, 11259$, and $f = 1$ is the bisector joint itself. The parabolic strands are the same local quadratic phase that produces the Cornu-like arcs of the curlicue literature of §15.3, here read off the joints rather than the partial sums.

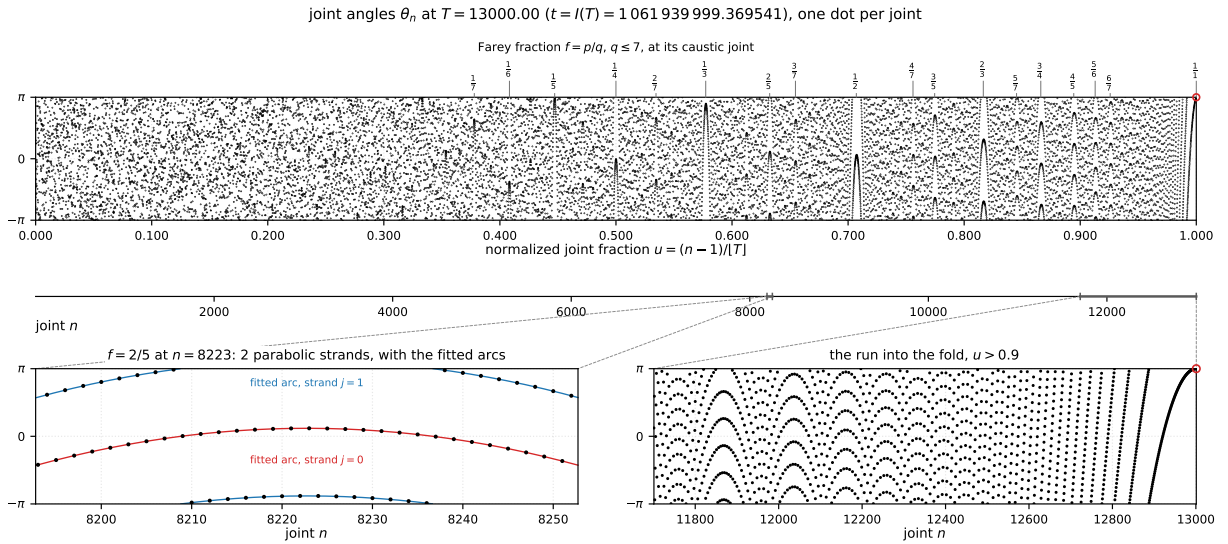


Figure 62: The joint-angle strip at $T = 13000$, where $t = I(T) = 1,061,939,999.369541$, one dot per joint of the forward chain, $n = 2, \dots, [T] + 1$, with θ_n of (214) on the vertical axis. The two edges $\pm\pi$ are identified, so the strip is a cylinder. *Top*: all 13000 joints, carrying three horizontal scales: the Farey fractions $f = p/q$ with $q \leq 7$ above, each at its caustic joint (215) and staggered in two rows where they crowd; the normalized joint fraction $u = (n-1)/[T]$ below, so the bisector joint sits at $u = 1$; and the joint index n below that. The red circle at the right edge is the bisector joint $n = [T] + 1$, where $\theta = -\pi$ exactly. *Bottom left*: a window at $f = \frac{2}{5}$, showing its $p = 2$ parabolic strands as dots, each with its fitted arc (219) drawn through them; the two arcs sit half a turn apart, so strand 1 has its vertex wrapped to the opposite edge. *Bottom right*: the last tenth of the strip, where the strands steepen and run into that bisector joint on the right in the figure. Dashed lines carry each window back to the stretch of the joint- n scale it magnifies, marked there by a bracket. Generated by `fig_joint_angles.py`.

The fitted curve. The strands are not merely parabola-like; they can be written down. Let $\tilde{\theta}(n) = -I(T) \log \frac{n}{n-1}$ be the unwrapped angle, so θ_n is $\tilde{\theta}(n)$ reduced to $[-\pi, \pi]$, and differentiate it three times:

$$\tilde{\theta}'(n) = \frac{I(T)}{n(n-1)}, \quad \tilde{\theta}''(n) = -\frac{I(T)(2n-1)}{[n(n-1)]^2}, \quad \tilde{\theta}'''(n) = \frac{2I(T)(3n^2-3n+1)}{[n(n-1)]^3}. \quad (216)$$

At a caustic these have closed forms in the fraction alone, since (215) fixes $n_f(n_f - 1) = f I(T)/2\pi$ and $2n_f - 1 = \sqrt{1 + 2f I(T)/\pi}$ exactly:

$$\begin{aligned}\tilde{\theta}'(n_f) &= \frac{2\pi}{f} = \frac{2\pi q}{p}, & \tilde{\theta}''(n_f) &= -\frac{4\pi^2}{f^2 I(T)} \sqrt{1 + \frac{2f I(T)}{\pi}} \approx -\frac{4\pi}{f^{3/2} T}, \\ \tilde{\theta}'''(n_f) &= \frac{16\pi^3}{f^3 I(T)^2} \left(1 + \frac{3f I(T)}{2\pi}\right) \approx \frac{12\pi}{f^2 T^2}.\end{aligned}\tag{217}$$

Write $\delta = n - n_f$ for the offset in joints from the caustic. The linear term $\tilde{\theta}'(n_f) \delta = (2\pi q/p) \delta$ is a carrier sweeping several radians per joint, and it is what interleaves the dots into strands in the first place. Along one strand, though, δ advances by p at a time, and $(2\pi q/p) pm = 2\pi q m$ drops out: the carrier contributes nothing but a constant. Absorbing it together with $\tilde{\theta}(n_f)$, strand j carries the phase constant

$$C_j = \tilde{\theta}(n_f) + \frac{2\pi q}{p} ([n_f] + j - n_f), \quad j = 0, \dots, p-1,\tag{218}$$

with $[\cdot]$ the nearest integer, and the strand itself is that cubic reduced to $[-\pi, \pi]$

$$\rho_j(\delta) = C_j + \frac{1}{2} \tilde{\theta}''(n_f) \delta^2 + \frac{1}{6} \tilde{\theta}'''(n_f) \delta^3 \quad (\text{reduced to } [-\pi, \pi]).\tag{219}$$

Because $\gcd(p, q) = 1$ the constants C_j run through all p residues of $2\pi/p$, so the p arcs are one parabola stacked at equal spacing around the cylinder. Third order is the right order: the second derivative supplies the parabola, the third its visible skew, and the fourth is smaller by a further $\delta/n_f \sim T^{-1/2}$.

The arcs drawn in Figure 62 are (219) at $f = \frac{2}{5}$ and $T = 13000$, where $n_f = 8222.74$, $\tilde{\theta}''(n_f) = -3.8208 \times 10^{-3}$ and $\tilde{\theta}'''(n_f) = 1.3941 \times 10^{-6}$ radians per joint squared and cubed, and the carrier is $2\pi q/p = 5\pi \equiv \pi$, which is why the two strands sit half a turn apart. Over the ± 30 joints shown they track the dots to 2.2×10^{-5} radians, that being the fourth-order term. Nothing is fitted in the statistical sense, and (219) is evaluated exactly. Two limits are worth noting. Toward small f the parabola stiffens like $f^{-3/2}$, which is why the left end of the strip looks like noise at this scale: consecutive joints there are many turns apart and only the caustics of very small f remain resolved. At the other end $f = 1$ closes the family, with $p = q = 1$: a single strand, a carrier of one full turn per joint, and $C_0 = -\pi$. That is the bisector joint of (98) again, so the bisector joint this paper is built on is the last caustic of the sequence, and the right-hand panel of Figure 62 is the run of the high-denominator caustics into it.

14.5 Incremental change

The strip is a snapshot. Advance the index and every dot moves, though not by the same amount. Differentiating the unwrapped angle in T at a fixed joint,

$$\begin{aligned}\frac{\partial \tilde{\theta}}{\partial T}(n) &= -I'(T) \log \frac{n}{n-1}, & L &= \log \left(1 + \frac{1}{T}\right), \\ I'(T) &= \frac{\pi}{L^2} \left(2L + \frac{2T+1}{T(T+1)}\right) = 4\pi T + 2\pi + O(T^{-1}),\end{aligned}\tag{220}$$

which in cycles is a drift rate

$$\mu(n) = \frac{I'(T)}{2\pi} \log \frac{n}{n-1} \approx \frac{2T+1}{n-\frac{1}{2}} \approx \frac{2}{u}, \quad u = \frac{n-1}{[T]}.\tag{221}$$

A value of μ is easiest to read as a lap count: $\mu(n) = 3$ means that dot laps the strip three times, falling from $+\pi$ to $-\pi$, wrapping, and repeating, while you advance T by one. Nothing here needs differencing: $I'(T)$ is available in closed form as $\pi(2L + (2T + 1)/T(T + 1))/L^2$ with $L = \log(1 + 1/T)$, which is $4\pi T + 2\pi$ to leading order, about 163,369 at $T = 13000$.

The sign is the same everywhere, so the whole strip drifts downward; the magnitude is not, and that is the point. Since $\mu(n) \approx (2T + 1)/(n - \frac{1}{2}) \approx 2/u$, the rate is essentially a $1/u$ hyperbola, equal to exactly 2 at the bisector joint $u = 1$ and growing leftward to 1.8×10^4 cycles per unit index at $n = 2$. Incidentally, the $2/u$ form is no longer any good there, reading 26,000, since $\log \frac{n}{n-1}$ and $1/(n - \frac{1}{2})$ part company at the first few joints. The strip does not translate as the index advances, it shears. Read the other way, $1/\mu(n)$ is the index step that returns joint n to the same angle: a dot at $u = \frac{1}{2}$ comes back around every $\Delta T = 0.25$, one at $u = 0.1$ every $\Delta T = 0.05$, and one at the bisector joint every half unit.

At the right edge the value is exact and already familiar: the middle expression of (221) at $n = \lfloor T \rfloor + 1$ is $(2T + 1)/(T + \frac{1}{2}) = 2$, and the exact rate is 2.000000001 at $T = 13000$. Two cycles per unit index is one more than (98) reports for the joint angle, and the discrepancy is the joint itself moving: the bisector joint sits at $n = \lfloor T \rfloor + 1$, which advances one joint per unit index, and one joint there is worth $\nu = 1$ turn, so an observer riding the bisector joint sees $\mu - \nu = 1$ cycle, namely $\frac{d}{dT}\pi(2T + 1) = 2\pi$. The two statements differ only in whether one rides the bisector joint or stands at a fixed joint.

The pattern the dots drift through, by contrast, hardly moves at all. Differentiating (215) gives $dn_f/dT = f I'(T)/2\pi(2n_f - 1) \approx \sqrt{f}$, so a caustic creeps outward by a fraction of a joint per unit index, and its normalized position $u_f = (n_f - 1)/\lfloor T \rfloor \approx \sqrt{f}$ does not depend on T at all: the spines stand still while the joints stream through them. Their drift rate follows from pairing (221) with the cycles per joint $\nu = I(T)/2\pi n(n - 1)$ of (214), which gives $\mu \approx 2\sqrt{\nu}$: at the caustic of Farey fraction f , where $\nu = 1/f$, the dots slide at $2/\sqrt{f}$ cycles per unit index, four at $f = \frac{1}{4}$ and two at $f = 1$. Only at an integer index does the frame change, when a new link is appended, a new joint appears at the right edge with $\theta = -\pi$, and u rescales by one part in $\lfloor T \rfloor$.

Figure 63 shows one step, $\Delta T = 0.02$, which advances t by $\Delta I = 3267.38$. The travel is $0.04/u$ cycles: 14.4° at the bisector joint, 28.8° at $u = \frac{1}{2}$, 143.9° at $u = 0.1$, half a turn at $u = 0.0800$, joint $n = 1041$, and a full turn at $u = 0.0400$, joint $n = 521$. Since the motion is downward everywhere, the arrows in the figure are drawn downward and wrapped at the lower edge, which is where they land for joints left of $n = 1041$. Left of $n = 521$ the joint has gone round at least once, and rather than draw what is left over the arrow there sweeps the whole strip from π to $-\pi$; at $n = 2$ the travel is 2264 radians, some 360 turns. This is the sense in which the left end of the strip is a chirp in the index as well as in the joint number: it is not only that neighboring joints are many turns apart at fixed T , but that one joint sweeps many turns over a step in T too small to move the bisector joint by a quarter of a radian. The bottom panel of Figure 63 is μ on a logarithmic axis, and the middle panel is a picture of the same quantity at $\Delta T = 0.02$.

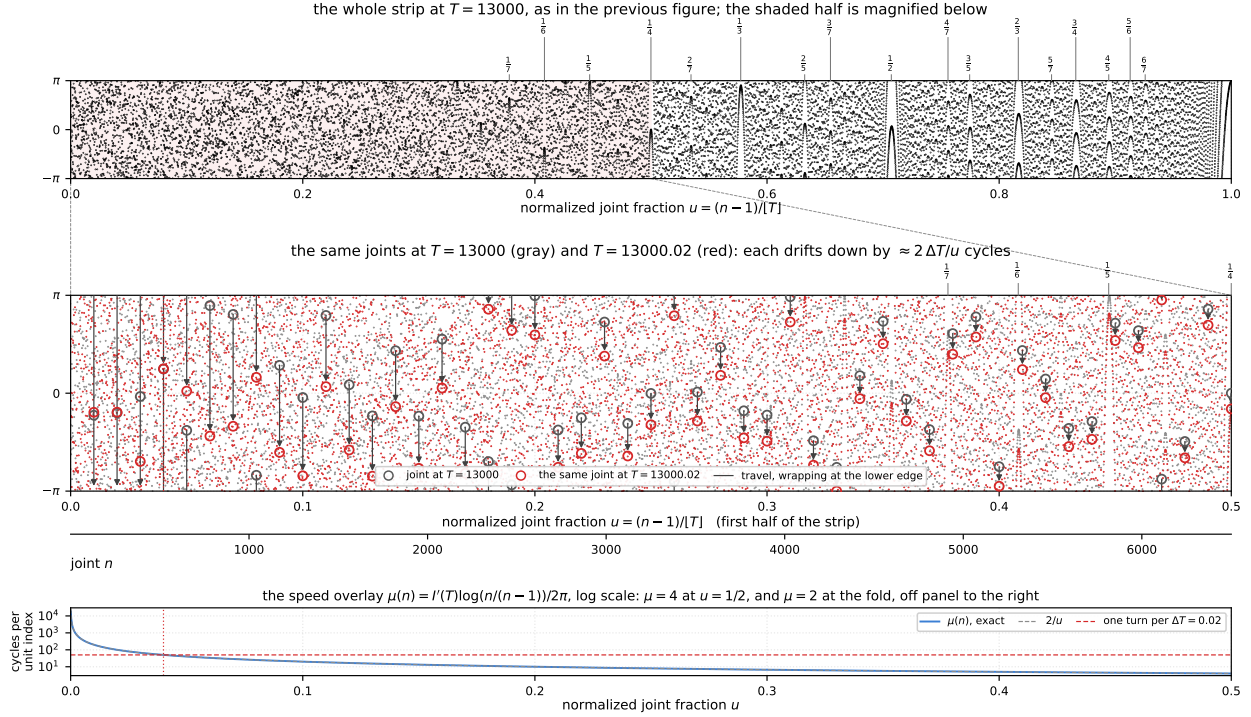


Figure 63: One increment of the index, at $T = 13000$. *Top*: the whole joint-angle strip as in Figure 62, with the shaded first half magnified below. *Middle*: that half twice over, gray at $T = 13000$ and red at $T = 13000.02$, every dot having drifted downward by $\mu(n) \Delta T \approx 0.04/u$ cycles of (221). Fifty joints spaced evenly across the panel are circled in both colors, with an arrow along the path traveled; since the drift is downward at every joint the arrows run downward and wrap at the lower edge, and where the travel exceeds a full turn the arrow sweeps the whole strip, from π to $-\pi$. The travel lengthens leftward like $1/u$, passing half a turn at $n = 1041$ and a full turn at $n = 521$, marked by the dotted vertical line. Both strips carry the Farey fractions $f = p/q$, $q \leq 7$, each at its caustic joint (215); only $f \leq \frac{1}{4}$ falls in the magnified half, since $u_f \approx \sqrt{f}$. *Bottom*: the drift rate itself on a logarithmic axis: exact (blue) against the $2/u$ form (gray), the two separating only at the first few joints, with the dashed line at one full turn per $\Delta T = 0.02$. Generated by `fig_incremental_change.py`.

15 Prior literature

15.1 Nickel's Argand-diagram geometry

Nickel [16], with a condensed sequel [17], starts from the same object this paper starts from, the Argand diagram of the steps n^{-s} , and reaches a construction close enough to ours that the two are worth setting side by side. They agree in shape, and they differ in the one respect this paper is about.

Dictionary. His center of symmetry is the step $n_p = \lfloor \sqrt{t/2\pi} \rfloor$, with fractional part $p = \sqrt{t/2\pi} - n_p$. Since $\sqrt{t/2\pi} \approx T + \frac{1}{2}$ (§10.1), his index is $n_p \approx \lfloor T + \frac{1}{2} \rfloor$ and $p \approx \{T + \frac{1}{2}\}$, so it agrees with our $m = \lfloor T \rfloor$ on the first half of each unit interval and exceeds it by one on the second half: the same half-unit offset already recorded for Siegel's cutoff in §7.1. His factor $Q(s)$, defined

by $\zeta(s) = Q(s)\zeta(1-s)$, is our $\chi(s)$; the displayed form $Q = n_p^{1-2s}e^{i(t+\pi/4)}$ is the familiar $e^{-2i\vartheta(t)}$ asymptotic, which reproduces χ to a relative 1.5×10^{-5} at $T = 20.3$ and 1.6×10^{-7} at $T = 200.3$ once the smooth $\sqrt{t/2\pi}$ is used in place of the rounded n_p . His symmetry angle Θ is our $\frac{\psi}{2} = \frac{1}{2} \arg \chi$.

What matches. Under that dictionary his Riemann–Siegel equation $\zeta = P(s) + Q(s)P(1-s)$ is our two-leg decomposition $\zeta = B_1 + B_2$ with $B_2 = \chi \overline{B_1}$ (Appendix B); his polar form $\zeta = 2 \cos(\phi - \Theta) P e^{i\Theta}$ is the shape of our $R = 2d_1 \cos(\omega + \frac{\psi}{2}) e^{i\psi/2}$; and his observation that the two magnitudes are equal exactly when $\sigma = \frac{1}{2}$ is Corollary 9.1. He also writes the classical remainder as $R = 2L \cos(\theta_L - \Theta)$, where $L = L e^{i\theta_L}$ runs from the joint at step n_p to his center: that is exactly the role played here by the pair $(d_1, -\omega)$. Where his index agrees with ours the two projected remainders converge, the ratio $2L \cos(\theta_L - \Theta) / 2d_1 \cos(\omega + \frac{\psi}{2})$ falling through 2.24, 1.34, 1.11, 1.03, 1.01 at $T = 6.3, 20.3, 60.3, 200.3, 600.3$.

Two centers, one line. His $P(s)$ and our B_1 are not the same point, and they are not meant to be. The difference lies almost entirely along the direction in which ζ cannot see it: at $T = 600.3$ we find $|P - B_1| = 0.064$, of which only 1.2×10^{-4} lies along $e^{i\psi/2}$. That is his own remark that ζ is unchanged when $P(s)$ is translated along the symmetry axis. In the notation of §9.2 the admissible centers are the solutions X of $\zeta - X = \chi \overline{X}$, a line in the plane, and every one of them projects onto $\zeta/2$, which is Corollary 9.2. The freedom he notes and the projection we prove are the same fact.

The sequel. Two things in [17] touch this paper further. It traces $P(s)$ and $\zeta(s)$ as two curves over a range of t , joined at regularly spaced values of t to show the two segments, which is the figure the conveyor belt draws with his pendant center in place of B_1 . And it counts zeros: it reads the count off Θ through the Gram points and tests that reading against the first 100,000 ordinates, finding one zero per Gram point on average with the zeros grouping between successive Gram points. The count there is the classical one, exact in the mean and never landing on an ordinate; the curves of §13 land on every one.

What differs. The correction itself. Nickel’s is a lowest-order determination, carrying a factor $1/(2n_p^\sigma \cos 2\pi p)$ that grows without bound as $\cos 2\pi p \rightarrow 0$, and he is explicit that its error matters for the distribution of $P(s)$ even though it cancels in ζ ; its direction, $e^{-i(t \ln n_p + 2\pi p)}$, is the step direction at n_p turned by the fractional offset $2\pi p$, not a summand direction of the chain. Our R_{1ps} and R_{2ps} are exact, and their directions are exactly those of the $(m+1)$ st summand of each chain. That is what lets the serial chain of §6.2 close on ζ identically rather than asymptotically, and it is the content of the “one more partial summand” reading: the correction is not merely near the next link, it lies along it.

15.2 Levinson’s G

The companion $B_1^* = \zeta + \zeta'/2\vartheta'$ of (190), whose argument carries the counting curve N_* of §13.2, is the function Levinson [14] calls G . That is the paper in which he proves that more than a third of the zeros of ζ are on $\sigma = \frac{1}{2}$. Figure 64 draws the function itself on the critical line, in the plane and in the frame where its winding is the count.

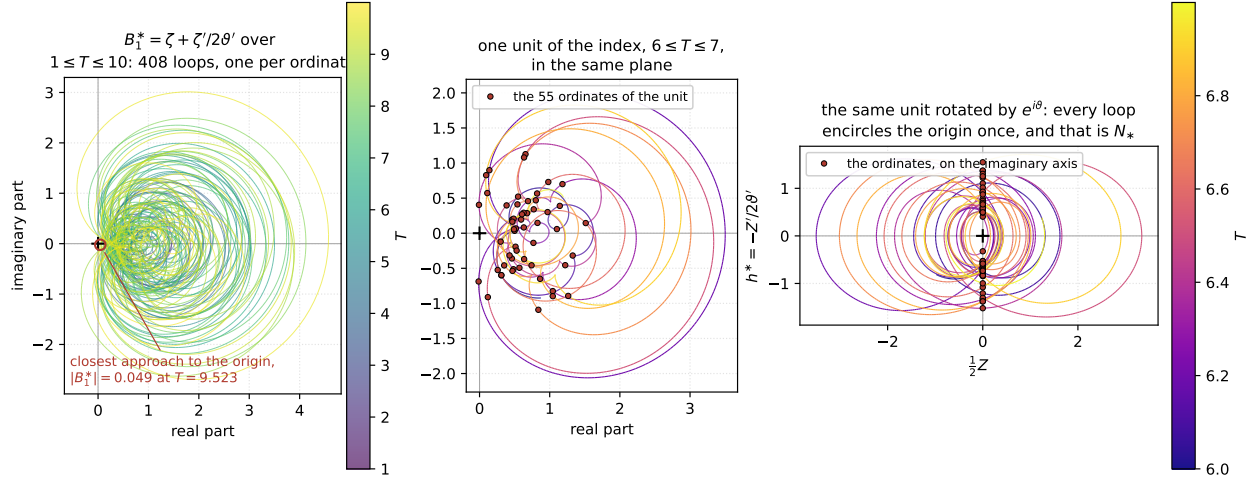


Figure 64: Levinson's G , which is the companion $B_1^* = \zeta + \zeta'/2\vartheta'$ of (190), drawn as a path in the plane on the critical line. *Left:* the whole of $1 \leq T \leq 10$, that is $13.6 \leq t \leq 692.2$, the color running with T : 408 loops, one for each ordinate of the range. The path never reaches the origin, which is what lets $\arg G$ carry a count, and its closest approach, $|G| = 0.048$ at $T = 9.523$, falls at the tightest pair of ordinates in the range, γ_{363} and γ_{364} , whose gap is 0.24 of the mean spacing. There $Z' = 0$ and $|G| = \frac{1}{2}|Z|$: it is a Lehmer-style near-miss, not a small value of ζ on its own, that brings the curve in. *Middle:* one unit of the index, $6 \leq T \leq 7$, in the same plane, with its 55 ordinates marked. *Right:* the same unit rotated by $e^{i\vartheta}$, the frame of (192) in which the real part is $\frac{1}{2}Z$ and the imaginary part is the offset h^* . Every loop encircles the origin once and crosses the imaginary axis exactly at an ordinate. Counting those crossings is Levinson's reading of the angle; following the turn continuously is N_* . Computed from $e^{i\vartheta}G = \frac{1}{2}Z - iZ'/2\vartheta'$, which agrees with $\zeta + \zeta'/2\vartheta'$ to 18 digits. Generated by `fig_b1_star_path.py`.

Dictionary. He writes the gamma factor as $h(s) = \pi^{-s/2}\Gamma(s/2) = e^{f(s)}$, so the functional equation reads $h(s)\zeta(s) = h(1-s)\zeta(1-s)$. Differentiating it and using it again to remove $\zeta(1-s)$ leaves him with the argument of

$$G(s) = \zeta(s) + \frac{\zeta'(s)}{f'(s) + f'(1-s)} \quad (222)$$

to control, which is his (1.9). His weight is ours, in two steps. From $f(s) = -\frac{s}{2} \log \pi + \log \Gamma(s/2)$,

$$f'(s) = -\frac{1}{2} \log \pi + \frac{1}{2} \frac{\Gamma'}{\Gamma} \left(\frac{s}{2} \right),$$

and $\chi = h(1-s)/h(s)$ has $\log \chi = f(1-s) - f(s)$, so differentiating in s gives

$$\frac{\chi'}{\chi}(s) = -f'(1-s) - f'(s), \quad \text{that is} \quad f'(s) + f'(1-s) = -\frac{\chi'}{\chi}(s), \quad (223)$$

at every s . That is the weight of the off-line form (196), so $G = B_1^*$ in the whole plane. On the critical line the same weight turns real: with $s = \frac{1}{2} + it$ we have $1-s = \bar{s}$ and $f'(1-s) = \overline{f'(s)}$, so the sum is twice a real part,

$$f'(s) + f'(1-s) = 2 \operatorname{Re} f' \left(\frac{1}{2} + it \right) = \operatorname{Re} \frac{\Gamma'}{\Gamma} \left(\frac{1}{4} + \frac{it}{2} \right) - \log \pi = 2\vartheta'(t),$$

the last equality being the exact ϑ' recorded above (190). Substituting that into (222) returns $\zeta + \zeta'/2\vartheta'$, which is (190). So the two are the same function and not two versions of one: at $t = 20$, to take one height, the two weights are both 1.157750994844834 and the two companions agree to the forty digits carried. Levinson himself trades the weight for $\log(t/2\pi)$ at his (1.4), which costs $O(1/t)$; we keep the exact $2\vartheta'$. In the language of §9.2, G is a leg: the first leg of the split whose second leg $\zeta - G$ is the velocity of the endpoint of the chain along the line, quarter-turned and halved by the local rate of the phase.

The same angle. His (1.8) locates the ordinates by the condition $\arg(h(s)G(s)) \equiv \frac{\pi}{2} \pmod{\pi}$, and on the critical line $\arg h$ is exactly the Riemann–Siegel phase, $\arg h(\frac{1}{2} + it) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi = \vartheta(t)$. So the angle he watches is the one (193) is built from,

$$\arg(hG) = \vartheta + \arg B_1^* = \pi(N_* - \frac{3}{2}) \pmod{2\pi},$$

and his condition on it is precisely the statement that N_* is an integer. The two readings of the same angle differ in what is asked of it. Levinson needs only how often the condition holds, so what he must control is how far $\arg G$ can move: he counts zeros of G in a rectangle with the critical line for its left edge and $\sigma = 3$ for its right, where $|G - 1| < \frac{1}{3}$ pins the argument down, and a mollifier and Littlewood’s lemma bound the count in between. His remark that ζ would have essentially all of its zeros on the line if $\arg G$ never moved at all is the extreme case of the same reading. Here the angle is wanted between the ordinates as well as at them. The exact weight gives $\operatorname{Re}(e^{i\vartheta}G) = \frac{1}{2}Z$ by (192), so the angle advances by exactly π from one ordinate to the next, and N_* is a continuous curve, integer-valued at the ordinates and nowhere else, carrying in its slope the size of $|Z|$ (Remark 13.5). One further consequence of (222) is common to both readings, the corollary Levinson credits to Montgomery: a zero of ζ of multiplicity m leaves G a zero of multiplicity $m - 1$.

15.3 Spirals of exponential sums

For quadratic phases this geometry is rigorous ground. The partial sums $S_N(\tau) = \sum_{n \leq N} e^{i\pi n^2 \tau}$ draw what Berry and Goldberg [2] call curlicues: replacing the sum by an integral produces the Cornu spiral exactly, and a renormalization then carries the whole pattern onto a magnified and rotated copy of a shorter sum of the same kind. Coutsiias and Kazarinoff [4, 5] work that picture the way this paper works ours. They give the polyline a discrete radius of curvature

$$R_N = \frac{1}{2} |\csc(\psi_N/2)|, \quad \psi_N = \pi\tau(2N + 1), \quad (224)$$

ψ_N being the turn between consecutive terms, so that inflections of the pattern sit where ψ_N is nearest a multiple of 2π and cusps where it turns around. They then cut the sum *at an inflection*, and replace each whole spiral of the first sum by a single vector carrying a scale and a phase, the Fresnel length $\sqrt{|\tau|}$ at $\pi/4$ to the spiral’s mid-vector. What comes out is Hardy and Littlewood’s approximate functional formula for the theta function, sharpened to an error uniform in τ and stated as a Diophantine inequality that contains Gauss’s sum formula. Cutting at the bisector and standing one link in for a whole spiral is the reading of §4; their $e^{i\pi/4}$ is the Gaussian integral’s, the same factor that leaves the $-\pi/8$ in ϑ ; and (224) is the bisector construction of §9.1 for unit links, the distance from a step to the meeting of the angle bisectors at its two ends, which is Nickel’s L once the length $n^{-\sigma}$ is restored.

The arithmetic is what does not carry over. Their turn ψ_N is linear in N , so the second difference is the constant $2\pi\tau$, the arcs are Cornu spirals outright, and the theory turns on the continued fraction of τ : the pattern is a self-similar hierarchy of curlicues within curlicues. Our

turn is $\theta_n = -t \log(1 + \frac{1}{n})$ and its second difference falls like t/n^2 , passing 2π once. That is why the chain has a single distinguished inflection, at $n \approx \sqrt{t/2\pi}$, instead of a hierarchy of them, and why its arcs are Cornu spirals only locally. The geometric machinery transfers; the renormalization does not.

One further reading of the same diagram deserves mention. Kapitonets [11] draws his axis of symmetry through the *midpoint* of the remainder vector at $\sigma = \frac{1}{2}$, observes that the remainder is then perpendicular to that axis, and concludes both that $\arg \zeta$ is the normal direction and that the two chains project onto the axis with canceling sums. That is the half-split of §7.2 together with Corollary 9.2, read off the picture. His other construction, recovering ζ by repeatedly replacing a polygon of partial sums by the polygon of their midpoints, is unrelated to the bisector point: iterated midpointing is the binomial transform, and the limit is Cesàro summation of the Dirichlet series, as he says himself.

16 Statements left unproved

The body of the paper mixes identities that are proved, identities that are written down and then checked, and readings that are only measurements. This section collects the ones we would like to prove and have not.

Possible, but longer

These need a real argument, not just a census.

1. Every forward link is crossed by a reverse link, and the partner is the summand $n' = a^2/n$ of (137) (§12.1). The product law is motivated by a stationary-phase sweep and then checked on 118,007 samples. Proposition 12.1 settles the direction: the named partner is never parallel to the link and never more than a quarter turn from antiparallel, because nn' is an integer. What is missing is the position. A theorem would have to show that the two polylines meet, not only that the named integer usually works, and that needs the sweep (145) with an effective error term rather than as a stationary-phase heuristic.
2. The pairing is an involution: $C_s(C_s(n, T), T) = n$. Claimed as the symmetry of (137); if the previous item is proved, this is immediate.
3. A retrograde stretch of ϑ_2 is an equal-leg oval, one to one (§13.3). The correspondence is exact on $5.9 \leq T \leq 6.7$, and it survives a much wider scan: over twelve windows spread from $T = 1$ to $T = 50$, holding 344 ordinates between them, the 45 retrograde stretches, the 45 off-line ovals and the 45 retrograde ordinates match one for one, with the stretch and the oval agreeing end to end to between 10^{-5} and 10^{-3} of T (`check_retro_ovals.py`). The only other off-line features met are the pole bands at $\{T\} \approx \frac{1}{4}, \frac{3}{4}$, which cross at every σ instead of closing. The geometric story is written (h small, the legs nearly folded, ϑ_2 stalls). Making that story a local lemma is one thing; the global one-to-one is still a scan.
4. Off the critical line, collinearity of R_{1ps} , R_{1rs} , and R_{1ak} fails except on isolated curves, and a hypothetical off-line zero can be connected to that structure (§14.3). The on-line identity is Proposition 14.6; these are the two steps that remain if collinearity is to say anything about zeros.

Observational

These are measurements. We do not have a mechanism that would turn them into theorems, and we do not claim one.

1. Five off-line collinearity bands per unit of T , two of them tracking the poles of d_1 (§14.3). The pattern is read from the heatmaps.
2. Scoring large values of $|Z|$ from the first few links, the rank correlation of the slope ρ of N_* against $|Z|$, and the grids on which N_* stays increasing (Remark 14.5, Remark 13.5, Remark 13.4).

17 Glossary

The paper uses a geometric language for objects that other writers have named differently, or not named at all. This section collects those words in one place. Each entry is a definition in ordinary language, a short formula when the formula is short, a pointer to the section that introduces the object, and, where we know one, the name another author has used for the same thing or a neighboring one. The names are those already used above; this list is only a map.

The list is not alphabetical. It runs from the parameters and the polyline itself to the more specialized constructions built on them, and the entries in each group belong with one another.

The parameters

Index T , and $I(T)$. The real parameter that replaces Siegel’s pair (t, m) : $t = I(T) = \pi(2T + 1)/\log(1 + 1/T)$ and $m = \lfloor T \rfloor$ (§3). T is called the index even though it is not an integer. The map is chosen so that integer T lands on a handoff (§10.1).

Unit of the index T . One stretch of T between consecutive integers: the interval $(m, m + 1)$ with $m = \lfloor T \rfloor$, for example T from 6 to 7. The chains keep the same number of partial-sum links on that stretch. The ends of the unit are $\{T\}$ near 0 and near 1; the middle is $\{T\}$ around $\frac{1}{2}$ (§12.1).

$\chi(s)$. The completion factor of the functional equation, $\zeta(s) = \chi(s)\zeta(1 - s)$ (§2). Nickel writes $Q(s)$ for the same factor.

ω . The absolute direction of the next forward summand: $\omega = I(T) \log \lceil T \rceil$, so that $(m + 1)^{-s}$ points along $e^{-i\omega}$ (§4).

a, a^2 . Siegel’s cutoff $a = \sqrt{t/2\pi}$ and its square $a^2 = t/2\pi = I(T)/2\pi$. Crossing partners multiply to a^2 . Under $t = I(T)$ one has $a^2 \approx (T + \frac{1}{2})^2$.

The chain: joints and links

Polyline. The drawing: a path of straight segments laid end to end, which is how a partial sum appears in the plane once each summand is drawn as a vector from the end of the last one (§6). The word is about the picture and not about any new object. The forward and reverse spirals are the two polylines of this paper, and statements like “the two polylines meet” are statements about the drawn segments rather than about the sums.

Sum chain, or link chain. The partial-sum polyline in the plane: joints connected by links, one Dirichlet summand per link (§6). “Sum chain” reads it as a running total; “link chain” reads it as a serial chain of segments. The two names are the same object. We also call it a spiral, though only the later links wind. Nickel [16] draws it as the Argand diagram of the steps n^{-s} . Erickson [8] identifies the later coils as Cornu spirals and labels their centers C_n . Kapitonets [11] calls the polyline the Riemann spiral. Berry and Goldberg [2] call the quadratic-phase analogue a curlicue; our turn $\theta_n = -t \log(1 + 1/n)$ is not quadratic, so the coils are Cornu only locally (§15.3). The 3×3 matrices of §6.2 are the planar Denavit–Hartenberg encoding of a serial chain [7].

Joint. A vertex of a chain: the value of the partial sum after an integer number of summands. Joint n of the forward chain is $J_n = \sum_{n' \leq n} n'^{-s}$, so joint 0 is the origin and joint m is Σ_1

itself (§6, §12.1). Erickson's centers C_n stand in one-to-one correspondence with the joints, though a center is not itself a joint (§9.1; see *Spiral center*).

Link. The segment between two consecutive joints: one Dirichlet summand, drawn as a vector. Link k of the forward chain runs from joint k to joint $k+1$ and carries the summand $n = k+1$ (§6). Nickel calls the same object a step. Coutsiias and Kazarinoff [4] treat the same polyline as a discrete curve with a radius of curvature at each turn.

Spiral center. The point a coil of the chain winds around, and the clearing it leaves at the middle of that coil (§9, §12.1, Figure 19). The link nearest it is $L_N(T, S_n)$ of (76), where the turn per link $t \log(1 + 1/n)$ is the odd multiple $\pi(2S_n + 1)$: successive links are antiparallel there, so the chain sweeps back and forth across the center and the joints settle onto a ring around it of radius about half a link length, which is the lens-shaped clearing the near-diameter links envelope. The center need not lie on any link or joint, not even on the nearest link. The centers are not scattered. Numbering them by the spiral number $S_n = c$ outward from the end of the chain, center $c = 0$ is the last spiral, the largest, and sits at ζ to within that radius; each further center is one sweep back: by (145) the step from center c to center $c-1$ is the c -th summand reflected in the fixed direction Θ , of length $c^{-1/2}$. Read outward from ζ the centers therefore reproduce the head of the chain reflected, which is Nickel's center-to-center distances and angles reproducing the initial steps. In summands, center c is near $n = a^2/(c + \frac{1}{2})$, interleaving with the sweeps at $n = a^2/c$ of (144). Erickson writes C_n for these points.

Σ_1 , **the forward chain (or forward spiral).** The sum chain of the first Dirichlet sum, $\Sigma_1 = \sum_{n=1}^m n^{-s}$ (§2, §6). The chain is a polyline of summands, not a placement: it may be drawn in any frame.

Σ_2 , **the reverse chain (or reverse spiral).** The sum chain of the reflected Dirichlet sum, $\Sigma_2 = \chi(s) \sum_{n=1}^m n^{s-1}$ (§2). In the plane the reverse chain is usually drawn from ζ , running counter to the forward chain; as a free vector it can sit anywhere. On the critical line it is the functional-equation image of the forward chain: each summand of Σ_2 is χ times the conjugate of the matching summand of Σ_1 . Nickel's second half $Q(s)P(1-s)$ is the same reflection, with his Q equal to our χ (§15.1).

The remainder on the chain

R. Siegel's unsplit remainder integral, so that $\zeta = \Sigma_1 + \Sigma_2 + R$ (§2). Every named split below recovers this same R , exactly or approximately.

Split. The choice of meeting point: a writing $R = R_1 + R_2$ (or $\zeta = B_1 + B_2$) that picks where the two halves join (§7.2). The two pieces leave the same pair of joints: R_1 starts at Σ_1 , and R_2 arrives at $\Sigma_1 + R$. What a split chooses is only the point $\Sigma_1 + R_1$ between those joints, its own bisector point. The three named splits are the partial-summand split ps , the half-split rs , and the Kuznetsov / Siegel- f split ak . The rs split just halves R along itself. The ak split does not follow the summand directions.

Cone. What the ps split looks like in the plane: the two halves run along two different link directions, $R = d_1 e^{-i\omega} + d_2 e^{i(\omega+\psi)}$, with real weights (§4, (23)). Those two rays form an angle, and R sits in that angle when both weights are positive. The same picture at a general forward link, with W_k in place of R , is (166). A cone is a split; not every split is a cone.

R_{ps}, R_{1ps}, R_{2ps} . The partial-summand split of R : $R = R_{ps} = R_{1ps} + R_{2ps}$, with $R_{1ps} = d_1 e^{-i\omega}$ and $R_{2ps} = d_2 e^{i(\omega + \arg \chi)}$ (§4). Each piece is the next Dirichlet term of its sum, shortened by a real weight. The subscript ps distinguishes this exact split from the approximate remainders of other authors.

Fractional summand. A remainder that is exactly the next Dirichlet term, scaled by a real factor: $R_{1ps} = \hat{d}_1 (m+1)^{-s}$ and $R_{2ps} = \hat{d}_2 \chi (m+1)^{s-1}$ (§5). The word “fractional” means a fraction of one summand, not a summand with a non-integer index.

d_1, d_2 . The real lengths of the two fractional summands: $|R_{1ps}| = d_1$ and $|R_{2ps}| = d_2$ (§4). Each is the distance, in the plane, from the bisector joint to the bisector point, measured along its own bisector link. On the critical line, $d_1 = d_2$. The sine formulas are (19)–(20).

$\hat{d}_1, \hat{d}_2$ (**fractional amounts**). The same weights read as fractions of a link rather than as lengths: $\hat{d}_1 = [T]^\sigma d_1$ and $\hat{d}_2 = [T]^{1-\sigma} d_2 / |\chi|$ (§5). $\hat{d}_1$ is the crossing fraction of the bisector link. At the running example $\sigma = \frac{1}{2}$, $T = 6.18$, one has $d_1 \approx 0.0875$ and $\hat{d}_1 \approx 0.231$.

R_{ak}, R_{1ak}, R_{2ak} . The Kuznetsov / Siegel- f split: $R \sim R_{ak} = R_{1ak} + R_{2ak}$, where $R_{1ak} = f(s) - \Sigma_1$ (§7.1, §7.2). Approximate, and not a single fractional summand.

R_{rs}, R_{1rs}, R_{2rs} . The half-split $R_{1rs} = R_{2rs} = \frac{1}{2}R$ (§7.2). Kapitons’ midpoint of the remainder vector is this split (§15.3).

The bisector

Bisector link. Link $m = [T]$ of a chain: the link that sits at the boundary between the nearly-straight early chain and the coils (§9.1). Each chain has one. The forward bisector link runs from Σ_1 along the $(m+1)$ st summand; the reverse bisector link runs from $\Sigma_1 + R$ along its $(m+1)$ st summand. On $\sigma = \frac{1}{2}$ the bisector line cuts this link at the bisector point B_1 : that is where the forward chain meets the bisector line, at crossing fraction $\hat{d}_1$ along the link, not (except at a handoff) at the bisector joint itself. This is the neighborhood of Siegel’s cutoff $m = \lfloor \sqrt{t/2\pi} \rfloor$ (§2), of Nickel’s center-of-symmetry step $n_p = \lfloor \sqrt{t/2\pi} \rfloor$ (offset from our m by about half a unit, §15.1), and of the inflection at which Coutsiass and Kazarinoff cut their sum (§15.3).

Bisector joint. Joint m of the forward chain, the integer vertex at which the bisector link begins: the point Σ_1 . The matching joint of the reverse chain is $\Sigma_1 + R$. All three remainder splits of §7.2 leave from this pair of joints.

Bisector point. The crossing of the two bisector links, $B_1 = \Sigma_1 + R_{1ps} = \Sigma_1 + R - R_{2ps}$ (§9.1). On the critical line it is the apex of the isosceles triangle with base from the origin to ζ . Nickel’s center $P(s)$ plays the same role and is not the same point: the two differ mostly along the symmetry axis, which ζ cannot see (§15.1). Kapitons draws his axis through the midpoint of the remainder vector R ; that midpoint is the half-split R_{rs} , not B_1 (§15.3).

Bisector line. The line through the bisector point B_1 and $\zeta/2$ (§9.2, Figure 18). On $\sigma = \frac{1}{2}$ it is the perpendicular bisector of the segment from the origin to ζ , and the axis of symmetry of the isosceles triangle $O-B_1-\zeta$ whose equal sides are the two legs. It cuts the bisector link at B_1 . Off the critical line the legs are unequal, so there is no isosceles triangle and the line is no longer a symmetry axis, but we keep the same name for the line through B_1 and $\zeta/2$.

B_1, B_2 . The two half-sums that add to ζ : $B_1 = \Sigma_1 + R_{1ps}$ and $B_2 = \Sigma_2 + R_{2ps}$, so $\zeta = B_1 + B_2$ (§6). B_1 is the bisector point. On the critical line, $B_2 = \chi \overline{B_1}$. Nickel's Riemann–Siegel equation $\zeta = P(s) + Q(s)P(1-s)$ is this two-leg split with a different choice of the meeting point (§15.1).

Frames

Link frame. The coordinate frame of an arbitrary forward link k : translate joint k to the origin and divide by the link vector $(k+1)^{-s}$, so that link k becomes the unit interval $[0, 1]$ (§12.2, Figure 45).

Bisector frame. The link frame at $k = m$: it sends the forward bisector link to $[0, 1]$ on the real axis (§10.1, §11). In that frame the reverse bisector link revolves as T advances, and the crossing with the axis is the number d_1 (or the crossing fraction $\hat{d}_1$, once the link is read at unit length).

ϑ -frame. The rotation that makes ζ real on the critical line: multiply every vector by $e^{i\vartheta(t)}$, where ϑ is the Riemann–Siegel theta function, so that $Z = e^{i\vartheta}\zeta$ lies on the real axis (§13.1, Corollary 9.2). The base $O \rightarrow \zeta$ is then the real axis, and a split point reads as $\frac{1}{2}Z + ih$. This is the same rotation as $e^{-i\psi/2}$ with $\vartheta = -\psi/2$, since $\chi = e^{-2i\vartheta}$ on the line. Z is Hardy's Z -function.

Legs

Legs. The two segments that add to ζ : Leg 1 is the vector B_1 from the origin to the bisector point, and Leg 2 is the vector $B_2 = \zeta - B_1$ from the bisector point to ζ (§9.2). Their lengths are $L_1 = |B_1|$ and $L_2 = |B_2|$. Nickel's two vectors $P(s)$ and $Q(s)P(1-s)$ are legs in the same sense, meeting at his center rather than at B_1 .

ϑ_1, ϑ_2 . The two leg angles: $\vartheta_1 = \arg B_1$ is the heading of Leg 1 from the positive real axis, and $\vartheta_2 = \arg(B_2/B_1)$ is the turn from Leg 1 to Leg 2 (§9.2). Then $\zeta = L_1 e^{i\vartheta_1} + L_2 e^{i(\vartheta_1 + \vartheta_2)}$. These are not the Riemann–Siegel theta function $\vartheta(t)$ of (183), which is the phase of $\chi^{1/2}$ and counts zeros; the clash of names is unfortunate and the two are kept apart by the subscripts.

Ordinate. Always the ordinate of a zeta zero: the imaginary part γ of a zero of ζ , so $\zeta(\frac{1}{2} + i\gamma_n) = 0$ with $\gamma_1 < \gamma_2 < \dots$ (§13). On the index, ordinate γ_n sits at $T = I^{-1}(\gamma_n)$. The word is never used here in its generic sense, the y -coordinate of an arbitrary point.

Retrograde, prograde. The two directions of the fold angle ϑ_2 (§13, Figure 53). Prograde is the usual way: ϑ_2 keeps going through π at an ordinate and does not come back. Retrograde is the other way: ϑ_2 rises through π , turns around, and comes back down through it (or the reverse). The boundary between the two is where the derivative vanishes, $\vartheta'_2 = 0$. A retrograde passage is where N_{ps} miscounts; ϑ_2^* of the velocity split never retrogrades, which is why N_* counts correctly (§13.3, Figure 56).

Equal legs. The condition $L_1 = L_2$. On the critical line it holds identically (Corollary 9.1). Off the line it is a locus of ovals and thin bands in the strip (Figure 21). A zero of ζ requires equal legs *and* $\vartheta_2 = \pi$. Nickel notes that the two magnitudes are equal exactly when $\sigma = \frac{1}{2}$ (§15.1).

$\mathcal{E}_{\text{ps}}, \mathcal{E}_{\text{rs}}$. The equal-leg loci of two different splits of the remainder. Each is the set of (σ, T) at which the two legs have the same length, $|B_1| = |\zeta - B_1|$, for $B_1 = B_{1\text{ps}}$ and $B_1 = B_{1\text{rs}}$ respectively (§9.4). A zero forces equal legs for every split, so every zero lies in the intersection $\mathcal{E}_{\text{ps}} \cap \mathcal{E}_{\text{rs}}$. The whole critical line is in both. Off the line, a meeting of the two oval families would be necessary for an off-line zero (not sufficient). If they never meet off $\sigma = \frac{1}{2}$, there is no off-line zero.

Fold, folded. Two related uses, both meaning that two segments occupy the same line and point opposite ways. At an integer T the neighboring link *folds back* onto the bisector link, the joint angle is an odd multiple of π , and the bisector point hands off to the next link (§10.1, §11): that is a *handoff*, and for zeta every handoff is of this kind. Separately, a *folded-leg* point in the strip is a height where $\vartheta_2 = \pi$, so that Leg 2 folds back onto Leg 1 (Figure 21); a zero is an equal-leg point that is also folded-leg.

How the picture moves with T

Handoff. The instant, at integer T , when the bisector point moves from link $m - 1$ onto link m (§8, §10.1). For zeta the two links fold back onto one another and the point slides across the joint with no jump.

Event. Any instant at which the bisector point moves from one link to the next (§10.2). For zeta every event is a handoff by fold-back. Dirichlet L -functions have a second kind.

Folded event. An event at which the turning angle at the current joint is $\pm\pi$, so the chain reverses and the bisector point slides across the joint (§10.2). For zeta, every handoff is a folded event, once per unit of T . For $L(s, \chi_p)$ these occur at $T \equiv 0 \pmod{p-1}$.

Bisector event. An event at which the bisector *line* (the perpendicular bisector of the segment from 0 to the L -value) sweeps through a joint and carries the crossing onto the neighboring link (§10.2). Zeta has none. For $L(s, \chi_p)$ they occupy the remaining residues modulo $p - 1$.

Phantom link. A zero-length summand of $L(s, \chi_p)$, where $\chi_p(n) = 0$ (§10.2). Phantoms receive no index T ; only the nonzero steps are links.

Crossings along the chain

Crossing (the point). Where a forward link and a reverse link meet in the plane. The distinguished one is the bisector point B_1 ; every other forward link k has one as well, written P_k (§12.1, §12.4).

Crossing partner. The reverse link that crosses forward link k . The two summands are reciprocal about the Riemann–Siegel cutoff $a = \sqrt{t/2\pi}$: $nn' = a^2 = I(T)/2\pi$, with $n = k + 1$ (§12.1). This is the same saddle that Siegel’s contour is built around.

Crossing fraction. How far along a link the crossing sits, from the left joint toward the right, as a number in $[0, 1]$ (§12.3, §12.4). On forward link k it is $\hat{d}_1(k)$; on reverse link k it is $\hat{d}_2(k)$, the T of $\hat{d}_1(k, T)$ and $\hat{d}_2(k, T)$ being suppressed where it is fixed. At the bisector, $\hat{d}_1(m) = \hat{d}_1$. Away from the fold the other $\hat{d}_1(k)$ sit near $\frac{1}{2}$. This is a fraction of the straight link, not of a curved arc.

Σ_{1x}, Σ_{2x} . The two Dirichlet sums re-jointed at the crossings instead of at the integers (§12.4). Σ_{1x} walks $P_{-1} = 0, P_0, \dots, P_m = B_1$ and totals $B_1 = \Sigma_1 + R_{1ps}$; Σ_{2x} does the same from the origin along Σ_2 and totals B_2 . The subscript x is for crossing, not a dummy index. The step lengths are the crossing fractions $\hat{d}_1(k)$ and $\hat{d}_2(k)$.

Yin and yang

Yin, yang. The two ends of the reverse bisector link, watched in the bisector frame as T advances (§11). Each traces a teardrop; the pair resembles a yin-yang. Yin is the joint- m end, yang the joint- $(m+1)$ end. The chord from yin to yang crosses the real axis at the bisector point, and that crossing is how d_1 and d_2 were found (§11.2–§11.3). The same pair exists for every forward link k , once the crossing partner is known (§12.2). The names are ours.

Y_{inf} (**yinfinity**). The common limit of the yin curve as $T \rightarrow \infty$ at fixed fractional part $q = \{T\}$: $Y_{\text{inf}}(q) = 1 - \Psi(q) e^{-2\pi i(q^2 - 1/16)}$ (§11.6). We call this curve yinfinity. Yang follows the same curve half a unit behind. For $0 < T < 1$ the yin path is ζ itself (Erickson's C_0).

Ψ . The classical Riemann–Siegel correction $\Psi(x) = \cos(2\pi(x^2 - x - 1/16)) / \cos(2\pi x)$, as in Titchmarsh [22, §15.3] and Pugh [19, §5.4.1]. It enters this paper as the amplitude of Y_{inf} , not as a remainder of its own.

The far chain (links near ζ)

Conveyor belt. The stretch of the reverse chain between the last two spirals (spirals 1 and 0 of (76)), the links near ζ . The name is ours.

Last link. The link nearest the center of the last spiral: $L_N(T, 0) = \lfloor I(T)/\pi \rfloor = \lfloor t/\pi \rfloor \approx 2T^2$ (§9, (76)). It is the link closest to ζ , far past the bisector; ζ itself is not typically on it. The same formula with spiral number S_n names the link nearest the center of any spiral, $L_N(T, S_n) = \lfloor I(T)/(\pi(2S_n + 1)) \rfloor$. Consecutive links there are antiparallel.

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Statement on the use of AI. Cursor and various AIs, including Anthropic’s Claude (Opus and Fable), OpenAI’s ChatGPT (5.6 Sol), and Grok 4.5, were used in preparing this paper. The AI wrote the programs (in Python) that, when run, generated the figures; the name of the program is listed in the caption of each figure created this way, and the programs themselves are available in the Zest repository at <https://github.com/paul-stahura/zest>. AI was also used to write the $\text{\LaTeX}$ source of the paper from the author’s text and instructions. AI was also used to generate some of the code for the Zest webapp (which was based on the human-created Zest Unity app) and it was used to generate the Lean code from the author’s proofs; that code was checked both by the AI and by the author at <https://live.lean-lang.org/>. An AI also assisted in creating the literature review and the glossary. Throughout, the AI was prompted in detail by the author to complete, fill in, or explain proofs, and its output was checked, edited, and verified by the author.

A Lean formalization of $R = R_{1ps} + R_{2ps}$

This appendix reproduces the Lean file formalizing the algebraic content of Theorem 4.1. The file contains no `axiom` and no `sorry`: every step is proved, and the only axioms the final theorem depends on are Lean’s own three (`propext`, `Classical.choice`, `Quot.sound`), as reported by `#print axioms`. It was checked against Lean 4.32.2 with Mathlib at commit 905b95818e. The Lean code was written with the assistance of AI coding assistants (Anthropic’s Claude).

A.1 Correspondence with the written proof

The file follows the proof of Theorem 4.1 step for step. The definitions `d_j1` and `d_j2` are the sine weights (19)–(20), and `R_1ps` and `R_2ps` are Step 1, transcribed from (17)–(18) in the $[T]^{\mp iI(T)}$ form rather than the $e^{\mp i\omega}$ form. Two small bridges convert them: `ceil_cpow` proves $[T]^{ir} = e^{ir \log[T]}$ for $T > 0$, which with $\omega(T) = I(T) \log[T]$ (the definition `omg`) is exactly $[T]^{iI(T)} = e^{i\omega}$; and `div_norm_eq_exp_arg` proves $z/|z| = e^{i \arg z}$ for $z \neq 0$, applied to χ to give $\chi/|\chi| = e^{i\psi}$. Then `factoring_lemma` is the regrouping of Step 2, pulling out the common factor $|R|/\sin(2\omega + \psi)$, and `key_algebraic_identity` is the content of Steps 3–5, namely

$$e^{-i\omega} \sin(\omega - \phi + \psi) + e^{i(\omega+\psi)} \sin(\omega + \phi) = e^{i\phi} \sin(2\omega + \psi); \quad (225)$$

it is proved by writing each sine as $(e^{i\theta} - e^{-i\theta})/2i$ (`sin_to_exp`, itself proved from Mathlib’s $e^{i\theta} = \cos \theta + i \sin \theta$), clearing the denominator, merging the resulting products of exponentials into single exponentials, and letting `ring_nf` cancel the two copies of $e^{i(\psi-\phi)}$. The closing lines are Step 6: cancel $\sin(2\omega + \psi)$ and recognize $|R|e^{i\phi} = R$, which is Mathlib’s `Complex.norm_mul_exp_arg_mul_I`.

A.2 What is proved and what is hypothesized

The theorem `R_1ps_plus_R_2ps_eq_R` carries three hypotheses. Two are needed for the objects to exist at all: $\chi(\sigma + iI(T)) \neq 0$, without which $\chi/|\chi|$ is meaningless, and $\sin(2\omega + \psi) \neq 0$, without which d_1 and d_2 of (19)–(20) are undefined. The third, $T > 0$, makes $[T]$ a positive base, so that its complex power can be written as an exponential. Nothing is assumed about whether T is an integer.

One convention has to be read carefully. Steps 2–6, that is `factoring_lemma` and `key_algebraic_identity`, hold for arbitrary real ω, ϕ, ψ and make no reference to T ; the only T -dependent ingredient is the bridge `ceil_cpow`, which holds for every $T > 0$ with ω defined as $I(T) \log[T]$. That is the paper’s ω whenever T is not an integer, since then $[T] = \lfloor T + 1 \rfloor$. At integer T the two part company, and the sign bookkeeping in the last paragraph of the proof of Theorem 4.1 turns on the factor $(-1)^{[T]}$ sitting inside Siegel’s R ; that factor is invisible here, because R enters as a free complex parameter, so what the file proves at integer T is the same identity read with $[T]$ throughout.

What remains outside the formalization is not algebra but analysis. The functional-equation factor χ , the index map I , and Siegel’s remainder R enter as free parameters, so what is machine-checked is that any R written in polar form splits as $R_{1ps} + R_{2ps}$ with the weights (19)–(20), which is exactly Steps 1–6. That R is Siegel’s remainder, and that I is the map of §10.1, are inputs to the theorem rather than consequences of it.

A.3 Source listing

```
1 /-
2   Formalization of Theorem (R = R_1ps + R_2ps) from
3   "Remainder Terms of the Riemann Zeta Function".
```



```

4
5   Contains no 'axiom' and no 'sorry': every step is proved.
6   -/
7   import Mathlib.Analysis.SpecialFunctions.Complex.Circle
8   import Mathlib.Analysis.SpecialFunctions.Pow.Complex
9
10  open Complex Real
11
12  /-! ### Phases -/
13
14  /-- The phase of Siegel's remainder, ' $\varphi = \arg R$ '. -/
15  noncomputable def phiR (R : ℂ) : ℝ := Complex.arg R
16
17  /-- The phase of the functional-equation factor, ' $\psi = \arg \chi(\sigma + i I(T))$ '. -/
18  noncomputable def psi (χ : ℂ → ℂ) (σ T : ℝ) (If : ℝ → ℝ) : ℝ :=
19    Complex.arg (χ (σ + If T * I))
20
21  /-- The turn angle ' $\omega(T) = I(T) \cdot \log[T]$ ', so that ' $[T]^{i I(T)} = e^{i\omega}$ '. -/
22  noncomputable def omg (T : ℝ) (If : ℝ → ℝ) : ℝ := If T * Real.log ([T] : ℝ)
23
24  /-! ### The two fractional link lengths -/
25
26  noncomputable def d_j1 (R : ℂ) (w f p : ℝ) : ℂ :=
27    (([R] * (Real.sin (w - f + p) / Real.sin (2 * w + p)) : ℝ) : ℂ)
28
29  noncomputable def d_j2 (R : ℂ) (w f p : ℝ) : ℂ :=
30    (([R] * (Real.sin (w + f) / Real.sin (2 * w + p)) : ℝ) : ℂ)
31
32  /-! ### The two partial-summand remainders, exactly as in the paper -/
33
34  noncomputable def R_1ps (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ) : ℂ :=
35    d_j1 R (omg T If) (phiR R) (psi χ σ T If) * (([T] : ℝ) : ℂ) ^ (-(I * (If T : ℂ)))
36
37  noncomputable def R_2ps (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ) : ℂ :=
38    d_j2 R (omg T If) (phiR R) (psi χ σ T If) * (([T] : ℝ) : ℂ) ^ (I * (If T : ℂ))
39    * (χ (σ + If T * I) / ([χ (σ + If T * I)] : ℂ))
40
41  /-! ### Step 0: elementary bridges -/
42
43  /-- ' $[T]^{i r} = e^{i r \log[T]}$ ' for ' $T > 0$ ', with no hypothesis on the
44  integrality of ' $T$ '. -/
45  lemma ceil_cpow (T r : ℝ) (hT : 0 < T) :
46    ((([T] : ℝ)) : ℂ) ^ (I * (r : ℂ))
47    = Complex.exp (I * ((r * Real.log ([T] : ℝ) : ℝ) : ℂ)) := by
48    have hpos : (0 : ℝ) < ([T] : ℝ) := lt_of_lt_of_le hT (Int.le_ceil T)
49    have hne : ((([T] : ℝ)) : ℂ) ≠ 0 := by
50      simp using (ne_of_gt hpos)
51    rw [Complex.cpow_def_of_ne_zero hne, ← Complex.ofReal_log hpos.le]
52    congr 1
53    push_cast
54    ring
55
56  /-- ' $z/[z] = e^{i \arg z}$ ' for ' $z \neq 0$ '. -/
57  lemma div_norm_eq_exp_arg (z : ℂ) (hz : z ≠ 0) :
58    z / ([z] : ℂ) = Complex.exp (I * (Complex.arg z : ℂ)) := by
59    have hn : (([z] : ℝ) : ℂ) ≠ 0 := by
60      simp using norm_ne_zero_iff.mpr hz
61    rw [div_eq_iff hn, mul_comm I ((Complex.arg z : ℝ) : ℂ),
62      mul_comm (Complex.exp (((Complex.arg z : ℝ) : ℂ) * I)) ([z] : ℝ) : ℂ)]

```

```

63   exact (Complex.norm_mul_exp_arg_mul_I z).symm
64
65 /-- Exponential form of the sine of a real argument, valued in 'C'. -/
66 lemma sin_to_exp (θ : ℝ) :
67   (Real.sin θ : ℂ)
68   = (Complex.exp (I * (θ : ℂ)) - Complex.exp (-I * (θ : ℂ))) / (2 * I) := by
69   have h1 : Complex.exp (I * (θ : ℂ)) = Complex.cos (θ : ℂ) + Complex.sin (θ : ℂ) * I := by
70     rw [mul_comm]; exact Complex.exp_mul_I _
71   have h2 : Complex.exp (-I * (θ : ℂ)) = Complex.cos (θ : ℂ) - Complex.sin (θ : ℂ) * I := by
72     have h : (-I * (θ : ℂ)) = (-θ : ℂ) * I := by ring
73     rw [h, Complex.exp_mul_I, Complex.cos_neg, Complex.sin_neg]; ring
74   rw [h1, h2, Complex.ofReal_sin]
75   have hI : (2 * I : ℂ) ≠ 0 := by simp [Complex.I_ne_zero]
76   field_simp
77   ring
78
79 /-! ### Steps 3-5: the key algebraic identity -/
80
81 /-- The heart of the theorem: the two sine-weighted phasors combine into a
82 single phasor along 'e^{iφ}'. -/
83 theorem key_algebraic_identity (w f p : ℝ) :
84   Complex.exp (-I * (w : ℂ)) * (Real.sin (w - f + p) : ℂ) +
85   Complex.exp (I * ((w + p : ℝ) : ℂ)) * (Real.sin (w + f) : ℂ) =
86   Complex.exp (I * (f : ℂ)) * (Real.sin (2 * w + p) : ℂ) := by
87   rw [sin_to_exp, sin_to_exp, sin_to_exp]
88   have hI : (2 * I : ℂ) ≠ 0 := by simp [Complex.I_ne_zero]
89   field_simp
90   simp only [mul_sub, ← Complex.exp_add]
91   push_cast
92   ring_nf
93
94 /-! ### Step 2: factoring -/
95
96 /-- Pulling the common factor '|R|/sin(2ω+ψ)' out front. -/
97 lemma factoring_lemma (absR w f p : ℝ) :
98   ((absR * (Real.sin (w - f + p) / Real.sin (2 * w + p)) : ℝ) : ℂ) *
99   Complex.exp (-I * (w : ℂ)) +
100   ((absR * (Real.sin (w + f) / Real.sin (2 * w + p)) : ℝ) : ℂ) *
101   Complex.exp (I * (w : ℂ)) * Complex.exp (I * (p : ℂ)) =
102   ((absR : ℝ) : ℂ) / (Real.sin (2 * w + p) : ℂ) *
103   (Complex.exp (-I * (w : ℂ)) * (Real.sin (w - f + p) : ℂ) +
104   Complex.exp (I * ((w + p : ℝ) : ℂ)) * (Real.sin (w + f) : ℂ)) := by
105   have hsplit : I * ((w + p : ℝ) : ℂ) = I * (w : ℂ) + I * (p : ℂ) := by
106     push_cast; ring
107   rw [hsplit, Complex.exp_add]
108   push_cast
109   ring
110
111 /-! ### The theorem -/
112
113 /-- **R = R1ps + R2ps.** The only hypotheses are that the index is positive,
114 that 'χ' does not vanish at the point in question, and that the denominator
115 'sin(2ω+ψ)' is nonzero (without which 'd1, d2' are undefined). In particular no
116 assumption is made about 'T' being an integer or not. -/
117 theorem R_1ps_plus_R_2ps_eq_R (σ T : ℝ) (R : ℂ) (χ : ℂ → ℂ) (If : ℝ → ℝ)
118   (hT : 0 < T)
119   (hχ : χ (σ + If T * I) ≠ 0)
120   (h_sin : Real.sin (2 * omg T If + psi χ σ T If) ≠ 0) :
121   R_1ps σ T R χ If + R_2ps σ T R χ If = R := by

```

```

122 have hsinC : (Real.sin (2 * omg T If + psi χ σ T If) : ℂ) ≠ 0 := by
123   exact_mod_cast h_sin
124 -- e^{iψ} for the normalized functional-equation factor
125 have hchi : χ (σ + If T * I) / (||χ (σ + If T * I)|| : ℂ)
126   = Complex.exp (I * (psi χ σ T If : ℂ)) := div_norm_eq_exp_arg _ hχ
127 -- the two cpow factors become e^{±iω}
128 have hpow : ((([T] : ℝ)) : ℂ) ^ (I * (If T : ℂ))
129   = Complex.exp (I * (omg T If : ℂ)) := by
130     rw [ceil_cpow T (If T) hT]; rfl
131 have hpow' : ((([T] : ℝ)) : ℂ) ^ (-(I * (If T : ℂ)))
132   = Complex.exp (-(I * (omg T If : ℂ))) := by
133     have h : -(I * (If T : ℂ)) = I * (-(If T) : ℝ) : ℂ := by push_cast; ring
134     rw [h, ceil_cpow T (-(If T)) hT]
135     congr 1
136     simp only [omg]
137     push_cast
138     ring
139 -- |R| e^{iφ} = R
140 have hR : ((||R|| : ℝ) : ℂ) * Complex.exp (I * (phiR R : ℂ)) = R := by
141   rw [mul_comm I ((phiR R : ℝ) : ℂ)]
142   exact Complex.norm_mul_exp_arg_mul_I R
143 unfold R_1ps R_2ps d_j1 d_j2
144 rw [hchi, hpow, hpow', factoring_lemma, key_algebraic_identity]
145 -- cancel sin(2ω+ψ)
146 field_simp
147 linear_combination hR

```

B Lean formalization of the critical line: $d_1 = d_2$, $|R_{1ps}| = |R_{2ps}|$, and equal legs

This appendix reproduces the Lean file establishing the three critical-line statements of the paper: Corollary 8.1 ($d_1 = d_2$ and $|R_{1ps}| = |R_{2ps}|$), Corollary 9.1 ($L_1 = L_2$), and Corollary 9.2 (the bisector point projects onto $\frac{\zeta}{2}$). As in Appendix A the file contains no `axiom` and no `sorry`, and each result depends only on Lean’s own three axioms (`propext`, `Classical.choice`, `Quot.sound`) as reported by `#print axioms`; it was checked against Lean 4.32.2 with Mathlib at commit 905b95818e. The Lean code was written with the assistance of AI coding assistants (Anthropic’s Claude).

B.1 The chain of implications

All three statements descend from a single fact, and the file is organized around it.

Step 0: the two partial sums are conjugate. At $\sigma = \frac{1}{2}$ one has $\overline{-s} = s - 1$, so $n^{s-1} = \overline{n^{-s}}$ for every n ; summing over $n \leq m$ gives $\Sigma_2 = \chi \overline{\Sigma_1}$. This is `partial_sum_reflect`, proved from Mathlib’s `Complex.cpow_conj`.

Step 1: the rotated remainder is real. Write $\psi = \arg \chi$ and let $u = e^{i\psi/2}$, a square root of χ . With $R = \zeta - \Sigma_1 - \Sigma_2$,

$$e^{-i\psi/2}R = \underbrace{e^{-i\psi/2}\zeta}_{\text{real}} - \underbrace{(e^{-i\psi/2}\Sigma_1 + e^{i\psi/2}\overline{\Sigma_1})}_{= 2\Re(e^{-i\psi/2}\Sigma_1)} \in \mathbb{R}, \quad (226)$$

the first term being real because the functional equation at $\sigma = \frac{1}{2}$ reads $\zeta = \chi \overline{\zeta}$, so that $e^{-i\psi/2}\zeta = \overline{e^{-i\psi/2}\zeta}$. In the file this is `rotated_remainder_self_conj` (the algebra, from $u\overline{u} = 1$ and $u^2 = \chi$) and `rotated_remainder_im_eq_zero` (its imaginary part vanishes). It is the fact quoted as $\arg R = \frac{1}{2} \arg \chi$ in §8.1. Stated as the reality of $e^{-i\psi/2}R$ it is exact; stated through the argument it holds modulo π , which is visible in $R = 2d_1 \cos(\omega + \frac{\psi}{2})e^{i\psi/2}$, whose argument is $\frac{\psi}{2} + \pi$ wherever the cosine is negative. The modulo- π version is all the sequel uses.

Step 2: $d_1 = d_2$. Since $\sin A - \sin B = 2 \sin \frac{A-B}{2} \cos \frac{A+B}{2}$, the two numerators of (19)–(20) differ by

$$\sin(\omega - \phi + \psi) - \sin(\omega + \phi) = 2 \sin\left(\frac{\psi}{2} - \phi\right) \cos\left(\omega + \frac{\psi}{2}\right), \quad (227)$$

and Step 1 kills the first factor: `sin_half_sub_arg_eq_zero` turns the reality of $e^{-i\psi/2}R$ into $\sin(\frac{\psi}{2} - \arg R) = 0$, using $\cos \arg R = \Re R/|R|$ and $\sin \arg R = \Im R/|R|$. Hence `d1_eq_d2_of_im_eq_zero` and its critical-line specialization `d1_eq_d2`. Note that ω is an arbitrary real throughout: the equality of the weights depends on the direction of R relative to χ and not at all on the turn angle.

Step 3: $|R_{1ps}| = |R_{2ps}|$. Both phases are unimodular, so this is $|d_1| = |d_2|$: `norm_R1ps_eq_norm_R2ps`. The sharper form (78), $R_{2ps} = \chi \overline{R_{1ps}}$, is `R2ps_eq_chi_conj_R1ps`, and it is what feeds Step 4; it needs only that d_1 is real and $d_1 = d_2$.

Step 4: the legs. Combining Steps 0 and 3, $B_2 = \Sigma_2 + R_{2ps} = \chi \overline{\Sigma_1 + R_{1ps}} = \chi \overline{B_1}$ (`leg2_eq_chi_conj_leg1`), and since $|\chi| = 1$ on the line the two legs have the same length (`legs_norm_eq`). Finally, from $\zeta = B_1 + \chi \overline{B_1}$,

$$e^{-i\psi/2}\zeta = e^{-i\psi/2}B_1 + \overline{e^{-i\psi/2}B_1} = 2\Re(e^{-i\psi/2}B_1), \quad (228)$$

which says that the perpendicular projection of the bisector point onto the line $\mathbb{R} e^{i\psi/2}$ carrying ζ lands exactly on $\frac{\zeta}{2}$: the apex of the isosceles triangle sits over the midpoint of its base. This is `proj_eq_half_zeta`, and it is the dotted bisector line drawn in Figures 18 and 20.

B.2 What is proved and what is hypothesized

Two analytic facts about χ on the critical line enter as hypotheses: $|\chi| = 1$, used in the form $\chi = e^{i \arg \chi}$ (eq_exp_arg_of_norm_one), and the functional equation $\zeta = \chi \bar{\zeta}$, which is $\zeta(s) = \chi(s) \zeta(1-s)$ together with $1-s = \bar{s}$ at $\sigma = \frac{1}{2}$. Everything else above is derived, including the reflection $\Sigma_2 = \chi \bar{\Sigma}_1$ of Step 0, which is proved rather than assumed. The projection statement additionally takes $\zeta = B_1 + B_2$, which is Theorem 4.1 of Appendix A.

The remainders are written here in the form $R_{1ps} = d_1 e^{-i\omega}$, $R_{2ps} = d_2 e^{i(\omega+\psi)}$ of (24) rather than the $[T]^{\mp iI(T)}$ form of (17)–(18); the bridge between them is `ceil_cpow` of Appendix A. As there, ζ , χ , Σ_1 and ω are free parameters, so what is checked is the geometry of the critical line and not the identity of the functions involved.

B.3 Source listing

```

1  /-
2    Critical-line consequences of the decomposition R = R1ps + R2ps:
3      (1) e^{-i\psi/2} R is real, i.e. arg R = \psi/2 modulo \pi,
4      (2) d_1 = d_2,
5      (3) |R1ps| = |R2ps|, indeed R2ps = \chi \cdot conj R1ps,
6      (4) the two legs have equal length, and the bisector point projects onto \zeta/2.
7
8    Contains no 'axiom' and no 'sorry': every step is proved.
9  -/
10 import Mathlib.Analysis.SpecialFunctions.Complex.Circle
11 import Mathlib.Analysis.SpecialFunctions.Pow.Complex
12
13 open Complex Real ComplexConjugate
14
15 /-! ### Unit phases -/
16
17 /-- A purely imaginary exponent gives a unit phase. -/
18 lemma norm_exp_imag (x : ℝ) : ||Complex.exp (I * (x : ℂ))|| = 1 := by
19   rw [Complex.norm_exp]
20   simp
21
22 lemma norm_exp_neg_imag (x : ℝ) : ||Complex.exp (-(I * (x : ℂ)))|| = 1 := by
23   rw [Complex.norm_exp]
24   simp
25
26 /-- Conjugation flips the sign of a purely imaginary exponent. -/
27 lemma conj_exp_imag (x : ℝ) :
28   conj (Complex.exp (I * (x : ℂ))) = Complex.exp (-(I * (x : ℂ))) := by
29   rw [← Complex.exp_conj]
30   congr 1
31   simp [Complex.ext_iff]
32
33 lemma conj_exp_neg_imag (x : ℝ) :
34   conj (Complex.exp (-(I * (x : ℂ)))) = Complex.exp (I * (x : ℂ)) := by
35   rw [← Complex.exp_conj]
36   congr 1
37   simp [Complex.ext_iff]
38
39 /-- A number of unit modulus is the exponential of its argument: this is the
40 form in which '|χ| = 1' on the critical line will be used. -/
41 lemma eq_exp_arg_of_norm_one {χ : ℂ} (hχ : ||χ|| = 1) :
42   χ = Complex.exp (I * ((Complex.arg χ : ℝ) : ℂ)) := by

```

```

43   have h := Complex.norm_mul_exp_arg_mul_I χ
44   rw [hχ] at h
45   rw [mul_comm I _]
46   simpa using h.symm
47
48 /-! ### Step 0: the second partial sum is the reflection of the first
49
50 At 'σ = 1/2' we have 'n^{s-1} = conj (n^{-s})' for every 'n', because
51 'conj(-s) = s - 1' there. Summing, 'Σ2 = χ · conj Σ1', which is the hypothesis
52 the results below are stated with. -/
53
54 lemma conj_natCast_cpow (n : ℕ) (z : ℂ) :
55   conj ((n : ℂ) ^ z) = (n : ℂ) ^ (conj z) := by
56   have harg : ((n : ℂ)).arg ≠ π := by
57     rw [show ((n : ℂ)) = (((n : ℝ)) : ℂ) by push_cast; ring,
58         Complex.arg_ofReal_of_nonneg (by positivity)]
59     exact Ne.symm Real.pi_ne_zero
60   rw [Complex.cpow_conj _ _ harg, Complex.conj_natCast]
61
62 /- On the critical line the two Dirichlet partial sums are conjugate. -/
63 theorem partial_sum_reflect (m : ℕ) (t : ℝ) :
64   ∑ n ∈ Finset.Icc 1 m, ((n : ℂ)) ^ (((1 / 2 : ℂ) + (t : ℂ) * I) - 1)
65   = conj (∑ n ∈ Finset.Icc 1 m, ((n : ℂ)) ^ (-((1 / 2 : ℂ) + (t : ℂ) * I))) := by
66   rw [map_sum]
67   refine Finset.sum_congr rfl fun n _ => ?_
68   rw [conj_natCast_cpow]
69   congr 1
70   simp [Complex.ext_iff]
71   norm_num
72
73 /-! ### Step 1: the rotated remainder is real
74
75 On the critical line the functional equation reads 'ζ = χ · conj ζ', and the
76 second partial sum is the reflection of the first, 'S2 = χ · conj S1'. With
77 'u = e^{iψ/2}' a square root of 'χ', the remainder 'R = ζ - S1 - S2' satisfies
78 'conj (conj u * R) = conj u * R', that is, 'e^{-iψ/2} R' is real. -/
79
80 lemma rotated_remainder_self_conj (ζ S1 χ u : ℂ)
81   (hu : u * conj u = 1) (hu2 : u ^ 2 = χ)
82   (hζ : ζ = χ * conj ζ) :
83   conj (conj u * (ζ - S1 - χ * conj S1)) = conj u * (ζ - S1 - χ * conj S1) := by
84   simp only [map_sub, map_mul, Complex.conj_conj]
85   rw [← hu2] at hζ
86   simp only [map_pow]
87   linear_combination (-conj u) * hζ
88   + (-u * conj ζ + u * conj S1 - conj u * S1) * hu
89
90 lemma rotated_remainder_im_eq_zero (ζ S1 χ : ℂ) (p : ℝ)
91   (hp : χ = Complex.exp (I * (p : ℂ)))
92   (hζ : ζ = χ * conj ζ) :
93   (Complex.exp (-I * ((p / 2 : ℝ) : ℂ))) * (ζ - S1 - χ * conj S1).im = 0 := by
94   have hu : Complex.exp (I * ((p / 2 : ℝ) : ℂ)) * conj (Complex.exp (I * ((p / 2 : ℝ) : ℂ))) = 1
95     := by
96       rw [conj_exp_imag, ← Complex.exp_add]
97     simp
98   have hu2 : Complex.exp (I * ((p / 2 : ℝ) : ℂ)) ^ 2 = χ := by
99     rw [hp, pow_two, ← Complex.exp_add]
100    congr 1
101    push_cast

```

```

101   ring
102   have hc : conj (Complex.exp (I * ((p / 2 : ℝ) : ℂ)))
103     = Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) := conj_exp_imag _
104   rw [← hc]
105   exact Complex.conj_eq_iff_im.mp
106     (rotated_remainder_self_conj ζ S₁ χ _ hu hu2 hζ)
107
108 /-! ### Step 2: reality of 'e^{-ip/2} R' says 'arg R = p/2' modulo 'π' -/
109
110 lemma sin_half_sub_arg_eq_zero (R : ℂ) (p : ℝ) (hR : R ≠ 0)
111   (him : (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * R).im = 0) :
112     Real.sin (p / 2 - Complex.arg R) = 0 := by
113   have hexp : Complex.exp (-(I * ((p / 2 : ℝ) : ℂ)))
114     = ((Real.cos (p / 2) : ℝ) : ℂ) - ((Real.sin (p / 2) : ℝ) : ℂ) * I := by
115     have h : -(I * ((p / 2 : ℝ) : ℂ)) = ((-p / 2) : ℝ) : ℂ * I := by
116       push_cast; ring
117     rw [h, Complex.exp_mul_I, ← Complex.ofReal_cos, ← Complex.ofReal_sin,
118       Real.cos_neg, Real.sin_neg]
119     push_cast
120     ring
121   have hnorm : ‖R‖ ≠ 0 := norm_ne_zero_iff.mpr hR
122   rw [hexp] at him
123   simp only [Complex.mul_im, Complex.sub_re, Complex.sub_im, Complex.mul_re,
124     Complex.ofReal_re, Complex.ofReal_im, Complex.I_re, Complex.I_im] at him
125   rw [Real.sin_sub, Complex.cos_arg hR, Complex.sin_arg]
126   field_simp
127   linear_combination -him
128
129 /-! ### The weights and the two fractional remainders -/
130
131 /- 'd₁ = |R| sin(ω - arg R + ψ)/sin(2ω + ψ)' -/
132 noncomputable def d1 (R : ℂ) (w p : ℝ) : ℝ :=
133   ‖R‖ * (Real.sin (w - Complex.arg R + p) / Real.sin (2 * w + p))
134
135 /- 'd₂ = |R| sin(ω + arg R)/sin(2ω + ψ)' -/
136 noncomputable def d2 (R : ℂ) (w p : ℝ) : ℝ :=
137   ‖R‖ * (Real.sin (w + Complex.arg R) / Real.sin (2 * w + p))
138
139 /- 'R₁ps = d₁ e^{-iω}' -/
140 noncomputable def R_1ps (R : ℂ) (w p : ℝ) : ℂ :=
141   ((d1 R w p : ℝ) : ℂ) * Complex.exp (-(I * ((w : ℝ) : ℂ)))
142
143 /- 'R₂ps = d₂ e^{i(ω+ψ)}' -/
144 noncomputable def R_2ps (R : ℂ) (w p : ℝ) : ℂ :=
145   ((d2 R w p : ℝ) : ℂ) * Complex.exp (I * ((w + p : ℝ) : ℂ))
146
147 /-! ### Result 1: the two weights are equal
148
149 Note that 'ω' is arbitrary here: the equality of the weights depends only on the
150 direction of 'R' relative to 'χ', not on the turn angle. -/
151
152 theorem d1_eq_d2_of_im_eq_zero (R : ℂ) (w p : ℝ)
153   (him : (Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * R).im = 0) :
154     d1 R w p = d2 R w p := by
155     rcases eq_or_ne R 0 with h | h
156     · simp [d1, d2, h]
157     · have hs : Real.sin (p / 2 - Complex.arg R) = 0 :=
158       sin_half_sub_arg_eq_zero R p h him
159       have hnum : Real.sin (w - Complex.arg R + p) = Real.sin (w + Complex.arg R) := by

```

```

160     have hd := Real.sin_sub_sin (w - Complex.arg R + p) (w + Complex.arg R)
161     have harg : (w - Complex.arg R + p - (w + Complex.arg R)) / 2
162       = p / 2 - Complex.arg R := by ring
163     rw [harg, hs, mul_zero, zero_mul] at hd
164     linarith
165     rw [d1, d2, hnum]
166
167 /-- **d1 = d2 on the critical line.** -/
168 theorem d1_eq_d2 (ζ S1 χ : ℂ) (w p : ℝ)
169   (hp : χ = Complex.exp (I * (p : ℂ)))
170   (hζ : ζ = χ * conj ζ) :
171   d1 (ζ - S1 - χ * conj S1) w p = d2 (ζ - S1 - χ * conj S1) w p :=
172   d1_eq_d2_of_im_eq_zero _ w p (rotated_remainder_im_eq_zero ζ S1 χ p hp hζ)
173
174 /-! ### Result 2: the two fractional remainders have equal length -/
175
176 /-- **|R1ps| = |R2ps|.** -/
177 theorem norm_R1ps_eq_norm_R2ps (R : ℂ) (w p : ℝ) (hd : d1 R w p = d2 R w p) :
178   ||R1ps R w p|| = ||R2ps R w p|| := by
179   rw [R1ps, R2ps, norm_mul, norm_mul, norm_exp_neg_imag, norm_exp_imag, hd]
180
181 /-- The sharper statement: 'R2ps' is the mirror image of 'R1ps' across the real
182 axis, followed by the rotation 'χ'. -/
183 theorem R2ps_eq_chi_conj_R1ps (R χ : ℂ) (w p : ℝ)
184   (hp : χ = Complex.exp (I * (p : ℂ)))
185   (hd : d1 R w p = d2 R w p) :
186   R2ps R w p = χ * conj (R1ps R w p) := by
187   have hexp : Complex.exp (I * ((w + p : ℝ) : ℂ))
188     = Complex.exp (I * (p : ℂ)) * Complex.exp (I * ((w : ℝ) : ℂ)) := by
189     rw [← Complex.exp_add]
190     congr 1
191     push_cast
192     ring
193   rw [R1ps, R2ps, map_mul, Complex.conj_ofReal, conj_exp_neg_imag, hd, hexp, hp]
194   ring
195
196 /-! ### Result 3: the two legs have equal length -/
197
198 /-- With 'B1 = S1 + R1ps' and 'B2 = S2 + R2ps = χ conj S1 + R2ps', the second leg
199 is the reflection of the first. -/
200 theorem leg2_eq_chi_conj_leg1 (S1 R χ : ℂ) (w p : ℝ)
201   (hp : χ = Complex.exp (I * (p : ℂ)))
202   (hd : d1 R w p = d2 R w p) :
203   χ * conj S1 + R2ps R w p = χ * conj (S1 + R1ps R w p) := by
204   rw [map_add, mul_add, R2ps_eq_chi_conj_R1ps R χ w p hp hd]
205
206 /-- **L1 = L2.** -/
207 theorem legs_norm_eq (S1 R χ : ℂ) (w p : ℝ) (hχ : ||χ|| = 1)
208   (hp : χ = Complex.exp (I * (p : ℂ)))
209   (hd : d1 R w p = d2 R w p) :
210   ||χ * conj S1 + R2ps R w p|| = ||S1 + R1ps R w p|| := by
211   rw [leg2_eq_chi_conj_leg1 S1 R χ w p hp hd, norm_mul, hχ, one_mul,
212     RCLike.norm_conj]
213
214 /-! ### Result 4: the bisector point projects onto ζ/2 -/
215
216 /-- If 'B2 = χ conj B1' and 'ζ = B1 + B2', then the orthogonal projection of the
217 bisector point 'B1' onto the line 'ℝ·eiψ/2' through 'ζ' is exactly 'ζ/2'. The
218 apex of the isosceles triangle sits over the midpoint of its base. -/

```



```

219 theorem proj_eq_half_zeta (B1 ζ χ : ℂ) (p : ℝ)
220   (hp : χ = Complex.exp (I * (p : ℂ)))
221   (hzeta : ζ = B1 + χ * conj B1) :
222   (((Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) * B1).re : ℝ) : ℂ) *
223     Complex.exp (I * ((p / 2 : ℝ) : ℂ))
224     = ζ / 2 := by
225   set em := Complex.exp (-(I * ((p / 2 : ℝ) : ℂ))) with hem
226   set ep := Complex.exp (I * ((p / 2 : ℝ) : ℂ)) with hep
227   have hu : ep * em = 1 := by
228     rw [hep, hem, ← Complex.exp_add]
229     simp
230   have hu2 : ep ^ 2 = χ := by
231     rw [hep, hp, pow_two, ← Complex.exp_add]
232     congr 1
233     push_cast
234     ring
235   have hconj : conj em = ep := by rw [hem, hep, conj_exp_neg_imag]
236   set z := em * B1 with hz
237   have key : ((z.re : ℝ) : ℂ) * 2 = z + conj z := by
238     rw [Complex.add_conj]
239     push_cast
240     ring
241   have hcz : conj z = ep * conj B1 := by rw [hz, map_mul, hconj]
242   rw [eq_div_iff (two_ne_zero : (2 : ℂ) ≠ 0), hzeta]
243   linear_combination ep * key + ep * hz + B1 * hu + ep * hcz + conj B1 * hu2

```

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